SRR-CWDA-2012-00106
Revision 1

Tanks 5 and 6 Special Analysis for the Performance Assessment for the F-Tank Farm at the Savannah River Site

January 2013

Prepared by: Savannah River Remediation LLC
Closure and Waste Disposal Authority
Aiken, SC 29808

Prepared for U.S. Department of Energy Under Contract No. DE-AC09-09SR22505

REVISION SUMMARY

REV. #	DESCRIPTION	DATE OF ISSUE
0	Initial Issue to DOE-SR	December 2012
1	Incorporation of comments from DOE-SR	January 2013

TABLE OF CONTENTS

TABLE OF CONTENTS 3
LIST OF FIGURES 5
LIST OF TABLES 8
ACRONYMS/ABBREVIATIONS 10
EXECUTIVE SUMMARY 11
1.0 SPECIAL ANALYSIS PURPOSE 15
2.0 BASIS 15
3.0 TANKS 5 AND 6 BACKGROUND INFORMATION 16
3.1 Savannah River Site Characteristics 16
3.2 F-Tank Farm Facility Description 16
3.3 Tanks 5 and 6 19
4.0 F-TANK FARM PERFORMANCE ASSESSMENT BACKGROUND INFORMATION 22
5.0 TANKS 5 AND 6 RESIDUAL WASTE INFORMATION 24
6.0 F-TANK FARM PERFORMANCE ASSESSMENT EVALUATION 27
6.1 F-Tank Farm Facility Characteristics 27
6.1.1 Site Characteristics 27
6.1.2 Principal Facility Design Features 27
6.1.3 Stabilized Contaminant Characteristics 27
6.2 The F-Tank Farm Analysis of Performance 34
6.2.1 Overview of Analysis 34
6.2.2 Integrated Site Conceptual Model of Facility Performance 34
6.2.3 Modeling Codes 34
6.2.4 Closure System Modeling 35
6.2.5 Airborne and Radon Analysis 35
6.2.6 Biotic Pathways 35
6.2.7 Dose Analysis 35
6.2.8 RCRA/CERCLA Risk Evaluation 36
6.3 FTF Results of Analysis 36
6.3.1 Source Term (Analyses Results) 36
6.3.2 Environmental Transport of Radionuclides 36
6.3.3 Air Pathway and Radon Analysis 41
6.3.4 Biotic Pathways 44
6.3.5 Dose Analysis of MOP Exposure 44
6.3.6 Uncertainty and Sensitivity Analysis 64
6.3.7 RCRA/CERCLA Risk Analysis 89

6.3.8 ALARA Analysis 93
6.4 FTF Inadvertent Intruder Analysis 93
6.4.1 Groundwater Concentrations at One Meter 93
6.4.2 Acute Exposure Scenarios 93
6.4.3 Chronic Exposure Scenarios 93
6.4.4 Groundwater Doses at One Meter 93
6.4.5 Inadvertent Intruder Uncertainty/Sensitivity Analysis 96
6.5 F-Tank Farm Interpretation of Results 97
6.5.1 100-Meter (Water from Well) Groundwater Pathways Doses 97
6.5.2 Water at the Stream Groundwater Pathways Doses 97
6.5.3 All-Pathways Dose 97
6.5.4 Inadvertent Intruder Dose 98
6.5.5 Airborne Dose / Radon Flux 98
6.5.6 Composite Sensitivity Study 98
6.6 FTF Performance Evaluation 98
7.0 CONCLUSION 99
8.0 REFERENCES 101
APPENDIX A 104
APPENDIX B 131
APPENDIX C 158

LIST OF FIGURES

Figure ES-1: Comparison of the 100-Meter MOP Peak All-Pathways Dose within 10,000 Years 12
Figure ES-2: Comparison 100-Meter MOP Peak All-Pathways Dose within 100,000 Years..... 13
Figure 3.2-1: SRS Operational Area Location Map 17
Figure 3.2-2: Layout of the GSA 18
Figure 3.2-3: General Layout of F-Tank Farm 19
Figure 3.3-1: Sketch of Typical Type I Tank (Cross-Sectional View)..... 20
Figure 3.3-2: Steel Liner Construction 21
Figure 3.3-3: Secondary Containment Detail Showing Annulus Pan..... 22
Figure 6.3-1: PORFLOW FTF 1-Meter and 100-Meter Model Evaluation Sectors..... 37
Figure 6.3-2: PORFLOW FTF Seepline Evaluation Sectors..... 41
Figure 6.3-3: 100-Meter MOP Peak Groundwater Pathway Dose within 1,000 Years for the Five 100-Meter Sectors..... 46
Figure 6.3-4: 100-Meter MOP Peak Groundwater Pathway Dose within 10,000 Years for the Five 100-Meter Sectors..... 47
Figure 6.3-5: 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for the Five 100-Meter Sectors (linear scale)..... 47
Figure 6.3-6: 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for the Five 100-Meter Sectors (log scale)..... 48
Figure 6.3-7: Individual Radionuclide Contributors to the Sector E 100-Meter Peak Groundwater Pathway Dose, 1,000 years..... 50
Figure 6.3-8: Individual Radionuclide Contributors to the Sector E 100-Meter Peak Groundwater Pathway Dose, 10,000 years..... 50
Figure 6.3-9: Individual Radionuclide Contributors to the Sector E 100-Meter Peak Groundwater Pathway Dose, 100,000 Years (linear scale) 51
Figure 6.3-10: Individual Radionuclide Contributors to the Sector E 100-Meter Peak Groundwater Pathway Dose, 100,000 Years (log scale) 51
Figure 6.3-11: 100-Meter MOP Peak Groundwater Pathway Dose within 20,000 Years for Sector D – with Tank 5 Contribution..... 53
Figure 6.3-12: 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for Sector D – with Tank 5 Contribution..... 54
Figure 6.3-13: 100-Meter MOP Peak Groundwater Pathway Dose within 20,000 Years for Sector D – with Tank 6 Contribution..... 54
Figure 6.3-14: 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for Sector D – with Tank 6 Contribution..... 55
Figure 6.3-15: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 20,000 Years for All Sources and for All Sources except Tanks 5 and 6..... 55

Figure 6.3-16: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for All Sources and for All Sources Except Tanks 5 or Tank 6 (linear scale) 56
Figure 6.3-17: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for All Sources and for All Sources Except Tanks 5 or Tank 6 (log scale) 56
Figure 6.3-18: 100-Meter MOP Peak Water Ingestion Dose Results within 10,000 Years for the Five 100-Meter Sectors 59
Figure 6.3-19: 100-Meter Peak MOP Water Ingestion Dose Results within 100,000 Years for the Five 100-Meter Sectors 59
Figure 6.3-20: 100-Meter MOP Peak Vegetable Ingestion Dose Results within 10,000 Years for the Five 100-Meter Sectors 60
Figure 6.3-21: 100-Meter MOP Peak Vegetable Ingestion Dose Results within 100,000 Years for the Five 100-Meter Sectors 60
Figure 6.3-22: Peak Groundwater Pathway MOP Dose at the Stream Results within 10,000 Years for the Two Stream Sectors 62
Figure 6.3-23: Peak Groundwater Pathway MOP Dose at the Stream Results within 100,000 Years for the Two Stream Sectors 62
Figure 6.3-24: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 20,000 Years for Tank 5 Only (Tank 5 Final Inventory and One-Half Final Inventory) 66
Figure 6.3-25: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 20,000 Years for Tank 6 Only (Tank 6 Final Inventory and One-Half Final Inventory) 67
Figure 6.3-26: Individual Radionuclide Contributors to the Sector D 100-Meter Peak Groundwater Pathway Dose from Tank 5 Only, 20,000 years (Final Inventory)..... 67
Figure 6.3-27: Individual Radionuclide Contributors to the Sector D 100-Meter Peak Groundwater Pathway Dose from Tank 6 Only, 20,000 years (Final Inventory)..... 68
Figure 6.3-28: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for Tank 5 Only (Tank 5 Final Inventory and One-Half Final Inventory)..... 68
Figure 6.3-29: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for Tank 6 Only (Tank 6 Final Inventory and One-Half Final Inventory)..... 69
Figure 6.3-30: Individual Radionuclide Contributors to the Sector D 100-Meter Peak Groundwater Pathway Dose from Tank 5 Only, 100,000 years (Final Inventory)..... 69
Figure 6.3-31: Individual Radionuclide Contributors to the Sector D 100-Meter Peak Groundwater Pathway Dose from Tank 6 Only, 100,000 years (Final Inventory)..... 70
Figure 6.3-32: Composite Sensitivity Study of 100-Meter Peak All Pathways Dose within 100,000 Years for Sector E – All Waste Tanks 75
Figure 6.3-33: Composite Sensitivity Study of 100-Meter Peak All Pathways Dose within 100,000 Years for Sector E – All Waste Tanks (Expanded View)..... 76
Figure 6.3-34: Individual Radionuclide Contributors to the Composite Sensitivity Study of 100-Meter Peak Groundwater Pathway Dose within 100,000 Years for Sector E 77
Figure 6.3-35: All Cases Probabilistic Results (10,000 Year) Linear Scale..... 80
Figure 6.3-36: All Cases Probabilistic Results (10,000 Year) Log Scale..... 80

Figure 6.3-37: All Cases Probabilistic Results (100,000 Year) Linear Scale..... 81
Figure 6.3-38: All Cases Probabilistic Results (100,000 Year) Log Scale..... 81
Figure 6.3-39: Base Case Probabilistic Results (10,000 Year) Linear Scale..... 83
Figure 6.3-40: Base Case Probabilistic Results (10,000 Year) Log Scale..... 83
Figure 6.3-41: Base Case Probabilistic Results (100,000 Year) Linear Scale..... 84
Figure 6.3-42: Base Case Probabilistic Results (100,000 Year) Log Scale..... 84
Figure 6.3-43: Composite Analysis All Cases Probabilistic Results (10,000 Year) Linear Scale 86
Figure 6.3-44: Composite Analysis All Cases Probabilistic Results (10,000 Year) Log Scale .. 86
Figure 6.3-45: Composite Analysis All Cases Probabilistic Results (100,000 Year) Linear Scale 87
Figure 6.3-46: Composite Analysis All Cases Probabilistic Results (100,000 Year) Log Scale 87
Figure 6.3-47: Radionuclide Contributors for All Cases Probabilistic Results (100,000 Year). 88
Figure 6.3-48: Radionuclide Contributors for Base Case Probabilistic Results (100,000 Year) 88
Figure 6.3-49: Radionuclide Contributors for Composite Analysis All Cases Probabilistic Results (100,000 Year) 89
Figure 6.4-1: Annual Dose to the Chronic Intruder within 10,000 Years 94
Figure 6.4-2: Annual Dose to the Chronic Intruder within 20,000 Years 96

LIST OF TABLES

Table 4.0-1: Key Performance Objectives 23
Table 5.0-1: Tanks 5 and 6 Radionuclide Inventories (Ci) 25
Table 5.0-2: Tanks 5 and 6 Chemical Inventories (kg) 26
Table 6.1-1: Revised FTF Radionuclide Inventories (Ci) 28
Table 6.1-2: Revised FTF Chemical Inventories (kg) 32
Table 6.3-1: Determination of Sensitivity Run Radionuclides 39
Table 6.3-2: Summary of Projected Total FTF Inventory of Gaseous Radionuclides 42
Table 6.3-3: 100-Meter DRFs and 10,000-Year FTF Dose 42
Table 6.3-4: 1,600-Meter DRFs and 10,000-Year FTF Dose 43
Table 6.3-5: Summary of Total Projected FTF Inventory of Isotopes Producing Rn-222 43
Table 6.3-6: Peak Instantaneous Rn-222 Flux at Land Surface 44
Table 6.3-7: 100-Meter MOP Peak Groundwater Pathways Dose by Sector 45
Table 6.3-8: 100-Meter MOP Peak Groundwater Pathway Dose Individual Radionuclide Contributions at Year 10,000 (Peak Year) 52
Table 6.3-9: 100-Meter MOP Peak Groundwater Pathway Dose Individual Source Contributions at Peak Years within 20,000 Years for Sector D 53
Table 6.3-10: 100-Meter MOP Peak Dose Individual Groundwater Pathway Contributions for Sector E 57
Table 6.3-11: 100-Meter MOP Peak Water Ingestion Doses by Sector 58
Table 6.3-12: 100-Meter MOP Peak Vegetable Ingestion Doses by Sector 58
Table 6.3-13: Peak Groundwater Pathway MOP Doses at the Stream by Sector 61
Table 6.3-14: Individual Peak MOP Dose at the Stream Groundwater Pathway Contributions for UTR 63
Table 6.3-15: Individual Peak MOP Dose at the Stream Groundwater Pathway Contributions for Fourmile Branch 63
Table 6.3-16: 100-Meter Peak Annual MOP All-Pathways Dose Contributors 64
Table 6.3-17: Peak Annual MOP at the Stream All-Pathways Dose Individual Contributions 64
Table 6.3-18: Calculated Solubility and Controlling Phases of Radionuclides of Interest 71
Table 6.3-19: Calculated Solubility and Controlling Phases of Radionuclides of Interest 72
Table 6.3-20: Number of Pore Volumes of Reacting Fluid Required to Reach Each Stage of Grout Degradation 72
Table 6.3-21: Recommended K_d Values for the Vadose Zone 73
Table 6.3-22: Revised Inventory Multipliers 78
Table 6.3-23: Groundwater Radionuclide Concentrations at 1 Meter from FTF[a] 90
Table 6.3-24: Groundwater Chemical Concentrations at 1 Meter from FTF[a] 92
Table 6.3-25: Groundwater Concentrations of CMCOC at 100 Meters 92

Table 6.4-1: Chronic Intruder Peak Dose Contributors within 10,000 Years 95
Table 7.0-1: Key Limits from Regulatory Requirements 99
Table 7.0-2: Summary Radiological Results for F-Tank Farm 100

ACRONYMS/ABBREVIATIONS

ALARA	As Low As Reasonably Achievable
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act
CFR	Code of Federal Regulations
CMCOC	Contaminant Migration Constituents of Concern
CTS	Concentrate Transfer System
DB	Diversion Box
DCF	Dose Conversion Factor
DOE	U.S. Department of Energy
DRF	Dose Release Factors
FFA	Federal Facility Agreement
FTF	F-Tank Farm
GSA	General Separations Area
HTF	H-Tank Farm
ICM	Integrated Conceptual Model
MCL	Maximum Contaminant Level
MEI	Maximally Exposed Individual
MOP	Member of the Public
N/A	Not Applicable/Not Available
ND	Not Determined
NDAA	Ronald W. Reagan National Defense Authorization Act (NDAA) for Fiscal Year 2005
NRC	U.S. Nuclear Regulatory Commission
PA	Performance Assessment
PCA	Pollution Control Act
PP	Pump Pit
PRG	Preliminary Remediation Goal
RCRA	Resource Conservation and Recovery Act
RSL	Regional Screening Level
SCDHEC	South Carolina Department of Health and Environmental Control
SRS	Savannah River Site
TER	Technical Evaluation Report
UTR	Upper Three Runs
UTRA	UTR Aquifer
UTRA-LZ	UTRA-Lower Zone
UTRA-UZ	UTRA-Upper Zone
Note:	Various figures throughout this document abbreviate Special Analysis as SA

EXECUTIVE SUMMARY

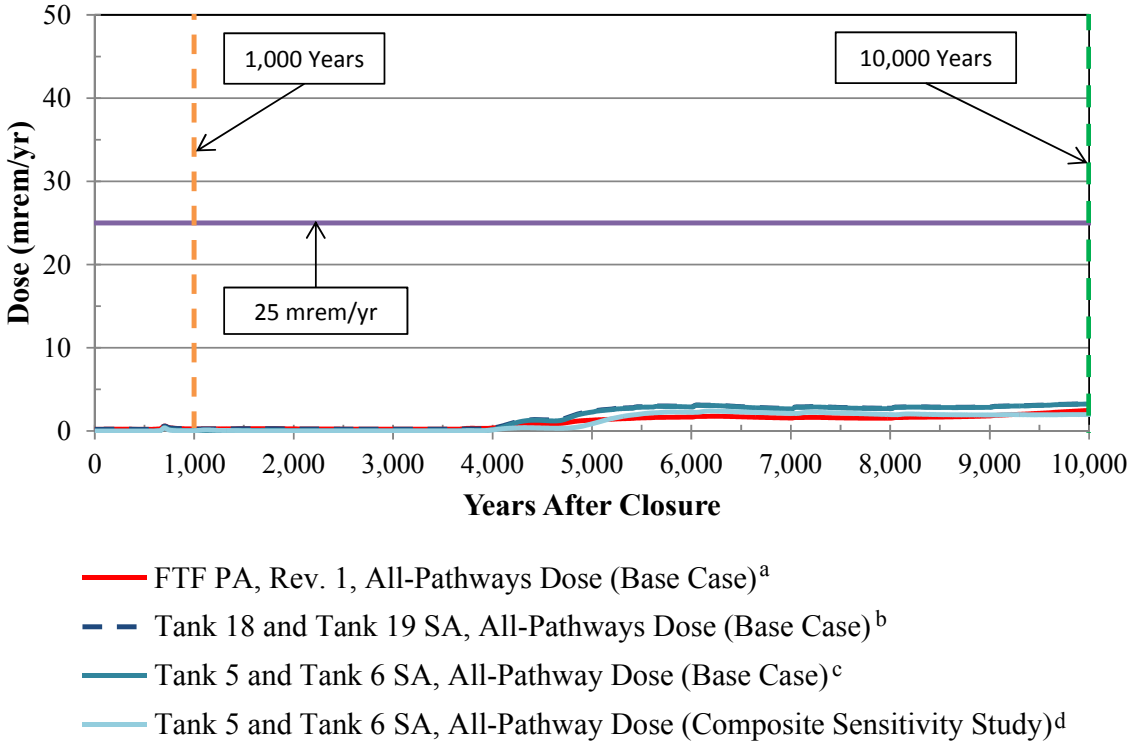
Special Analyses are performed to evaluate the significance of new information or new analytical methods to the results and associated conclusions of a performance assessment (PA). As waste tanks and ancillary equipment are cleaned at the F-Tank Farm (FTF) at the U.S. Depart of Energy's (DOE) Savannah River Site (SRS), final residual inventories will be used to update the FTF fate and transport modeling performed as part of SRS-REG-2007-00002, Revision 1 (hereinafter referred to as FTF PA). This allows for evaluation of the difference between the projected and final waste tank inventories to determine if the results of the FTF PA, and the conclusions reached based on the FTF PA information remain valid. The *Tank 18/Tank 19 Special Analysis for the Performance Assessment for the F-Tank Farm at the Savannah River Site* (SRR-CWDA-2010-00124), (hereinafter referred to as the Tank 18/19 Special Analysis) evaluated the impact of the final Tank 18 and 19 inventories and therefore represents the updated FTF base case dose results. The Tank 5 and 6 Special Analysis uses the FTF PA Base Case (Case A) model to evaluate the final residuals that are planned to be grouted in-place in Tanks 5 and 6 (using final residual characterization data). This Special Analysis progresses from the Tank 18/19 Special Analysis and takes advantage of new information gathered and generated since both the FTF PA and Tank 18/19 Special Analysis were developed. This includes information used to address some of the observations and recommendations found in the U.S. Nuclear Regulatory Commission (NRC) *Technical Evaluation Report for F-Area Tank Farm Facility* (ML112371715). The issuance of the Technical Evaluation Report (TER) culminated a multi-year consultation with NRC under the *Ronald W. Reagan National Defense Authorization Act for Fiscal Year 2005 (NDAA)* Section 3116. In particular, sensitivity studies were carried out in response to questions regarding waste-release solubility modeling, natural attenuation in soil, and groundwater conditioning impacts.

The results of this Special Analysis demonstrate that the conclusions reached based on the FTF PA and the Tank 18/19 Special Analysis remain valid. In addition, sensitivity analyses (using lessons learned from waste-release modeling since the FTF PA was issued) shows that peak dose, beyond 10,000 years after FTF closure, is more attenuated and occurs later in time than projected in the FTF PA Base Case. Further, the special analysis process has confirmed that there continues to be reasonable assurance that the Title 10 Code of Federal Regulations (CFR) Part 61 Subpart C performance objectives will be met for FTF during the reasonably foreseeable future.[1] A comparison was made of the peak all-pathways dose within 10,000 years, calculated using projected FTF waste tank inventories plus actual residual inventories for those FTF waste tanks that have been cleaned and characterized. As shown in Figure ES-1, the maximum peak annual all-pathways dose to a member of the public (MOP) for 10,000 years following FTF closure using the updated inventories (including actual Tank 5 and 6 inventories) is

[1] In accordance with DOE Manual 435.1-1, Chapter 4, the performance assessment considers a period of 1,000 years after the disposal facility has been closed, to assess compliance with the performance objectives. The Manual concludes that longer times of assessment are not used to assess compliance because of the inherently large uncertainties in extrapolating such calculations over long periods of time. Nevertheless, DOE considers results for time periods much longer than this in making risk-informed decisions related to the stabilization of waste tanks and ancillary structures.

approximately 3 mrem/yr and occurs at year 10,000. This maximum peak all-pathways dose in 10,000 years is essentially unchanged from the peak dose calculated in the Tank 18/19 Special Analysis (SRR-CWDA-2010-00124).

Figure ES-1: Comparison of the 100-Meter MOP Peak All-Pathways Dose within 10,000 Years



a. All-Pathways total dose results from the Base Case of the FTF PA, Rev.1. [SRS-REG-2007-00002]
b. All-Pathways total dose results from the Tank 18/ 19 Special Analysis applies the same Base Case assumptions used in the FTF PA with revised residual waste inventories for Tanks 18 and 19. [SRR-CWDA-2010-00124]
c. All-Pathways total dose results from this Tank 5/Tank 6 Special Analysis applies the same Base Case assumptions used in the FTF PA (discussed in Section 6.3) with revised residual waste inventories. [SRR-CWDA-2009-00045]
d. All-Pathways total dose results from this Tank 5/Tank 6 Special Analysis applies more realistic and probable assumptions for the waste release model and contaminant properties (discussed in Section 6.3.6.3) and the revised residual waste inventories. [SRR-CWDA-2009-00045]

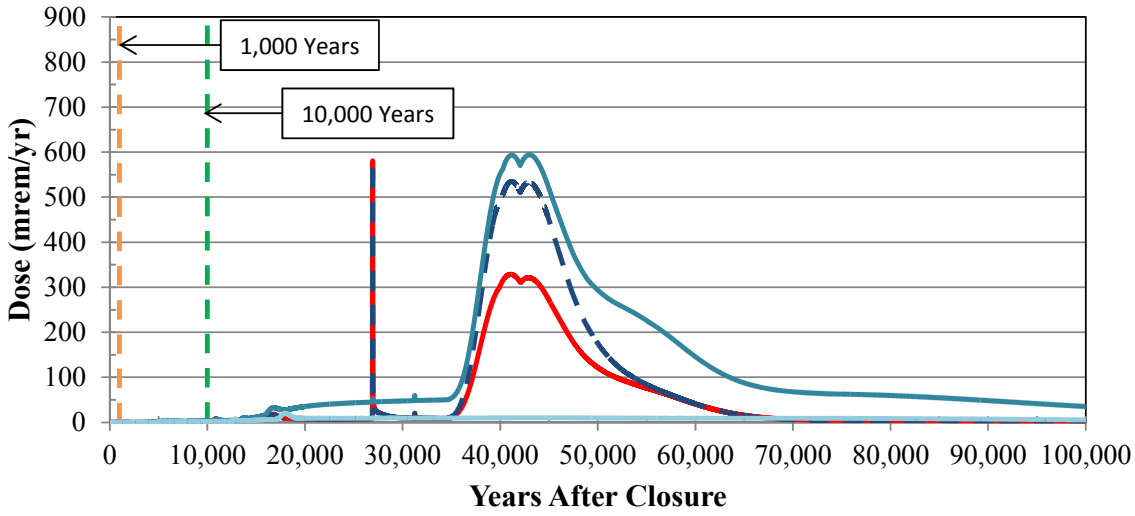
The horizontal purple line at 25 mrem/yr is provided for illustration only.

Figure ES-2(I) presents the doses to 100,000 years after FTF closure and demonstrates that the long-term doses remain less than the 620 mrem/yr average annual dose received by the average Unites States resident. [NCRP-160] The expanded scale in Figure ES-2(II) provides additional detail. These figures include the results of a Composite Sensitivity Study that was performed to evaluate the impact of an improved waste release model and updated radionuclide-specific transport values such as K_ds. The resulting peak dose remained below 25 mrem/yr out to 100,000 years. The Composite Sensitivity Study uses more probable and defensible inputs,

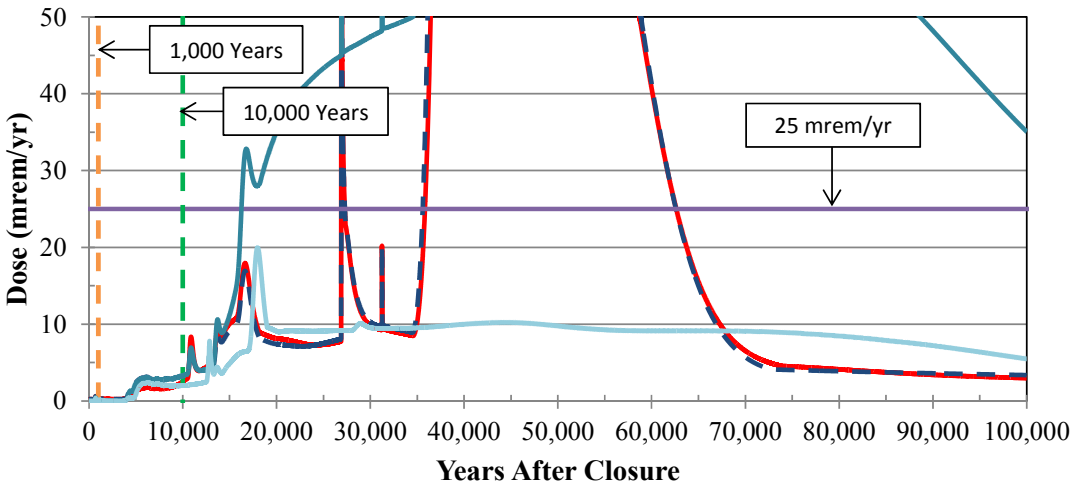
providing greater confidence that the FTF PA results and conclusions remain valid. This Composite Sensitivity Study is described on Section 6.3.6.3.

Figure ES-2: Comparison 100-Meter MOP Peak All-Pathways Dose within 100,000 Years

(I)



(II)



FTF PA, Rev. 1, All-Pathways Dose (Base Case) [a]

Tank 18 and Tank 19 SA, All-Pathways Dose (Base Case) [a]

Tank 5 and Tank 6 SA, All-Pathway Dose (Base Case) [b]

Tank 5 and Tank 6 SA, All-Pathway Dose (Composite Sensitivity Study) [c]

a. Peak near 27,000 years (associated with Tc-99 releases) occurs in both the FTF PA and the Tank 18/19 Special Analysis, but not in this Tank 5/6 Special Analysis.
b. Applies Base Case assumptions for the waste release model and contaminant properties.
c. Applies more realistic and probable assumptions for the waste release model and contaminant properties, as discussed in Section 6.3.6.3.

The horizontal purple line at 25 mrem/yr is provided for illustration only.

The maximum peak annual dose to a FTF intruder is 70 millirem at year 100 following FTF closure from a chronic scenario (i.e., drilling through a transfer line and using groundwater maximum concentrations at 1 meter from the FTF). The 70-millirem peak intruder annual dose value is essentially unchanged from the FTF PA.

Peak groundwater-radionuclide concentrations were also calculated using the FTF inventories and no maximum contaminant levels (MCLs) or preliminary remediation goals (PRGs) were exceeded at 100 meters from the FTF boundary within 1,000 years. All radionuclides were well below the MCL or PRG at the seepline. The peak concentrations for the chemicals of concern were also calculated, and all were less than the MCL or regional screening level (RSL) at a distance of 100 meters from the FTF boundary. These peak chemical concentration results are unchanged from the FTF PA.

In the NRC TER for FTF, ML112371715, the NRC stated:

> *"However, considering the fact that unacceptably high peak doses could occur within the 10,000 year period of compliance with only a factor of 3 or 4 faster time to collective failure of a combination of barriers for Tc or Pu, respectively, and considering the large uncertainty associated with predictions of long-term performance of engineered barriers, NRC staff are not convinced that the high peak doses currently presented in DOE's PA (or lower peak doses of unknown magnitude that might be associated with a more realistic model) could not be realized within a 10,000 year compliance period."*

The peak dose of NRC concern relating to Tc-99 was an approximately 600 millirem annual dose occurring approximately 30,000 years following the FTF closure. This peak dose was associated with the estimated residual Tc-99 inventory in the Type I tanks. Based on the revised FTF Tc-99 inventory projections (informed by actual Tank 5 and Tank 6 Tc-99 inventory data) this concern is alleviated, as illustrated in Figure ES-2. The peak dose associated with the residual Tc-99 inventories in the Type I tanks is significantly lowered based on factoring in actual waste tank data that is now available for the FTF Type I waste tanks. Concerns regarding the peak dose associated with Pu-239 from Tank 18 have been addressed previously in SRR-CWDA-2010-00124.

In addition to evaluating the impact of the updated tank inventory data (including final inventories for Tanks 5, 6, 17, 18, 19 and 20) on peak doses within 10,000 years following FTF closure, additional sensitivity analyses were performed. The results of these sensitivity analyses are used to assess the potential dose impacts of the final inventory, including evaluating out to 100,000 years after closure for the purposes of "chasing the peaks," to further understand system performance, and inform the closure decision process. These additional sensitivity analyses placed emphasis on understanding releases associated with the residual inventory remaining in Tanks 5 and 6 at closure.

Tanks 5 and 6 are Type I tanks, whose liners are projected in the FTF PA Base Case to remain intact until approximately 13,000 years after closure; contaminants are not expected to be released until the waste tank liners fail. Additionally, the chemical and physical properties of Nb-93m (a decay product from Zr-93) and Pu-239 act as barriers to release, preventing the dose peaks associated with Tanks 5 and 6 from occurring earlier or with an increased magnitude.

Initial inventory estimates in the Base Case (for the FTF PA) indicated that Nb-93m would not be a significant dose contributor, such that conservative assumptions regarding Nb-93m were appropriate. For example, soil acts as a natural retardant to Nb-93m transport; however, the Base Case conservatively assumes that Nb-93m transport will not be retarded by soil.

Using the revised inventory estimates based on Tank 5 and Tank 6 sampling, however, shows that more realistic and probable modeling assumptions are needed to achieve an appropriate understanding of the behavior of this contaminant. A Composite Sensitivity Study (described in Section 6.3.6.3) demonstrates that using more appropriate modeling parameters for modeling transport (such as soil retardation and solubility controls) reduces the magnitude of Nb-93m doses. Studies regarding the release and transport of Pu-239 are discussed at length in SRR-CWDA-2010-00124.

Based on the above discussion, the Special Analysis results continue to provide reasonable assurance that compliance is maintained with the specific requirements of 10 CFR 61 as referenced in NDAA Section 3116, DOE Manual 435.1-1 (DOE M 435.1-1), and the MCLs. The conclusions that were made based on the FTF PA regarding the stabilization of Tanks 5 and 6 have not been changed by the new information regarding the final residual inventories that are planned to be grouted in-place in Tanks 5 and 6.

1.0 SPECIAL ANALYSIS PURPOSE

The purpose of this Special Analysis is to evaluate the information regarding the final residual inventories that are planned to be grouted in-place in Tanks 5 and 6. This new inventory information was used to update the FTF fate and transport modeling performed in the FTF PA and the Tank 18/19 Special Analysis. The potential impacts of the new inventory information on the assumptions from the FTF PA and the Tank 18/19 Special Analysis were also considered. Because the FTF PA and Tank 18/19 Special Analysis analyzed projected inventories for Tanks 5 and 6, this report focuses on the impact of the final residual waste data on the information previously presented. Because there is the possibility of a significant dose associated with some Tanks 5 and 6 radionuclides (e.g., Nb-93m and Pu-239) occurring beyond 10,000 years, sensitivity analyses were performed as part of this Special Analysis. These analyses demonstrate that the FTF Base Case model incorporates conservative approaches/inputs. Based on lessons learned regarding waste release modeling and distribution coefficients, projected peak doses will be significantly lower than the Base Case modeling results.

It is not intended that information previously provided in the FTF PA and Tank 18/19 Special Analysis that is unaffected by the new residual waste data be duplicated in this report. The Special Analysis results will be used to inform decisions documented in Tanks 5 and 6 closure documents.

2.0 BASIS

The *Maintenance Guide for U.S. Department of Energy Low-Level Waste Disposal Facility Performance Assessments and Composite Analyses* (DOE_11-10-1999) recognizes that conduct of a PA is not a static process, and states that, "*Special analyses are expected to be needed as part of the routine maintenance of the performance assessment.*" As described in the

maintenance guide, "*special analyses are analyses performed to evaluate the significance of new information or new analytical methods to the results of the performance assessment, or to supplement or amend the analyses performed in the original performance assessment. A special analysis is not the same as a revision to the performance assessment, but the results of the special analysis may be used to determine whether a performance assessment revision is needed.*" As stated in the maintenance guide, a number of different factors may prompt a special analysis, including "*wastes that exceed the concentrations analyzed for performance assessment-significant radionuclides.*"

The guide also states that, "*the purpose of conducting special analyses can be thought of as similar to the process for resolving unreviewed safety questions described in the DOE Order 5480.21, Unreviewed Safety Questions. The intent of the process is to provide flexibility in day-to-day operations and to require those issues with a significant impact on the performance assessment's conclusions, and therefore the projected compliance with performance objectives, to be brought to the proper level for attention.*" [DOE_11-10-1999]

3.0 TANKS 5 AND 6 BACKGROUND INFORMATION

3.1 Savannah River Site Characteristics

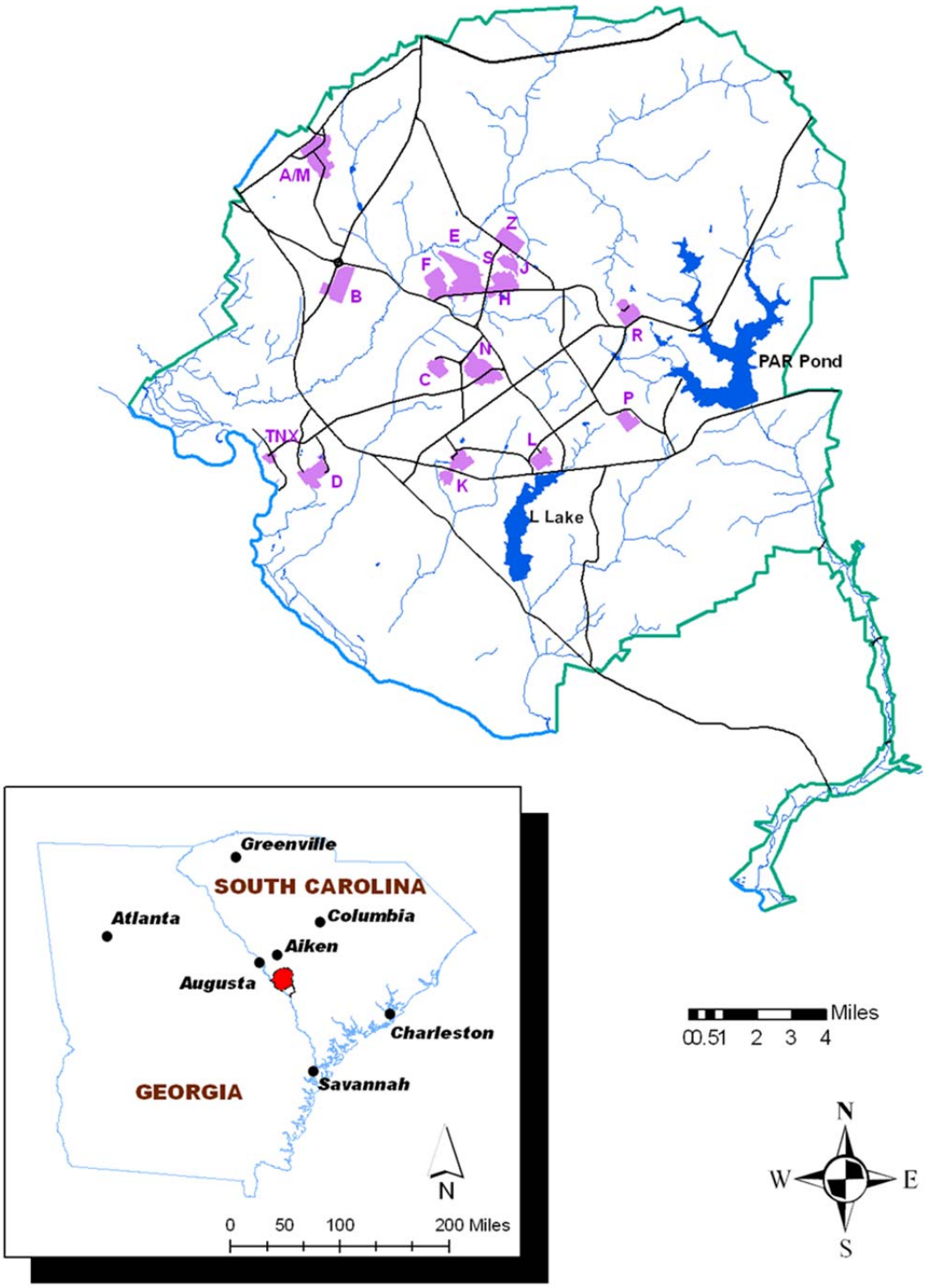
The SRS is located in south-central South Carolina, approximately 100 miles from the Atlantic Coast. The major physical feature at SRS is the Savannah River, approximately 20 miles of which serves as the southwestern boundary of the site and the South Carolina-Georgia border. The SRS encompasses portions of Aiken, Barnwell, and Allendale counties in South Carolina. The SRS occupies approximately 310 square miles, or more than 198,000 acres, and contains operations, service, and research and development areas. The developed areas occupy less than 10 % of the SRS footprint while the remainder of the site is undeveloped forest or wetlands. [SRS-REG-2007-00002]

Additional site characteristics are addressed in Section 3.1 of FTF PA. Topics addressed in Section 3.1 of FTF PA include geography, demography, meteorology, climatology, ecology, geology, seismology, hydrogeology, geochemistry, and natural resources.

3.2 F-Tank Farm Facility Description

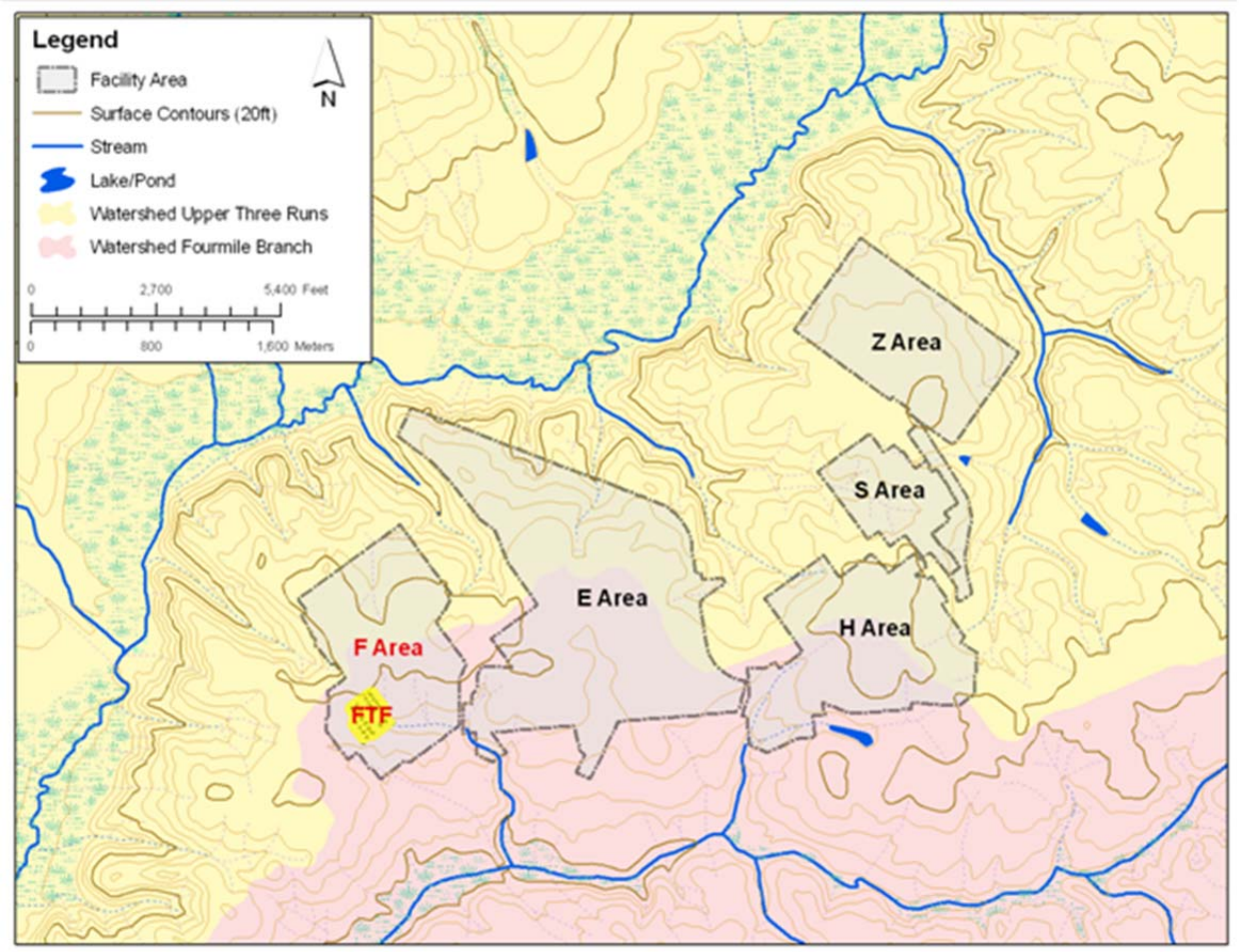
A legacy of the SRS mission was the generation of liquid waste from chemical separations processes in both F and H Areas. Since the beginning of SRS operations, an integrated waste management system has evolved, which consists of several facilities designed for the overall processing of liquid waste. Two of the major components of this system are the FTF and H-Tank Farm (HTF) (located in F Area and H Area, respectively), near the center of the site (Figure 3.2-1). In F Area, plutonium, uranium, and other radionuclides were separated from target assemblies using chemical separations processes. The tank farms, which store and process waste from the chemical separations operations, include waste tanks, evaporators, transfer line systems, and other ancillary structures. Additional FTF facility characteristics are addressed in Section 3.2 of FTF PA.

Figure 3.2-1: SRS Operational Area Location Map



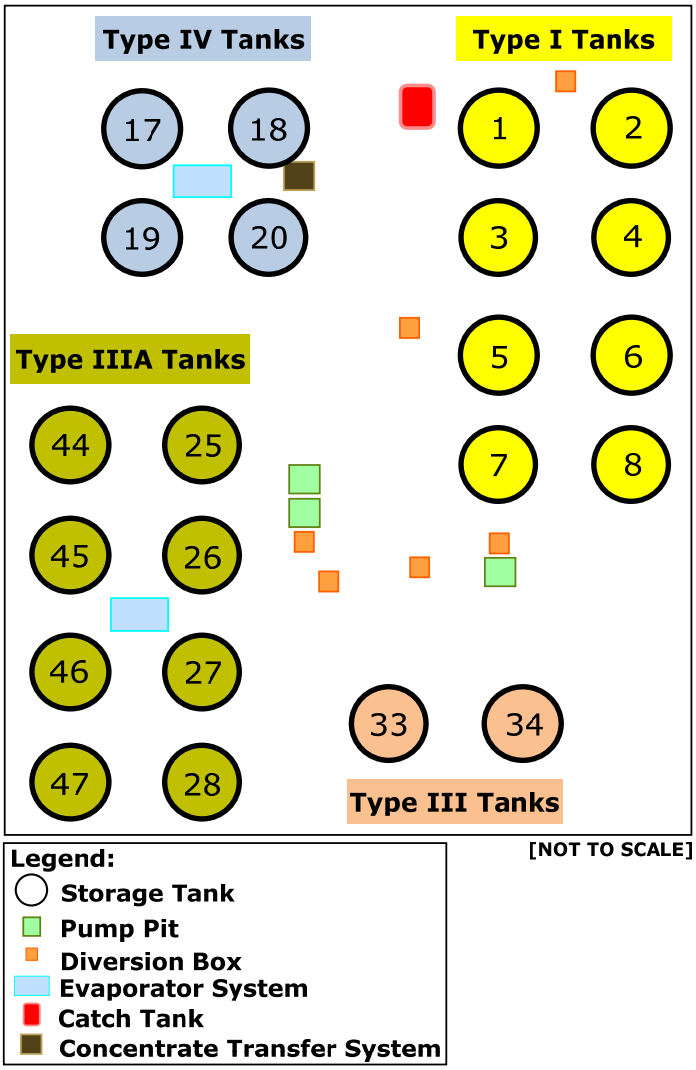
The FTF site was chosen because of its favorable terrain, proximity to the F-Canyon Separations Facility (the major waste generation source), and isolation distance from the SRS boundaries. Figure 3.2-1 shows the distance of the FTF from the SRS boundary. Figure 3.2-2 shows the setting of F Area and FTF within the General Separations Area (GSA).

Figure 3.2-2: Layout of the GSA



The FTF is a 22-acre site, which consists of 22 waste tanks and supporting ancillary structures. The major FTF ancillary structures are transfer lines, two evaporator systems, six diversion boxes (DBs), one catch tank, one concentrate transfer system (CTS) tank, and three pump pits (PPs). There are three major waste tank types in FTF with nominal operating capacities ranging from 750,000 gallons (Type I tanks) to 1.3 million gallons (Type III, IIIA, and IV tanks). Figure 3.2-3 shows the general layout of FTF. The waste tanks have varying degrees of secondary containment (liner) and in-tank structural features (e.g., cooling coils and columns). All FTF waste tank types have primary liners constructed of carbon steel. The FTF was constructed to receive waste generated by various SRS production, processing, and laboratory facilities. The use of FTF safely isolated these wastes from the environment, SRS workers, and the public. The FTF PA provides extensive descriptions of the FTF and waste processing facilities. [SRS-REG-2007-00002]

Figure 3.2-3: General Layout of F-Tank Farm



3.3 Tanks 5 and 6

Tanks 5 and 6 are Type I tanks. There are eight Type I tanks in FTF (Tanks 1 through 8). The FTF Type I tanks were constructed between 1951 and 1953 as part of the original site construction activities. Figure 3.3-1 shows a typical Type I tank. These waste tanks have a carbon-steel primary liner and a 5-foot high carbon-steel annulus pan (secondary liner) that forms an annular space 2.5 feet wide between the secondary and primary liners (see Figure 3.3-2 and 3.3-3). Type I tanks are 75 feet in diameter and 24.5 feet in height with a nominal operating capacity of 750,000 gallons. Twelve concrete and steel columns support the roof of a Type I tank. Type I tanks have 34 vertical cooling coil arrays and 2 horizontal cooling coil arrays that extend across the bottom of the waste tanks. Section 3.2.1.1 of the FTF PA provides additional details of the Type I tanks. [SRS-REG-2007-00002]

Figure 3.3-1: Sketch of Typical Type I Tank (Cross-Sectional View)

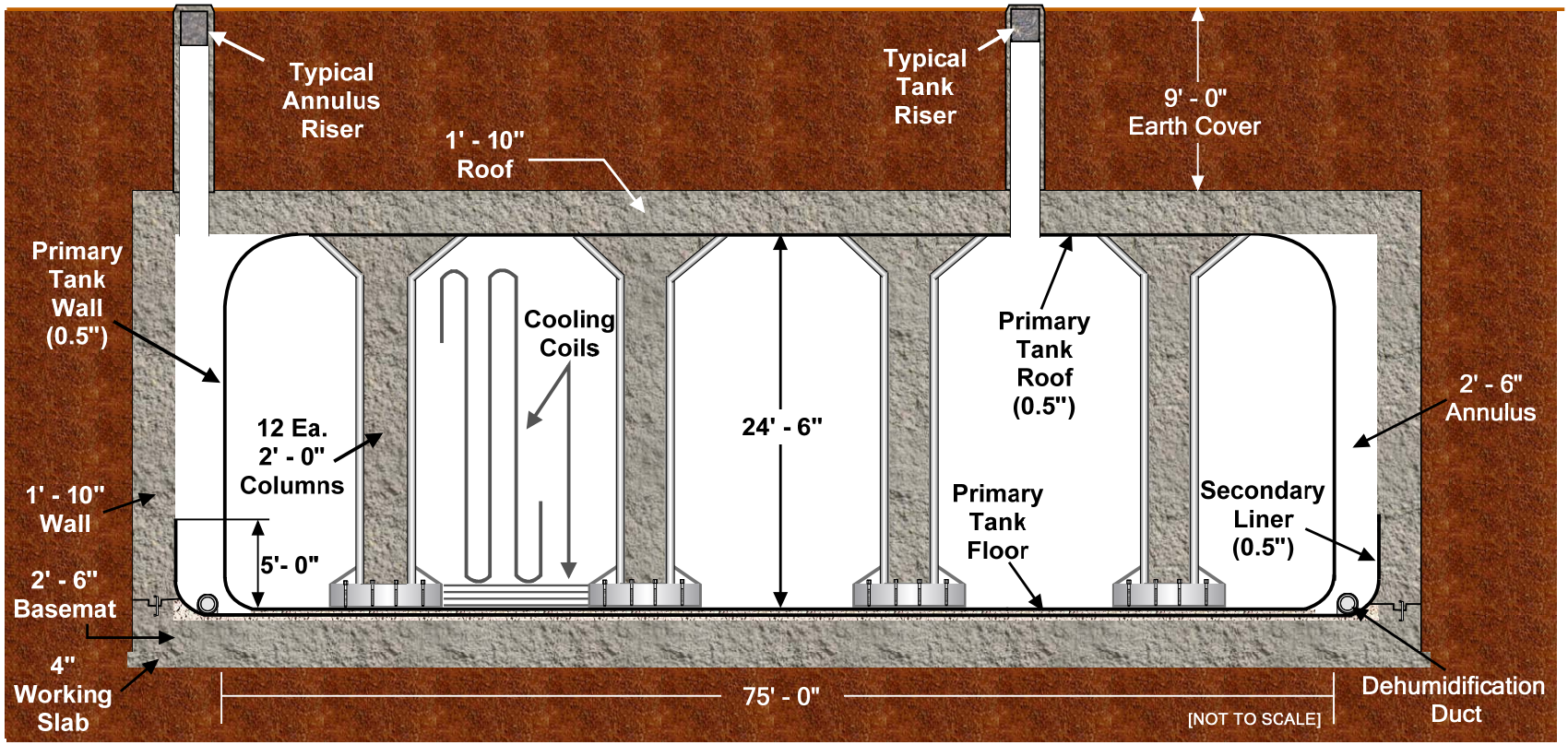
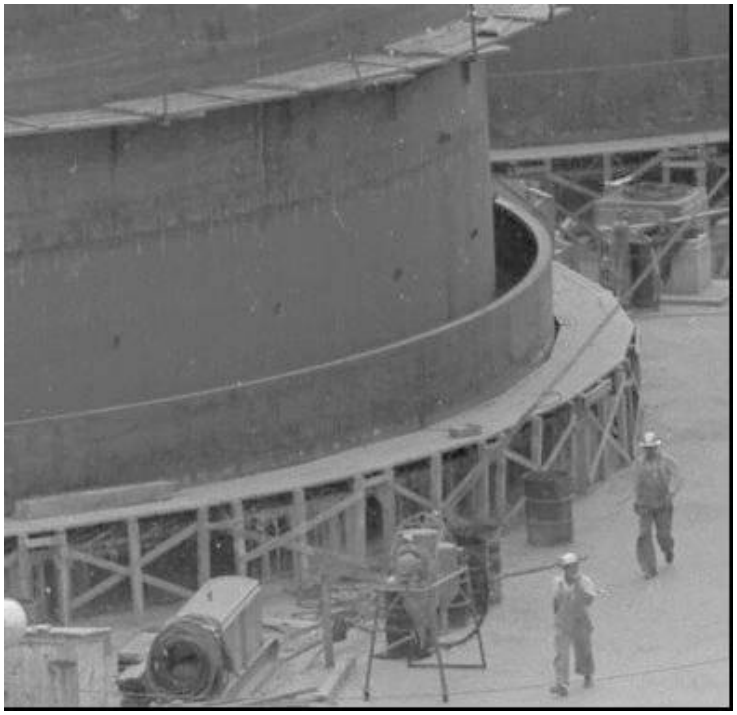


Figure 3.3-2: Steel Liner Construction



Figure 3.3-3: Secondary Containment Detail Showing Annulus Pan



4.0 F-TANK FARM PERFORMANCE ASSESSMENT BACKGROUND INFORMATION

This Special Analysis is being prepared based on the information presented in the FTF PA and the Tank 18/19 Special Analysis. The FTF PA was prepared to support the eventual operational closure of the FTF underground waste tanks and ancillary equipment. The Tank 18/19 Special Analysis evaluated the impact of the Tank 18 and Tank 19 final inventories and updated parameter sensitivity studies on the results and conclusion of the FTF PA. The FTF PA provides the technical basis and results to be used in subsequent documents to demonstrate compliance with the pertinent requirements from the documents identified below for final closure of FTF as indicated in Table 4.0-1:

- DOE M 435.1-1
- NDAA Section 3116
- *Bureau of Water Permit to Construct, F and H-Area High-Level Radioactive Waste Tank Farms, Construction Permit #17,424-IW* (DHEC_01-25-1993)
- *Federal Facility Agreement* (FFA) *for the Savannah River Site* (WSRC-OS-94-42)

The key requirements from the preceding documents necessitate development and calculation of the following for the FTF: potential radiological doses to a hypothetical MOP; potential radiological doses to a hypothetical inadvertent intruder; radiological dose to a human receptor via the air pathway, radon flux and water concentrations. All of these calculations were performed to provide results over a minimum of 10,000 years. The water concentrations were

calculated for both radioactive and non-radioactive contaminants at multiple locations outside FTF.

Table 4.0-1: Key Performance Objectives

Requirement	All-Pathways Dose	Intruder Dose	Air Pathway Dose	Radon Flux	Groundwater Protection
NDAA Section 3116: 10 CFR 61.41 and 61.42	25 mrem/yr	500 mrem/yr	N/A	N/A	N/A
DOE M 435.1-1	25 mrem/yr	500 mrem – acute 100 mrem/yr – chronic	10 mrem/yr	20 $pCi/m^2/s$ at ground surface	< MCL
SCDHEC Primary Drinking Water Regulations (SCDHEC R.61-58)	N/A	N/A	N/A	N/A	< MCL

N/A Not applicable
MCL Maximum Contaminant Level

In accordance with the FFA requirements for high-level radioactive waste tank system(s), a construction and operating permit (hereinafter referred to as Construction Permit #17,424-IW) was obtained for the SRS tank farms' waste tank systems from the South Carolina Department of Health and Environmental Control (SCDHEC). [DHEC_01-25-1993] The FFA requires that waste tank system(s) that have been issued an industrial wastewater operating permit under the Pollution Control Act (PCA), shall be removed from service in accordance with S.C. Code Ann., Section 48-1-10, et seq. (1985) and all applicable regulations promulgated pursuant to the PCA. [Title 48_Chapter 1_SC_Laws] Applicable regulations include SCDHEC Regulation 61-67, *Standards for Wastewater Facility Construction* and Regulation 61-82, *Proper Closeout of Wastewater Treatment Facilities*. [WSRC-OS-94-42 Section IX.E (4)] The SCDHEC has advised that this process will involve two bureaus (Bureau of Water and Bureau of Land and Waste Management).

The regulatory process to complete closure of the FTF requires the development of multiple detailed technical documents with reviews and approvals by state and federal agencies. The documents involved include DOE's final revision of the FTF basis for Section 3116 determination for facility closure, which will be used to demonstrate that the criteria in NDAA Section 3116 are met. [NDAA_3116] DOE's *Basis for Section 3116 Determination for Closure of F-Tank Farm at the Savannah River Site* (hereinafter referred to as the FTF 3116 Basis Document) provides a basis upon which the Secretary of Energy (in consultation with the NRC) has determined that the criteria in NDAA Section 3116 is met and that the stabilized residual waste at closure is not high-level waste. [DOE/SRS-WD-2012-001] The criteria in NDAA Section 3116 provide that the waste will be disposed of in compliance with the performance objectives in 10 CFR Part 61, Subpart C. The current revision of the FTF PA (SRS-REG-2007-00002) provides the technical basis that has been used to demonstrate compliance with 10 CFR 61.41 (*Protection of the General Population from Releases of Radioactivity*) and 61.42 (*Protection of Individuals from Inadvertent Intrusion*) performance objectives, as described in

the FTF 3116 Basis Document. [10 CFR 61] These performance objectives are used in concert with the comparable performance objectives from DOE M 435.1-1.
Compliance with the SCDHEC requirements will be demonstrated using two primary documents that are supported by FTF PA. The first is the *Industrial Wastewater General Closure Plan for F-Area Waste Tank Systems, Industrial Wastewater Construction Permit #17,424-IW* (LWO-RIP-2009-00009), which establishes the general protocols, requirements, and processes for operational closure of FTF. The second set of documents are waste tank-specific closure modules that authorize the operational closure and grouting of a specific waste tank, group of waste tanks, or ancillary equipment.

The FTF PA provides the technical information at different points of assessment that can be used in subsequent decision documents (e.g., waste tank-specific closure modules). The FTF PA provides groundwater radionuclide concentrations at one meter, 100 meters, and exposure points at the two seeplines approximately 1,600 meters from FTF. The groundwater concentrations are provided for each of the three aquifers as applicable as a part of the FTF groundwater modeling. FTF PA also provides groundwater concentrations for chemical contaminants at one meter and 100 meters. In addition to doses determined from these groundwater concentrations, the FTF PA provides inadvertent intruder doses consistent with the requirements for 10 CFR 61.42, as well as analyses for the air pathways and radon ground-surface flux, as required by DOE M 435.1-1.

5.0 TANKS 5 AND 6 RESIDUAL WASTE INFORMATION

As part of the Tanks 5 and 6 operational closure process, actual residual inventories at operational closure have been determined for Tanks 5 and 6 using samples from the tanks after waste removal activities were completed. The waste characterization methodology used to develop the Tanks 5 and 6 characterizations summed the inventory from discrete areas of each waste tank. Each area's residual inventory was determined by taking its material concentration and scaling it by the corresponding residual volume. The calculation was repeated for each constituent (radionuclides and chemicals). Tables 5.0-1 and 5.0-2 list Tanks 5 and 6 residual inventories at final facility closure for the FTF radionuclides and chemicals, respectively. [SRR-CWDA-2009-00045]

Note that the combined closure inventories of Tc-99 from the Type I tanks (including Tanks 5 and 6) were reduced from a projected 632 total curies to less than 8 total curies based on actual Tank 5 and Tank 6 characterization. This reduction is important as Tc-99 releases from the Type I tanks contributed to a significant future dose peak in the FTF PA, but do not contribute significantly in this special analysis, as shown in Section 6.3.5.2. [SRS-REG-2007-00002]

Table 5.0-1: Tanks 5 and 6 Radionuclide Inventories (Ci)

Element	Tank 5 Estimate from PA	Tank 5 Actual	Tank 6 Estimate from PA	Tank 6 Actual
Ac-227	1.0E-03	6.9E-04	1.0E-03	8.1E-05
Al-26	1.0E+00	1.2E-01	1.0E+00	ND
Am-241	6.0E+02	6.9E+02	6.0E+02	1.3E+03
Am-242m	1.0E+00	1.7E+00	1.0E+00	1.8E+00
Am-243	1.4E+00	5.3E+00	1.4E+00	3.0E+01
Ba-137m	8.7E+03	3.3E+03	8.7E+03	6.3E+03
C-14	1.0E+00	7.1E-03	1.0E+00	3.1E-01
Cf-249	1.0E+00	7.0E-02	1.0E+00	6.4E-02
Cl-36	1.0E-03	2.8E-02	1.0E-03	ND
Cm-243	1.0E+00	4.5E-01	1.0E+00	6.2E+00
Cm-244	1.2E+02	2.2E+01	1.2E+02	7.3E+02
Cm-245	1.0E+00	5.4E-03	1.0E+00	1.0E-01
Cm-247	1.0E-03	9.7E-07	1.0E-03	2.4E-06
Cm-248	1.0E-03	1.3E-04	1.0E-03	1.2E-04
Co-60	1.8E+01	2.2E+01	1.8E+01	4.2E+01
Cs-135	1.0E+00	2.2E-02	1.0E+00	4.2E-02
Cs-137	9.2E+03	3.5E+03	9.2E+03	6.7E+03
Eu-152	1.9E+01	9.0E-01	1.9E+01	1.4E+00
Eu-154	1.3E+02	1.5E+02	1.3E+02	1.6E+02
H-3	1.0E+00	4.9E-02	1.0E+00	1.8E-01
I-129	1.0E-03	2.2E-03	1.0E-03	3.0E-03
K-40	1.0E-03	7.9E-03	1.0E-03	ND
Nb-93m	1.0E-03	ND	1.0E-03	ND
Nb-94	1.0E+00	1.1E-02	1.0E+00	1.5E-02
Ni-59	6.3E+00	6.0E+01	6.3E+00	7.5E+01
Ni-63	4.9E+02	3.1E+03	4.9E+02	5.2E+03
Np-237	2.3E-01	2.6E-01	2.3E-01	4.7E-01
Pa-231	1.0E-03	1.4E-03	1.0E-03	6.0E-03
Pd-107	1.0E-03	5.0E-02	1.0E-03	ND
Pt-193	1.0E-03	3.6E-03	1.0E-03	ND
Pu-238	1.4E+02	2.5E+01	1.4E+02	4.8E+01
Pu-239	3.2E+01	8.5E+01	3.2E+01	4.5E+01
Pu-240	7.2E+00	2.0E+01	7.2E+00	2.8E+01
Pu-241	3.2E+01	5.3E+01	3.2E+01	6.8E+01
Pu-242	1.0E+00	3.9E-03	1.0E+00	3.3E-02
Pu-244	1.0E-03	7.1E-06	1.0E-03	8.5E-06
Ra-226	1.0E-03	6.5E-03	1.0E-03	3.8E-02
Sb-126	9.9E-02	ND	1.2E-01	ND
Sb-126m	7.1E-01	ND	8.4E-01	ND
Se-79	4.5E+00	1.3E-01	4.5E+00	3.0E-01
Sm-151	1.2E+04	7.3E+03	1.2E+04	2.9E+03

Table 5.0-1: Tanks 5 and 6 Radionuclide Inventories (Ci) (Continued)

	Tank 5 Estimate from PA	Tank 5 Actual	Tank 6 Estimate from PA	Tank 6 Actual
Sn-126	7.1E-01	7.7E+00	8.4E-01	9.3E+00
Sr-90	1.3E+05	9.7E+04	1.3E+05	2.0E+05
Tc-99	7.9E+01	1.0E-01	7.9E+01	1.7E+00
Th-229	2.4E-01	2.9E-04	2.4E-01	1.5E-03
Th-230	1.0E-03	2.2E-02	1.0E-03	1.5E-02
U-232	1.0E+00	4.1E-05	1.0E+00	1.1E-03
U-233	2.3E-01	2.2E-03	2.3E-01	5.2E-02
U-234	1.7E-01	4.5E-02	1.7E-01	1.5E-01
U-235	5.8E-03	2.0E-03	5.8E-03	6.8E-03
U-236	1.0E+00	2.5E-03	1.0E+00	8.3E-03
U-238	1.7E-01	4.7E-02	1.7E-01	2.5E-01
Y-90	1.3E+05	9.7E+04	1.3E+05	2.0E+05
Zr-93	1.0E-03	3.0E+01	1.0E-03	2.2E+01

[SRR-CWDA-2009-00045, SRS-REG-2007-00002]

ND Not determined as described in Section 4.3 of SRR-CWDA-2009-00045.

Note: Green indicates that the projected estimates were higher than the actual values. Pink indicates that the projected estimates were lower than the actual values.

Table 5.0-2: Tanks 5 and 6 Chemical Inventories (kg)

Chemical	Tank 5 Estimate from PA	Tank 5 Actual	Tank 6 Estimate from PA	Tank 6 Actual
Ag	5.9E+00	3.0E+00	5.9E+00	1.7E+00
As	4.5E-02	9.1E-02	4.5E-02	7.9E-02
Ba	1.3E+01	2.1E+01	1.3E+01	2.0E+01
Cd	4.7E+00	9.0E-01	4.7E+00	1.7E+00
Cr	9.7E+00	1.1E+01	9.7E+00	8.2E+00
Cu	3.9E+00	7.5E+00	3.9E+00	1.7E+01
F	2.9E+00	3.5E-01	2.9E+00	5.9E+00
Fe	8.1E+02	4.9E+03	8.1E+02	3.3E+03
Hg	6.3E+00	2.4E+01	6.3E+00	6.1E+01
Mn	5.7E+02	3.4E+02	5.7E+02	3.0E+02
Ni	3.1E+02	5.8E+02	3.1E+02	8.5E+02
NO_2	4.3E+01	5.0E-01	4.3E+01	4.9E+02
NO_3	9.0E+02	2.0E+00	9.0E+02	4.9E+02
Pb	1.3E+01	3.6E+01	1.3E+01	8.2E+00
Sb	1.9E+00	7.4E+00	1.9E+00	1.5E+01
Se	3.6E-02	1.9E-02	3.6E-02	1.6E-01
U	5.1E+02	9.3E+01	5.1E+02	7.1E+02
Zn	7.1E+00	4.9E+00	7.1E+00	1.6E+01

[SRR-CWDA-2009-00045, SRS-REG-2007-00002]

Note: Green indicates that the projected estimates were higher than the actual values. Pink indicates that the projected estimates were lower than the actual values.

6.0 F-TANK FARM PERFORMANCE ASSESSMENT EVALUATION

This section will evaluate the new information (i.e., final characterization) on Tanks 5 and 6 residual wastes at operational closure presented in Section 5.0, as well as the impact that this new characterization has on the residual waste estimates for other waste tanks. This new inventory information is used to update the FTF fate and transport modeling performed as part of FTF PA and the Tank 18/19 Special Analysis. The potential impacts of the new inventory information on the assumptions from the FTF PA and the Tank 18/19 Special Analysis are also considered. As part of this evaluation, the final Tanks 5 and 6 characterization data was used to update the inventory projection for all of the FTF Types I, III, and IIIA tanks. [SRR-CWDA-2009-00045]

6.1 F-Tank Farm Facility Characteristics

This section will discuss the impact of the new residual waste information on the FTF characteristics information presented in Section 3.0 of the FTF PA.

6.1.1 Site Characteristics

Section 3.1 of FTF PA contains a description of SRS site characteristics. The site characteristics information presented in the FTF PA and the Tank 18/19 Special Analysis are not affected by the new residual waste information.

6.1.2 Principal Facility Design Features

Section 3.2 of FTF PA contains a description of the FTF facility design features. The FTF facility design features information presented in the FTF PA and the Tank 18/19 Special Analysis are not affected by the new residual waste information.

6.1.3 Stabilized Contaminant Characteristics

Section 3.3 of FTF PA contains a description of the FTF stabilized contaminant characteristics, which is affected by the new residual waste information as detailed below.

Section 3.3.1 of FTF PA describes the stabilized contaminant characterization screening process. The screening process is not affected by the new residual waste information.

Section 3.3.2 of FTF PA includes the individual waste tank inventories, which are an integral part of the modeling process. The updated FTF waste tank radionuclide and chemical inventories are presented in Tables 6.1-1 and 6.1-2. The font colors are provided to highlight differences in these inventories relative to those used in the Tank 18/19 Special Analysis. Additional details regarding determination of the revised FTF waste tank projected inventories and inventories at operational closure (i.e., for Tanks 17 through 20 and Tanks 5 and 6) can be found in the FTF inventory report (SRR-CWDA-2009-00045).

Table 6.1-1: Revised FTF Radionuclide Inventories (Ci)

	1	2	3	4	5	6	7	8	17	18	19	20
Ac-227	1.0E+00	1.0E+00	1.0E+00	1.0E+00	6.9E-04	8.1E-05	1.0E+00	1.0E+00	ND	1.5E-04	9.6E-06	ND
Al-26	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.2E-01	ND	1.0E+00	1.0E+00	ND	1.9E-04	3.8E-05	ND
Am-241	7.3E+02	7.3E+02	7.3E+02	7.3E+02	6.9E+02	1.3E+03	7.3E+02	7.3E+02	8.4E+00	1.6E+02	2.6E+00	1.6E+00
Am-242m	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.7E+00	1.8E+00	1.0E+00	1.0E+00	ND	3.8E-02	2.5E-04	ND
Am-243	1.7E+01	1.7E+01	1.7E+01	1.7E+01	5.3E+00	3.0E+01	1.7E+01	1.7E+01	ND	2.3E+00	6.8E-03	ND
Ba-137m	8.7E+03	8.7E+03	8.7E+03	8.7E+03	3.3E+03	6.3E+03	8.7E+03	8.7E+03	1.0E+01	8.7E+03	4.0E+03	2.3E+01
C-14	1.0E+00	1.0E+00	1.0E+00	1.0E+00	7.1E-03	3.1E-01	1.0E+00	1.0E+00	3.1E-03	9.0E-01	4.1E+00	6.6E-04
Cf-249	1.0E+00	1.0E+00	1.0E+00	1.0E+00	7.0E-02	6.4E-02	1.0E+00	1.0E+00	ND	2.3E-03	5.2E-04	ND
Cl-36	9.0E-04	9.0E-04	9.0E-04	9.0E-04	2.8E-02	ND	9.0E-04	9.0E-04	ND	2.8E-04	9.1E-05	ND
Cm-243	1.0E+00	1.0E+00	1.0E+00	1.0E+00	4.5E-01	6.2E+00	1.0E+00	1.0E+00	ND	1.8E-02	1.7E-03	ND
Cm-244	4.1E+02	4.1E+02	4.1E+02	4.1E+02	2.2E+01	7.3E+02	4.1E+02	4.1E+02	2.9E-04	9.8E+01	2.7E-01	ND
Cm-245	9.0E-02	9.0E-02	9.0E-02	9.0E-02	5.4E-03	1.0E-01	9.0E-02	9.0E-02	4.4E-10	1.2E-02	1.6E-03	ND
Cm-247	9.0E-04	9.0E-04	9.0E-04	9.0E-04	9.7E-07	2.4E-06	9.0E-04	9.0E-04	ND	2.1E-06	1.3E-06	ND
Cm-248	9.0E-04	9.0E-04	9.0E-04	9.0E-04	1.3E-04	1.2E-04	9.0E-04	9.0E-04	ND	9.5E-05	5.8E-05	ND
Co-60	2.4E+01	2.4E+01	2.4E+01	2.4E+01	2.2E+01	4.2E+01	2.4E+01	2.4E+01	3.4E-02	3.2E-01	1.2E-02	4.8E-03
Cs-135	9.0E-02	9.0E-02	9.0E-02	9.0E-02	2.2E-02	4.2E-02	9.0E-02	9.0E-02	1.7E-04	3.0E-02	5.4E-02	3.6E-05
Cs-137	9.2E+03	9.2E+03	9.2E+03	9.2E+03	3.5E+03	6.7E+03	9.2E+03	9.2E+03	1.1E+01	9.2E+03	4.2E+03	2.4E+01
Eu-152	1.9E+01	1.9E+01	1.9E+01	1.9E+01	9.0E-01	1.4E+00	1.9E+01	1.9E+01	ND	4.7E-03	1.7E-04	ND
Eu-154	1.3E+02	1.3E+02	1.3E+02	1.3E+02	1.5E+02	1.6E+02	1.3E+02	1.3E+02	2.3E-02	2.1E-01	3.8E-03	1.6E-01
H-3	1.0E+00	1.0E+00	1.0E+00	1.0E+00	4.9E-02	1.8E-01	1.0E+00	1.0E+00	5.8E+00	8.0E-03	2.5E-03	ND
I-129	2.0E-03	2.0E-03	2.0E-03	2.0E-03	2.2E-03	3.0E-03	2.0E-03	2.0E-03	1.3E-06	2.7E-04	2.2E-04	2.6E-07
K-40	4.5E-04	4.5E-04	4.5E-04	4.5E-04	7.9E-03	ND	4.5E-04	4.5E-04	ND	1.6E-02	1.0E-03	ND
Nb-94	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.1E-02	1.5E-02	1.0E+00	1.0E+00	ND	5.5E-04	1.0E-04	ND
Ni-59	5.4E+01	5.4E+01	5.4E+01	5.4E+01	6.0E+01	7.5E+01	5.4E+01	5.4E+01	1.8E-01	3.3E-01	3.5E-04	3.9E-02
Ni-63	2.9E+03	2.9E+03	2.9E+03	2.9E+03	3.1E+03	5.2E+03	2.9E+03	2.9E+03	ND	1.6E+01	1.3E-02	ND
Np-237	2.7E-01	2.7E-01	2.7E-01	2.7E-01	2.6E-01	4.7E-01	2.7E-01	2.7E-01	1.4E-02	1.9E-01	1.5E-03	7.2E-04
Pa-231	9.0E-04	9.0E-04	9.0E-04	9.0E-04	1.4E-03	6.0E-03	9.0E-04	9.0E-04	ND	4.6E-02	6.9E-05	ND
Pd-107	9.0E-02	9.0E-02	9.0E-02	9.0E-02	5.0E-02	ND	9.0E-02	9.0E-02	ND	1.2E-01	2.0E-01	ND
Pt-193	9.0E-02	9.0E-02	9.0E-02	9.0E-02	3.6E-03	ND	9.0E-02	9.0E-02	ND	3.6E-03	1.5E-03	ND

Table 6.1-1: Revised FTF Radionuclide Inventories (Ci) (Continued)

	1	2	3	4	5	6	7	8	17	18	19	20
Pu-238	1.4E+02	1.4E+02	1.4E+02	1.4E+02	2.5E+01	4.8E+01	2.6E+03	2.6E+03	5.4E+01	1.3E+03	3.4E+00	6.1E+00
Pu-239	7.6E+01	7.6E+01	7.6E+01	7.6E+01	8.5E+01	4.5E+01	5.2E+01	5.2E+01	1.5E+01	2.8E+02	4.0E+00	8.5E-01
Pu-240	1.8E+01	1.8E+01	1.8E+01	1.8E+01	2.0E+01	2.8E+01	2.6E+01	2.6E+01	3.4E+00	6.5E+01	9.8E-01	1.8E-01
Pu-241	4.8E+01	4.8E+01	4.8E+01	4.8E+01	5.3E+01	6.8E+01	5.0E+02	5.0E+02	9.3E+01	2.7E+02	3.9E+00	1.6E+01
Pu-242	9.0E-04	9.0E-04	9.0E-04	9.0E-04	3.9E-03	3.3E-02	9.0E-04	9.0E-04	5.3E-03	2.7E-02	1.7E-03	1.6E-03
Pu-244	9.0E-04	9.0E-04	9.0E-04	9.0E-04	7.1E-06	8.5E-06	9.0E-04	9.0E-04	ND	6.2E-06	5.3E-06	ND
Ra-226	9.0E-03	9.0E-03	9.0E-03	9.0E-03	6.5E-03	3.8E-02	9.0E-03	9.0E-03	ND	3.4E-03	4.1E-03	ND
Se-79	4.5E+00	4.5E+00	4.5E+00	4.5E+00	1.3E-01	3.0E-01	4.5E+00	4.5E+00	1.6E-02	4.8E-04	4.6E-04	3.2E-03
Sm-151	1.2E+04	1.2E+04	1.2E+04	1.2E+04	7.3E+03	2.9E+03	1.2E+04	1.2E+04	ND	3.7E+01	1.5E-01	ND
Sn-126	6.9E+00	6.9E+00	6.9E+00	6.9E+00	7.7E+00	9.3E+00	6.9E+00	6.9E+00	2.8E-02	1.3E-02	3.3E-03	5.9E-03
Sr-90	1.3E+05	1.3E+05	1.3E+05	1.3E+05	9.7E+04	2.0E+05	1.3E+05	1.3E+05	6.6E+01	2.5E+03	6.9E+00	2.3E+01
Tc-99	9.6E-01	9.6E-01	9.6E-01	9.6E-01	1.0E-01	1.7E+00	9.6E-01	9.6E-01	9.0E-01	9.0E-01	3.8E-01	8.5E-01
Th-229	9.0E-04	9.0E-04	9.0E-04	9.0E-04	2.9E-04	1.5E-03	9.0E-04	9.0E-04	ND	8.9E-04	2.0E-04	ND
Th-230	9.0E-03	9.0E-03	9.0E-03	9.0E-03	2.2E-02	1.5E-02	9.0E-03	9.0E-03	ND	2.1E-03	1.1E-04	ND
U-232	9.0E-04	9.0E-04	9.0E-04	9.0E-04	4.1E-05	1.1E-03	9.0E-04	9.0E-04	3.7E-05	6.9E-04	9.5E-05	8.0E-06
U-233	2.3E-01	2.3E-01	2.3E-01	2.3E-01	2.2E-03	5.2E-02	2.3E-01	2.3E-01	ND	4.0E-02	4.3E-03	ND
U-234	1.7E-01	1.7E-01	1.7E-01	1.7E-01	4.5E-02	1.5E-01	1.7E-01	1.7E-01	ND	3.1E-01	4.8E-03	ND
U-235	5.8E-03	5.8E-03	5.8E-03	5.8E-03	2.0E-03	6.8E-03	5.8E-03	5.8E-03	3.0E-04	1.1E-02	1.7E-04	1.9E-05
U-236	9.0E-03	9.0E-03	9.0E-03	9.0E-03	2.5E-03	8.3E-03	9.0E-03	9.0E-03	ND	1.2E-02	2.5E-04	2.7E-05
U-238	1.7E-01	1.7E-01	1.7E-01	1.7E-01	4.7E-02	2.5E-01	1.7E-01	1.7E-01	6.4E-03	2.8E-01	5.4E-03	5.6E-04
Y-90	1.3E+05	1.3E+05	1.3E+05	1.3E+05	9.7E+04	2.0E+05	1.3E+05	1.3E+05	6.6E+01	2.5E+03	6.9E+00	2.3E+01
Zr-93	2.7E+01	2.7E+01	2.7E+01	2.7E+01	3.0E+01	2.2E+01	2.7E+01	2.7E+01	ND	8.6E-02	1.8E-02	ND
Total Curies by Tank	2.9E+05	2.9E+05	2.9E+05	2.9E+05	2.1E+05	4.2E+05	3.0E+05	3.0E+05	3.3E+02	2.5E+04	8.2E+03	1.2E+02

Table 6.1-1: Revised FTF Radionuclide Inventories (Ci) (Continued)

	25	26	27	28	33	34	44	45	46	47	Total Curies by Radionuclide
Ac-227	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.6E+01
Al-26	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.6E+01
Am-241	5.7E+00	5.7E+00	5.7E+00	5.7E+00	6.3E+01	1.6E+03	5.7E+00	5.7E+00	5.7E+00	5.7E+00	8.3E+03
Am-242m	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	2.0E+01
Am-243	1.0E-03	1.0E-03	1.0E-03	1.0E-03	1.0E-03	1.0E-03	1.0E-03	1.0E-03	1.0E-03	1.0E-03	1.4E+02
Ba-137m	4.9E+03	4.9E+03	4.9E+03	4.9E+03	9.0E+02	3.7E+03	4.9E+03	4.9E+03	4.9E+03	4.9E+03	1.2E+05
C-14	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	2.1E+01
Cf-249	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.6E+01
Cl-36	1.2E-03	1.2E-03	1.2E-03	1.2E-03	1.2E-04	1.2E-04	1.2E-03	1.2E-03	1.2E-03	1.2E-03	4.4E-02
Cm-243	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	2.3E+01
Cm-244	1.0E-03	1.0E-03	1.0E-03	1.0E-03	1.0E+00	1.0E+00	1.0E-03	1.0E-03	1.0E-03	1.0E-03	3.3E+03
Cm-245	1.2E-01	1.2E-01	1.2E-01	1.2E-01	1.2E-02	1.2E-02	1.2E-01	1.2E-01	1.2E-01	1.2E-01	1.6E+00
Cm-247	1.2E-03	1.2E-03	1.2E-03	1.2E-03	1.2E-04	1.2E-04	1.2E-03	1.2E-03	1.2E-03	1.2E-03	1.5E-02
Cm-248	1.2E-03	1.2E-03	1.2E-03	1.2E-03	1.2E-04	1.2E-04	1.2E-03	1.2E-03	1.2E-03	1.2E-03	1.6E-02
Co-60	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.7E+01	4.7E+01	1.0E+00	1.0E+00	1.0E+00	1.0E+00	2.8E+02
Cs-135	1.2E-01	1.2E-01	1.2E-01	1.2E-01	1.2E-02	1.2E-02	1.2E-01	1.2E-01	1.2E-01	1.2E-01	1.7E+00
Cs-137	5.2E+03	5.2E+03	5.2E+03	5.2E+03	9.5E+02	3.9E+03	5.2E+03	5.2E+03	5.2E+03	5.2E+03	1.3E+05
Eu-152	1.0E+00	1.0E+00	1.0E+00	1.0E+00	3.5E+00	1.3E+01	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.4E+02
Eu-154	2.9E+00	2.9E+00	2.9E+00	2.9E+00	4.3E+01	1.4E+02	2.9E+00	2.9E+00	2.9E+00	2.9E+00	1.3E+03
H-3	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	2.2E+01
I-129	1.2E-04	1.2E-04	1.2E-04	1.2E-04	3.2E-05	1.7E-04	1.2E-04	1.2E-04	1.2E-04	1.2E-04	1.9E-02
K-40	5.8E-04	5.8E-04	5.8E-04	5.8E-04	5.8E-05	5.8E-05	5.8E-04	5.8E-04	5.8E-04	5.8E-04	3.2E-02
Nb-94	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.6E+01
Ni-59	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.8E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	4.7E+02
Ni-63	2.4E+01	2.4E+01	2.4E+01	2.4E+01	3.8E+01	1.6E+02	2.4E+01	2.4E+01	2.4E+01	2.4E+01	2.6E+04
Np-237	1.0E-03	1.0E-03	1.0E-03	1.0E-03	2.5E-02	6.8E-02	1.0E-03	1.0E-03	1.0E-03	1.0E-03	2.7E+00
Pa-231	1.2E-03	1.2E-03	1.2E-03	1.2E-03	1.2E-04	1.2E-04	1.2E-03	1.2E-03	1.2E-03	1.2E-03	6.9E-02
Pd-107	1.2E-01	1.2E-01	1.2E-01	1.2E-01	1.2E-02	1.2E-02	1.2E-01	1.2E-01	1.2E-01	1.2E-01	1.9E+00
Pt-193	1.2E-01	1.2E-01	1.2E-01	1.2E-01	1.2E-02	1.2E-02	1.2E-01	1.2E-01	1.2E-01	1.2E-01	1.5E+00

Table 6.1-1: Revised FTF Radionuclide Inventories (Ci) (Continued)

	25	26	27	28	33	34	44	45	46	47	Total Curies by Radionuclide
Pu-238	1.2E+02	1.2E+02	1.2E+02	1.2E+02	3.6E+01	1.0E+00	1.2E+02	1.2E+02	1.2E+02	1.2E+02	8.2E+03
Pu-239	2.2E+01	2.2E+01	2.2E+01	2.2E+01	2.2E+01	1.4E+01	2.2E+01	2.2E+01	2.2E+01	2.2E+01	1.0E+03
Pu-240	4.8E+00	4.8E+00	4.8E+00	4.8E+00	3.9E+00	3.2E+00	4.8E+00	4.8E+00	4.8E+00	4.8E+00	2.9E+02
Pu-241	5.4E+01	5.4E+01	5.4E+01	5.4E+01	5.5E+01	3.1E+01	5.4E+01	5.4E+01	5.4E+01	5.4E+01	2.2E+03
Pu-242	1.2E-03	1.2E-03	1.2E-03	1.2E-03	1.2E-04	1.2E-04	1.2E-03	1.2E-03	1.2E-03	1.2E-03	8.8E-02
Pu-244	1.2E-03	1.2E-03	1.2E-03	1.2E-03	1.2E-04	1.2E-04	1.2E-03	1.2E-03	1.2E-03	1.2E-03	1.5E-02
Ra-226	1.2E-02	1.2E-02	1.2E-02	1.2E-02	1.2E-03	1.2E-03	1.2E-02	1.2E-02	1.2E-02	1.2E-02	2.0E-01
Se-79	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	1.3E+00	1.0E+00	1.0E+00	1.0E+00	1.0E+00	3.8E+01
Sm-151	7.1E+01	7.1E+01	7.1E+01	7.1E+01	9.3E+02	4.0E+03	7.1E+01	7.1E+01	7.1E+01	7.1E+01	8.8E+04
Sn-126	2.4E-03	4.2E-03	2.0E-03	4.2E-03	5.5E-01	2.4E+00	4.2E-03	4.2E-03	4.2E-03	4.2E-03	6.1E+01
Sr-90	1.0E+03	1.0E+03	1.0E+03	1.0E+03	1.4E+04	5.5E+04	1.0E+03	1.0E+03	1.0E+03	1.0E+03	1.2E+06
Tc-99	2.2E-02	2.2E-02	2.2E-02	2.2E-02	1.1E-01	4.9E-01	2.2E-02	2.2E-02	2.2E-02	2.2E-02	1.1E+01
Th-229	1.2E-03	1.2E-03	1.2E-03	1.2E-03	1.2E-04	1.2E-04	1.2E-03	1.2E-03	1.2E-03	1.2E-03	1.8E-02
Th-230	1.2E-02	1.2E-02	1.2E-02	1.2E-02	1.2E-03	1.2E-03	1.2E-02	1.2E-02	1.2E-02	1.2E-02	1.9E-01
U-232	1.2E-03	1.2E-03	1.2E-03	1.2E-03	1.2E-04	1.2E-04	1.2E-03	1.2E-03	1.2E-03	1.2E-03	1.7E-02
U-233	1.0E-03	1.0E-03	1.0E-03	1.0E-03	2.5E-02	6.8E-02	1.0E-03	1.0E-03	1.0E-03	1.0E-03	1.6E+00
U-234	2.6E-02	2.6E-02	2.6E-02	2.6E-02	7.9E-02	8.8E-02	2.6E-02	2.6E-02	2.6E-02	2.6E-02	1.9E+00
U-235	1.0E-03	1.0E-03	1.0E-03	1.0E-03	1.0E-03	1.2E-03	1.0E-03	1.0E-03	1.0E-03	1.0E-03	6.5E-02
U-236	1.2E-02	1.2E-02	1.2E-02	1.2E-02	1.2E-03	1.2E-03	1.2E-02	1.2E-02	1.2E-02	1.2E-02	1.8E-01
U-238	2.6E-02	2.6E-02	2.6E-02	2.6E-02	7.9E-02	8.8E-02	2.6E-02	2.6E-02	2.6E-02	2.6E-02	2.0E+00
Y-90	1.0E+03	1.0E+03	1.0E+03	1.0E+03	1.4E+04	5.5E+04	1.0E+03	1.0E+03	1.0E+03	1.0E+03	1.2E+06
Zr-93	2.7E+01	2.7E+01	2.7E+01	2.7E+01	2.7E+01	2.7E+01	2.7E+01	2.7E+01	2.7E+01	2.7E+01	4.8E+02
Total Curies by Tank	1.2E+04	1.2E+04	1.2E+04	1.2E+04	3.1E+04	1.2E+05	1.2E+04	1.2E+04	1.2E+04	1.2E+04	**2.7E+06**

[SRR-CWDA-2009-00045]

ND Not determined as described in Section 4.3 of SRR-CWDA-2009-00045.

Notes: Green font indicates that the revised value is lower than the inventory used in the Tank 18/19 Special Analysis. Red font indicates that the revised value is higher than the inventory used in the Tank 18/19 Special Analysis. [SRR-CWDA-2010-00124]

Shaded cells indicate actual inventories rather than projected inventories.

Table 6.1-2: Revised FTF Chemical Inventories (kg)

	1	2	3	4	5	6	7	8	17	18	19	20
Ag	5.9E+00	5.9E+00	5.9E+00	5.9E+00	3.0E+00	1.7E+00	5.9E+00	5.9E+00	6.6E+00	4.5E+00	7.1E-01	3.1E+00
As	8.2E-02	8.2E-02	8.2E-02	8.2E-02	9.1E-02	7.9E-02	8.2E-02	8.2E-02	ND	8.5E-02	2.9E-02	NE
Ba	1.9E+01	1.9E+01	1.9E+01	1.9E+01	2.1E+01	2.0E+01	1.9E+01	1.9E+01	3.9E+00	5.4E+00	8.0E+00	1.8E+00
Cd	4.7E+00	4.7E+00	4.7E+00	4.7E+00	9.0E-01	1.7E+00	4.7E+00	4.7E+00	1.8E+01	1.6E+02	1.2E+00	1.8E+00
Cr	9.9E+00	9.9E+00	9.9E+00	9.9E+00	1.1E+01	8.2E+00	9.9E+00	9.9E+00	4.7E+00	1.2E+01	3.0E+00	2.5E+00
Cu	9.6E+00	9.6E+00	9.6E+00	9.6E+00	7.5E+00	1.7E+01	9.6E+00	9.6E+00	3.3E+00	1.8E+00	5.0E-01	1.5E+00
F	3.3E+00	3.3E+00	3.3E+00	3.3E+00	3.5E-01	5.9E+00	3.3E+00	3.3E+00	3.5E+00	8.3E+00	1.5E+01	2.4E+01
Fe	4.4E+03	4.4E+03	4.4E+03	4.4E+03	4.9E+03	3.3E+03	4.4E+03	4.4E+03	5.4E+02	1.9E+03	4.0E+02	2.5E+02
Hg	3.4E+01	3.4E+01	3.4E+01	3.4E+01	2.4E+01	6.1E+01	3.4E+01	3.4E+01	1.4E+00	1.7E+01	3.8E+00	6.3E-01
Mn	5.7E+02	5.7E+02	5.7E+02	5.7E+02	3.4E+02	3.0E+02	5.7E+02	5.7E+02	4.8E+01	2.4E+02	1.6E+01	1.2E+01
Ni	5.2E+02	5.2E+02	5.2E+02	5.2E+02	5.8E+02	8.5E+02	5.2E+02	5.2E+02	8.3E-01	2.4E+01	2.3E+00	8.0E-01
NO_2	4.3E+01	4.3E+01	4.3E+01	4.3E+01	5.0E-01	4.9E+02	4.3E+01	4.3E+01	ND	9.6E+00	8.8E+01	NE
NO_3	9.0E+02	9.0E+02	9.0E+02	9.0E+02	2.0E+00	4.9E+02	9.0E+02	9.0E+02	ND	8.2E+00	1.7E+02	1.7E+01
Pb	3.2E+01	3.2E+01	3.2E+01	3.2E+01	3.6E+01	8.2E+00	3.2E+01	3.2E+01	5.5E+00	1.2E+01	3.5E+00	2.6E+00
Sb	8.5E+00	8.5E+00	8.5E+00	8.5E+00	7.4E+00	1.5E+01	8.5E+00	8.5E+00	ND	8.8E+00	3.5E+00	NE
Se	9.0E-02	9.0E-02	9.0E-02	9.0E-02	1.9E-02	1.6E-01	9.0E-02	9.0E-02	ND	1.7E-01	8.8E-03	NE
U	5.1E+02	5.1E+02	5.1E+02	5.1E+02	9.3E+01	7.1E+02	5.1E+02	5.1E+02	5.6E+01	9.0E+02	1.6E+01	1.7E+01
Zn	9.0E+00	9.0E+00	9.0E+00	9.0E+00	4.9E+00	1.6E+01	9.0E+00	9.0E+00	6.6E+00	3.9E+00	6.9E-01	3.1E+00

Table 6.1-2: Revised FTF Chemical Inventories (kg) (Continued)

	25	26	27	28	33	34	44	45	46	47
Ag	3.7E+00	3.7E+00	3.7E+00	3.7E+00	9.5E-01	6.0E-01	3.7E+00	3.7E+00	3.7E+00	3.7E+00
As	1.5E-02	1.5E-02	1.5E-02	1.5E-02	7.4E-03	1.2E-02	1.5E-02	1.5E-02	1.5E-02	1.5E-02
Ba	2.2E+00	2.2E+00	2.2E+00	2.2E+00	1.6E+00	4.5E+00	2.2E+00	2.2E+00	2.2E+00	2.2E+00
Cd	1.6E+00	1.6E+00	1.6E+00	1.6E+00	7.8E-01	1.3E+00	1.6E+00	1.6E+00	1.6E+00	1.6E+00
Cr	2.6E+00	2.6E+00	2.6E+00	2.6E+00	1.4E+00	3.3E+00	2.6E+00	2.6E+00	2.6E+00	2.6E+00
Cu	1.8E+00	1.8E+00	1.8E+00	1.8E+00	6.4E-01	9.7E-01	1.8E+00	1.8E+00	1.8E+00	1.8E+00
F	1.9E+00	1.9E+00	1.9E+00	1.9E+00	4.7E-01	1.8E-01	1.9E+00	1.9E+00	1.9E+00	1.9E+00
Fe	3.0E+02	3.0E+02	3.0E+02	3.0E+02	1.2E+02	2.4E+02	3.0E+02	3.0E+02	3.0E+02	3.0E+02
Hg	7.6E-01	7.6E-01	7.6E-01	7.6E-01	6.9E-01	2.2E+00	7.6E-01	7.6E-01	7.6E-01	7.6E-01
Mn	5.7E+01	5.7E+01	5.7E+01	5.7E+01	1.3E+00	1.2E-01	5.7E+01	5.7E+01	5.7E+01	5.7E+01
Ni	6.8E+02	6.8E+02	6.8E+02	6.8E+02	1.1E+01	6.8E+01	6.8E+02	6.8E+02	6.8E+02	6.8E+02
NO_2	2.0E+01	2.0E+01	2.0E+01	2.0E+01	7.0E+00	1.1E+01	2.0E+01	2.0E+01	2.0E+01	2.0E+01
NO_3	6.8E+02	6.8E+02	6.8E+02	6.8E+02	3.8E+01	6.3E+01	6.8E+02	6.8E+02	6.8E+02	6.8E+02
Pb	3.9E+01	3.9E+01	3.9E+01	3.9E+01	1.7E+00	4.3E+00	3.9E+01	3.9E+01	3.9E+01	3.9E+01
Sb	6.4E-01	6.4E-01	6.4E-01	6.4E-01	3.1E-01	5.1E-01	6.4E-01	6.4E-01	6.4E-01	6.4E-01
Se	1.2E-02	1.2E-02	1.2E-02	1.2E-02	5.9E-03	9.6E-03	1.2E-02	1.2E-02	1.2E-02	1.2E-02
U	7.7E+01	7.7E+01	7.7E+01	7.7E+01	2.3E+02	2.5E+02	7.7E+01	7.7E+01	7.7E+01	7.7E+01
Zn	3.7E+00	3.7E+00	3.7E+00	3.7E+00	1.2E+00	1.5E+00	3.7E+00	3.7E+00	3.7E+00	3.7E+00

[SRR-CWDA-2009-00045]

ND Not determined as described in Section 4.3 of SRR-CWDA-2009-00045.

Notes: Green font indicates that the revised value is lower than the inventory used in the Tank 18/19 Special Analysis. Red font indicates that the revised value is higher than the inventory used in the Tank 18/19 Special Analysis. [SRR-CWDA-2010-00124]
Shaded cells indicate actual inventories rather than projected inventories.

The impacts of these new FTF inventories on various PA analyses (e.g., public dose analyses) are presented later in Section 6.3 of this Special Analysis.

Section 3.3.3 of FTF PA describes the FTF ancillary equipment inventory. The ancillary equipment inventory is based on the residual inventories at those specific locations (e.g., evaporator vessels, pump tanks) and is not affected by the new residual waste information.

6.2 The F-Tank Farm Analysis of Performance

This section will discuss the impact of the new residual waste information on the FTF analysis of performance information presented in Section 4.0 of the FTF PA and Section 6.2 of the Tank 18/19 Special Analysis.

6.2.1 Overview of Analysis

Section 4.1 of FTF PA contains an overview of analysis. The overview of analysis information presented in the FTF PA is not affected by the new residual waste information.

6.2.2 Integrated Site Conceptual Model of Facility Performance

Section 4.2 of FTF PA describes the Integrated Conceptual Model (ICM), which is used to simulate the release of radiological and chemical contaminants from the 22 underground waste tanks and associated ancillary equipment in FTF. The Section 4.2 ICM information presented in the FTF PA is not affected by the new residual waste information.

Section 4.2.2 (Source Term Release) of FTF PA includes a discussion of the waste release model, which is an integral part of the modeling process. The information regarding the residual waste characteristics presented in the Tanks 5 and 6 waste characterization reports (SRR-CWDA-2012-00027, SRR-CWDA-2012-00075) is summarized in Section 5.0 of this Special Analysis, is consistent with the source term release assumptions, and does not impact the waste release model.

6.2.3 Modeling Codes

Section 4.3 of FTF PA contains a discussion of the modeling codes. The FTF PA Base Case (i.e., deterministic) modeling was performed using the same flow and transport model that was used in the FTF PA, Rev. 1 and the Tank 18/19 Special Analysis. The FTF PORFLOW flow and transport model was not changed for the modeling runs performed for this Special Analysis but the model was rerun with the updated FTF inventories. For the PORFLOW deterministic sensitivity runs, some additional inputs were updated, as discussed in detail in Section 6.3.6.

FTF probabilistic modeling for this Special Analysis was performed using the GoldSim model: SRS FTF v3.0. This GoldSim model is an update of the model used for FTF PA and the Tank 18/19 Special Analysis with multiple changes and improvements, as detailed in the report *F-Area Tank Farm Stochastic Fate and Transport Model* (SRR-CWDA-2012-00119). SRS FTF v3.0 identified and corrected errors from the previous GoldSim model and an improved modeling structure was applied. The updated FTF inventories were used in new the modeling runs. [SRR-CWDA-2009-00045] Quality assurance (e.g., verification of modeling inputs used) for this model revision is documented in the *Tanks 5 and 6 Special Analysis for the Performance Assessment for the F-Tank Farm at the Savannah River Site: Quality Assurance Report* (SRR-CWDA-2012-00134). The modeling codes information presented in FTF PA is not affected by the new residual waste information.

6.2.4 Closure System Modeling

Section 4.4 of FTF PA describes how the FTF design elements and their associated properties were represented in the computer modeling codes. The closure system modeling information presented in the FTF PA is not affected by the new residual waste information.

6.2.5 Airborne and Radon Analysis

Section 4.5 of FTF PA contains a discussion of the airborne and radon analysis methodology. The air and radon pathway conceptual model and analysis approach presented in the FTF PA is affected by the new residual waste information and a revised analysis methodology. In Section 4.5 of the FTF PA, the airborne and radon pathway analysis conservatively assumed that the concrete roof, reducing grout, and stabilized contaminant layer were dry (i.e., 0 % saturation). When applying the revised inventory data, this assumption proved to be overly conservative to the extent that the results were not realistic.

An alternative assumption of 50 % saturation for the concrete roof, reducing grout, and stabilized contaminant layer was applied for this Special Analysis. Given the short curing time and material water-retention properties of these materials, the modeled environment is expected to be at or near saturation. [WSRC-STI-2007-00369] Assuming 50 % saturation maintains conservatism, but more closely reflects expected conditions. This revised methodology for determining airborne and radon pathway releases is consistent with the approach used for more recent PAs (e.g., HTF PA). [SRR-CWDA-2010-00128]

6.2.6 Biotic Pathways

Section 4.6 of FTF PA documents the bioaccumulation factors and human health exposure parameters used in the FTF PA modeling effort. The bioaccumulation factors and human health exposure parameters presented in the FTF PA are not affected by the new residual waste information.

6.2.7 Dose Analysis

Section 4.7 of FTF PA contains a discussion of the dose analysis approach used and presents the set of dose conversion factors (DCFs) used in the dose calculations modeling effort methodology. The dose analysis information presented in FTF PA is not affected by the new residual waste information.

6.2.8 RCRA/CERCLA Risk Evaluation

Section 4.8 of FTF PA contains a discussion of the Resource Conservation and Recovery Act/Comprehensive Environmental Response, Compensation, and Liability Act (RCRA/CERCLA) risk evaluation methodology. The RCRA/CERCLA risk evaluation approach presented in the FTF PA is not affected by the new residual waste information.

6.3 FTF Results of Analysis

This section will discuss the impact of the new residual waste information on the FTF results presented in Section 5.0 of FTF PA and Section 6.3 of the Tank 18/19 Special Analysis.

6.3.1 Source Term (Analyses Results)

Section 5.1 of FTF PA presents the peak stabilized contaminant release rates from the FTF waste tanks and ancillary equipment. The release rates (fluxes) from PORFLOW were not presented herein, since the flows did not change and the groundwater concentrations at 100 meters from the FTF boundary provide analogous information (Section 6.3.2.1).

6.3.2 Environmental Transport of Radionuclides

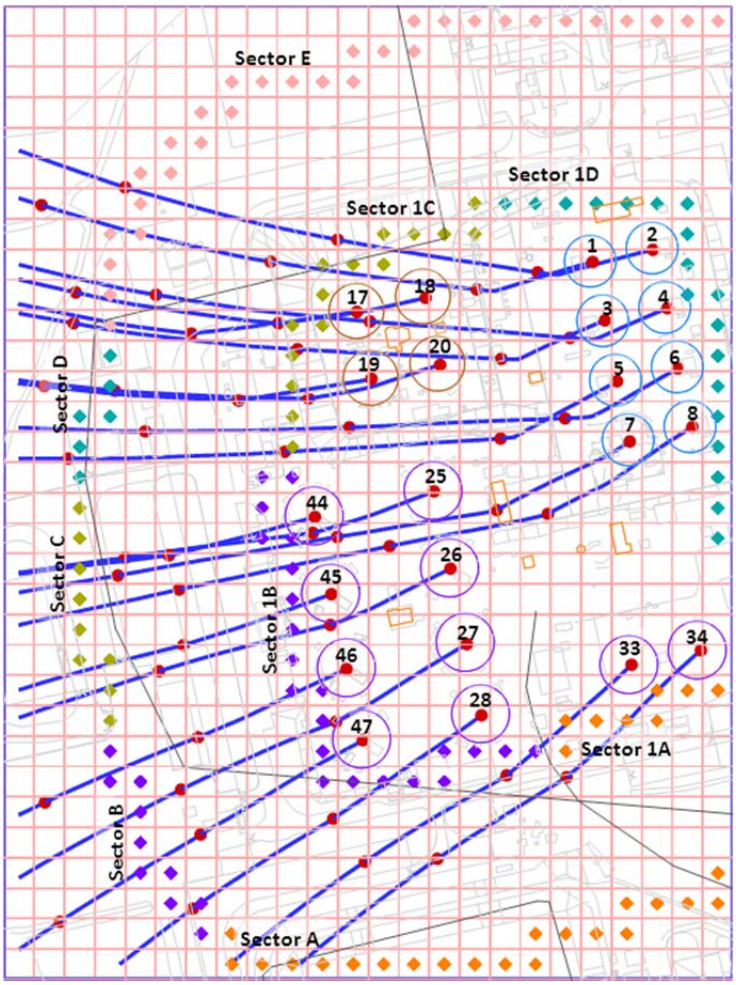
Section 5.2 (Environmental Transport of Radionuclides) of FTF PA presents the groundwater concentrations for the FTF radionuclides and chemicals. Maximum groundwater concentrations are presented for two exposure points: 1) 100 meters from the FTF and 2) at the seeplines for Upper Three Runs (UTR) and Fourmile Branch. Results are presented in FTF PA for the three distinct aquifers modeled, UTR Aquifer-Upper Zone (UTRA-UZ), UTRA-Lower Zone (UTRA-LZ), and Gordon Aquifer. The groundwater concentrations at 100 meters and at the seepline were recalculated using the PORFLOW FTF model for the FTF PA Base Case using the revised FTF inventories presented in Section 6.1.3 of this Special Analysis.

6.3.2.1 Groundwater Concentrations

Groundwater concentrations at 100 meters are calculated in the FTF PA using the PORFLOW FTF model, which divides the area around FTF into computational cells. Calculation of the 100-meter groundwater concentrations using the PORFLOW FTF model is discussed in more detail in Section 5.2.2 of the FTF PA. The PORFLOW 100-meter groundwater concentrations are calculated for five sectors (Sectors A through E) as shown on Figure 6.3-1. The peak concentration values for the 100-meter results are recorded for the three aquifers of concern (i.e., UTRA-UZ, UTRA-LZ, and Gordon Aquifer). The five sectors are analyzed for each radionuclide and chemical to find the maximum groundwater concentrations at 100 meters from the FTF.

The PORFLOW one-meter concentrations are calculated for four sectors (Sector 1A through 1D), as shown in Figure 6.3-1. Using the highest groundwater concentration from any sector to determine the calculated peak doses overstates the dose results. This is because the concentration for each radionuclide is determined independent of the location (i.e., horizontal mesh and aquifer) within the given sector.

Figure 6.3-1: PORFLOW FTF 1-Meter and 100-Meter Model Evaluation Sectors



Note The individual sectors are indicated by unique diamond colors.

Tables A-1 through A-3 (Appendix A) present the updated peak 100-meter radionuclide concentrations within 1,000 years in each sector for the three aquifers. These radionuclide concentrations reflect the peak concentrations for each radionuclide in the sector.

Tables A-1 through A-3 (Appendix A) also list the MCL for each constituent with the derived values for beta-gamma and photon emitters from Table II-3 of FR-00-9654. The MCLs provided in the reference are based on a beta-gamma dose of 4 mrem/yr. The peak concentration of each beta-gamma emitter is compared to a specific MCL to determine their fraction. To determine if the 4 mrem/yr beta-gamma limit is met, the sum of the fractions must be less than 1.0. The total alpha MCL includes Ra-226, but does not include radon or uranium. The radium MCL includes both Ra-226 and Ra-228. [SCDHEC R.61-58]

Tables A-4 through A-6 (Appendix A) show the peak updated 100-meter chemical concentrations within 1,000 years in each sector for the three aquifers. Similarly, Appendix B presents peak radionuclide concentrations within 10,000 years.

6.3.2.2 Sensitivity Run Radionuclide Determination

Section 5.2.2 of the FTF PA presents the methodology used to determine which radionuclides were most significant and to document which radionuclides would be considered a "sensitivity run radionuclide." While all radionuclides identified in the FTF waste tank inventory were included in 100-meter groundwater modeling efforts, narrowing the catalog of radionuclides down to a sensitivity run radionuclide list allowed the analysis to concentrate on the few radionuclides which posed more risk and concentrated modeling efforts on the areas of greatest concern. Only the sensitivity run radionuclides were included in the PORFLOW seepline modeling runs, with doses associated with seepline concentrations for the other radionuclides calculated by using 5 % of the applicable 100-meter groundwater concentrations. The 5 % factor is based on the ratio calculated between the bounding concentrations for the sensitivity run radionuclides at 100 meters and the seepline.

The sensitivity run radionuclides were recalculated based on the peak 100-meter groundwater concentrations determined using the revised FTF inventories. Any radionuclide with a dose (assuming FTF PA Base Case pathways and assumptions) greater than 0.1 mrem/yr was considered a sensitivity run radionuclide. The sensitivity run, radionuclide determination was conducted based on the peak 100-meter groundwater dose within 20,000 years.

The new screening conclusions are provided in Table 6.3-1. The resulting list of sensitivity run radionuclides (Am-241, Am-243, C-14, Cl-36, Cm-244, Cs-135, I-129, Nb-93m, Nb-94, Np-237, Pa-231, Pu-238, Pu-239, Pu-240, Pu-241, Ra-226, Tc-99, Th-230, U-234, U-235, U-238, and Zr-93) is different from the FTF PA, in that Th-229 and U-233 were deleted and Cl-36, Nb-93m, Nb-94, Pu-241, U-238, Zr-93 were added. This revised set is intended to capture those radionuclides with the greatest potential to influence dose results, based on the revised inventories.

Table 6.3-1: Determination of Sensitivity Run Radionuclides

Radionuclide	Sector A 20k Peak Dose (mem/yr)	Sector B 20k Peak Dose (mem/yr)	Sector C 20k Peak Dose (mem/yr)	Sector D 20k Peak Dose (mem/yr)	Sector E 20k Peak Dose (mem/yr)	Key Cause
Ac-227	0.00	0.00	0.00	0.00	0.00	N/A
Al-26	0.00	0.00	0.00	0.00	0.00	N/A
Am-241	0.00	0.00	0.00	0.00	0.00	Np-237 parent
Am-242m	0.00	0.00	0.00	0.00	0.00	N/A
Am-243	0.00	0.00	0.00	0.00	0.00	Pu-239 parent
C-14	0.12	0.24	0.23	1.38	0.46	Dose > 0.1
Cf-249	0.00	0.00	0.00	0.00	0.00	N/A
Cl-36	0.15	0.42	1.43	2.07	0.95	Dose > 0.1
Cm-243	0.00	0.00	0.00	0.00	0.00	N/A
Cm-244	0.00	0.00	0.00	0.00	0.00	Pu-240 parent
Cm-245	0.00	0.00	0.00	0.00	0.00	N/A
Cm-247	0.00	0.00	0.00	0.00	0.00	N/A
Cm-248	0.00	0.00	0.00	0.00	0.00	N/A
Co-60	0.00	0.00	0.00	0.00	0.00	N/A
Cs-135	0.67	0.68	0.44	0.43	0.43	Dose > 0.1
Cs-137	0.00	0.00	0.00	0.00	0.00	N/A
Eu-152	0.00	0.00	0.00	0.00	0.00	N/A
Eu-154	0.00	0.00	0.00	0.00	0.00	N/A
H-3	0.00	0.00	0.00	0.00	0.00	N/A
I-129	0.06	0.21	0.89	1.06	0.96	Dose > 0.1
K-40	0.01	0.01	0.02	0.05	0.08	N/A
Mo-93m	0.00	0.00	0.00	0.00	0.00	N/A
Nb-93m	13.38	28.45	29.12	25.34	23.78	Dose > 0.1
Nb-94	0.38	0.74	0.72	0.46	0.56	Dose > 0.1
Ni-59	0.01	0.01	0.02	0.03	0.02	N/A
Ni-63	0.00	0.00	0.00	0.00	0.00	N/A
Np-237	1.30	3.43	13.97	16.71	15.03	Dose > 0.1
Pa-231	0.08	0.19	0.29	0.70	1.27	Dose > 0.1
Pb-210	0.01	0.03	0.08	0.08	0.05	N/A
Pd-107	0.00	0.00	0.00	0.00	0.00	N/A
Pu-238	0.00	0.00	0.00	0.00	0.00	U-234 parent
Pu-239	0.00	0.01	0.05	0.83	1.13	Dose > 0.1
Pu-240	0.00	0.00	0.04	0.60	0.64	Dose > 0.1
Pu-241	0.00	0.00	0.00	0.00	0.00	Np-237 parent
Pu-242	0.00	0.00	0.00	0.02	0.02	N/A
Pu-244	0.00	0.00	0.00	0.00	0.00	N/A
Ra-226	1.58	5.03	12.35	12.56	7.68	Dose > 0.1
Ra-228	0.00	0.00	0.00	0.00	0.00	N/A

Table 6.3-1: Determination of Sensitivity Run Radionuclides (Continued)

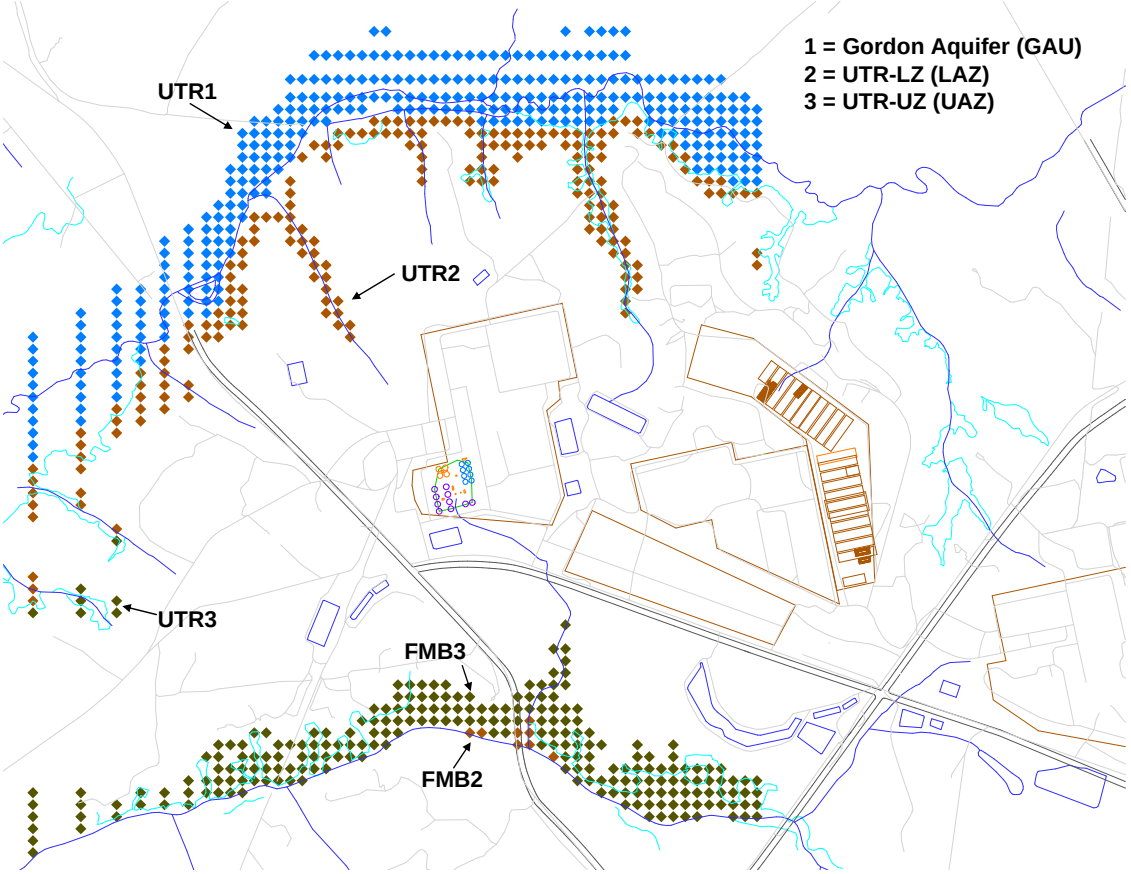
Radionuclide	Sector A 20k Peak Dose (mem/yr)	Sector B 20k Peak Dose (mem/yr)	Sector C 20k Peak Dose (mem/yr)	Sector D 20k Peak Dose (mem/yr)	Sector E 20k Peak Dose (mem/yr)	Key Cause
Rn-222	0.04	0.12	0.31	0.31	0.19	Excluded [a]
Se-79	0.00	0.00	0.00	0.00	0.00	N/A
Sm-151	0.00	0.00	0.00	0.00	0.00	N/A
Sn-126	0.00	0.00	0.00	0.00	0.00	N/A
Sr-90	0.00	0.00	0.00	0.00	0.00	N/A
Tc-99	0.05	0.10	0.10	0.29	0.36	Dose > 0.1
Th-228	0.00	0.00	0.00	0.00	0.00	N/A
Th-229	0.00	0.00	0.00	0.05	0.09	N/A
Th-230	0.00	0.00	0.00	0.04	0.08	Ra-226 parent
Th-232	0.00	0.00	0.00	0.00	0.00	N/A
U-232	0.00	0.00	0.00	0.00	0.00	N/A
U-233	0.00	0.00	0.00	0.04	0.07	N/A
U-234	0.00	0.00	0.02	0.70	1.15	Dose > 0.1
U-235	0.00	0.00	0.00	0.00	0.00	Pa-231 parent
U-236	0.00	0.00	0.00	0.02	0.03	N/A
U-238	0.00	0.00	0.00	0.00	0.00	U-234 parent
Zr-93	0.00	0.00	0.00	0.00	0.00	Nb-93m parent

N/A Not Applicable
Note Sensitivity Run Radionuclides are shaded in gray.
a Although the dose contribution from Rn-222 exceeds 0.1 mrem/yr, this radionuclide is excluded from further groundwater analysis as this radionuclide is short-lived with a half-life of less than four days.

6.3.2.3 Groundwater Concentrations at the Seeplines

The seepline groundwater concentrations were calculated using the PORFLOW FTF model. Figure 6.3-2 shows that the PORFLOW seepline concentrations are provided for five evaluation sectors (UTR1, UTR2, UTR3, FMB1, and FMB2). These evaluation sectors are based on stream locations (UTR or FMB) and aquifer depths (1 = Gordon, 2 = UTR-LZ, and 3 = UTR-UZ) that represent the nearest downstream seeplines to the FTF. The peak concentration values for the seepline results were recorded for each evaluation sector.

Figure 6.3-2: PORFLOW FTF Seepline Evaluation Sectors



Tables A-7 (UTR) and A-8 (Fourmile Branch) in Appendix A present the updated peak seepline, radionuclide concentrations out to 1,000 years. These radionuclide concentrations reflect the peak concentrations for each radionuclide in the highest sector. These values are conservatively high for the radionuclides present in multiple decay chains because the totals are simply the sum of the individual peaks within that sector for a given radionuclide without regard to time or location. Similarly, Tables B-7 (UTR) and B-8 (Fourmile Branch) in Appendix B presents peak radionuclide concentrations within 10,000 years.

6.3.3 Air Pathway and Radon Analysis

Section 5.3 of FTF PA presents the air pathway and radon analysis results. Section 4.5 of FTF PA describes the method used to conservatively bound the dose from airborne radionuclides and the results in that section provides a dose to maximally exposed individual (MEI) per curie of inventory. The approach described in the FTF PA assumed 0 % saturation of the concrete tank roofs, tank grout, and stabilized contaminant layers. For this Special Analysis, an alternative assumption of 50 % saturation more closely reflects expected conditions. This alternative assumption is consistent with more recent PAs (e.g., the HTF PA). [SRR-CWDA-2010-00128]

Table 6.3-2 summarizes the total waste tank and ancillary equipment inventory of selected potentially airborne (i.e., gaseous) isotopes, including the Tanks 5, 6, 17, 18, 19, and 20

actual residual inventories at operational closure presented in Section 6.1.3. Table 6.3-3 presents FTF inventories, specific SRS 100-meter dose release factors (DRFs) and the calculated exposure levels for the 100-year to 10,000-year MEI at 100 meters. Table 6.3-4 presents specific SRS 1,600-meter (seepline) DRFs and the calculated exposure levels for the 10,000-year MEI at 1,600 meters. The contribution of Sb-125 to the air pathways dose is insignificant based on the revised waste tank inventory and the short half-life of Sb-125, and is not included in Tables 6.3-3 and 6.3-4.

Table 6.3-2: Summary of Projected Total FTF Inventory of Gaseous Radionuclides

Inventory Location	C-14 (Ci)	Cl-36 (Ci)	H-3 (Ci)	I-129 (Ci)	Se-79 (Ci)	Sn-126 (Ci)	Tc-99 (Ci)
All Waste Tanks	2.0E+01	4.3E-02	2.2E+01	1.9-02	3.8E+01	6.2E+01	1.1E+01
Transfer Lines	1.5E-03	ND	1.1E-01	2.5E-06	3.0E-02	5.6E-02	5.3E-01
FPT-1	8.0E-06	ND	1.5E-04	1.3E-08	1.6E-04	3.0E-04	2.8E-03
FPT-2	8.0E-06	ND	1.5E-04	1.3E-08	1.6E-04	3.0E-04	2.8E-03
FPT-3	8.0E-06	ND	1.5E-04	1.3E-08	1.6E-04	3.0E-04	2.8E-03
FTF Catch Tank	3.8E-05	ND	8.6E-04	7.2E-08	8.7E-04	1.6E-03	1.5E-02
242-3F CTS	ND	ND	6.7E-02	ND	1.8E-06	ND	1.1E-01
Evaporator Vessel - 242-F	ND	ND	ND	ND	7.7E-09	ND	1.3E-03
Evaporator Vessel - 242-16F	ND	ND	ND	ND	7.7E-09	ND	1.3E-03
Total FTF Inventory	**2.1E+01**	**4.3E-02**	**2.2E+01**	**1.9E-02**	**3.8E+01**	**6.2E+01**	**1.2E+01**

ND Not determined

Table 6.3-3: 100-Meter DRFs and 10,000-Year FTF Dose

Radionuclide	Peak Flux (Ci/yr/Ci)[a]	SRS 100-meter DRF[b] (mrem/Ci)	Dose to MEI at 100-meter Boundary[c] (mrem/yr/Ci)	FTF Inventory (Ci)	Dose to MEI at 100-meter Boundary (mrem/yr)
C-14	1.62E-05	1.3E-02	2.2E-07	2.1E+01	4.6E-06
Cl-36	9.53E-19	2.9E-02	2.7E-20	4.3E-02	1.2E-21
H-3	6.33E-10	2.8E-04	1.8E-13	2.2E+01	3.9E-12
I-129	4.11E-22	2.0E+01	8.4E-21	1.9E-02	1.6E-22
Se-79	5.51E-12	3.8E-02	2.09E-13	3.8E+01	7.9E-12
Sn-126	3.88E-61	1.8E+01	6.8E-60	6.2E+01	4.2E-58
Tc-99	2.82E-47	1.1E-01	3.0E-48	1.2E+01	3.6E-47
				Total Dose	**4.6E-06**

a From SRR-CWDA-2010-00128, Table 4.5-7.
b From WSRC-STI-2007-00343, Table 7.
c Dose to MEI = Peak Flux × DRF

Table 6.3-4: 1,600-Meter DRFs and 10,000-Year FTF Dose

Radionuclide	Peak Flux (Ci/yr/Ci)[a]	SRS 1,600-meter DRF[b] (mrem/Ci)	Dose to MEI at 1,600-meter Boundary[c] (mrem/yr/Ci)	FTF Inventory (Ci)	Dose to MEI at 1,600-meter Boundary (mrem/yr)
C-14	1.62E-05	2.4E-03	3.9E-08	2.1E+01	8.3E-07
Cl-36	9.53E-19	6.2E-03	5.9E-21	4.3E-02	2.6E-22
H-3	6.33E-10	4.9E-05	3.1E-14	2.2E+01	6.9E-13
I-129	4.11E-22	2.3E+00	9.5E-22	1.9E-02	1.8E-23
Se-79	5.51E-12	9.4E-03	5.2E-14	3.8E+01	2.0E-12
Sn-126	3.88E-61	4.4E+00	1.7E-60	6.2E+01	1.1E-58
Tc-99	2.82E-47	2.6E-02	7.3E-49	1.2E+01	8.8E-48
				Total Dose	**8.3E-07**

a From SRR-CWDA-2010-00128, Table 4.5-7.
b From WSRC-STI-2007-00343, Table 7.
c Dose to MEI = Peak Flux × DRF.

The instantaneous radon flux is determined by multiplying the peak flux by the total inventory divided by the total area in the FTF. Table 6.3-5 summarizes the total inventory of isotopes contributing to the radon flux. The peak dose of radon for the period is assumed to be at 10,000 years. These results are highly conservative because the entire inventory is assumed concentrated in a 1-foot layer in a Type I tank. As shown in Table 6.3-6, the peak instantaneous radon flux using the entire FTF inventory (presented in Section 6.1.3) is 1.6E-16 pCi/m^2/s.

Table 6.3-5: Summary of Total Projected FTF Inventory of Isotopes Producing Rn-222

Inventory Location	Pu-238 (Ci)	U-238 (Ci)	U-234 (Ci)	Th-230 (Ci)	Ra-226 (Ci)
Total Waste Tank	8.1E+03	2.0E+00	1.9E+00	1.9E-01	2.0E-01
Transfer Lines	6.5E+00	5.0E-03	8.5E-03	5.1E-03	5.1E-03
FPT-1	3.4E-02	2.6E-05	4.5E-05	2.7E-05	2.7E-05
FPT-2	3.4E-02	2.6E-05	4.5E-05	2.7E-05	2.7E-05
FPT-3	3.4E-02	2.6E-05	4.5E-05	2.7E-05	2.7E-05
FTF Catch Tank	1.0E-01	9.0E-05	9.0E-05	1.3E-04	1.3E-04
242-3F CTS	9.2E-01	3.9E-04	2.1E-03	N/A	N/A
242-F Evaporator	4.7E-03	7.5E-06	7.1E-06	N/A	N/A
242-16F Evaporator	4.7E-03	7.5E-06	7.1E-06	N/A	N/A
Total FTF Inventory	**8.1E+03**	**2.0E+00**	**1.9E+00**	**1.9E-01**	**2.1E-01**

Table 6.3-6: Peak Instantaneous Rn-222 Flux at Land Surface

Parent Source	FTF Inventory (Ci)	FTF Inventory (Ci/m^2) [a]	Peak Instantaneous Rn-222 flux at Land Surface	
			($pCi/m^2/sec$) / (Ci/m^2)[b]	($pCi/m^2/sec$)
Pu-238	8.1E+03	9.8E-02	5.01E-16	4.91E-17
U-238	2.0E+00	2.4E-05	1.72E-14	4.12E-19
U-234	1.9E+00	2.3E-05	1.42E-12	3.27E-17
Th-230	1.9E-01	2.3E-06	1.19E-11	2.79E-17
Ra-226	2.1E-01	2.5E-06	2.08E-11	5.19E-17
			Total	**1.6E-16**

a Total area of FTF is 82,910 square meters
b From SRR-CWDA-2010-00128, Table 4.5-15.

6.3.4 Biotic Pathways

Section 5.4 of FTF PA describes how the biotic pathways doses to the MOP are calculated for the receptor with 100-meter well water as a primary water source and for the receptor with groundwater from a stream as a primary water source. The information regarding biotic pathways calculations presented in the FTF PA is not affected by the new residual waste information.

6.3.5 Dose Analysis of MOP Exposure

Section 5.5 (Dose Analysis) of FTF PA contains calculations of the peak total doses for a) the MOP at the 100-meter well and b) the MOP at applicable streams (either UTR or Fourmile Branch). The peak doses have been recalculated using the peak groundwater concentrations identified in Section 6.3.2 of this Special Analysis (i.e., the groundwater concentrations at 100 meters and at the seepline). The doses were recalculated using the PORFLOW FTF model for the FTF PA Base Case using the revised FTF inventories.

6.3.5.1 Member of the Public at 100-Meter Groundwater Pathway Dose Results

Table 6.3-7 shows a comparison of the 100-meter peak groundwater pathway doses (recalculated using the concentration associated with the revised FTF inventories) for the different 100-meter sectors within 1,000 and 10,000 years. In calculating the peak groundwater pathway dose, the highest radionuclide concentration within each sector's vertical computational mesh from each of the three distinct aquifers modeled (UTRA-UZ, UTRA-LZ, and the Gordon Aquifer), is used.

Table 6.3-7: 100-Meter MOP Peak Groundwater Pathways Dose by Sector

Sector	Highest Peak Dose in 1,000 Years	Highest Peak Dose in 10,000 Years
A	**0.1 mrem/yr (year 752)** Principal Pathways: Water Ingestion Vegetable Ingestion Principal Radionuclide: Tc-99	**0.1 mrem/yr (year 752)** Principal Pathways: Water Ingestion Vegetable Ingestion Principal Radionuclide: Tc-99
B	**0.1 mrem/yr (year 754)** Principal Pathways: Water Ingestion Vegetable Ingestion Principal Radionuclide: Tc-99	**0.1 mrem/yr (year 754)** Principal Pathways: Water Ingestion Vegetable Ingestion Principal Radionuclide: Tc-99
C	**0.1 mrem/yr (year 740)** Principal Pathways: Water Ingestion Vegetable Ingestion Principal Radionuclides: Tc-99	**0.2 mrem/yr (year 4,306)** Principal Pathways: Water Ingestion Vegetable Ingestion Principal Radionuclides: C-14, Ra-226
D	**0.3 mrem/yr (year 704)** Principal Pathways: Water Ingestion Vegetable Ingestion Principal Radionuclides: Tc-99	**1.8 mrem/yr (year 6,056)** Principal Pathways: Water Ingestion Vegetable Ingestion Principal Radionuclides: Np-237, Pa-231, Ra-226
E	**0.4 mrem/yr (year 704)** Principal Pathways: Water Ingestion Vegetable Ingestion Principal Radionuclides: Tc-99	**3.3 mrem/yr (year 10,000)** Principal Pathways: Water Ingestion Vegetable Ingestion Principal Radionuclides: Np-237, Pa-231, Ra-226

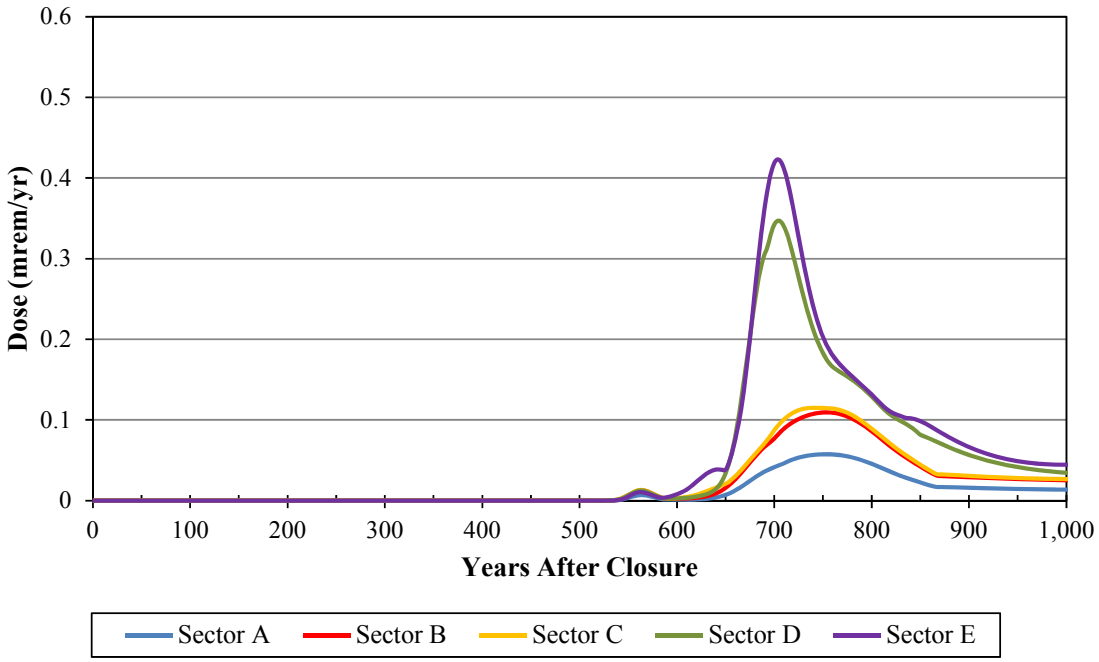
Note: Dose values are shown to two significant figures to illustrate changes/trends. The use of two significant figures is not meant to imply this level of precision for these dose projections.

The highest peak-groundwater pathway dose in 10,000 years remains associated with Sector E. Sector E is the sector associated most closely with the Type IV tanks, which are the waste tanks whose liners are modeled as failing earlier than any other waste tank type (and within 10,000 years).

Figures 6.3-3 and 6.3-4 show the peak doses to the MOP receptor for the five 100-meter sectors in 1,000 years and in 10,000 years, respectively. The highest 100-meter MOP-groundwater pathway peak dose in 10,000 years is a 3.3 mrem/yr[2] dose at year 10,000. The peak dose in 10,000 years unchanged from the updated Base Case results previously reported in the Tank 18/19 Special Analysis. [SRR-CWDA-2010-00124] The Tank 18/19 Special Analysis Base Case results differed from the FTF PA results because they reflected Tank 18 and Tank 19 final inventories at closure rather than inventory projections.

Figure 6.3-5 shows the 100-meter MOP receptor doses within 100,000 years for the five 100-meter sectors. The 100-meter MOP peak groundwater pathway peak dose within 100,000 years for the peak sector (Sector E) is shown in Figure 6.3-5 (linear y-axis scale) and Figure 6.3-6 (logarithmic y-axis scale).

Figure 6.3-3: 100-Meter MOP Peak Groundwater Pathway Dose within 1,000 Years for the Five 100-Meter Sectors



[2] Dose values are shown to two significant figures to illustrate changes/trends. The use of two significant figures is not meant to imply this level of precision for these dose projections.

Figure 6.3-4: 100-Meter MOP Peak Groundwater Pathway Dose within 10,000 Years for the Five 100-Meter Sectors

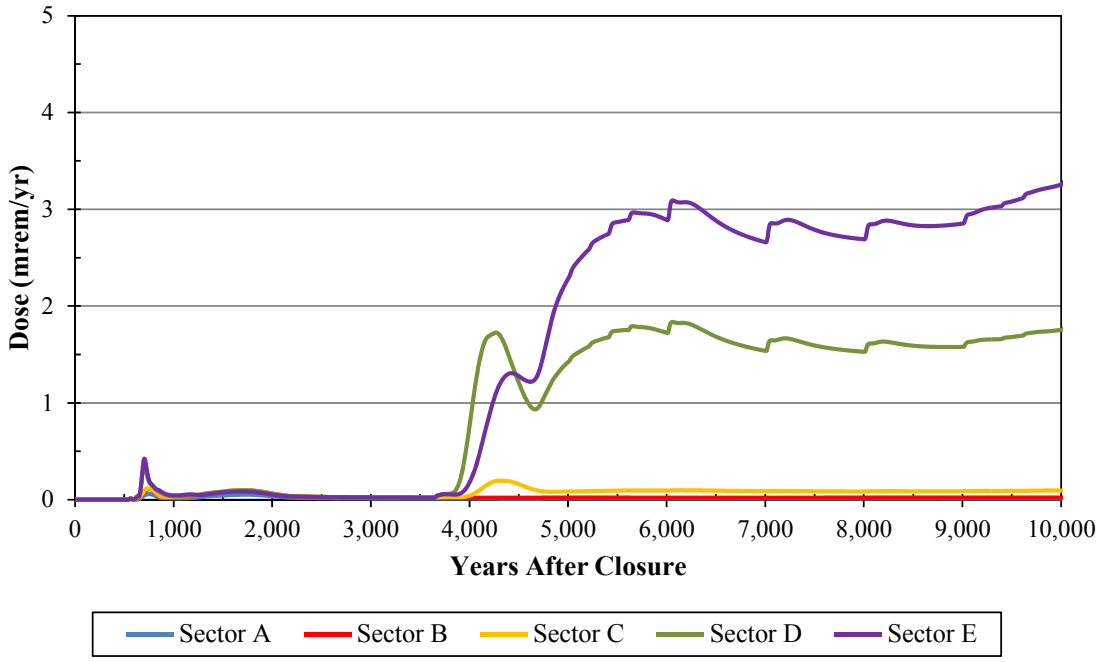


Figure 6.3-5: 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for the Five 100-Meter Sectors (linear scale)

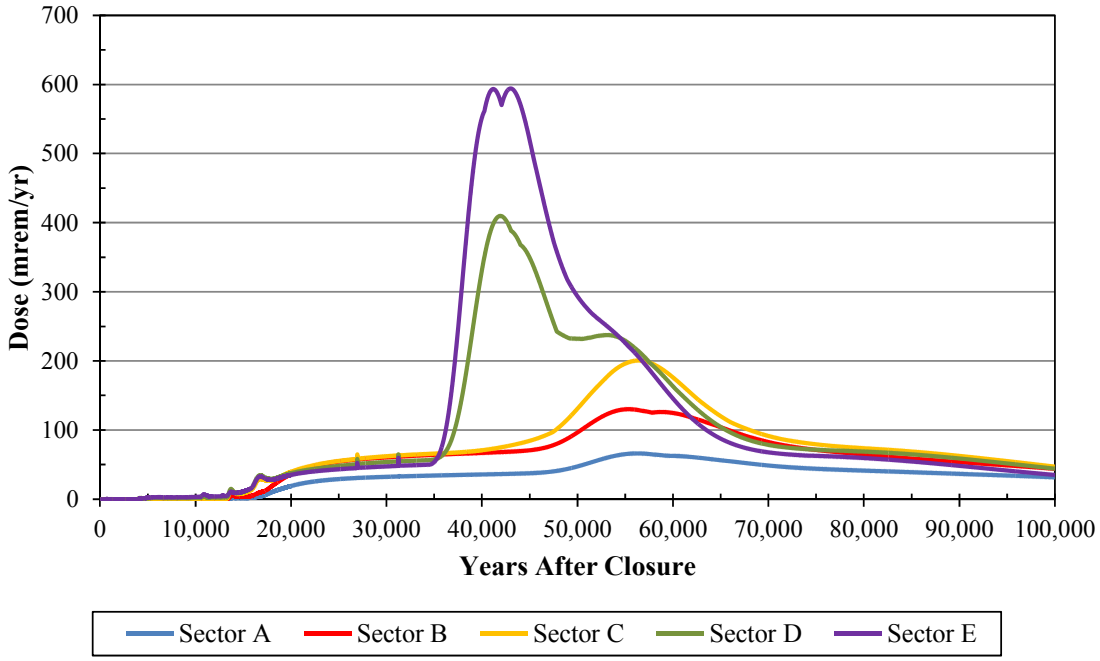
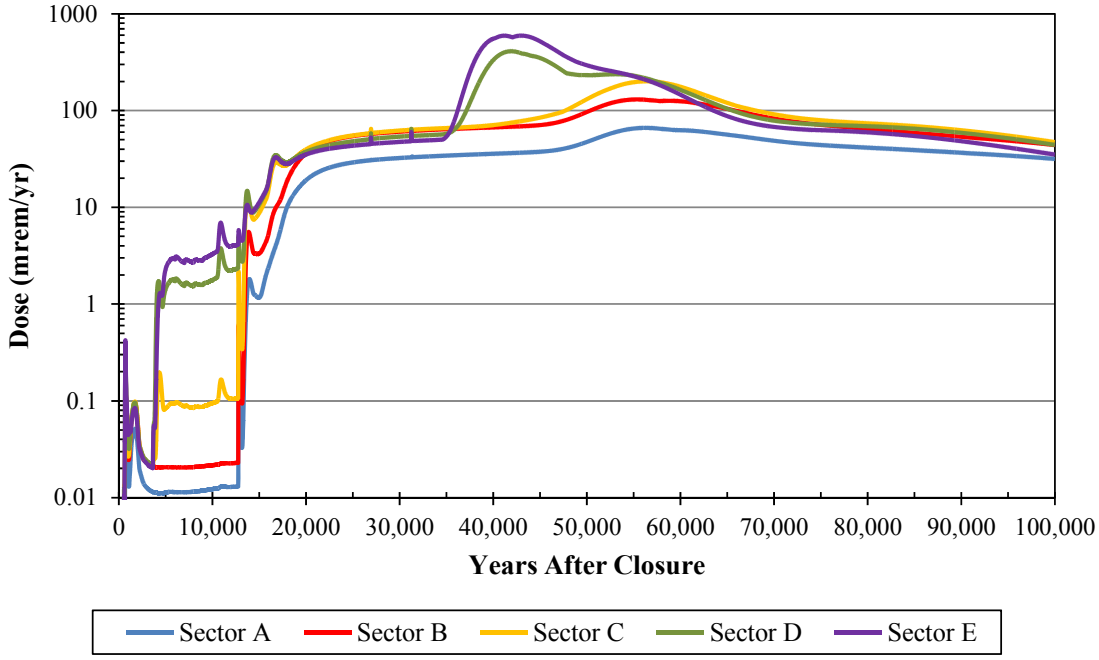


Figure 6.3-6: 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for the Five 100-Meter Sectors (log scale)



Using the actual residual inventories at operational closure for Tanks 5, 6, 17, 18, 19, and 20, as well as the revised estimates for the other waste tanks, resulted in no significant change to the peak groundwater pathway doses in 10,000 years, relative to the Tank 18/19 Special Analysis. The peak groundwater pathway dose in 100,000 years increased relative to the Tank 18/19 Special Analysis, which reported a peak of approximately 500 mrem/yr at approximately 40,000 years. The increased Zr-93 inventory projected in the Type I tanks resulted in an increased dose contribution from the Nb-93m (a decay product of Zr-93), thus increasing the peak 100,000-year dose to nearly 600 mrem/yr.

This dose represents conservative modeling assumptions that were incorporated into the FTF PA Base due to uncertainty inherent in a number of the modeling parameters. Since the development of the FTF PA Base Case, additional studies have been performed that have improved understanding of the waste release model and of specific parameters related to contaminant transport. [SRNL-STI-2012-00404, SRNL-STI-2011-00011, SRNL-STI-2011-00672, ML073510127] This more recent data was incorporated into a Composite Sensitivity Study (see Section 6.3.6.3). This Composite Sensitivity Study shows that the dose results from both Pu-239 and Nb-93m are conservative in the Base Case due to modeling assumptions. The updated parameters result in a lower peak 100,000-year dose (with a peak of only about 20 mrem/yr), driven by Np-237 and Ra-226 releases.

6.3.5.2 Individual Radionuclide Contributions to the MOP 100-Meter Peak Annual Groundwater Pathway Dose

Figures 6.3-7 through 6.3-10 show the relative contributions from individual radionuclides to the Sector E 100-meter groundwater pathway dose over time. Table 6.3-8 presents radionuclide contributions to the 3.3 mrem/yr[3] peak groundwater pathway dose. The peak groundwater pathway dose to the MOP at 100-meters in 10,000 years is primarily associated with Ra-226 (52 %), Np-237 (17 %), Pa-231 (15 %), and U-234 (13 %). The individual contributions to the peak groundwater pathway dose in 10,000 years are essentially unchanged (relative to the Tank 18/19 Special Analysis) when the actual residual inventories at operational closure are used for Tanks 5, 6, 17, 18, 19, and 20.

The increase to the peak dose in the first 10,000 years, relative to the FTF PA, associated with Ra-226 is due to the Pu-238 inventory at operational closure in Tank 18 (Pu-238 is a parent to Ra-226), which is higher than the projected inventory used in the FTF PA. The peak dose in 10,000 years associated with Pa-231 is driven by the Pa-231 inventory at operational closure in Tank 18 differing from the projected inventory.

As shown in Figure 6.3-9, the Tc-99 contribution to the peak dose is significantly less than it was in the FTF PA and the Tank 18/19 Special Analysis due to the smaller Tc-99 inventory in the Type I waste tanks. At approximately 18,000 years after closure (and continuing throughout the duration of the model), the Nb-93m released from the Type I waste tanks contributes 20 to 50 mrem/yr. Additional sensitivity analyses were performed as part of this Special Analysis to better assess the potential impacts associated with Nb-93m (see Section 6.3.6.3). The Composite Sensitivity Study used more probable and defensible inputs, providing greater confidence that the FTF PA results and conclusions remain valid to demonstrate that the FTF PA Base Case incorporates conservative approaches/inputs related to the release and transport of Nb-93m and Pu-239.

[3] Dose values are shown to two significant figures to illustrate changes/trends. The use of two significant figures is not meant to imply this level of precision for these dose projections.

Figure 6.3-7: Individual Radionuclide Contributors to the Sector E 100-Meter Peak Groundwater Pathway Dose, 1,000 years

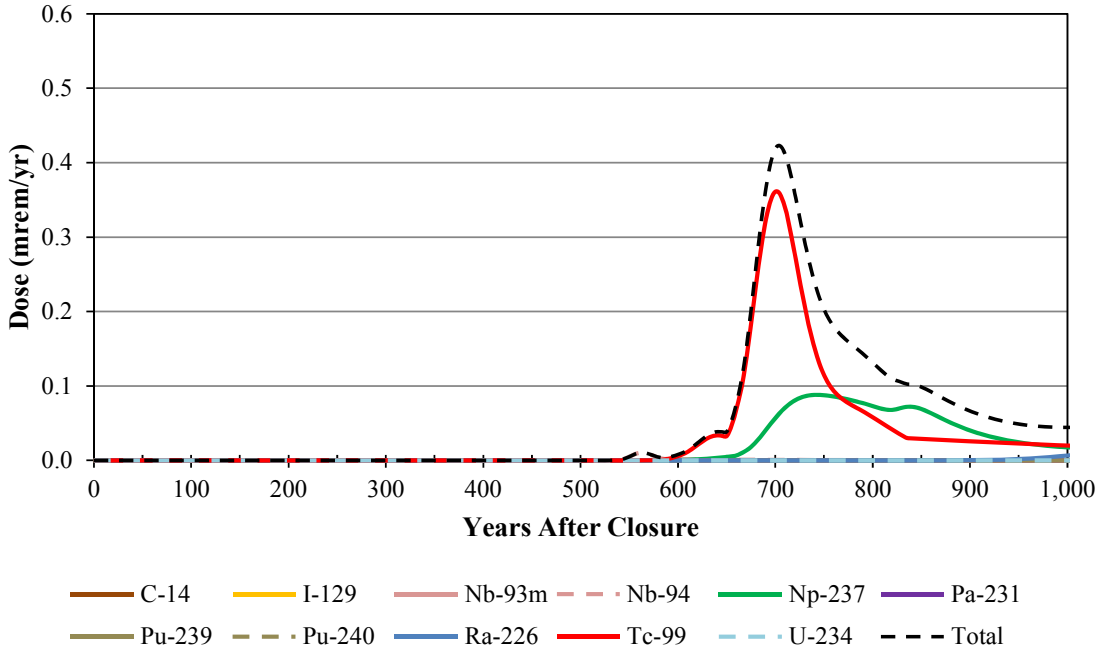


Figure 6.3-8: Individual Radionuclide Contributors to the Sector E 100-Meter Peak Groundwater Pathway Dose, 10,000 years

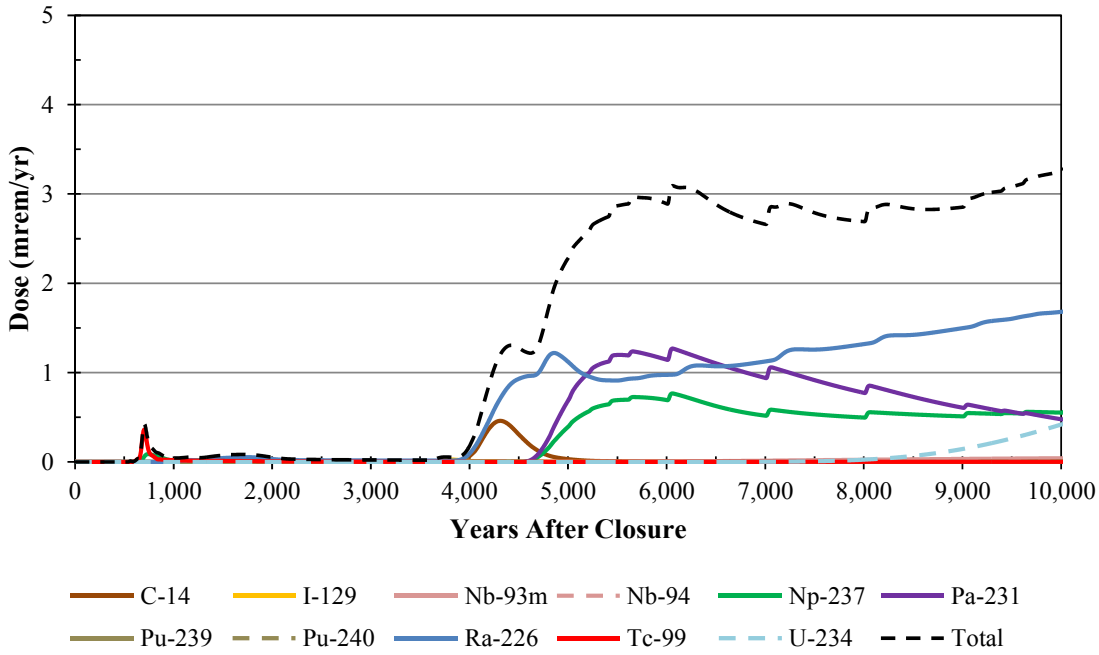


Figure 6.3-9: Individual Radionuclide Contributors to the Sector E 100-Meter Peak Groundwater Pathway Dose, 100,000 Years (linear scale)

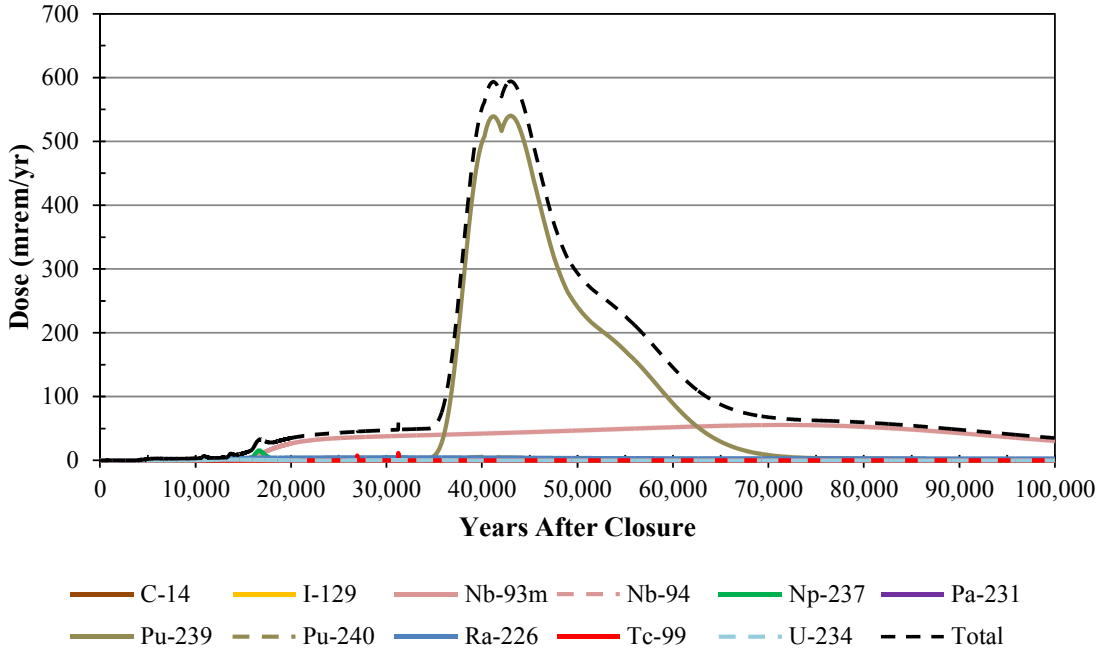


Figure 6.3-10: Individual Radionuclide Contributors to the Sector E 100-Meter Peak Groundwater Pathway Dose, 100,000 Years (log scale)

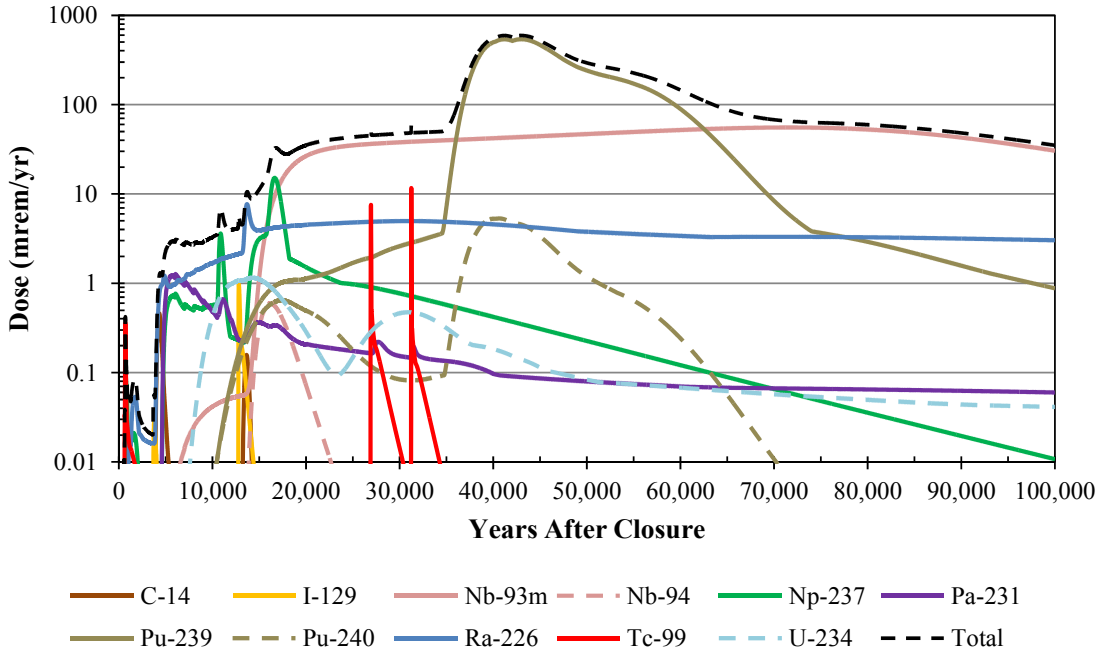


Table 6.3-8: 100-Meter MOP Peak Groundwater Pathway Dose Individual Radionuclide Contributions at Year 10,000 (Peak Year)

Radionuclide	Contribution to Sector E Peak dose at year 10,000 (mrem/yr)	Percentage of Total Peak Dose
Am-241	<0.1	<1 %
Am-243	<0.1	<1 %
C-14	<0.1	<1 %
Cl-36	<0.1	<1 %
Cm-244	<0.1	<1 %
Cs-135	<0.1	<1 %
I-129	<0.1	<1 %
Nb-93m	<0.1	1 %
Nb-94	<0.1	<1 %
Np-237	0.55	17 %
Pa-231	0.48	15 %
Pu-238	<0.1	<1 %
Pu-239	<0.1	<1 %
Pu-240	<0.1	<1 %
Pu-241	<0.1	<1 %
Ra-226	1.68	52 %
Tc-99	<0.1	<1 %
Th-230	<0.1	<1 %
U-234	0.42	13 %
U-235	<0.1	<1 %
U-238	<0.1	<1 %
Zr-93	<0.1	<1 %
Total	**3.3**	**100 %**

6.3.5.3 Individual Tank Contributions to MOP 100-Meter Peak Annual Groundwater Pathway Dose

The following describes the impacts of the updated inventories for Tanks 5 and 6. Although Sector E provides the highest doses when considering all of the sources within the FTF (see Section 6.3.5.1), Tanks 5 and 6 provide greater impacts on Sector D, therefore this discussion focuses on the Sector D dose results. Since the liners for Tanks 5 and 6 are not projected to fail in the Base Case within 10,000 years, the contributors to peak dose within 10,000 years are unchanged from previous analyses. [SRR-CWDA-2010-00124]

Table 6.3-9 presents the relative contributions from those residual inventories (Tanks 5 and 6) which will contribute to the Sector D 100-meter MOP groundwater pathway dose. Tanks 5 and 6 are contributors (approximately 28 % and 16 %, respectively) to the 100-meter peak groundwater-pathway dose in Sector D within 20,000 years. Figures 6.3-11 and 6.3-12 show the Sector D 100-meter MOP peak groundwater-pathway dose associated with the Tank 5 residual inventory for 20,000 years and 100,000 years, respectively. Similarly, Figures 6.3-13 and 6.3-14 show the Sector D 100-meter MOP peak groundwater-pathway dose associated with the Tank 6 residual inventory for 20,000 years and 100,000 years, respectively. Since Tanks 5 and 6 are modeled as having liners that remain intact during the first 10,000 years after FTF closure, the 20,000-year and 100,000-year timeframes were chosen to better understand barrier behavior and to illustrate peak doses.

Comparison of the Sector D 100-meter MOP peak groundwater pathway dose for all sources and for all sources except Tank 5 and Tank 6 is shown in Figure 6.3-15 (20,000 years), Figure 6.3-16 (100,000 years, linear y-axis scale) and Figure 6.3-17 (100,000 years, logarithmic y-axis scale).

Table 6.3-9: 100-Meter MOP Peak Groundwater Pathway Dose Individual Source Contributions at Peak Years within 20,000 Years for Sector D

Waste Source	Contribution (mrem/yr) to Sector D Peak Dose at Year 20,000	Percentage of Sector D Peak Dose at Year 20,000
Tank 5	10.4	28%
All Except Tank 5[a]	31.5	84%
Tank 6	6.0	16%
All Except Tank 6[a]	33.5	89%
Total[b]	**37.6**	**100%**

a. As with the Base Case MOP dose calculations for the 100-meter boundary, the "All Except Tank 5" and "All Except Tank 6" dose calculations also include the impacts from stream related activities (such as fish ingestion and swimming exposure). However, these dose calculations, conservatively used the seepline concentrations from the Base Case (All Sources) run (i.e., the dose contributions from the seepline include releases from Tanks 5 and 6).

b. During each PORFLOW run, concentrations are selected from the point of highest concentration within each sector for the specific run. These locations may vary between source runs, such that the sum of individual contaminant sources (e.g., Tank 5, Tank 6) may exceed the total calculated dose. This also means that the percentages reflected in this table reflect an overestimate, exceeding 100 %.

Figure 6.3-11: 100-Meter MOP Peak Groundwater Pathway Dose within 20,000 Years for Sector D – with Tank 5 Contribution

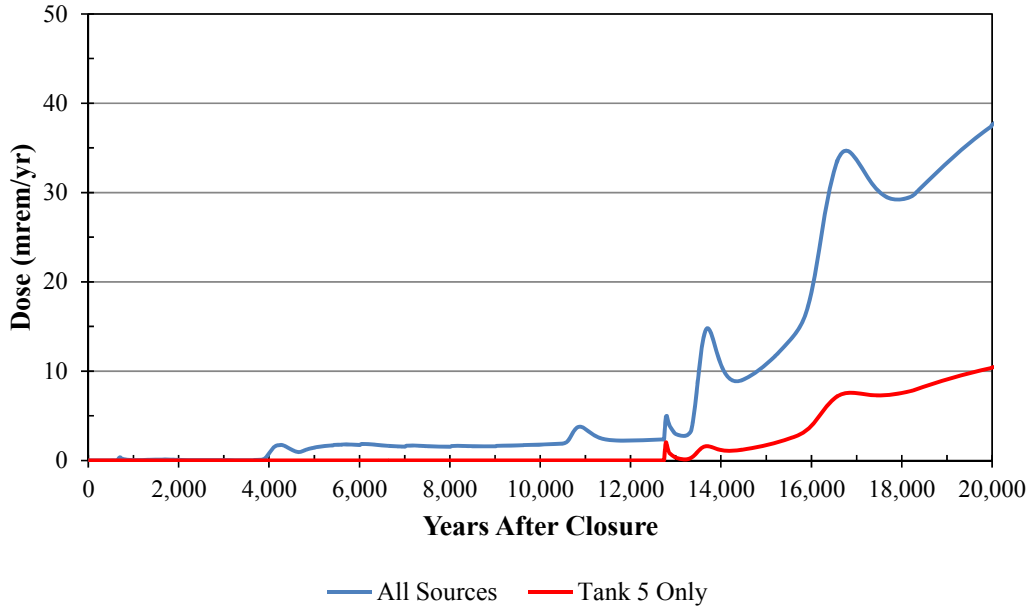


Figure 6.3-12: 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for Sector D – with Tank 5 Contribution

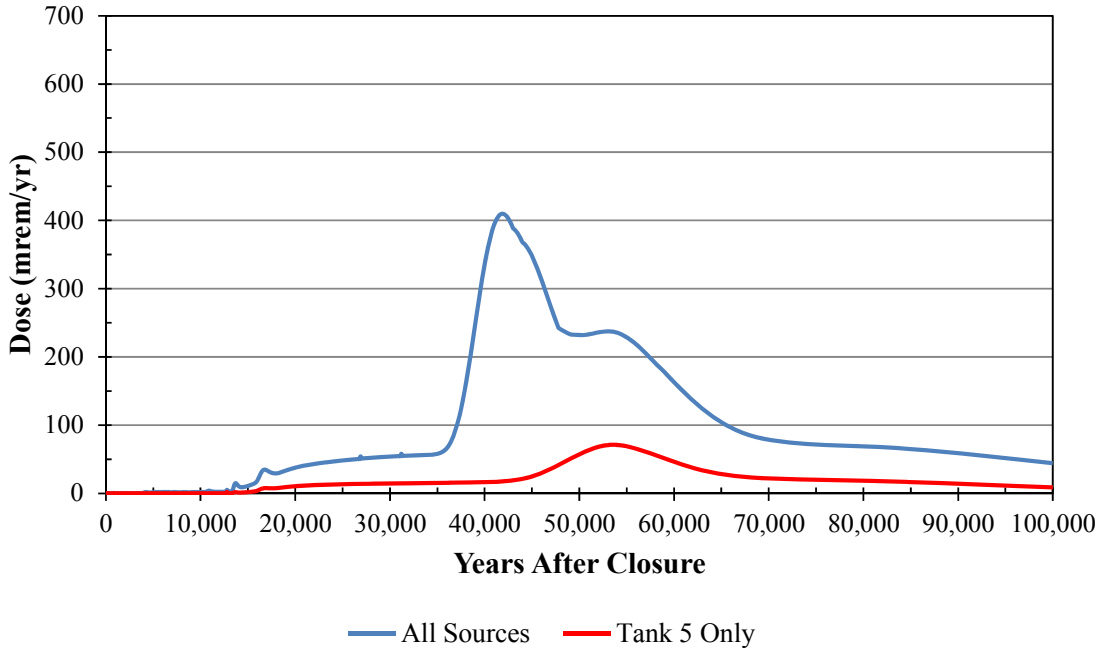


Figure 6.3-13: 100-Meter MOP Peak Groundwater Pathway Dose within 20,000 Years for Sector D – with Tank 6 Contribution

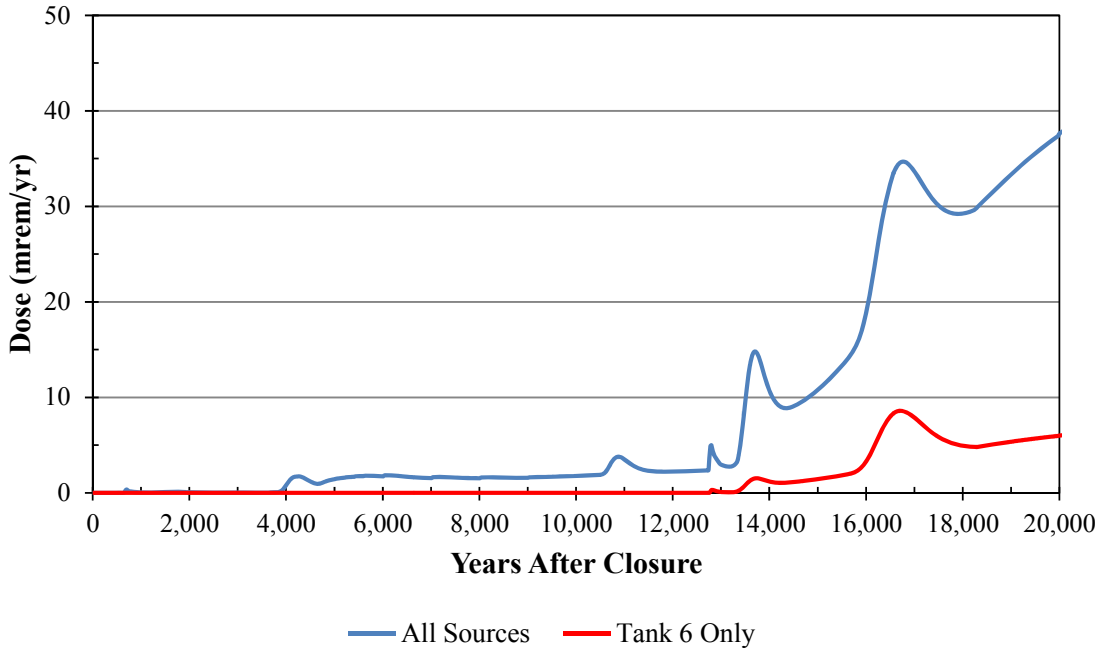


Figure 6.3-14: 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for Sector D – with Tank 6 Contribution

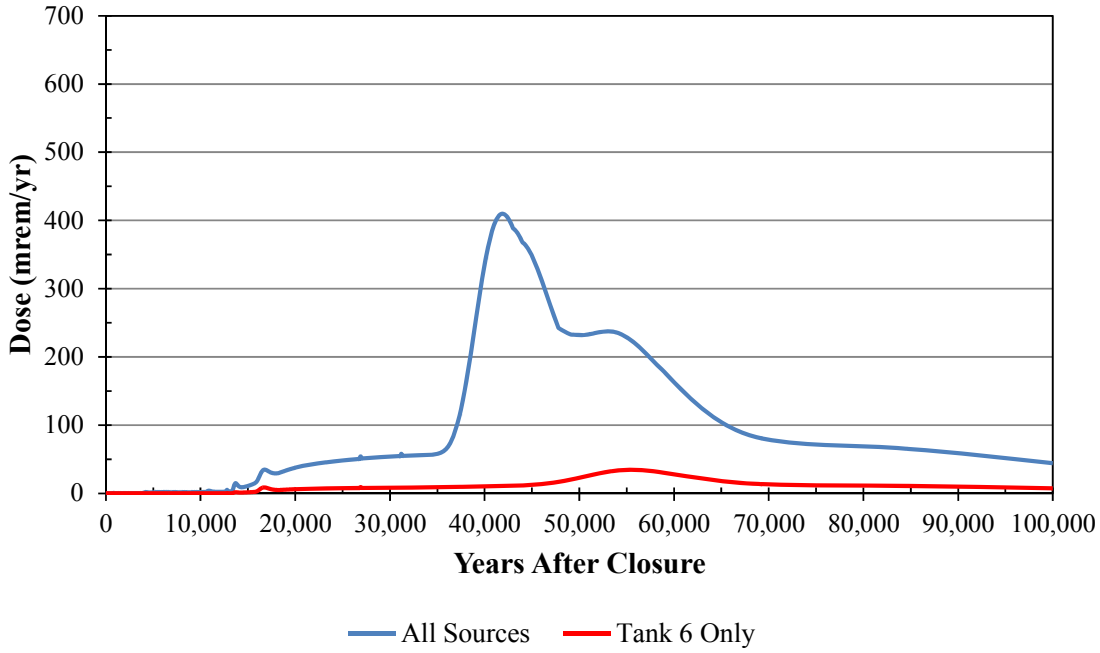


Figure 6.3-15: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 20,000 Years for All Sources and for All Sources except Tanks 5 and 6

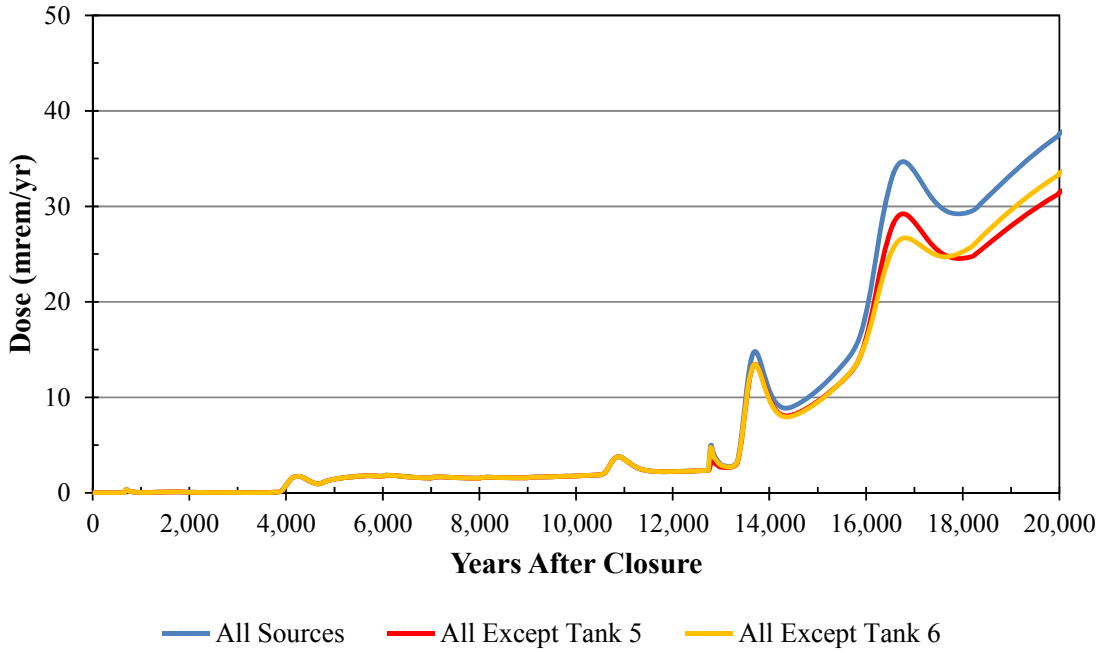


Figure 6.3-16: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for All Sources and for All Sources Except Tanks 5 or Tank 6 (linear scale)

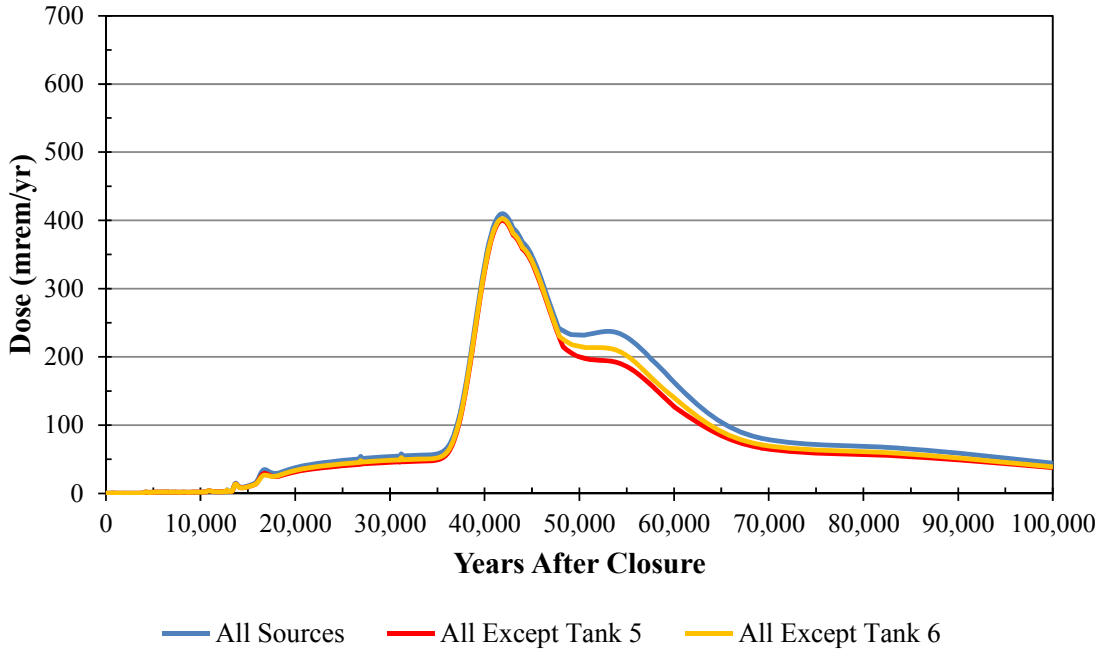
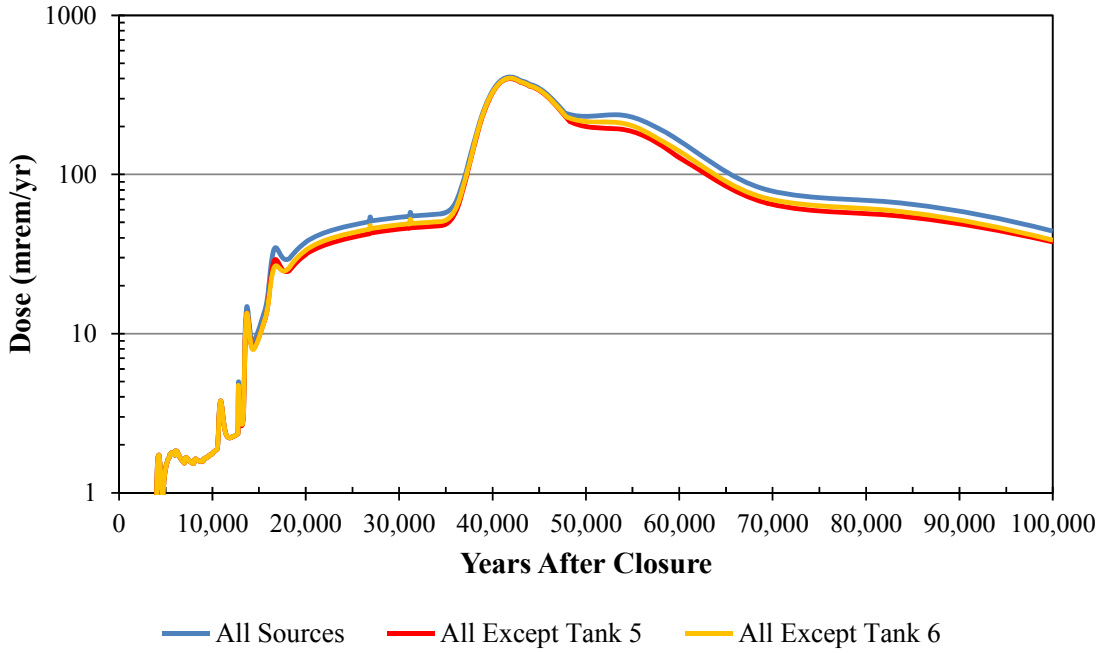


Figure 6.3-17: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for All Sources and for All Sources Except Tanks 5 or Tank 6 (log scale)



With respect to Tanks 5 and 6, the most significant peak dose within 100,000 years is associated with a combination of Pu-239 and Nb-93m from Tank 5. Although the peak dose in Sector D occurs around 42,000 years after closure, this dose is primarily driven by Pu-239 released from Tank 18. The maximum dose contributions from Tanks 5 and 6 both occur later than this, as seen in Figures 6.3-12 and 6.3-14. Contributions from Tank 5 peak at 71 mrem/yr, approximately 54,000 years after closure, at which time it comprises about 30 % of the total dose. Similarly, Tank 6 peaks at 34 mrem/yr at approximately 56,000 years, at which time it comprises about 15 % of the total dose.

To better assess the risk and uncertainty associated with these dose peaks, DOE developed a Composite Sensitivity Study that incorporated more recent data pertaining to the release and transport of contaminants, including Pu-239 and Nb-93m (see Section 6.3.6.3). Based on this Composite Sensitivity Study, the Base Case dose results from both Pu-239 and Nb-93m are artificially high due to modeling assumptions. The Composite Sensitivity Study shows the total dose (from all contaminants) peaks around 20 mrem/yr near 18,000 years (i.e., remains below 25 mrem/yr for 100,000 years).

6.3.5.4 Individual Pathway Contributions to MOP 100-Meter Peak Annual Groundwater Pathway Dose

As stated previously, the total peak groundwater-pathway dose results are the summation of the doses associated with all the individual 100-meter well pathways. Table 6.3-10 shows the relative contributions from the individual groundwater pathways to the recalculated (i.e., using the updated tank inventories) Sector E 100-meter MOP receptor dose at 10,000 years (the year of the peak dose). The primary contributors are water ingestion (65 % of peak dose) and vegetable ingestion (31 % of peak dose), with Ra-226 being the principal radionuclide contributor for all of the individual groundwater pathways. The individual pathway contributions are not significantly impacted by using the actual residual inventories at operational closure for Tanks 5 and 6 as compared to the Tank 18/19 Special Analysis.

Table 6.3-10: 100-Meter MOP Peak Dose Individual Groundwater Pathway Contributions for Sector E

Pathway	Associated Contribution at year 10,000 (mrem/yr)	Percentage of Total Peak Dose
Water Ingestion	2.1	65%
Vegetable Ingestion	1.0	31%
Irrigation Shine	<0.05	1%
Finfish Ingestion	<0.05	1%
Milk Ingestion	<0.05	1%
Total	**3.3**	**100%**

Table 6.3-11 shows a comparison of the 100-meter peak, water ingestion doses for the different 100-meter sectors within 1,000 and 10,000 years. Figure 6.3-18 shows the water ingestion doses to the 100-meter MOP receptor over 10,000 years for the five 100-meter sectors. The highest 100-meter MOP water ingestion dose in 10,000 years for Sector E is 2.1 mrem/yr, at year 10,000. Figure 6.3-19 shows the 100-meter MOP receptor, water ingestion doses within 100,000 years for the five 100-meter sectors. Table 6.3-12 and Figure 6.3-20 show the vegetable ingestion doses to the 100-meter MOP receptor for the five 100-meter sectors within 10,000 years. Figure 6.3-21 shows the 100-meter MOP receptor, vegetable ingestion doses within 100,000 years for the five 100-meter sectors.

Table 6.3-11: 100-Meter MOP Peak Water Ingestion Doses by Sector

Sector	Peak Water Ingestion Dose in 1,000 years (mrem/yr)	Principal Radionuclide	Peak Water Ingestion Dose in 10,000 Years (mrem/yr)	Principal Radionuclide
A	0.03 (year 750)	Tc-99	0.03 (year 750)	Tc-99
B	0.06 (year 752)	Tc-99	0.06 (year 752)	Tc-99
C	0.07 (year 738)	Tc-99	0.09 (year 4,310)	C-14
D	0.21 (year 704)	Tc-99	1.16 (year 6,056)	Pa-231
E	0.26 (year 704)	Tc-99	2.12 (year 10,000)	Ra-226

Table 6.3-12: 100-Meter MOP Peak Vegetable Ingestion Doses by Sector

Sector	Peak Veg. Ingestion Dose in 1,000 years (mrem/yr)	Principal Radionuclide	Peak Veg. Ingestion Dose in 10,000 Years (mrem/yr)	Principal Radionuclide
A	0.02 (year 750)	Tc-99	0.02 (year 750)	Tc-99
B	0.04 (year 752)	Tc-99	0.04 (year 752)	Tc-99
C	0.04 (year 738)	Tc-99	0.07 (year 4,306)	C-14
D	0.12 (year 704)	Tc-99	0.59 (year 6,056)	Pa-231
E	0.15 (year 704)	Tc-99	1.02 (year 10,000)	Ra-226

Figure 6.3-18: 100-Meter MOP Peak Water Ingestion Dose Results within 10,000 Years for the Five 100-Meter Sectors

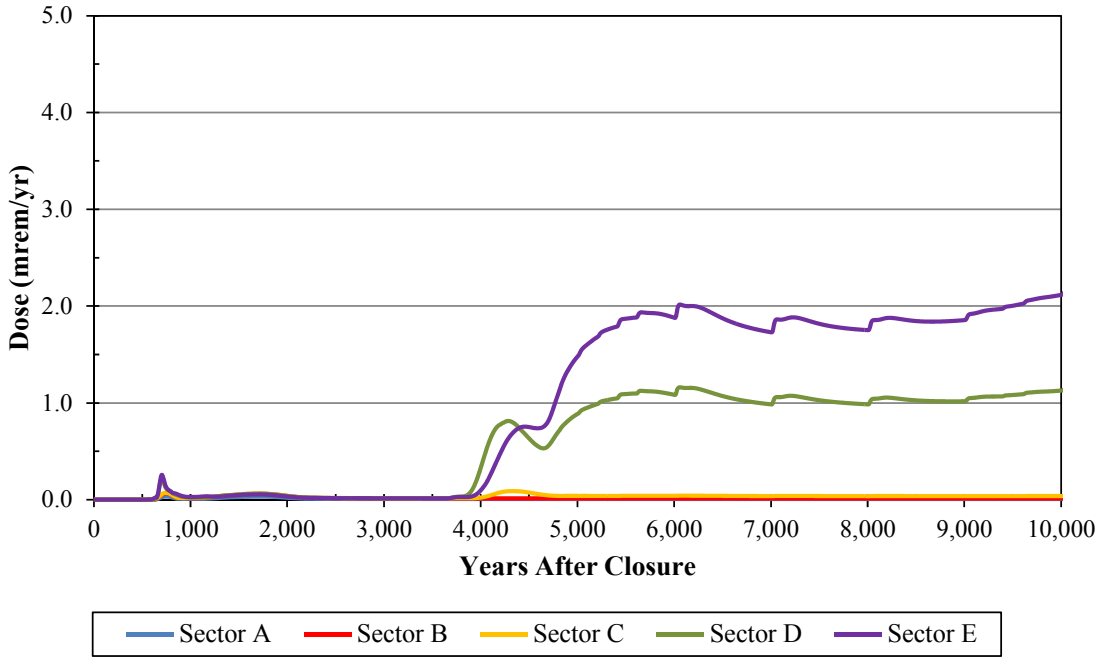


Figure 6.3-19: 100-Meter Peak MOP Water Ingestion Dose Results within 100,000 Years for the Five 100-Meter Sectors

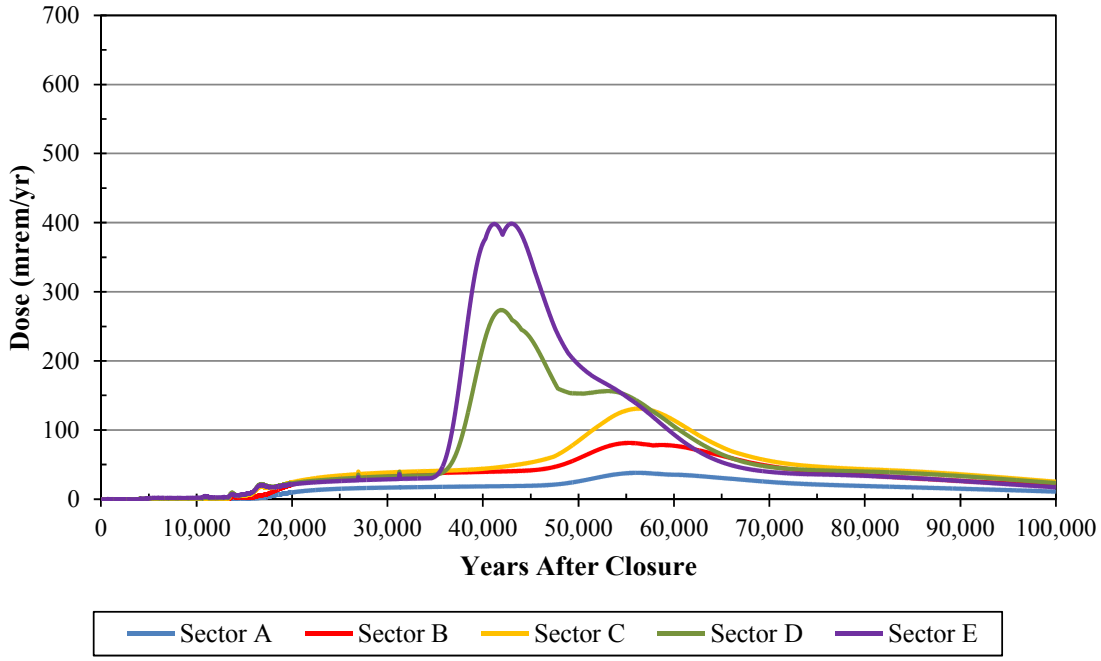


Figure 6.3-20: 100-Meter MOP Peak Vegetable Ingestion Dose Results within 10,000 Years for the Five 100-Meter Sectors

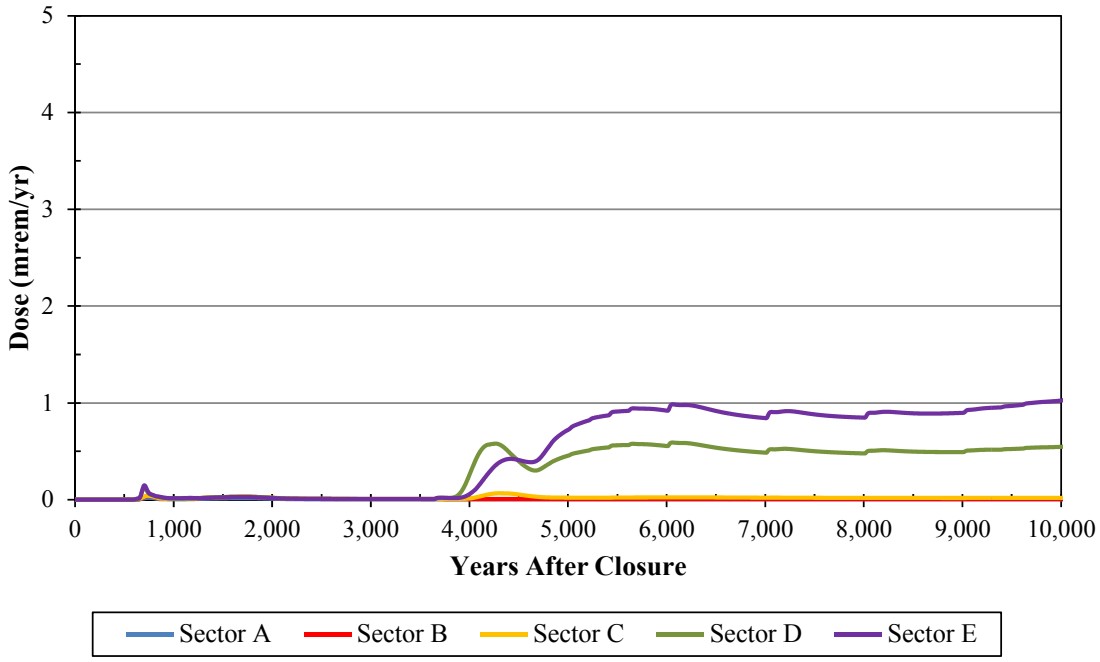
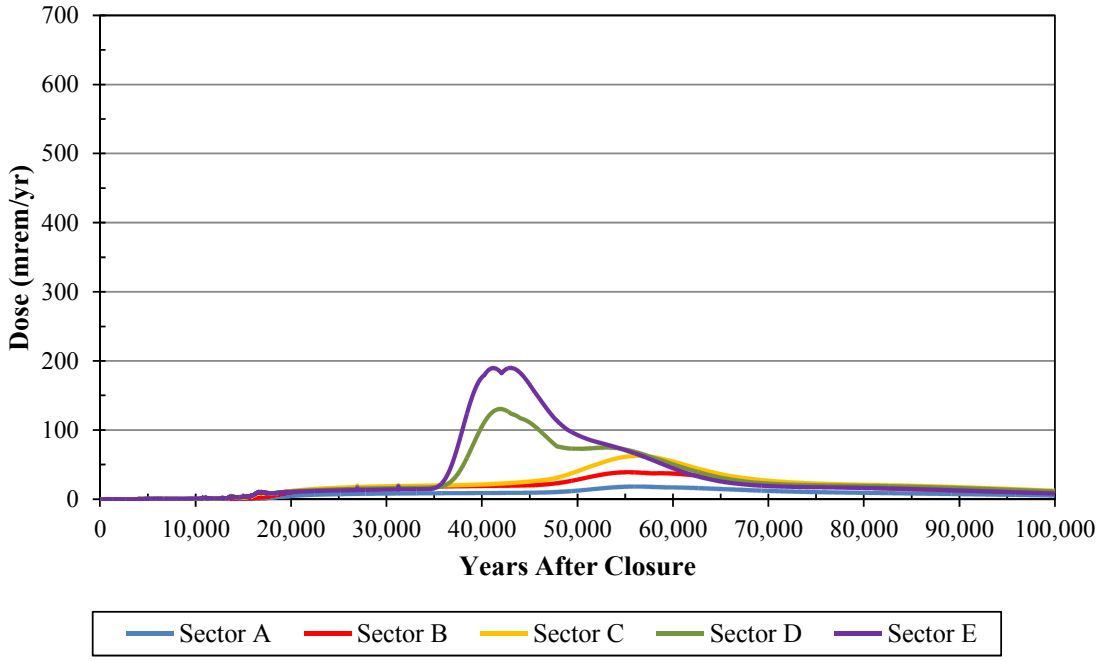


Figure 6.3-21: 100-Meter MOP Peak Vegetable Ingestion Dose Results within 100,000 Years for the Five 100-Meter Sectors



6.3.5.5 Peak Annual MOP Dose at the Stream

The peak groundwater-pathway doses for the two stream seeplines (Fourmile Branch and UTR) have been recalculated using the highest concentration for each radionuclide in the sector. Table 6.3-13 shows a comparison of the MOP, stream peak groundwater pathway doses for the two stream sectors. The highest peak groundwater-pathway dose within 10,000 years is associated with the UTR. Figure 6.3-22 shows the peak groundwater pathway doses over 10,000 years for the two streams (UTR and Fourmile Branch). The stream peak groundwater-pathway dose for a MOP in 10,000 years is 0.10 mrem/yr. Figure 6.3-23 shows the peak groundwater-pathway stream doses within 100,000 years.

Table 6.3-13: Peak Groundwater Pathway MOP Doses at the Stream by Sector

Sector	Peak Dose in 1,000 Years	Peak Dose in 10,000 Years
Fourmile Branch	**0.011 mrem/yr (year 826)** Principal Pathways: Water Ingestion Vegetable Ingestion Finfish Ingestion Principal Radionuclides: Tc-99, Np-237	**0.011 mrem/yr (year 826)** Principal Pathways: Water Ingestion Vegetable Ingestion Finfish Ingestion Principal Radionuclides: Tc-99, Np-237
UTR	**0.013 mrem/yr (year 804)** Principal Pathway: Water Ingestion Vegetable Ingestion Finfish Ingestion Principal Radionuclides: Tc-99, Np-237	**0.10 mrem/yr (year 5,550)** Principal Pathway: Water Ingestion Finfish Ingestion Vegetable Ingestion Principal Radionuclides: Ra-226, Pa-231

Figure 6.3-22: Peak Groundwater Pathway MOP Dose at the Stream Results within 10,000 Years for the Two Stream Sectors

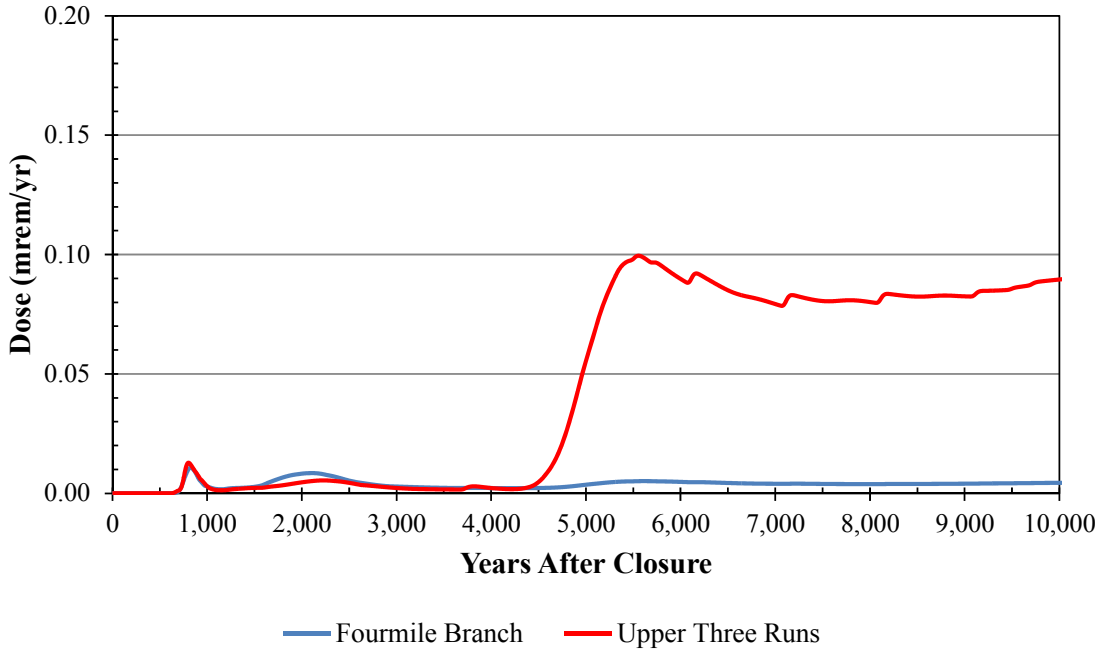
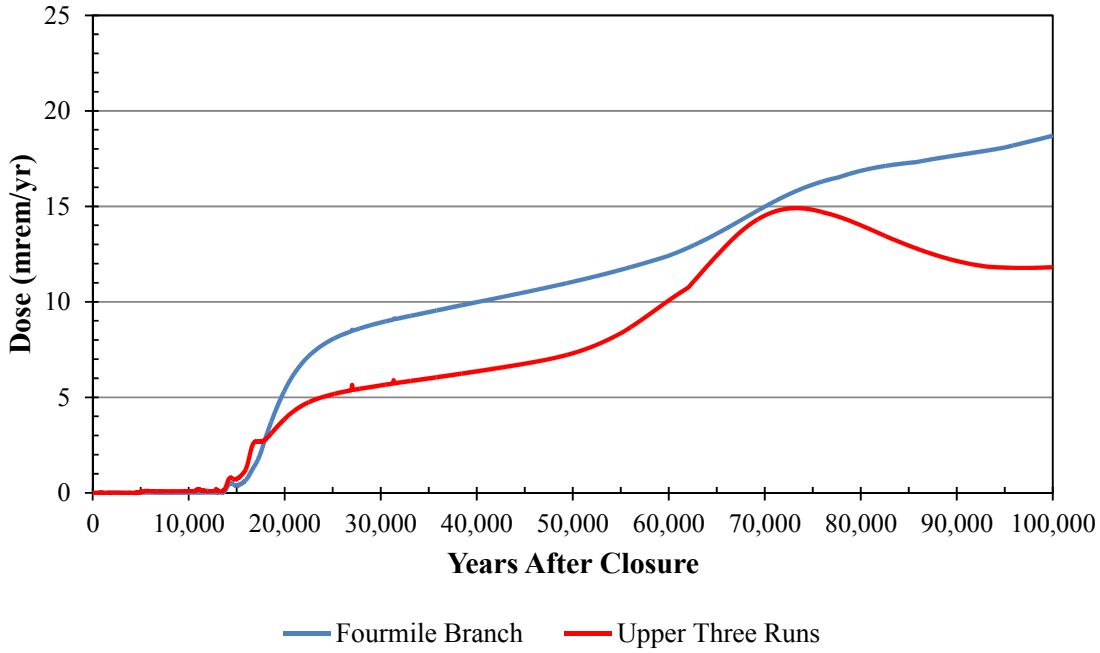


Figure 6.3-23: Peak Groundwater Pathway MOP Dose at the Stream Results within 100,000 Years for the Two Stream Sectors



6.3.5.6 Individual Pathway MOP Contributors at Stream

Table 6.3-14 shows the relative contributions from the individual groundwater pathways to the UTR MOP receptor dose at year 5,550 (the year of the peak UTR dose within 10,000 years). The primary contributor to the UTR peak is water ingestion, with additional contributions from finfish and vegetable ingestion. Table 6.3-15 shows the relative contributions from the individual groundwater pathways to the Fourmile Branch MOP receptor dose at 826 years (the year of the peak Fourmile Branch dose within 10,000 years). The primary contributor is water ingestion, with additional contributions from vegetable and finfish ingestion.

Table 6.3-14: Individual Peak MOP Dose at the Stream Groundwater Pathway Contributions for UTR

Pathway	Associated Contribution at year 5,550 (mrem/yr)	Total Peak Dose	Principal Radionuclide Pathway Dose
Water Ingestion	0.04	45 %	Pa-231
Finfish Ingestion	0.03	29 %	Ra-226
Vegetable Ingestion	0.02	23 %	Pa-231
All Others	< 0.01	3 %	- -
Total	**0.10**	**100 %**	

Table 6.3-15: Individual Peak MOP Dose at the Stream Groundwater Pathway Contributions for Fourmile Branch

Pathway	Associated Contribution at year 826 (mrem/yr)	Total Peak Dose	Principal Radionuclide Pathway Dose
Water Ingestion	0.005	46 %	Tc-99
Vegetable Ingestion	0.003	27 %	Tc-99
Finfish Ingestion	0.003	24 %	Tc-99
All Others	<0.001	3 %	- -
Total	**0.011**	**100 %**	

6.3.5.7 Member of the Public at 100-Meter Peak Annual All-Pathway Dose

The peak all-pathways annual dose for the MOP at 100-meters is calculated using the recalculated peak 100-meter groundwater-pathway dose results in 10,000 years (3.3 mrem/yr) in combination with the recalculated air pathway results (4.6E-06 mrem/yr). The peak all-pathways annual dose for the MOP is 3.3 mrem/yr[4] and is associated with Sector E. The breakdown of the individual dose contributors is provided in Table 6.3-16.

[4] Dose values are shown to two significant figures to illustrate changes/trends. The use of two significant figures is not meant to imply this level of precision for these dose projections.

Table 6.3-16: 100-Meter Peak Annual MOP All-Pathways Dose Contributors

Pathway	Associated Contribution at year 10,000 (mrem/yr)	Percentage of Total Peak Dose	Principal Radionuclide Pathway Dose
Water Ingestion	2.1	65 %	Ra-226
Vegetable Ingestion	1.0	31 %	Ra-226
Air Pathway	< 0.01	< 1 %	C-14
All Others	0.2	3 %	n/a
Total	**3.3**	**100 %**	

6.3.5.8 MOP at Stream Peak Annual All-Pathways Dose

The peak all-pathways annual dose for the MOP at the stream is calculated using the recalculated peak stream groundwater-pathway dose results in 10,000 years (0.10 mrem/yr) in combination with the air pathway results (8.3E-07 mrem/yr). The peak all-pathways annual dose for the MOP within 10,000 years is 0.1 mrem/yr and is associated with UTR. The breakdown of the individual dose contributors is provided in Table 6.3-17.

Table 6.3-17: Peak Annual MOP at the Stream All-Pathways Dose Individual Contributions

Pathway	Associated Contribution at Year 5,550 (mrem/yr)	Percentage of Total Peak Dose	Principal Radionuclide Pathway Dose
Water Ingestion	0.04	45 %	Pa-231
Finfish Ingestion	0.03	29 %	Ra-226
Vegetable Ingestion	0.02	23 %	Pa-231
Air Pathway	<0.01	<1 %	C-14
All Others	<0.01	3 %	n/a
Total	**0.10**	**100 %**	

6.3.6 Uncertainty and Sensitivity Analysis

Section 5.6 of the FTF PA considers the effects of uncertainties in the conceptual models used and sensitivities in the parameters used in the mathematical models. The uncertainty analyses and sensitivity analyses were primarily performed using a probabilistic model (i.e., the GoldSim FTF model), but some additional single parameter sensitivity analyses were performed through deterministic modeling using both the PORFLOW and GoldSim models.

The probabilistic model varies multiple parameters simultaneously, so concurrent effects of changes in the model can be analyzed, and the potential impact of changes can be assessed. This assessment allows for identification of parameters that are only of significance when varied simultaneously with another parameter. The deterministic model single parameter analysis provides a method to evaluate parametric effects in isolation, so the importance of the uncertainty around a parameter of concern can be more effectively evaluated. Using both probabilistic and deterministic models for sensitivity analysis versus a single approach provides additional information concerning which parameters are of most importance to the FTF PA modeling.

In general, the uncertainty analyses and sensitivity analyses information presented in the FTF PA and the Tank 18/19 Special Analysis are not affected by the new residual waste information, and the uncertainty analyses and sensitivity analyses insights remain unaffected. The impacts on the GoldSim benchmarking, as discussed in Section 5.6.2 of FTF PA, are not presented herein but are documented in the FTF GoldSim Model report. [SRR-CWDA-2012-00119] Note that the benchmarking efforts are not impacted by the changes to the Tank 5 and 6 inventories but are impacted by the improvements to the modeling methodology applied within GoldSim. The updated version (Version 3.0) of the FTF GoldSim Model is based on the version used for the Tank 18/19 Special Analysis. [SRR-CWDA-2010-00124] Several improvements were made to the model to allow for a more detailed analysis (SRR-CWDA-2012-00119). With respect to the use of FTF PA parameters presented in Section 5.6.3 of FTF PA, the parameter impacted by the residual inventories at operational closure is the radiological inventory, discussed in the Section 5.6.3.2 of FTF PA. Most of the Tanks 5 and 6 residual inventories are within the inventory distributions used in the probabilistic model (documented in the FTF PA Table 5.6-3, Inventory Multipliers), with the exceptions of Am-243 (Tank 6), Ni-59 (Tank 6), Ni-63 (Tank 6), Th-230 (Tanks 5 and 6), and Zr-93 (Tanks 5 and 6).

The following sections describe some additional uncertainty analyses and sensitivity analyses activities that were performed utilizing the Tanks 5 and 6 residual inventories at operational closure.

6.3.6.1 Deterministic Model Barrier Analysis

Section 5.6.7.3 of the FTF PA provided a comprehensive barrier analyses that identified barriers to waste migration and evaluated the capabilities of each barrier as understood from the results of the FTF PA. The barrier analyses assessed the contribution of individual barriers (e.g., closure cap, grout, contamination zone, waste tank liner, and waste tank concrete) by comparing contaminant flux results under various barrier conditions. No new barrier analyses were performed in support of this Special Analysis.

6.3.6.2 Deterministic Model Single Parameter Sensitivity Analysis

The impact of input variability on the peak dose was further evaluated in this Special Analysis by performing several single parameter sensitivity analyses using the Base Case PORFLOW FTF model with changes to select parameters (a detailed discussion of the PORFLOW FTF model and the individual FTF PA Base Case parameters is provided in the FTF PA).

Additional inventory sensitivity analyses were performed to provide insight into the potential effects of removing additional waste material from Tanks 5 and 6. These analyses compare dose results using the final FTF Tank 5 inventories to one-half of the final FTF Tank 5 inventories and by comparing dose results using the final FTF Tank 6 inventories to one-half of the final FTF Tank 6 inventories. Results are shown for the 100-meter MOP, peak groundwater-pathway dose within 20,000 years (Figures 6.3-24 through 6.3-27) and within 100,000 years (Figures 6.3-28 through 6.3-31).

This sensitivity study shows that the peak doses (primarily due to Nb-93m) can be reduced by reducing the total curies of Zr-93 (the parent radionuclide) in Tanks 5 and 6 and that this reduction approaches a linear function (except when the Nb-93m is overwhelmed by the dose contribution from Pu-239 around 50,000 to 60,000 years after closure).

The conservatisms surrounding the Pu-239 release and their effects on the FTF doses are discussed in the Tank 18/19 Special Analysis. It states that "doses associated with Pu-239 can be expected to occur later than currently reflected in the FTF PA Base Case as additional conservatisms are removed" and that the peak dose associated with Pu-239 occurs well beyond 10,000 years. [SRR-CWDA-2010-00124] Additionally, the Composite Sensitivity Study described in Section 6.3.6.3 further illustrates that the contributions from Nb-93m and Pu-239 may both be conservatively high due to base case modeling simplifications.

Figure 6.3-24: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 20,000 Years for Tank 5 Only (Tank 5 Final Inventory and One-Half Final Inventory)

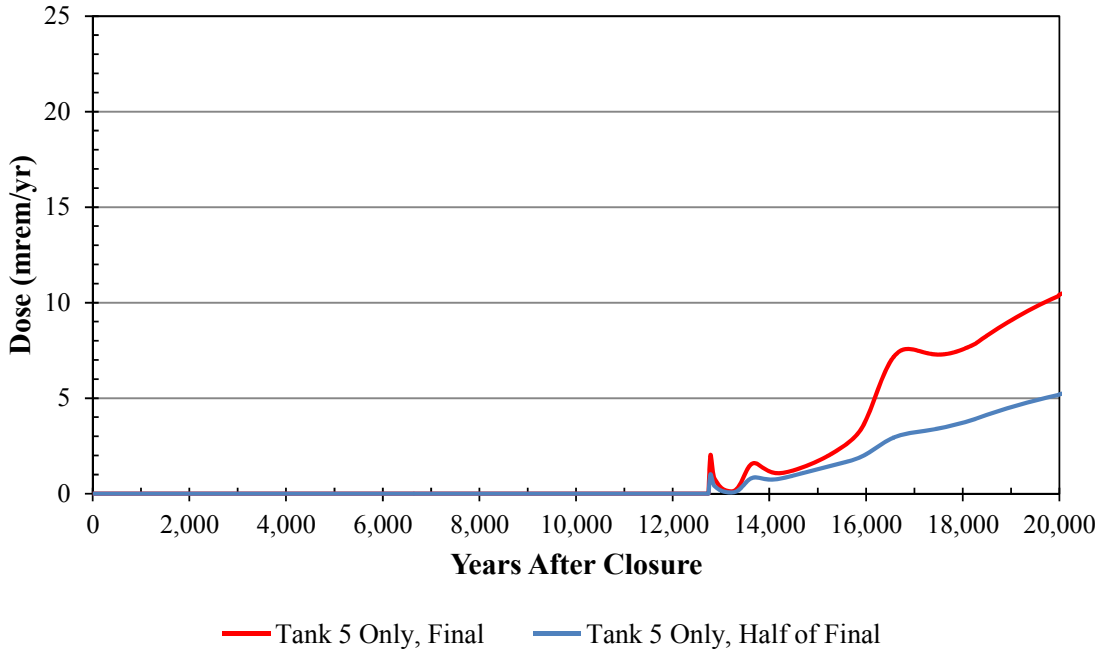


Figure 6.3-25: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 20,000 Years for Tank 6 Only (Tank 6 Final Inventory and One-Half Final Inventory)

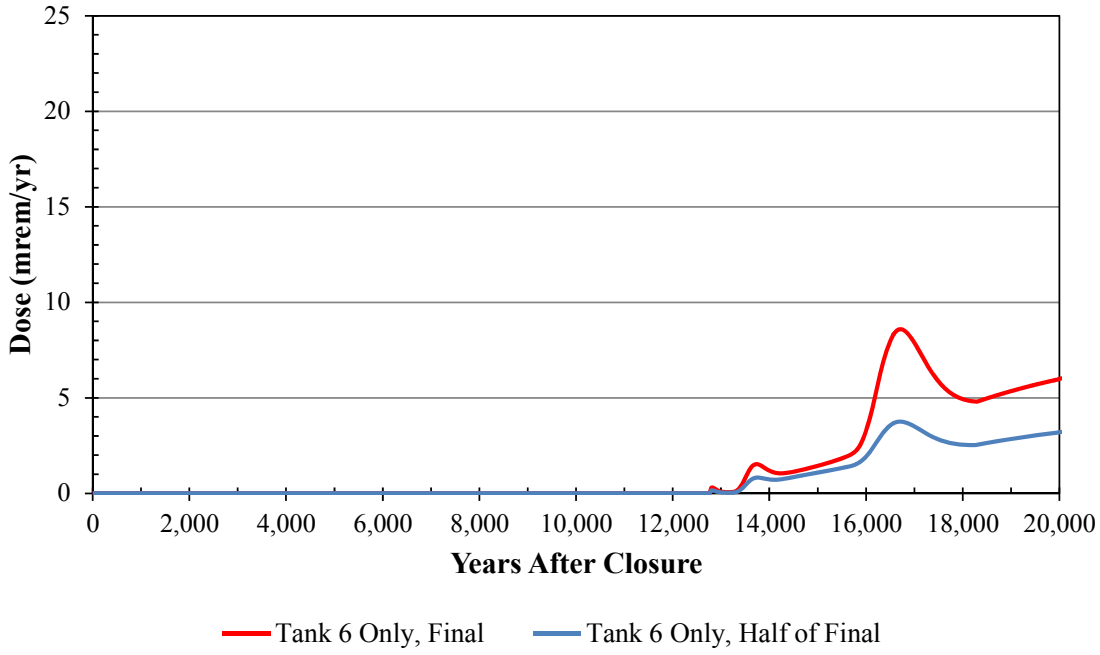


Figure 6.3-26: Individual Radionuclide Contributors to the Sector D 100-Meter Peak Groundwater Pathway Dose from Tank 5 Only, 20,000 years (Final Inventory)

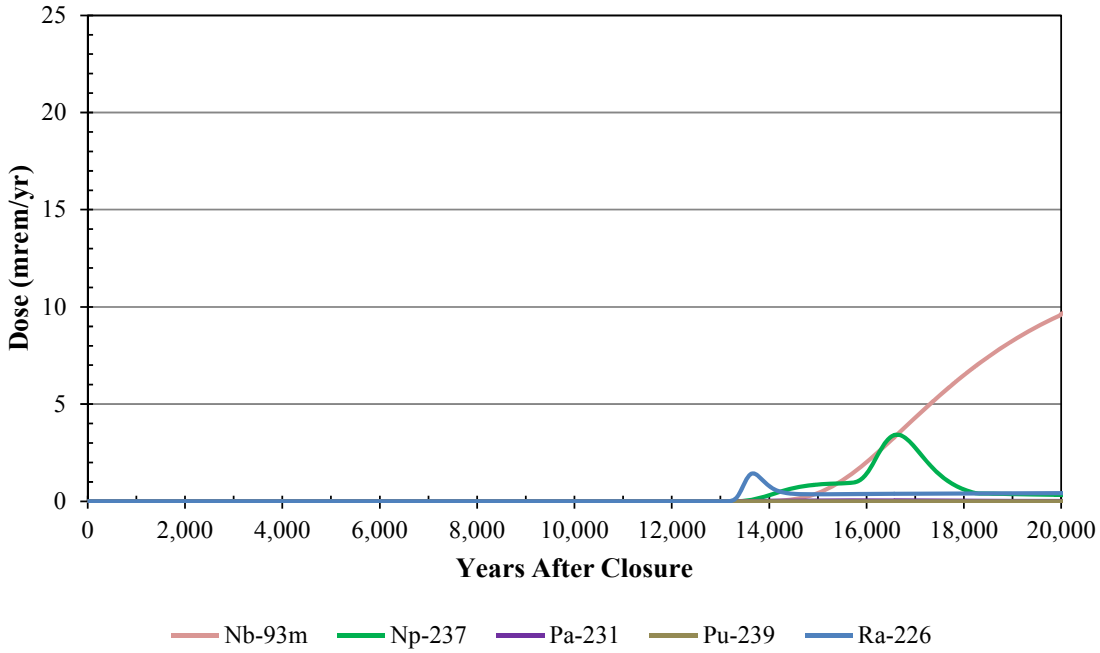


Figure 6.3-27: Individual Radionuclide Contributors to the Sector D 100-Meter Peak Groundwater Pathway Dose from Tank 6 Only, 20,000 years (Final Inventory)

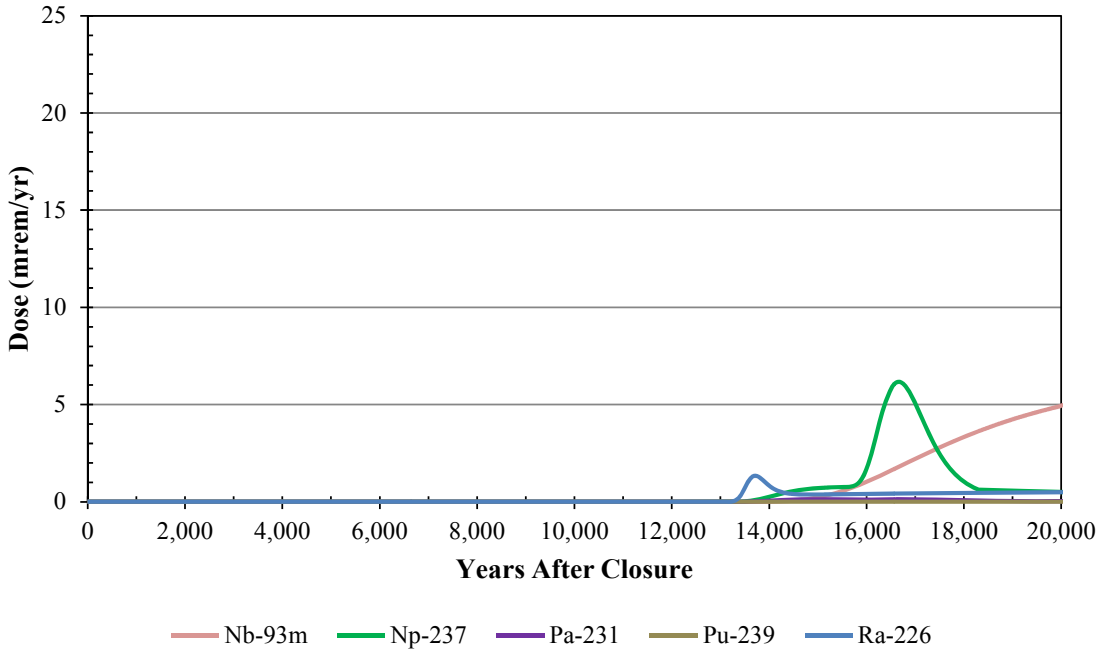


Figure 6.3-28: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for Tank 5 Only (Tank 5 Final Inventory and One-Half Final Inventory)

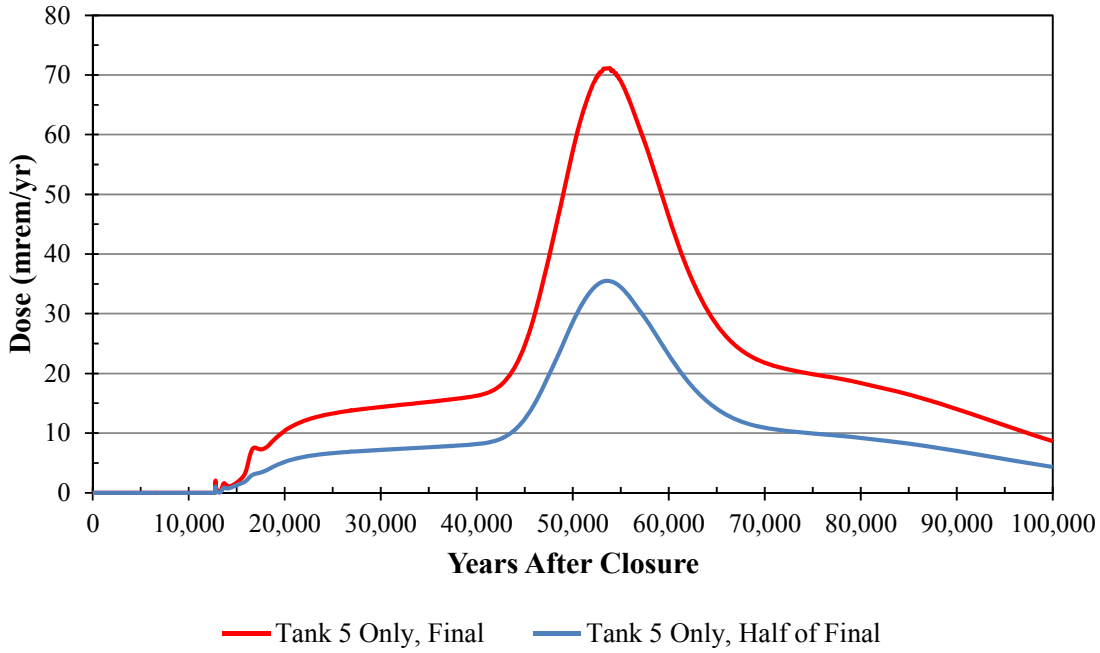


Figure 6.3-29: Sector D 100-Meter MOP Peak Groundwater Pathway Dose within 100,000 Years for Tank 6 Only (Tank 6 Final Inventory and One-Half Final Inventory)

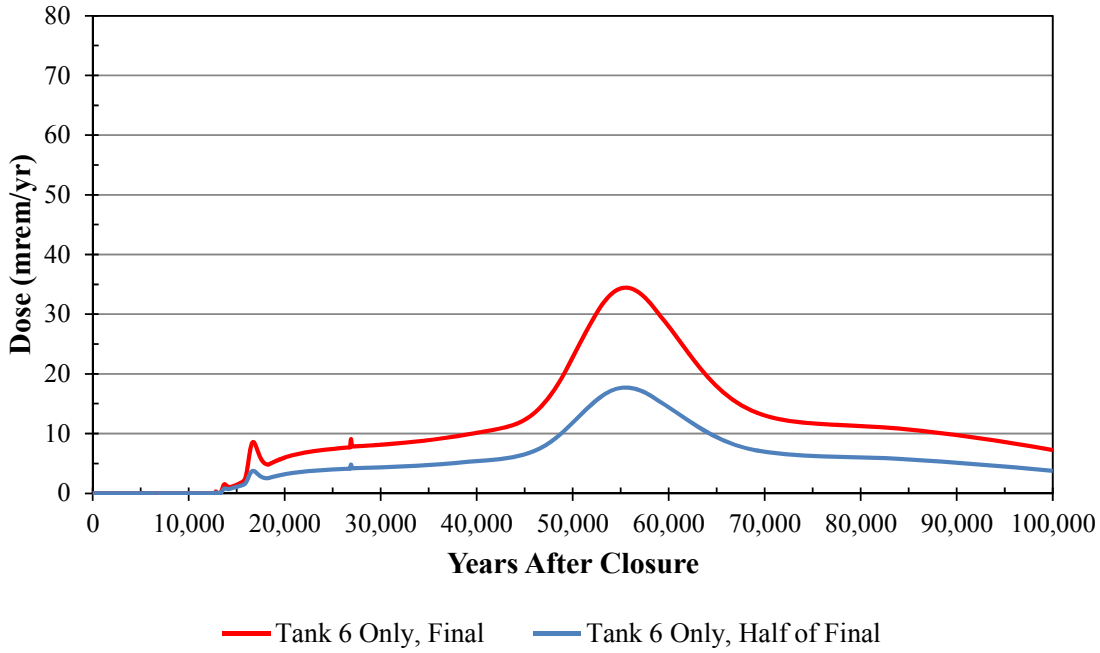


Figure 6.3-30: Individual Radionuclide Contributors to the Sector D 100-Meter Peak Groundwater Pathway Dose from Tank 5 Only, 100,000 years (Final Inventory)

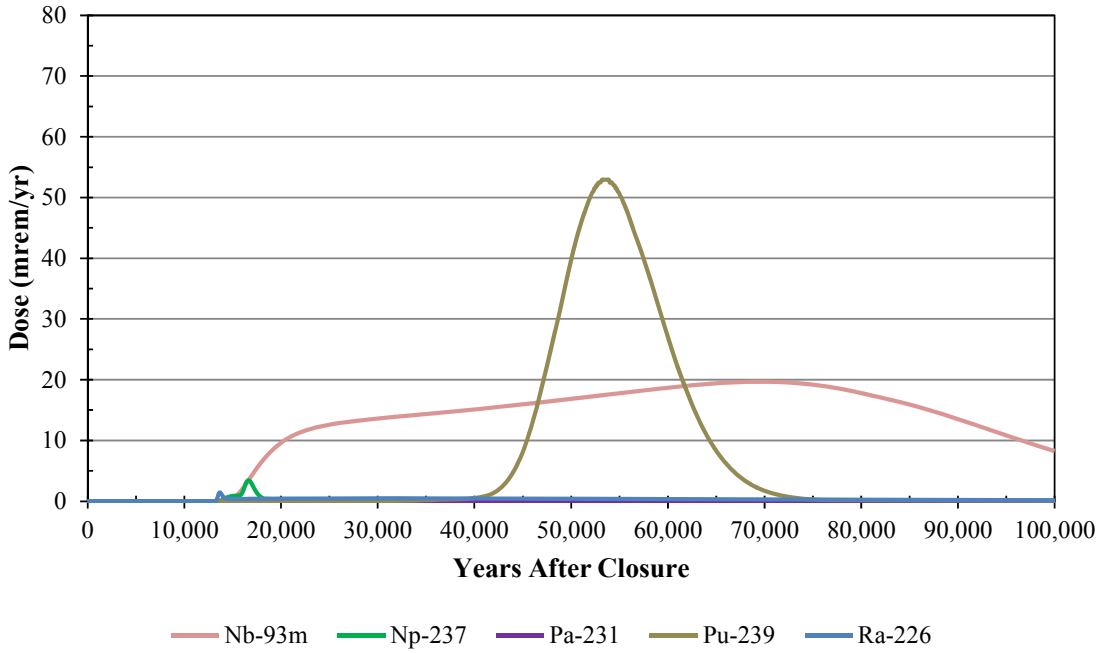
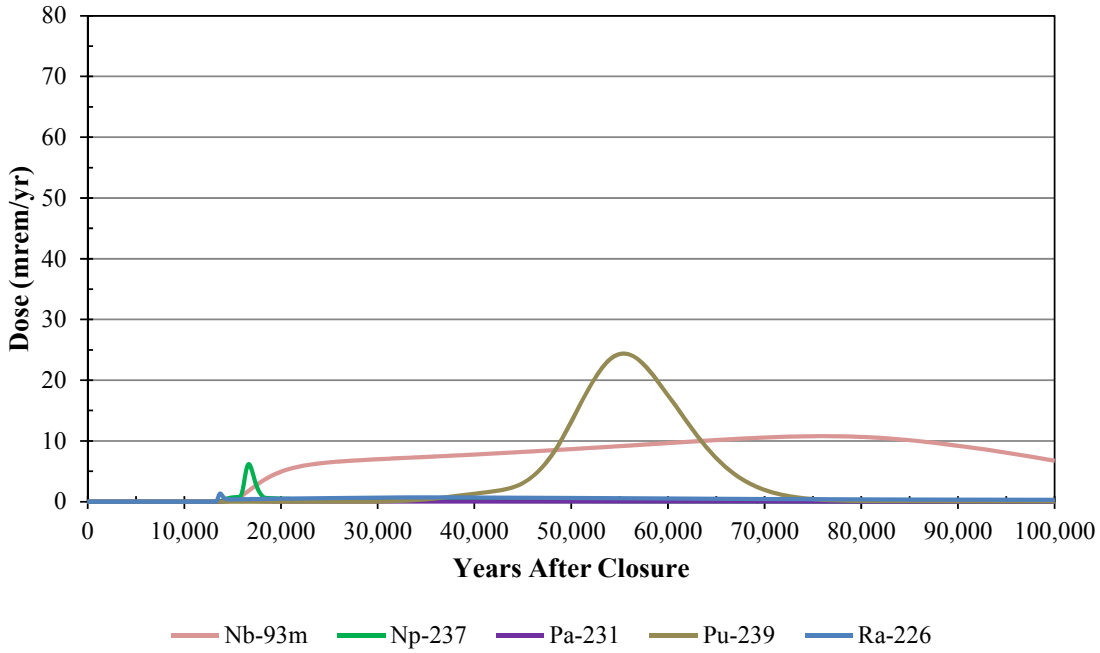


Figure 6.3-31: Individual Radionuclide Contributors to the Sector D 100-Meter Peak Groundwater Pathway Dose from Tank 6 Only, 100,000 years (Final Inventory)



6.3.6.3 Deterministic Model Composite Sensitivity Analyses

To evaluate the impact of uncertainty surrounding multiple parameters simultaneously, several new PORFLOW sensitivity analyses were performed incorporating lessons learned regarding waste release modeling and distribution coefficients into the Base Case modeling. The following tables represent Composite Sensitivity Study inputs based on lessons learned from data made available since the FTF PA and Tank 18/19 Special Analysis were released and based on inputs from the NRC Monitoring Plan.

Tables 6.3-18 and 6.3-19 provide updated solubility information presented in *Evolution of Chemical Conditions and Estimated Solubility Controls on Radionuclides in the Residual Waste Layer During Post-Closure Aging of High-Level Waste Tanks* (SRNL-STI-2012-00404), which was prepared subsequent to submittal of the FTF PA and the Tank 18/19 Special Analysis. This updated solubility report incorporated, and built upon new information, garnered from the plutonium solubility study described in Section 6.3.6 of the Tank 18/19 Special Analysis. [SRR-CWDA-2010-00124] In addition, the number of pore volumes of reacting fluid required to reach each stage of grout degradation has been updated in SRNL-STI-2012-00404, and this new information is provided in Table 6.3-20.

Table 6.3-18: Calculated Solubility and Controlling Phases of Radionuclides of Interest

Element	Reducing Region II		Oxidized Region II	
	Controlling Phase	Solubility (mol/L)	Controlling Phase	Solubility (mol/L)
Ac	$Ac(OH)_{3(am)}$	1E-09	$Ac(OH)_{3(am)}$	1E-09
Am	$Am(OH)_{3(am)}$	1E-09	$Am(OH)_{3(am)}$	1E-09
Ba	$BaSO_4$ (barite)	3E-05	$BaSO_4$ (barite)	3E-05
Bk	Short half-life	-	Short half-life	-
C	$CaCO_3$ (calcite)	2E-06	$CaCO_3$ (calcite)	2E-06
Cf	Small inventory	-	Small inventory	-
Cm	$Cm(OH)_{3(am)}$	1E-09	$Cm(OH)_{3(am)}$	1E-09
Co	CoS(beta)	3E-02	No solubility control	-
Cs	No solubility control	-	No solubility control	-
Eu	$Eu(OH)_{3(am)}$	8E-07	$Eu(OH)_{3(am)}$	8E-07
I	No solubility control	-	No solubility control	-
Nb	No solubility control	-	No solubility control	-
Ni	$NiS_{(c)}$ alpha	2E-09	$Ni(OH)_2$(beta)	1E-07
Np	$NpO_{2(am, hyd)}$	1E-09	$NpO_{2(am, hyd)}$	3E-07
Pa	No solubility control	-	No solubility control	-
Pu	$PuO_{2(am, hyd)}$	3E-11	$PuO_{2(am, hyd)}$	3E-11
Ra	$RaSO_4$	3E-05	$RaSO_4$	3E-05
Rh	Short half-life	-	Short half-life	-
Se	$FeSe_{2(cr)}$	2E-05	No solubility control	-
Sm	$Sm(OH)_{3(am)}$	1E-09	$Sm(OH)_{3(am)}$	1E-09
Sn	$SnO_{2(am)}$	4E-04	$SnO_{2(am)}$	4E-04
Sr	$SrCO_3$ (strontianite)	3E-03	$SrCO_3$ (strontianite)	3E-03
Tc	$TcO_2 \cdot 1.6H_2O$	1E-14	Fe co-precipitation	1E-13
Te	Short half-life	-	Short half-life	-
Th	ThO_2 (am, hyd, aged)	1E-09	$ThO_{2(am, hyd, aged)}$	1E-09
U	$UO_{2\ (am, hyd)}$	5E-09	$UO_3 \cdot 2H_2O$	5E-05
Y	$Y(OH)_{3(c)}$	4E-13[2]	$Y(OH)_{3(c)}$	4E-13
Zr	$Zr(OH)_4$	1E-07[1]	$Zr(OH)_4$	1E-07[1]

1 Zirconium solubility values from DOI: 10.1080/18811248.2007.9711260

2 Yttrium-90 is the yttrium isotope of concern and its transport is controlled by the transport and decay of Sr-90.

Table 6.3-19: Calculated Solubility and Controlling Phases of Radionuclides of Interest

Element	Oxidizing Region III	
	Controlling Phase	Solubility (mol/L)
Ac	$Ac(OH)_{3(am)}$	6E-08
Am	$Am(OH)_{3(am)}$	2E-09
Ba	$BaSO_4$ (barite)	1E-05
Bk	Short half-life	-
C	$CaCO_3$ (calcite)	1E-03
Cf	Small inventory	-
Cm	$CmCO_3OH{\cdot}0.5H_2O$	2E-09
Co	No solubility control	-
Cs	No solubility control	-
Eu	$EuOHCO_{3(cr)}$	3E-08
I	No solubility control	-
Nb	No solubility control	-
Ni	$NiCO_3$	1E-05
Np	$NpO_{2(am, hyd)}$	2E-06
Pa	No solubility control	-
Pu	$PuO_{2(am, hyd)}$	3E-11
Ra	$RaSO_4$	1E-05
Rh	Short half-life	-
Se	No solubility control	-
Sm	$SmCO_3OH{\cdot}0.5H_2O$	2E-09
Sn	$SnO_{2(am)}$	7E-07
Sr	SrCO3 (strontianite)	1E-04
Tc	Fe co-precipitation	2E-15
Te	Short half-life	-
Th	$ThO_{2(am, hyd, aged)}$	1E-09
U	$UO_3{\cdot}2H_2O$	4E-06
Y	$Y(OH)_{3(c)}$	2E-09
Zr	$Zr(OH)_4$	1E-08[1]

1 Zirconium solubility values from DOI: 10.1080/18811248.2007.9711260

2 Yttrium-90 is the yttrium isotope of concern and its transport is controlled by the transport and decay of Sr-90.

Table 6.3-20: Number of Pore Volumes of Reacting Fluid Required to Reach Each Stage of Grout Degradation

Chemical Stage	Pore Volumes Reacted
Reduced Region II	0 - 523
Oxidized Region II	524 - 2,119
Oxidized Region III	> 2,119

Table 6.3-21 provides updated K_d values for vadose zone components, including leachate impacted K_d values, which have been developed subsequent to submittal of the FTF PA. Note that in the FTF PA, a site-specific K_d value was not presented for niobium in soil (sand and clay); however, based on available research data in *Default Soil Solid/Liquid Partition*

Coefficients, K_ds, for Four Major Soil Types: A Compendium (PIT-MISC-0071) and in the NRC's *Recommended Site-Specific Sorption Coefficients for Reviewing Non-High-Level Waste Determinations at the Savannah River Site and Idaho National Laboratory* (ML073510127), niobium values are presented for sand and clay of 160 mL/g and 900 mL/g, respectively. The Base Case conservatively assumed no retardation of niobium (i.e., K_d value of zero mL/g), which attributed to the relatively large doses described in Section 6.3.5.

Table 6.3-21: Recommended K_d Values for the Vadose Zone

Element	Soils Media		Cement Leachate Impacted Soils Media	
	Backfill Soil (mL/g)*	Vadose Zone Soil (mL/g)**	Backfill Soil (mL/g)*	Vadose Zone Soil (mL/g)**
Ac	8,500	1,100	12,750	1,650
Ag	30[b]	10[b]	480	192
Al	1,300	1,300	1,950	1,950
Am	8,500	1,100	12,750	1,650
Ar	0	0	0	0
As	200	100	280	140
At	0.9	0.3	0.1	0
B[a]	0	0	0	0
Ba[c]	101	15	303	45
Bi	8,500	1,100	12,750	1,650
Bk	8,500	1,100	12,750	1,650
C	400	10	2,000	50
Ca	17	5	51	15
Cd	30	15	90	45
Ce	8,500	1,100	12,750	1,650
Cf	8,500	1,100	12,750	1,650
Cl	8[b]	1[b]	0[a]	0[a]
Cm	8,500	1,100	12,750	1,650
Co	100	40	320	128
Cr	400[b]	1,000[b]	14	6
Cs	50	10	50	10
Cu	70	50	224	160
Eu	8,500	1,100	12,750	1,650
F	0	0	0	0
Fe	400	200	600	300
Fr	50	10	50	10
Gd	8,500	1,100	12,750	1,650
H	0	0	0	0
Hg	1,000	800	3,200	2,560
I	0.9	0.3	0.1	0
K	25	5	25	5
Lu	8,500	1,100	12,750	1,650
Mn	200	15	280	21
Mo	1,000	1,000	1,400	1,400
N	0	0	0	0

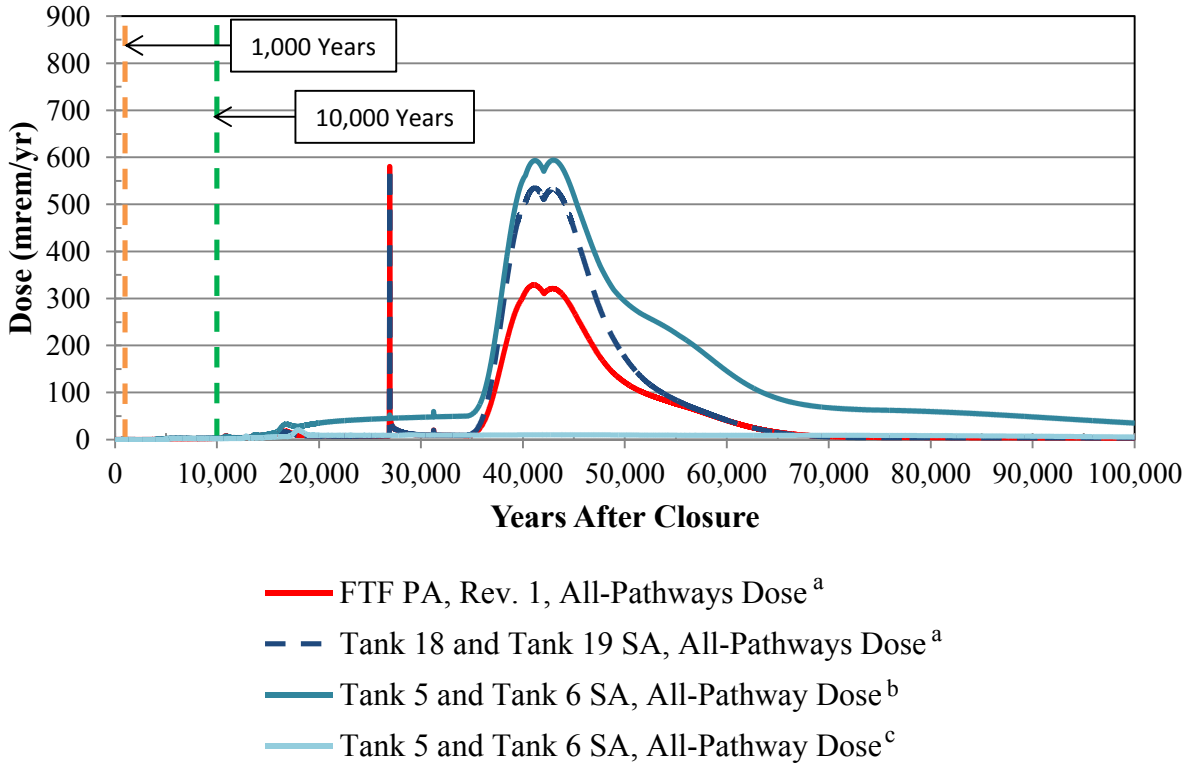
Table 6.3-21: Recommended K_d Values for the Vadose Zone (Continued)

Element	Soils Media		Cement Leachate Impacted Soils Media	
	Backfill Soil (mL/g)*	Vadose Zone Soil (mL/g)**	Backfill Soil (mL/g)*	Vadose Zone Soil (mL/g)**
Na	25	5	25	5
Nb	160[e]	900[e]	224[f]	1,260[f]
Ni	30	7	96	22
Np	9	3	14	5
Pa	9	3	14	5
Pb	5,000	2,000	16,000	6,400
Pd	30	7	96	22
Pm[a]	0	0	0	0
Po	5,000	2,000	10,000	4,000
Pr[a]	0	0	0	0
Pt	30	7	96	22
Pu	5,950	650[d]	11,900	580
Ra[b]	185	25	555	75
Rb	50	10	50	10
Re	1.8	0.6	0.2	0.1
Rh[a]	0	0	0	0
Rn	0	0	0	0
Ru[a]	0	0	0	0
Sb	2,500	2,500	3,500	3,500
Se	1,000	1,000	1,400	1,400
Sm	8,500	1,100	12,750	1,650
Sn	5,000	2,000	15,000	6,000
Sr	17	5	51	15
Tc	1.8	0.6	0.2	0.1
Te	1,000	1,000	1,400	1,400
Th	2,000	900	4,000	1,800
Tl	70[b]	25[b]	0[a]	0[a]
U	400[b]	300[b]	900	600
V[a]	0	0	0	0
Y	8,500	1,100	12,750	1,650
Zn	30	15	90	45
Zr	2,000	900	4,000	1,800

* Backfill soil represented by clayey sediment
** Vadose zone soil represented by sandy sediment
Note: Values from SRNL-STI-2009-00473 unless otherwise noted
a Assigned a value of zero
b SRNL-STI-2010-00493
c SRNL-STI-2011-00011
d SRNL-STI-2011-00672
e ML073510127
f Values based on leachate multiplication factor for soils of 1.4

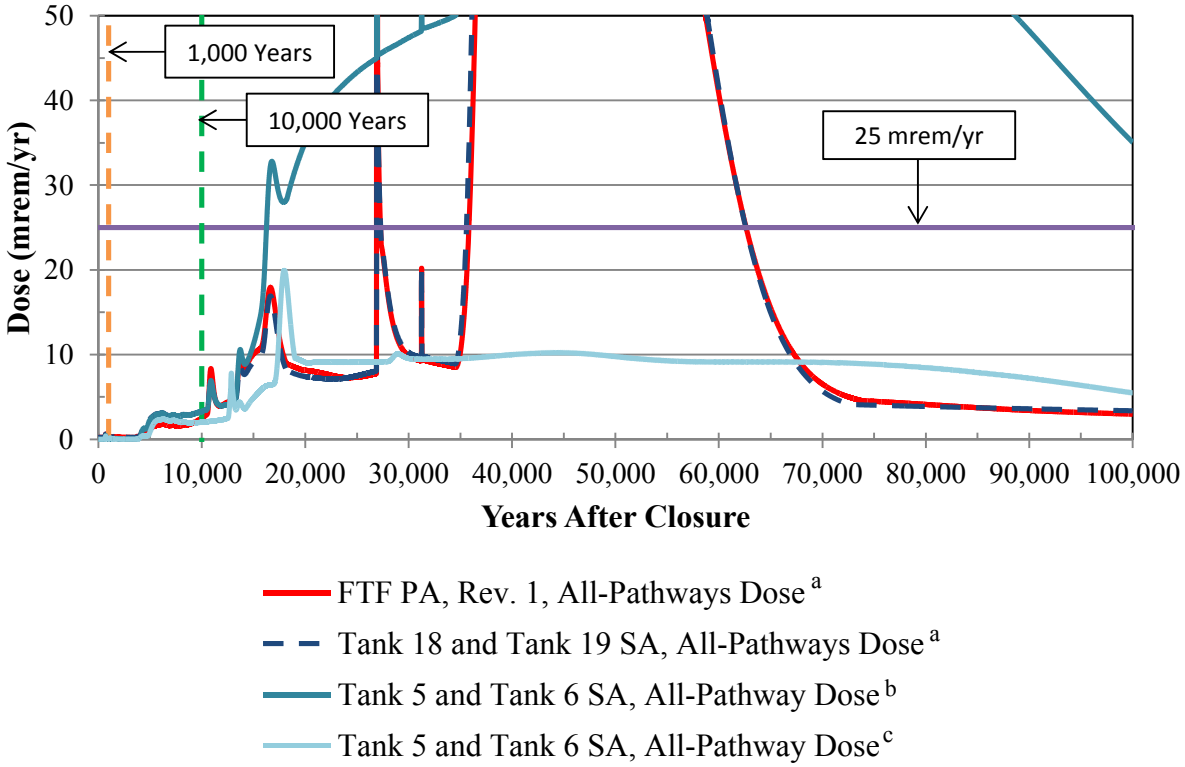
The new data from Tables 6.3-18 through 6.3-21 were used to run a sensitivity study, for comparison to the FTF PA Base Case. As shown in Figures 6.3-32 and 6.3-33, doses associated with the more realistic and probable modeling assumptions can be expected to be lower than currently reflected in the FTF PA Base Case deterministic model, with a peak dose in 100,000 years of only 20 mrem/yr. This 20 mrem/yr dose is approximately 15 times lower than the FTF PA Base Case peak dose, approximately 25 times lower than the Tank 18/19 Special Analysis Base Case peak dose, and approximately 30 times lower than the Tanks 5 and 6 SA Base Case peak dose.

Figure 6.3-32: Composite Sensitivity Study of 100-Meter Peak All Pathways Dose within 100,000 Years for Sector E – All Waste Tanks



a. Peak near 27,000 years (associated with Tc-99 releases) occurs in both the FTF PA and the Tank 18/19 Special Analysis, but not in this Tank 5/6 Special Analysis.
b. Applies Base Case assumptions for the waste release model and contaminant properties.
c. Applies more realistic and probable assumptions for the waste release model and contaminant properties.

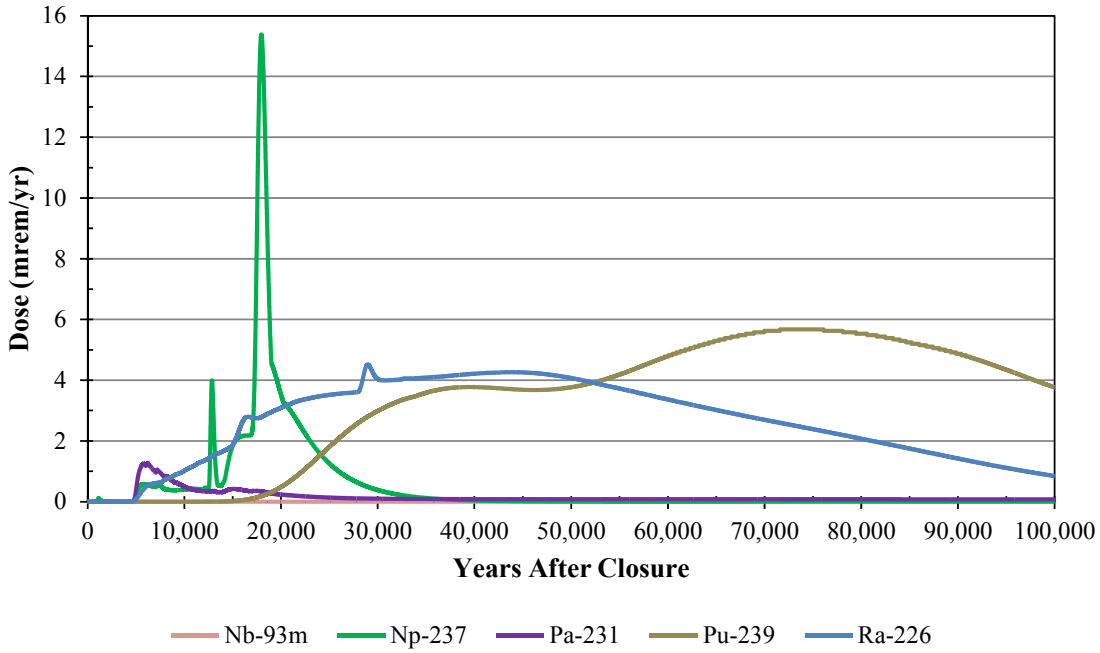
Figure 6.3-33: Composite Sensitivity Study of 100-Meter Peak All Pathways Dose within 100,000 Years for Sector E – All Waste Tanks (Expanded View)



a. Peak near 27,000 years (associated with Tc-99 releases) occurs in both the FTF PA and the Tank 18/19 Special Analysis, but not in this Tank 5/6 Special Analysis.
b. Applies Base Case assumptions for the waste release model and contaminant properties.
c. Applies more realistic and probable assumptions for the waste release model and contaminant properties.
The horizontal purple line at 25 mrem/yr is provided for illustration only.

Figure 6.3-34 shows the individual dose contributors to the Sector E groundwater pathway dose for this Composite Sensitivity Study. The Np-237 dose is associated with the timing of waste tank liner failures and subsequent chemical transitions of cementitious materials. Specifically, the solubility control associated with fill grout transitions results in a relatively acute peak dose shortly after Np-237 is released from the waste tank basemat. The transition from reducing to oxidizing is a change of two orders of magnitude (from 1E-09 mol/L to 3E-07 mol/L). The transition from Region II to Region III is a change of another order of magnitude (from 3E-07 mol/L to 2E-06 mol/L). The earlier Np-237 release is associated with the Type I waste tanks which have a thinner basemat (30 inches) than the Type III/IIIA waste tanks (41 to 42 inches).

Figure 6.3-34: Individual Radionuclide Contributors to the Composite Sensitivity Study of 100-Meter Peak Groundwater Pathway Dose within 100,000 Years for Sector E



6.3.6.4 Probabilistic Model Sensitivity Analysis

The FTF inventory distributions used in the probabilistic model have been revised to reflect the inventory uncertainty associated with the new FTF inventory information (SRR-CWDA-2009-00045), as shown in Table 6.3-22.

Table 6.3-22: Revised Inventory Multipliers

Radio-nuclide	Tank 5 [a]		Tank 6 [a]		Tank 18 [a]		Tank 19 [a]		All Other Waste Tanks	
	Mean	σ	Mean	σ	Mean	σ	Mean	σ	Min	Max
Ac-227	1	0.5	1	0.5	1	0.5	1	0.5	0.01	1
Al-26	1	0.5	0	0	1	0.5	1	0.5	0.01	1
Am-241	0.99	0.08	0.72	0.13	0.68	0.22	0.81	0.31	0.01	10
Am-242m	0.93	0.14	0.82	0.13	0.45	0.29	1	0.5	0.01	1
Am-243	0.97	0.08	0.66	0.16	0.56	0.38	0.77	0.35	0.01	10
C-14	1	0.5	0.8	0.23	0.69	0.44	0.71	0.44	0.01	1
Cf-249	1	0.5	1	0.5	1	0.5	1	0.5	0.01	1
Cl-36	1	0.5	0	0	1	0.5	1	0.5	0.01	1
Cm-243	1	0.5	1	0.5	1	0.5	1	0.5	0.01	1
Cm-244	0.93	0.1	0.74	0.12	0.74	0.51	0.84	0.32	0.01	10
Cm-245	1	0.5	0.73	0.04	1	0.5	1	0.5	0.01	1
Cm-247	1	0.5	1	0.5	1	0.5	1	0.5	0.01	1
Cm-248	1	0.5	1	0.5	1	0.5	1	0.5	0.01	1
Co-60	0.99	0.07	0.76	0.09	0.6	0.17	0.89	0.23	0.01	10
Cs-135	0.95	0.14	0.88	0.06	0.79	0.28	0.93	0.32	0.01	1
Cs-137	0.96	0.12	0.89	0.05	0.63	0.17	0.95	0.22	0.01	10
Eu-152	1	0.5	1	0.5	1	0.5	1	0.5	0.01	10
Eu-154	0.98	0.06	0.74	0.11	0.94	0.21	0.88	0.29	0.01	10
H-3	1	0.5	1	0.5	1	0.5	1	0.5	0.01	1
I-129	0.8	0.27	0.73	0.13	0.84	0.25	0.41	0.22	0.01	10
K-40	1	0.5	0	0	0.87	0.21	1	0.5	0.01	10
Mo-93m	1	0	0	0	0	0	0	0	0.01	1
Nb-93m	0.96	0.09	0.59	0.19	1	0.5	0.73	0.4	0.01	10
Nb-94	1	0.5	1	0.5	1	0.5	1	0.5	0.01	1
Ni-59	0.9	0.2	0.81	0.2	0.71	0.26	1	0.5	0.01	10
Ni-63	0.87	0.24	0.79	0.23	0.82	0.27	1	0.5	0.01	10
Np-237	0.87	0.26	0.57	0.23	0.64	0.25	0.92	0.21	0.01	10
Pa-231	1	0.5	1	0.5	0.24	0.14	1	0.5	0.01	1
Pb-210	1	0	0	0	0	0	0	0	0.01	1
Pd-107 [b]	0.75	0.4	0	0	0.79	0.3	1	0.5	0.01	1
Pu-238	0.98	0.11	0.87	0.08	0.4	0.19	0.96	0.23	0.01	10
Pu-239	0.94	0.11	0.87	0.07	0.72	0.27	0.9	0.25	0.01	10
Pu-240	0.96	0.12	0.86	0.08	0.71	0.26	0.88	0.26	0.01	10

Table 6.3-22: Revised Inventory Multipliers (Continued)

Radio-nuclide	Tank 5 [a]		Tank 6 [a]		Tank 18 [a]		Tank 19 [a]		All Other Waste Tanks	
	Mean	Σ	Mean	σ	Mean	σ	Mean	σ	Min	Max
Pu-241	1	0.5	0.88	0.07	0.67	0.18	0.93	0.25	0.01	10
Pu-242	0.94	0.14	0.9	0.07	1	0.5	1	0.5	0.01	1
Pu-244	1	0.5	1	0.5	1	0.5	1	0.5	0.01	1
Ra-226	1	0.5	1	0.5	1	0.5	1	0.5	0.01	1
Ra-228	1	0	0	0	0	0	0	0	0.01	1
Rn-222	1	0	0	0	0	0	0	0	0.01	1
Se-79	0.78	0.36	0.7	0.33	1	0.5	1	0.5	0.01	10
Sm-151	0.99	0.08	0.93	0.05	0.84	0.25	0.91	0.23	0.01	10
Sn-126	1	0.5	1	0.5	0.92	0.2	0.99	0.23	0.01	10
Sr-90	1	0.05	0.89	0.04	0.6	0.15	0.78	0.35	0.01	10
Tc-99	0.94	0.14	0.87	0.13	0.87	0.23	0.87	0.26	0.01	10
Th-228	1	0	0	0	0	0	0	0	0.01	1
Th-229 [b]	0.34	0.48	0.73	0.28	0.66	0.35	0.65	0.26	0.01	1
Th-230 [b]	0.74	0.51	0.81	0.19	0.68	0.37	0.59	0.16	0.01	1
Th-232	1	0	0	0	0	0	0	0	0.01	1
U-232	1	0.5	0.8	0.2	0.7	0.38	0.88	0.26	0.01	1
U-233	1	0.5	0.87	0.09	0.75	0.16	0.94	0.23	0.01	10
U-234	0.91	0.18	0.89	0.1	0.81	0.36	0.92	0.23	0.01	10
U-235	0.98	0.12	0.87	0.1	0.68	0.38	0.9	0.23	0.01	10
U-236	0.97	0.12	0.87	0.09	0.74	0.29	0.94	0.23	0.01	1
U-238	0.96	0.12	0.76	0.1	0.68	0.39	0.93	0.22	0.01	10
Zr-93	0.96	0.09	0.59	0.19	0.66	0.36	0.73	0.4	0.01	10

Notes Radionuclide inventory multipliers are sampled with log-uniform distributions except:
a: Tanks 5, 6, 18 and 19 are sampled along a normal distribution
b: Within Tank 5, Pd-107, Th-229, and Th-230 are independently sampled with a gamma distribution.

Note: All radionuclides truncated at a minimum of zero. Radionuclides with a mean of 1.0 were truncated at a maximum of 1.0.

σ standard deviation

A probabilistic uncertainty analysis has been performed (using the GoldSim FTF model version SRS FTF v3.0) to assess the impact of the revised FTF inventory distributions. In addition to the revised inventories and inventory distributions, this probabilistic model incorporates other improvements to the GoldSim FTF model from the FTF PA as discussed in the GoldSim FTF v3.0 model report (SRR-CWDA-2012-00119). This model was run twice, each for 1,000 realizations (for an All Cases probabilistic model and for the Base Case probabilistic model) to demonstrate the effects of the revised inventory data and modeling corrections. Additionally, a Composite Sensitivity Study was performed to incorporate revised waste release model information and revised K_d values. The results of these modeling runs are discussed below.

6.3.6.4.1 All Cases Probabilistic Model

Statistical data associated with the GoldSim SRS FTF v3.0 probabilistic peak dose results for 1,000 All Cases realizations is shown in Figure 6.3-35 (10,000 years linear scale), Figure 6.3-36 (10,000 years log scale), Figure 6.3-37 (100,000 years linear scale), and Figure 6.3-38 (100,000 years log scale).

Figure 6.3-35: All Cases Probabilistic Results (10,000 Year) Linear Scale

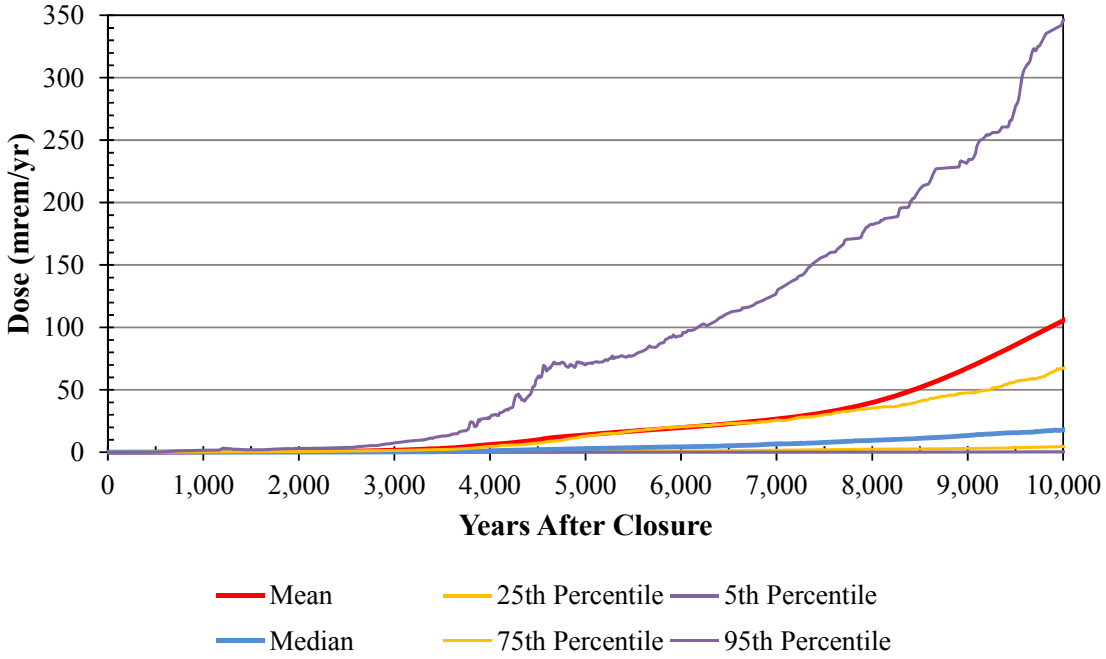


Figure 6.3-36: All Cases Probabilistic Results (10,000 Year) Log Scale

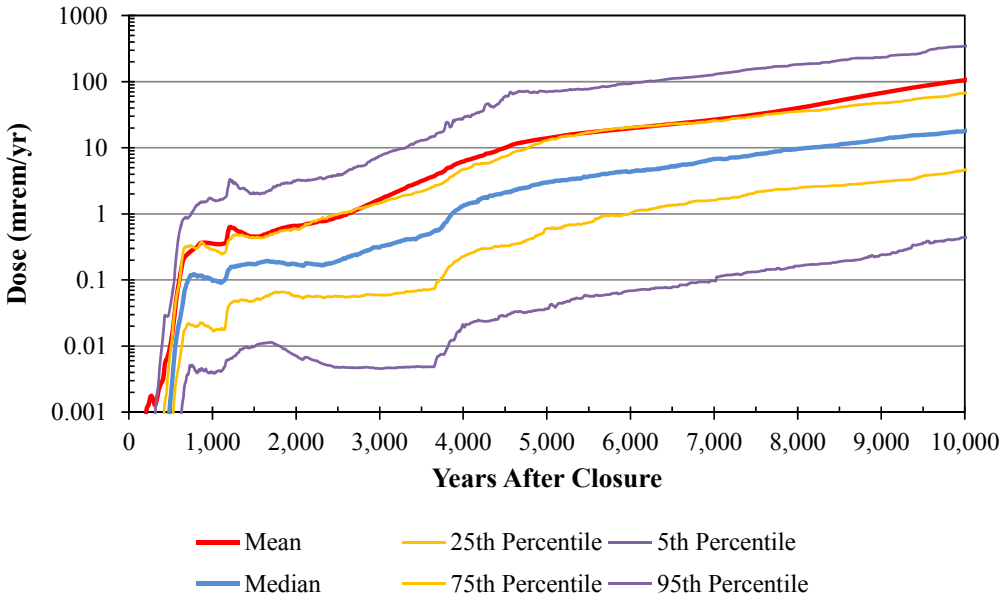


Figure 6.3-37: All Cases Probabilistic Results (100,000 Year) Linear Scale

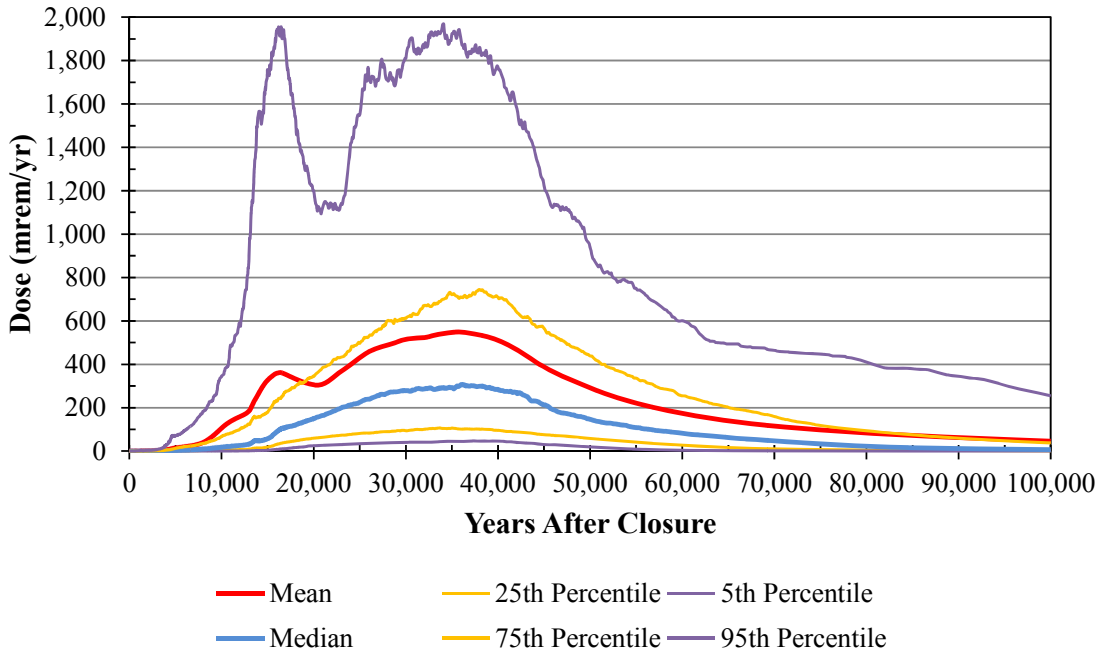
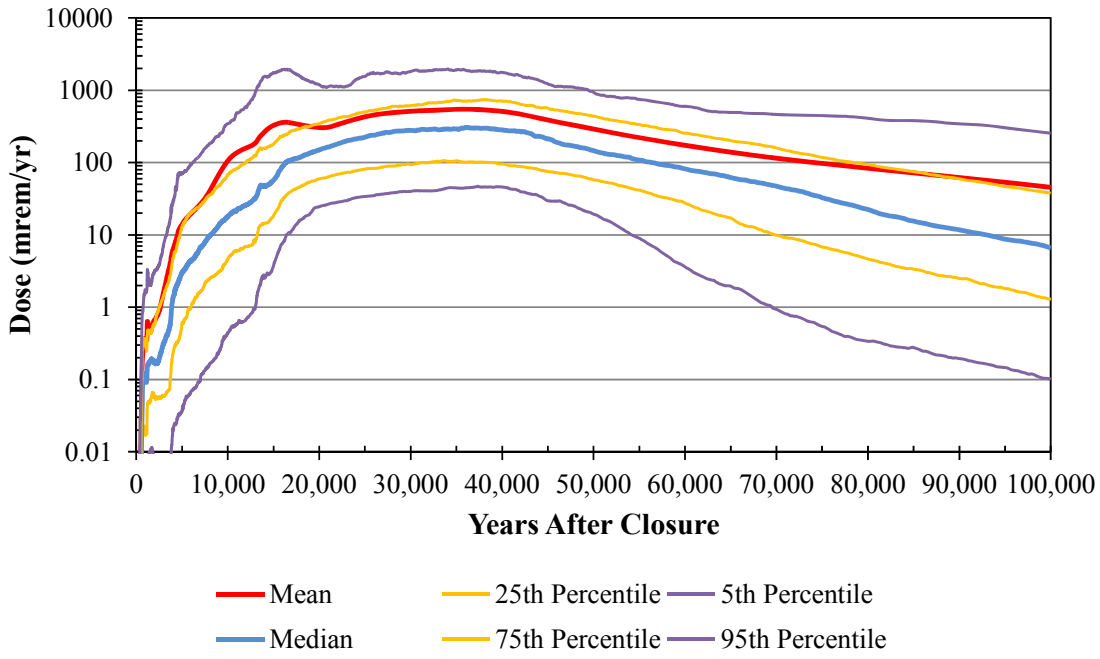


Figure 6.3-38: All Cases Probabilistic Results (100,000 Year) Log Scale



The probabilistic peak dose results (using the GoldSim SRS FTF v3.0 model) for 1,000 All Cases realizations in 10,000 years showed the peak of the mean to be 106 mrem/yr and the 95th percentile of the doses to be 346 mrem/yr. (In the FTF PA, the peak of the mean was 14 mrem/yr and the 95th percentile was 77 mrem/yr; and in the Tank 18/19 Special Analysis the peak of the mean was 13 mrem/yr and the 95th percentile was 74 mrem/yr). In 100,000 years the peak of the mean was 549 mrem/yr (versus 344 mrem/yr in the FTF PA and 344 mrem/yr in the Tank 18/19 Special Analysis) and the 95th percentile of the doses to be 1,969 mrem/yr (versus 1,919 mrem/yr in the FTF PA and 1,554 mrem/yr in the Tank 18/19 Special Analysis).

The 10,000-year dose results are significantly higher than those shown previously (i.e., in the FTF PA and in the Tank 18/19 Special Analysis) because 1) earlier releases of Pu-239 and 2) increased inventories in some of the dose-significant radionuclides (such as Zr-93, which decays in Nb-93m). Both of these factors are addressed by the Composite Sensitivity Study, which includes more probable and defensible assumptions regarding Pu-239 and Zr-93 release (see Section 6.3.6.3) than were assumed in the Base Case probabilistic model (described in Section 6.3.6.4.2).

6.3.6.4.2 Base Case Probabilistic Model

Statistical data associated with the GoldSim SRS FTF v3.0 probabilistic peak dose results for 1,000 Base Case realizations is shown in Figure 6.3-39 (10,000 years linear scale), Figure 6.3-40 (10,000 years log scale), Figure 6.3-41 (100,000 years linear scale), and Figure 6.3-42 (100,000 years log scale).

The probabilistic peak dose results (using the GoldSim SRS FTF v3.0 model) for 1,000 Base Case realizations in 10,000 years showed the peak of the mean to be 11.5 mrem/yr (versus 4.8 mrem/yr in the FTF PA and 4 mrem/yr in the Tank 18/19 Special Analysis). The primary reason for the higher 10,000-year dose is that, in some realizations, Pu-239 became available for earlier release in the GoldSim SRS FTF v3.0 model than was modeled in the FTF PA. The 95th percentile of the doses was 61 mrem/yr (versus 30 mrem/yr in the FTF PA and 35 mrem/yr in the Tank 18/19 Special Analysis). In 100,000 years, the peak of the mean was shown to be 542 mrem/yr and the 95th percentile of the doses to be 2,179 mrem/yr. Note that Base Case doses were not calculated out to 100,000 years in the FTF PA or the Tank 18/19 Special Analysis.

The fact that the 95th percentile value is higher than calculated previously is not unexpected. Although many of the reasonably conservative stochastic inventory distributions used in the FTF PA were tightened (as shown in Table 6.3-22), when the actual waste tank residual characterization data was used the Zr-93 inventory was substantially increased, leading to an increase in the dose associated with Nb-93m.

Figure 6.3-39: Base Case Probabilistic Results (10,000 Year) Linear Scale

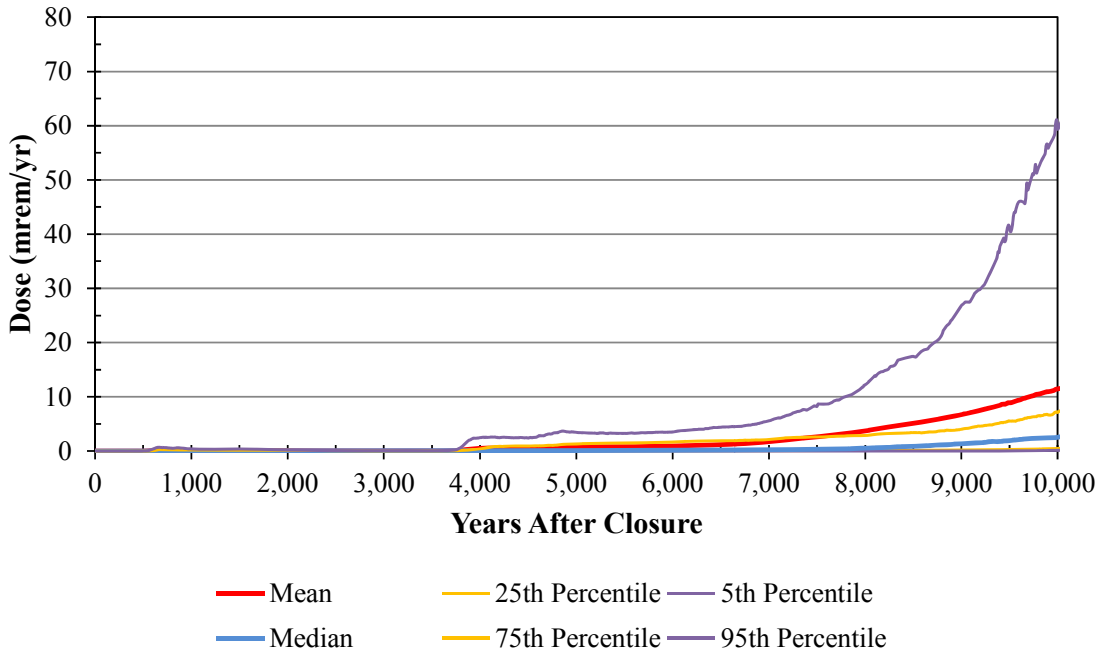


Figure 6.3-40: Base Case Probabilistic Results (10,000 Year) Log Scale

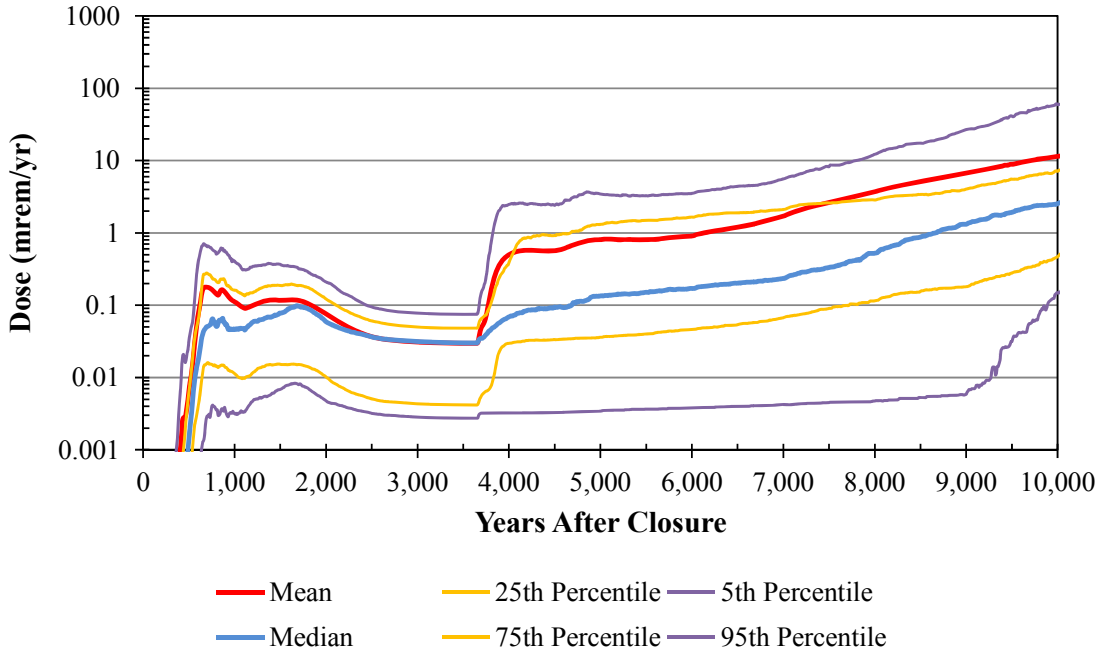


Figure 6.3-41: Base Case Probabilistic Results (100,000 Year) Linear Scale

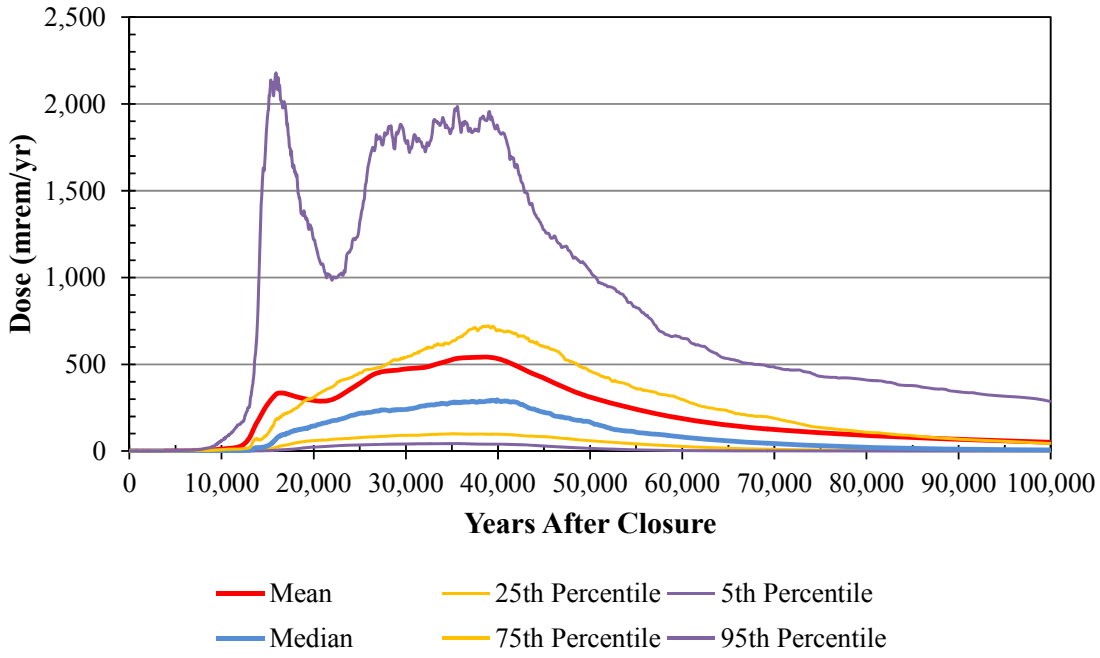
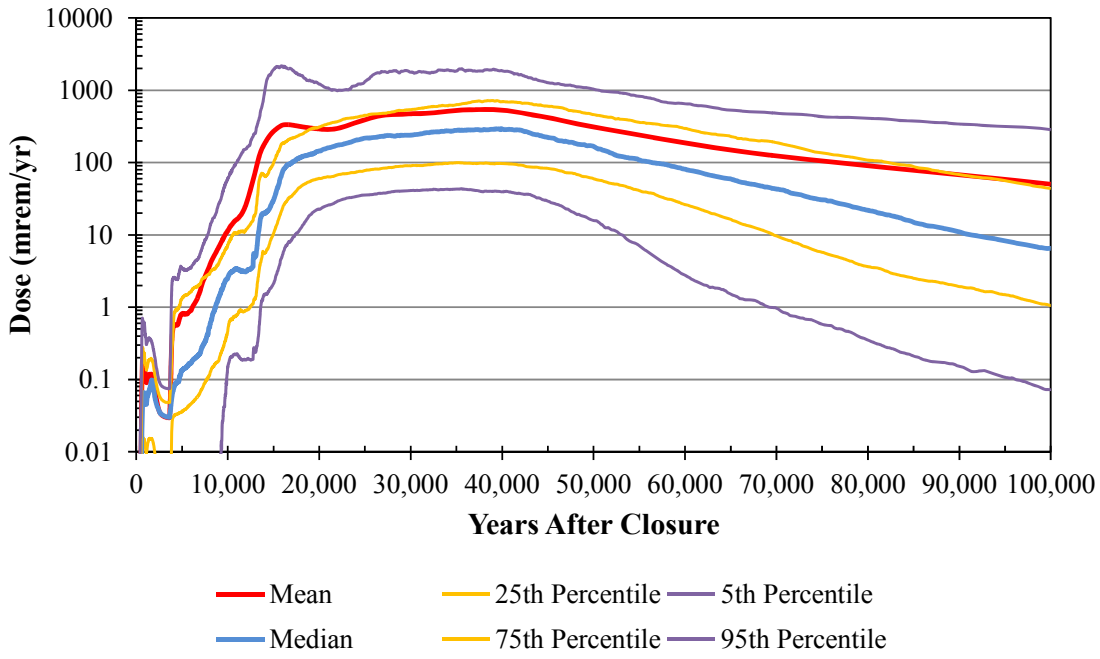


Figure 6.3-42: Base Case Probabilistic Results (100,000 Year) Log Scale



6.3.6.4.3 Composite Sensitivity Study, All Cases Probabilistic Model

New studies have increased DOE's understanding of the physical and chemical parameters that affect PA results. A final probabilistic run was performed that applied a revised waste release model (including updated solubility controls) and updated K_d values, in addition to the updated inventory parameters as described in Section 6.3.6.3. This Composite Sensitivity Study was run for 1,000 realizations, using the All Cases modeling configuration.

Statistical data associated with the GoldSim SRS FTF v3.0 probabilistic peak dose results for 1,000 Composite Sensitivity Study realizations is shown in Figure 6.3-43 (10,000 years linear scale), Figure 6.3-44 (10,000 years log scale), Figure 6.3-45 (100,000 years linear scale), and Figure 6.3-46 (100,000 years log scale).

The probabilistic peak dose results (using the GoldSim SRS FTF v3.0 model) for 1,000 All Cases realizations in 10,000 years showed the peak of the mean to be 2.5 mrem/yr and the 95th percentile of the doses to be 10.7 mrem/yr. In 100,000 years, the peak of the mean was shown to be 27.8 mrem/yr and the 95th percentile of the doses to be 90.7 mrem/yr. As previously described, the peak of the mean in the FTF PA was 14 mrem/yr and the 95th percentile was 77 mrem/yr, and in the Tank 18/19 Special Analysis the peak of the mean was 13 mrem/yr and the 95th percentile was 74 mrem/yr.

This composite analysis used input parameters that are less conservative but reflect assumptions that are more reasonable, relative to the modeling case discussed in Section 6.3.6.4.1. The most notable differences in the results were the reductions to the Nb-93m and Pu-239, which were driven by the application of more reasonable solubility controls and K_d values. As an example, Nb-93m (and the parent Zr-93) was previously modeled without solubility controls and without K_d values. This conservative assumption was used in the FTF PA, as the FTF PA inventory had underestimated the presence of these radionuclides, such that the dose contribution was not significant. However, given the increased inventory, a more reasonable modeling approach was needed to assess the impact of this inventory appropriately.

Figure 6.3-43: Composite Analysis All Cases Probabilistic Results (10,000 Year) Linear Scale

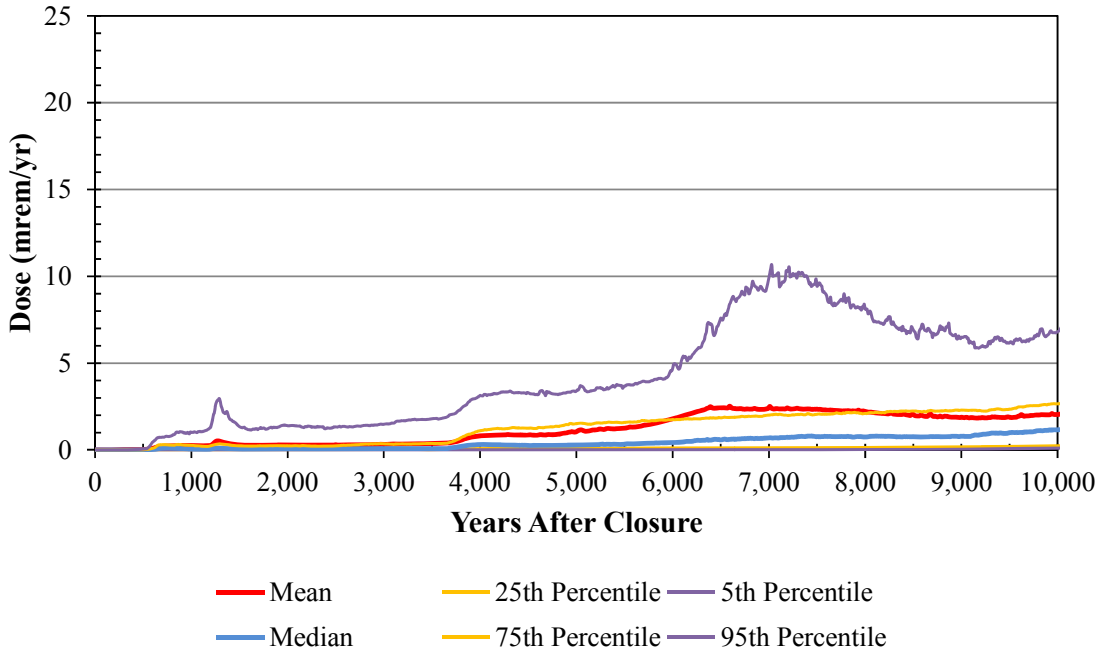


Figure 6.3-44: Composite Analysis All Cases Probabilistic Results (10,000 Year) Log Scale

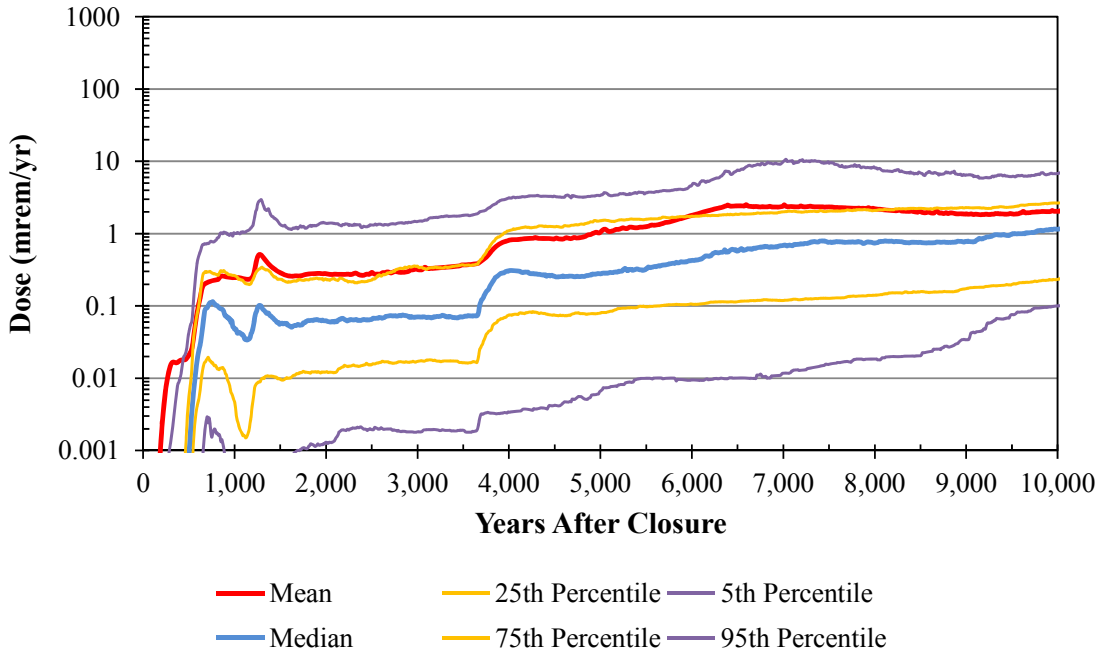


Figure 6.3-45: Composite Analysis All Cases Probabilistic Results (100,000 Year) Linear Scale

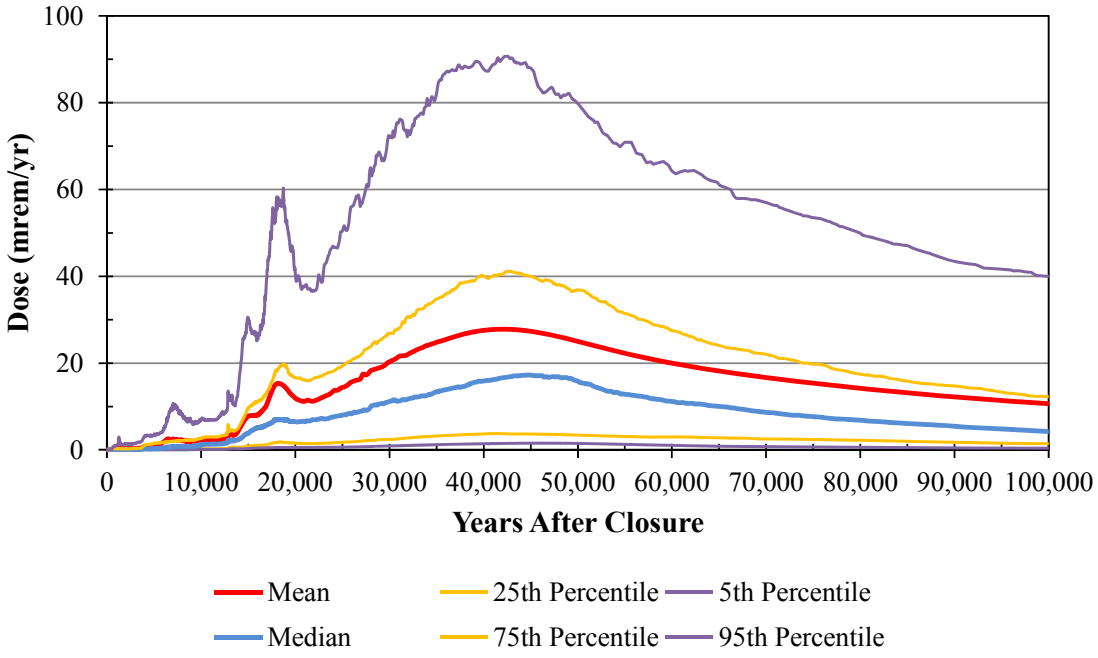
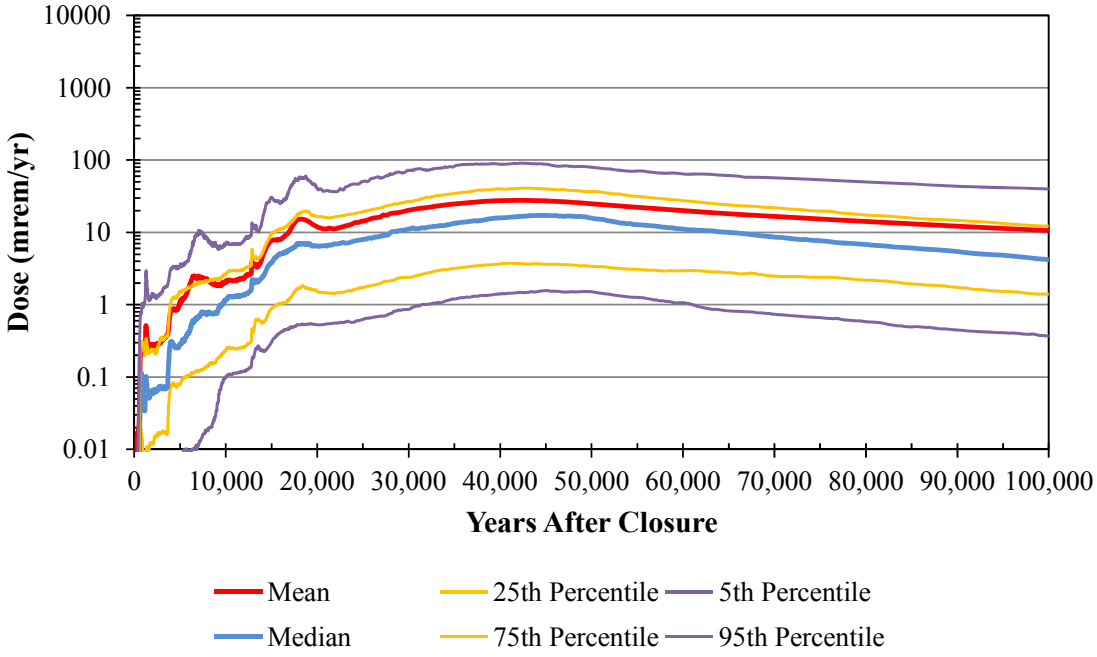


Figure 6.3-46: Composite Analysis All Cases Probabilistic Results (100,000 Year) Log Scale



6.3.6.4.4 Radionuclide Dose Comparisons from Probabilistic Models

Figures 6.3-47, 6.3-48, and 6.3-49 show the dose contributions from specific radionuclides (Nb-93m, Np-237, Pa-231, Pu-239, and Ra-226) for each of the three probabilistic models described above. These figures illustrate which radionuclides are most important in driving the doses in each of these models. These doses, presented for 100,000 years, represent the means of the dose contributions, taken from the sector, with the highest contribution for the given radionuclide at each time-step.

Figure 6.3-47: Radionuclide Contributors for All Cases Probabilistic Results (100,000 Year)

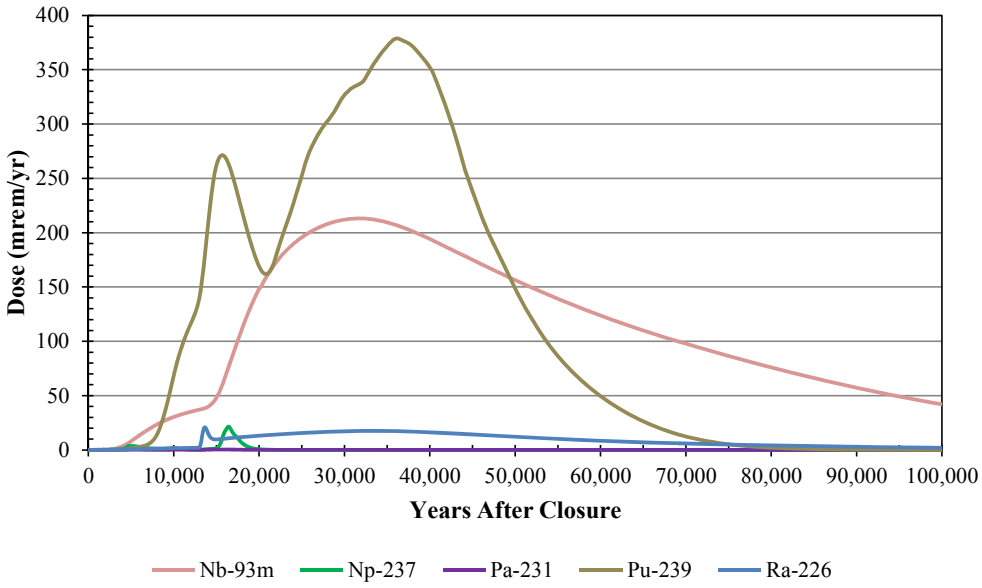


Figure 6.3-48: Radionuclide Contributors for Base Case Probabilistic Results (100,000 Year)

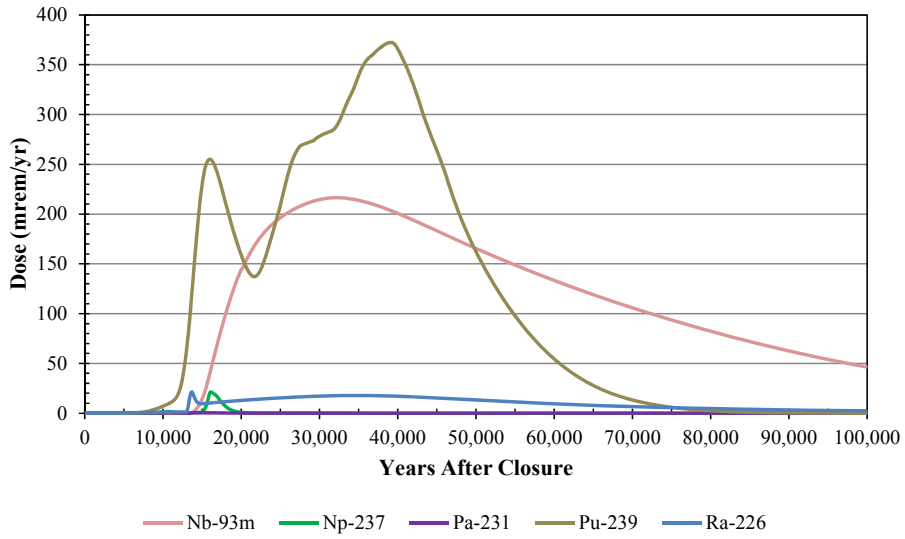
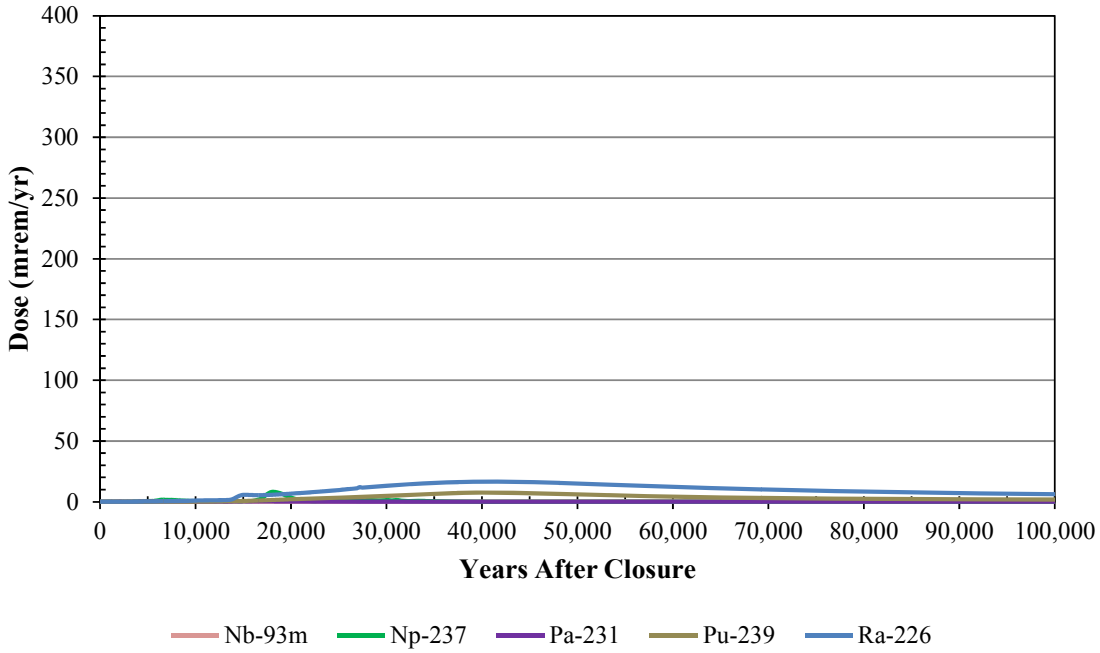


Figure 6.3-49: Radionuclide Contributors for Composite Analysis All Cases Probabilistic Results (100,000 Year)



The other modeling parameters described in the FTF PA Section 5.6.3 are unaffected by the residual inventories at operational closure. The uncertainty and sensitivity analyses results described in the remainder of FTF PA Sections 5.6.4 through 5.6.7 (aside from those discussed above) remain valid irrespective of the revised Tanks 5 and 6 residual inventories and do not require further analysis.

6.3.7 RCRA/CERCLA Risk Analysis

The RCRA/CERCLA risk assessment for the FTF final facility closure follows the current Area Completion Project protocols for human health and ecological risk assessments. The FTF Contaminant Migration Constituents of Concern (CMCOC) were identified through a system that is consistent with both the Area Completion Project protocols and FTF PA. The CMCOC were identified by modeling the release of contaminants and their travel through the vadose zone, using the deterministic Base Case model with the updated inventories. The concentrations of contaminants that are modeled to reach the water table are compared to MCL, RSLs, PRGs, or other appropriate standards in cases where the constituent does not have an MCL. Any constituents that are predicted to exceed these standards (i.e., fraction greater than 1.0) in the groundwater directly beneath FTF (one meter from boundary) are identified as CMCOC as shown in Tables 6.3-23 and 6.3-24, which reflects concentrations calculated using the revised FTF inventories. Based on the 1,000-year concentration curves, the CMCOC identified using the described protocols are Np-237 and Tc-99.

Table 6.3-23: Groundwater Radionuclide Concentrations at 1 Meter from FTF[a]

Radionuclide	MCL** (pCi/L)	Residential Tap Water PRG*** (pCi/L)	Peak Concentration (pCi/L) 1 to 1,000 Years	Fraction of MCL or PRG at 1 meter
Ac-227	N/A	2.4E-01	1.7E-07	7.2E-07
Al-26	N/A	2.8E+00	b	N/A
Am-241	N/A	4.6E-01	1.4E-23	3.0E-23
Am-242m	N/A	6.7E-01	b	N/A
Am-243	N/A	4.6E-01	3.6E-28	7.7E-28
Ba-137m*	N/A	Cs-137 daughter	N/A	N/A
Bk-249	N/A	4.3E+01	b	N/A
C-14	2.0E+03	MCL used	4.5E-03	2.2E-06
Ce-144	N/A	1.4E+00	b	N/A
Cf-249	N/A	3.8E-01	3.5E-30	9.4E-30
Cl-36	7.0E+02	MCL used	6.6E-06	9.4E-09
Cm-242	N/A	1.2E+00	b	N/A
Cm-243	N/A	5.0E-01	b	N/A
Cm-244	N/A	5.7E-01	b	N/A
Cm-245	N/A	4.6E-01	7.5E-29	1.6E-28
Cm-247	N/A	4.8E-01	b	N/A
Cm-248	N/A	5.0E-03	2.9E-30	5.8E-28
Co-60	1.0E+02	MCL used	b	N/A
Cs-134	N/A	1.1E+00	b	N/A
Cs-135	9.0E+02	MCL used	9.9E-07	1.1E-09
Cs-137	2.0E+02	MCL used	2.0E-11	9.9E-14
Eu-152	2.0E+02	MCL used	b	N/A
Eu-154	6.0E+01	MCL used	b	N/A
Eu-155	6.0E+02	2.5E+01	b	N/A
Gd-152	N/A	1.6E+00	b	N/A
H-3	2.0E+04	MCL used	1.9E-08	9.3E-13
I-129	1.0E+00	MCL used	3.9E-03	3.9E-03
K-40	N/A	1.9E+00	9.8E-05	5.1E-05
Mo-93m	N/A	1.5E+02	b	N/A
Na-22	N/A	5.0E+00	b	N/A
Nb-93m	1.0E+03	MCL used	2.3E-13	2.3E-16
Nb-94	N/A	6.1E+00	2.0E+00	3.3E-01
Ni-59	3.0E+02	MCL used	2.4E+00	8.0E-03
Ni-63	5.0E+01	MCL used	1.8E-01	3.7E-03
Np-237	N/A	7.7E-01	1.6E+00	2.0E+00
Pa-231	N/A	2.8E-01	2.4E-04	8.8E-04
Pb-210	N/A	5.4E+02	1.6E-03	2.9E-02
Pd-107	N/A	1.9E+02	3.8E-06	2.0E-08
Pm-147	N/A	2.8E+01	b	N/A

Table 6.3-23: Groundwater Radionuclide Concentrations at 1 Meter from FTF[a] (Continued)

Radionuclide	MCL** (pCi/L)	Residential Tap Water PRG*** (pCi/L)	Peak Concentration (pCi/L) 1 to 1,000 Years	Fraction of MCL or PRG at 1m
Pr-144	N/A	5.9E+02	b	N/A
Pu-238	N/A	3.6E-01	7.6E-26	2.1E-25
Pu-239	N/A	3.5E-01	1.8E-25	5.2E-25
Pu-240	N/A	3.5E-01	5.1E-25	1.4E-24
Pu-241	300	MCL used	5.7E-27	2.1E-28
Pu-242	N/A	3.7E-01	9.1E-27	2.4E-26
Pu-244	N/A	3.5E-01	b	N/A
Ra-226 + Ra-228	5.0E+00	MCL used	6.7E-01	1.3E-01
Ra-228	N/A	4.6E-02	3.6E-20	7.8E-19
Rh-106*	N/A	Ru-106 daughter	b	N/A
Ru-106 + D	N/A	1.1E+00	b	N/A
Sb-125	N/A	1.1E+01	b	N/A
Sb-126	N/A	4.3E+00	b	N/A
Sb-126m	N/A	7.2E+02	b	N/A
Se-79	N/A	6.5E+00	1.4E-17	2.1E-18
Sm-147	N/A	1.3E+00	b	N/A
Sm-151	1.0E+03	MCL used	1.8E-29	1.8E-32
Sn-126	N/A	1.9E+00	3.6E-26	2.0E-26
Sr-90	8.0E+00	MCL used	4.0E-05	5.0E-06
Tc-99	9.0E+02	MCL used	1.1E+03	1.2E+00
Te-125m	N/A	1.4E+01	b	N/A
Th-228	N/A	4.5E-01	4.8E-20	1.1E-19
Th-229	N/A	2.1E-01	2.5E-08	1.2E-07
Th-230	N/A	5.2E-01	2.2E-17	4.2E-17
Th-232	N/A	4.7E-01	3.6E-25	7.7E-25
U-232	N/A	1.6E-01	1.2E-20	7.2E-20
U-233	N/A	6.6E-01	5.6E-06	8.5E-06
U-234	N/A	6.7E-01	2.0E-13	3.0E-13
U-235	N/A	6.8E-01	1.4E-16	2.0E-16
U-236	N/A	7.1E-01	6.3E-16	8.8E-16
U-238	N/A	7.4E-01	1.1E-17	1.4E-17
Y-90*	N/A	2.6E+00	b	N/A
Zr-93	2.0E+03	MCL used	3.5E-27	1.8E-30

a CMCOC are shaded gray in the table.
b Values < 1.0E-30
* Daughters are assumed to be in equilibrium with the parent nuclide.
** MCL values for beta and photon emitters are calculated in Table II-3 of FR-00-9654 based on a beta-gamma dose of 4 mrem/yr.
*** Residential tap water PRGs are calculated based on a target cancer risk of 1.0E-06 (EPA-PRGs_11-13-2007)
N/A Not Available

Table 6.3-24: Groundwater Chemical Concentrations at 1 Meter from FTF[a]

Chemical	MCL* (μg/L)	Tap Water RSLs** (μg/L)	Peak Concentration (μg/L) 1 to1,000 Yrs	Fraction of MCL or RSL at 1 meter
Ag	1.0E+02***	MCL used	3.5E-06	3.5E-08
As	1.0E+01	MCL used	3.0E-19	3.0E-20
Ba	2.0E+03	MCL used	6.7E-02	3.3E-05
Cd	5.0E+00	MCL used	2.5E-01	4.9E-02
Total Chromium	1.0E+02	MCL used	9.2E-02	9.2E-04
Cu	1.3E+03	MCL used	3.8E-06	2.9E-09
F	4.0E+03	MCL used	5.2E-01	1.3E-04
Fe	3.0E+02***	MCL used	8.1E-20	2.7E-22
Hg	2.0E+00	MCL used	2.9E-18	1.4E-18
Mn	5.0E+01***	MCL used	1.2E-06	2.3E-08
Ni	N/A	7.3E+02	1.7E+00	2.4E-03
$NO_2 + NO_3$	1.0E+04	MCL used	2.0E+01	2.3E-03
Pb	1.5E+01	MCL used	4.8E-21	3.2E-22
Sb	6.0E+00	MCL used	b	b
Se	5.0E+01	MCL used	4.9E-23	9.8E-25
U	3.0E+01	MCL used	3.2E-17	1.1E-18
Zn	5.0E+03***	MCL used	1.5E-10	3.0E-14

a CMCOC are shaded gray in the table.
b Values < 1.0E-30
* EPA 816-F-09-0004
** EPA_RSL Tbl_11-2011
*** EPA 816-F-10-079
N/A Not Available

6.3.7.1 Neptunium-237

Neptunium-237 does not have an MCL so the peak concentration of 1.6 pCi/L at 1 meter is compared to the calculated PRG of 7.7E-01 pCi/L. This peak concentration occurs 722 years following FTF facility closure. As shown in Table 6.3-25, the peak concentration drops to 0.42 pCi/L at 100 meters, which is below the PRG value.

6.3.7.2 Technetium-99

Technetium-99 has a peak concentration of 1,100 pCi/L at 1 meter compared to the MCL of 900 pCi/L. This peak concentration occurs approximately 678 years following FTF facility closure. As shown in Table 6.3-25, the peak concentration drops to 270 pCi/L at 100 meters, which is below the MCL value.

Table 6.3-25: Groundwater Concentrations of CMCOC at 100 Meters

Contaminant	MCL (pCi/L)	Residential Tap Water PRG (pCi/L)	Peak Concentration at 100 meters (pCi/L) 1 to 1,000 Yrs	Fraction of MCL or PRG at 100 meters
Np-237	N/A	7.7E-01	4.2E-01	5.5E-01
Tc-99	9.0E+02	MCL used	2.7E+02	3.0E-01

N/A Not Available

6.3.8 ALARA Analysis

Section 5.8 of the FTF PA describes how the "as low as reasonably achievable" (ALARA) requirement of DOE O 435.1, Chg. 1 and 10 CFR 61.41 are implemented for FTF. The ALARA information presented in the FTF PA is not affected by the new residual waste information.

6.4 FTF Inadvertent Intruder Analysis

This section will discuss the impact of the new residual waste information on the FTF inadvertent intruder analysis information presented in Section 6.0 of the FTF PA.

6.4.1 Groundwater Concentrations at One Meter

Section 6.1 of the FTF PA presents the one-meter groundwater concentrations for the FTF radionuclides and chemicals. Maximum groundwater concentrations are given for the modeling cell adjoining the analyzed source terms. Results are presented for the three distinct aquifers modeled (UTRA-UZ, UTRA-LZ, and Gordon Aquifer).

These groundwater concentrations were recalculated using the PORFLOW FTF model for the FTF PA Base Case using the revised FTF residual inventories presented in Section 6.3.1. Tables A-9 through A-11 (see Appendix A) show peak one-meter radionuclides concentrations for the three aquifers in 1,000 years. These radionuclide concentrations reflect the peak concentrations for each radionuclide in the highest sector. The peak 1-meter chemical concentrations for the three aquifers in 1,000 years are shown Tables A-12 through A-14 (see Appendix A). These chemical concentrations also reflect the peak concentrations for the sector. Similarly, Appendix B presents peak radionuclide concentrations within 10,000 years.

6.4.2 Acute Exposure Scenarios

Section 6.2 of the FTF PA describes how the biotic pathways doses are calculated for the Acute Exposure Scenarios. The acute exposure scenarios information presented in the FTF PA is not significantly affected by the new residual waste information.

6.4.3 Chronic Exposure Scenarios

Section 6.3 of FTF PA describes how the biotic pathways doses are calculated for the chronic exposure scenarios. The chronic exposure scenarios information presented in the FTF PA is not affected by the new residual waste information.

6.4.4 Groundwater Doses at One Meter

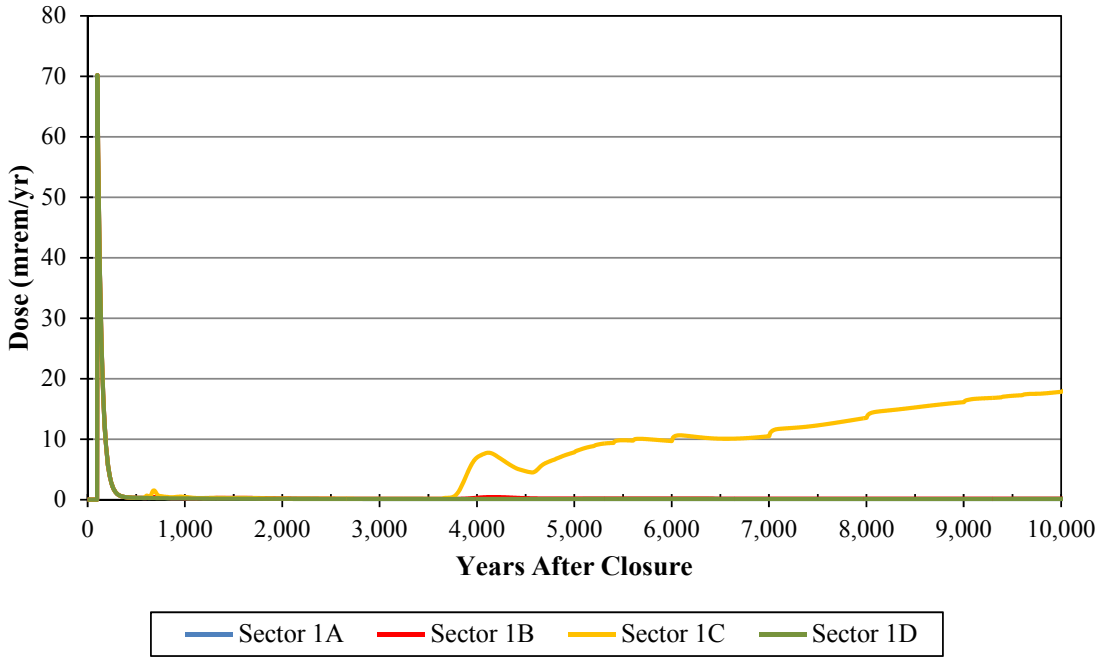
Section 6.4 of FTF PA contains calculations of the peak total intruder doses for the acute intruder scenario and for the chronic intruder agricultural (post-drilling) scenario. For the acute intruder, doses were calculated assuming the acute intruder drills into a three-inch diameter transfer line at any time after the 100-year period of institutional control following FTF facility closure. For the chronic intruder, annual doses were calculated assuming contamination from the drill cuttings, as well as from the use of water obtained from a well.

As described in Section 6.5.2.2 of the FTF PA, "the waste tank engineered barriers (e.g., closure cap erosion barrier, tank top concrete, and tank liner, where applicable), will prevent drilling into the waste inventory." [SRS-REG-2007-00002] Therefore, the acute intruder scenario is dependent on the transfer line inventory (i.e., it does not include a groundwater contribution) and is not affected by a revision to the FTF inventories. The inadvertent human intruder doses were negligibly affected by the correction to the timing input in the soil buildup calculation. The peak dose for the acute intruder in 10,000 years therefore remains 2 millirem at year 100, primarily due to exposure to drill cuttings.

The peak chronic doses have been recalculated using the peak groundwater concentrations identified in Section 6.3.2 of this Special Analysis (i.e., groundwater concentrations at 1-meter and at the seepline recalculated using the PORFLOW FTF model for the FTF PA Base Case using the revised FTF inventories).

For the chronic intruder, the contributions to the peak doses from the four 1-meter sectors were recalculated using the highest concentration for each radionuclide in the sector, with the recalculation based on the revised FTF inventories. These peak doses were the total dose associated with drill cuttings and all the individual 1-meter well pathways. Figure 6.4-1 graphically presents the annual dose to the chronic intruder for each of the four one-meter sectors for 10,000 years after FTF facility closure. As shown in Figure 6.4-1, the dose to the chronic intruder, within 10,000 years, is highest at 100 years after FTF final facility closure and is attributed to the earliest time after FTF final facility closure that an intruder is assumed to drill into the closure area.

Figure 6.4-1: Annual Dose to the Chronic Intruder within 10,000 Years



Note Peak at year 100 occurs for all sectors.

Table 6.4-1 presents the chronic intruder peak dose within 10,000 years and identifies the contribution from the significant pathways and their contributing radionuclides. The peak dose for the chronic intruder scenario in 10,000 years was 70 mrem/yr at year 100. This peak annual dose was almost entirely due to ingestion of vegetables contaminated with drill cuttings, with 96 % of the 70 mrem/yr being due to vegetable ingestion. The principal radionuclide contributors to this vegetable dose were the short-lived isotopes Sr-90/Y-90 and Cs-137/Ba-137m.

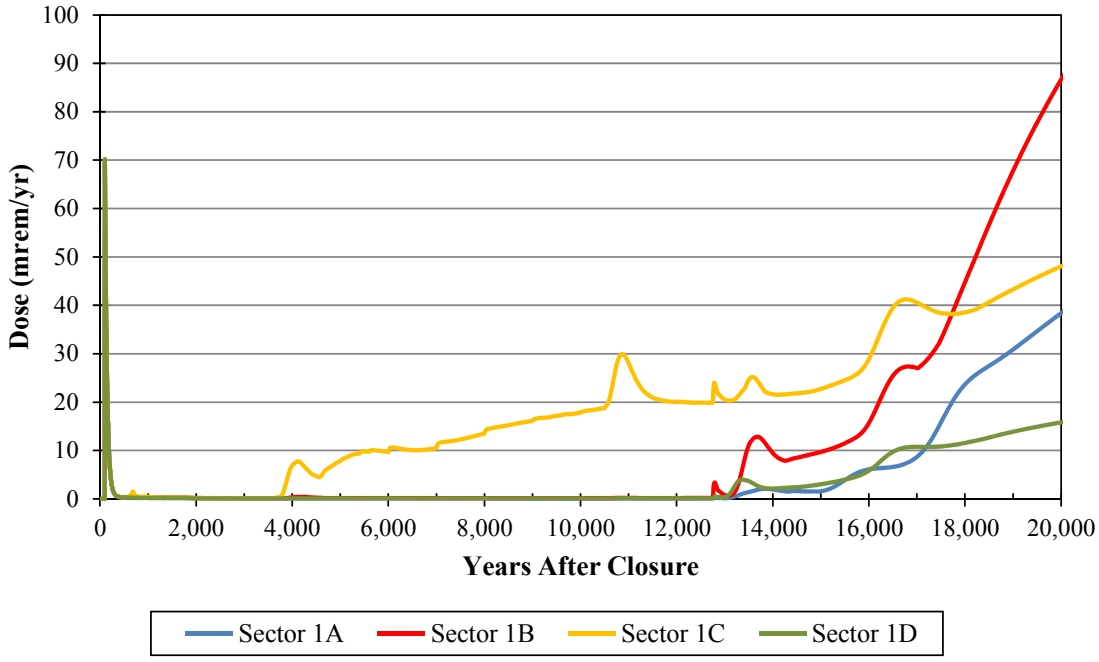
The chronic intruder-scenario peak dose at year 100 does not include a groundwater contribution because no liner failures are assumed to occur until approximately 500 years after facility closure (which are associated with the ancillary equipment and transfer lines). Thus, the peak dose to the chronic intruder in 10,000 years does not vary by FTF sector. Therefore, the chronic intruder peak dose is not impacted by using the actual residual inventories at operational closure for Tanks 5 and 6.

Table 6.4-1: Chronic Intruder Peak Dose Contributors within 10,000 Years

Chronic Intruder Pathway Contributors	Contribution to Peak (mrem/yr)	Principal Radionuclide Pathway Dose (%)
Vegetable Ingestion	69	Sr-90 / Y-90 (56 %) Cs-137 / Ba-137m (44 %)
Soil Ingestion	0.7	Am-241 (42 %) Sr-90 / Y-90 (36 %)
External Exposure	0.4	Cs-137 / Ba-137m (94 %)
Total	70	

As illustrated in Figure 6.4-2, the early peak dose to the chronic intruder (70 mrem/yr at year 100) remains the peak dose for all sectors within 20,000 years after FTF facility closure except for at Sector 1B where contributions from Nb-93m increase the chronic intruder dose to 90 mrem/yr by year 20,000. This radionuclide is a decay product from an initial Zr-93 inventory in Type I tanks. Prior to inventory updates, the groundwater pathway contribution to the peak dose was approximately 73 mrem/yr in FTF PA (i.e., the inventory updates had no significant impact on the chronic intruder peak dose).

Figure 6.4-2: Annual Dose to the Chronic Intruder within 20,000 Years



Note Peak at year 100 occurs for all sectors.

6.4.5 Inadvertent Intruder Uncertainty/Sensitivity Analysis

Section 6.5 of the FTF PA considers the effects on the inadvertent intruder analysis of uncertainties in the conceptual models used and sensitivities to the parameters used in the mathematical models. In general, the inadvertent intruder uncertainty and sensitivity analyses information presented in the FTF PA is not affected by the new residual waste information and the uncertainty and sensitivity analyses insights remain unaffected. The FTF PA sensitivity analysis most affected is the evaluation performed to investigate the effect of an intruder drilling into a waste tank (which is not considered a credible scenario).

Given the influence of the inventories from Tanks 5 and 6 on estimating other waste tank inventories, the uncertainty ranges applied to the inventory estimates were updated for this Special Analysis.

As described in the FTF PA, the chronic intruder dose is based on releases from FTF waste tanks using the same parameters as in the MOP analysis thus uncertainty analyses performed for the MOP, as related to tank releases, is also applicable to the chronic IHI. In the FTF PA, the peak of the mean from the chronic IHI dose (within the first 20,000-years) is 639 mrem/yr, with a median dose of 112 mrem/yr. [SRS-REG-2007-00002] The Base Case probabilistic model for the Tanks 5 and 6 Special Analysis (described in Section 6.3.6.4.2) showed chronic IHI doses with a peak of the mean of 965 mrem/yr, with a median dose of 73 mrem/yr (within the first 20,000 years of the model). The higher mean dose is associated with increased Pu-239 from Tank 18 and increased Zr-93 decaying into Nb-93m from the Type I tanks; whereas the lower median dose is associated with the reduced uncertainty in the inventory estimates.

6.5 F-Tank Farm Interpretation of Results

Section 7.0 of FTF PA summarizes the conservatisms used in modeling and provides a summary and interpretation of the results presented in Section 5.0 and Section 6.0 of the FTF PA. The FTF PA conservatisms information presented in the FTF PA is not affected by the new residual waste information. The integrated system behavior discussion provided in the Section 7.1.1 of FTF PA remains valid irrespective of the revised Tanks 5 and 6 residual inventories. The individual dose results provided in the Section 7.1.2 of FTF PA and in Section 6.3 of the Tank 18/19 Special Analysis have been updated based on the revised Base Case modeling using the updated FTF residual inventories at closure, and are summarized in the following sections. An additional sensitivity study was performed to incorporate a revised waste release model and updated transport parameters.

6.5.1 100-Meter (Water from Well) Groundwater Pathways Doses

The peak 100-meter groundwater pathway doses in 1,000 years is in Sector E (0.4 mrem/yr) and Sector D (0.3 mrem/yr). Similarly, in 10,000 years the peak 100-meter groundwater pathway doses are also in Sector E (3.3 mrem/yr[5]) and Sector D (2 mrem/yr), as expected, because these sectors are closest to the Type IV tanks, which are the only waste tanks projected in the Base Case to have their liner fail in less than 10,000 years. The primary pathway contributors to the peak 100-meter groundwater dose are water ingestion and vegetable ingestion. The 100-meter groundwater-pathway peak doses in 10,000 years are primarily associated with Ra-226, Pa-231, and Np-237. The peak 100-meter groundwater pathway doses within 100,000 years are also in Sectors D and E. The peak 100-meter groundwater pathway dose in 10,000 years increased relative to FTF PA when the actual residual inventories at facility closure are used for Tanks 5, 6, 17, 18, 19, and 20. The increase in peak dose in 10,000 years is associated with Ra-226 (increased due to an increase in the Pu-238 inventory in Tank 18) and Pa-231 (increased due to an increase in the Pa-231 inventory in Tank 18).

6.5.2 Water at the Stream Groundwater Pathways Doses

The peak-groundwater pathway dose at the stream in 10,000 years is associated with UTR. The MOP at the stream peak-groundwater pathway dose in 10,000 years is 0.1 mrem/yr at year 10,000. The primary contributor to the UTR peak dose is water ingestion, followed by fish ingestion.

6.5.3 All-Pathways Dose

The peak all-pathways annual dose for the MOP at 100 meters is calculated using the highest 100-meter groundwater-pathway dose results in 10,000 years in combination with the air pathway results. The peak all-pathways annual dose for the MOP is 3.3 mrem/yr and is associated with Sector E at 100 meters. The all-pathways dose was dominated by the groundwater pathway, with the airborne pathway adding an additional 5E-06 mrem/yr to the MOP. When the actual residual inventories at facility closure are used for Tanks 5, 6, 17, 18,

[5] Dose values are shown to two significant figures to illustrate changes/trends. The use of two significant figures is not meant to imply this level of precision for these dose projections.

19, and 20, the peak all-pathways annual dose for the MOP at 100 meters increased relative to the Tank 18/19 Special Analysis from 3.2 mrem/yr to 3.3 mrem/yr[6].

The peak all-pathways dose was 0.4 mrem/yr in 1,000 years (at approximately year 700) and 600 mrem/yr in 100,000 years (at year 43,000).

6.5.4 Inadvertent Intruder Dose

The peak dose for the acute intruder in 10,000 years is 2 millirem (primarily due to exposure to drill cuttings). The acute intruder scenario does not include a groundwater contribution and therefore does not vary by FTF sector. The peak dose for the chronic intruder scenario in 10,000 years is approximately 70 mrem/yr. This peak dose was almost entirely due to ingestion of vegetables contaminated with drill cuttings. The chronic intruder scenario peak dose is also driven by the drill cutting contributions. The peak inadvertent intruder doses in 10,000 years were not significantly impacted when the actual residual inventories at facility closure are used for Tanks 5 and 6 and are not changed from FTF PA or Tank 18/19 Special Analysis.

6.5.5 Airborne Dose / Radon Flux

The annual dose from airborne releases resulted in a total dose 5E-06 mrem/yr (principally from C-14) at 100 meters from the FTF in 10,000 years. These results were very conservative because the flux rates are based on simplified models. These simplified models also resulted in a peak flux of radon at the ground surface of 2E-16 pCi/m^2/s.

6.5.6 Composite Sensitivity Study

Recent laboratory studies and literature reviews have provided improved understanding of a number of important input parameters affecting the FTF PA. Specifically, the waste release model was updated, as were radionuclide-specific transport values such as K_ds. A Composite Sensitivity Study applied this new information and combined it with the updated inventory data to estimate potential doses, showing a peak dose in 100,000 years of 20 mrem/yr. In addition, the Composite Sensitivity Study assumptions were also applied to the probabilistic model, resulting in mean doses of less than 3 mrem/yr over 10,000 years and about 28 mrem/yr over 100,000 years. The composite sensitivity studies use more probable and defensible inputs, providing greater confidence that the results and conclusions remain valid.

6.6 FTF Performance Evaluation

Section 8.0 of the FTF PA describes intended use and future work to be done to support its maintenance. The FTF PA use and future work information presented in the FTF PA are not adversely impacted by the new residual waste information, and the new information documented in this Special Analysis will be used for further information during future FTF PA maintenance activities.

[6] Dose values are shown to two significant figures to illustrate changes/trends. The use of two significant figures is not meant to imply this level of precision for these dose projections.

As mentioned in Section 6.2.3, the GoldSim FTF model (SRS FTF v3.0) incorporates multiple changes and improvements. The intent of these changes was to simulate transport processes more accurately. This model was used for benchmarking and probabilistic simulation (see Section 6.3.6.4), and included major refinements, as detailed in the report for the FTF GoldSim model v3.0. [SRR-CWDA-2012-00119]

7.0 CONCLUSION

The FTF PA provides groundwater radionuclide concentrations at one meter, 100 meters, and exposure points at the two seeplines approximately 1,600 meters from FTF. The groundwater concentrations are provided for each of the three aquifers as applicable as a part of the FTF groundwater modeling. The FTF PA also provides groundwater concentrations for chemical contaminants at one meter and 100 meters. In addition, FTF PA provides intruder doses as well as analyses for the air pathways and radon ground surface flux. The FTF PA results can be used in subsequent documents to demonstrate compliance with the pertinent requirements identified below for final facility closure of FTF as indicated in Table 7.0-1.

Table 7.0-1: Key Limits from Regulatory Requirements

Requirement	All-Pathways Dose	Intruder Dose	Air Pathway Dose	Radon Flux	Groundwater Protection
NDAA Section 3116: 10 CFR 61.41 and 61.42	25 mrem/yr	500 mrem/yr	N/A	N/A	N/A
DOE M 435.1-1	25 mrem/yr	500 mrem – acute 100 mrem/yr – chronic	10 mrem/yr	20 $pCi/m^2/s$ at ground surface	< MCL
SCDHEC Primary Drinking Water Regulations (SCDHEC R.61-58)	N/A	N/A	N/A	N/A	< MCL

The key radiological results from the Tank 18/19 Special Analysis and this Special Analysis are shown in Table 7.0-2.

Table 7.0-2: Summary Radiological Results for F-Tank Farm

Location	Tank 18/19 Special Analysis Peak Within 10,000 Years			Tank 5/6 Special Analysis Peak Within 10,000 Years		
	All-Pathways Dose (mrem/yr)	Groundwater Pathway Dose (mrem/yr)	Air Pathway Dose (mrem/yr)	All-Pathways Dose (mrem/yr)	Groundwater Pathway Dose (mrem/yr)	Air Pathway Dose (mrem/yr)
100 meters from FTF	3.4 at ~ year 10,000	3.2 at ~ year 10,000	0.2	3.3 at ~ year 10,000	3.3 at ~ year 10,000	4.6E-06
At Seepline	0.12 at ~ year 5,600	0.07 at ~ year 5,600	0.05	0.10 at ~ year 5,550	0.10 at ~ year 5,550	8.3E-07

Note Dose values are shown to two significant figures to illustrate changes/trends. The use of two significant figures is not meant to imply this level of precision for these dose projections.

Note The FTF PA, Rev.1 peak intruder dose is 73 mrem/yr at year 101 from a chronic scenario, drilling through a transfer line and using groundwater concentrations at the maximum 1-meter FTF location. This value is not significantly changed (70 mrem/yr) in this Special Analysis.

Note The FTF PA peak radon flux at the ground surface is 3.6E-08 $pCi/m^2/s$ and the Special Analysis peak radon flux is decreased to 1.6E-16 $pCi/m^2/s$.

Note The decrease in the air pathway dose is due to using a less conservative and more probable assumption of 50 % saturation in the transport media (i.e., the concrete tank roofs, waste tank grout, and stabilized contaminant layers), as opposed to the 0 % saturation assumed in the FTF PA.

The peak groundwater radionuclide concentrations were calculated and only Np-237 and Tc-99 were above the respective PRG or MCL values at one meter; no PRG or MCL values were exceeded at 100 meters or at the seepline. The peak concentrations for the chemicals of concern were also calculated, and all were less than the MCL or RSL at a distance of 1 meter from FTF.

The Special Analysis results provide reasonable assurance that compliance is maintained with the specific requirements of NDAA Section 3116, DOE M 435.1-1, and the MCLs. The additional sensitivity and uncertainty analyses performed using the lessons learned regarding waste release modeling and distribution coefficients provide further confidence. The results presented in the FTF PA reflecting Tanks 5 and 6 final operational closure inventories are not significantly impacted by new information regarding the final residual inventories that are planned to be grouted in-place in Tanks 5 and 6.

8.0 REFERENCES

10 CFR 61, *Licensing Requirements for Land Disposal of Radioactive Waste*, U.S. Nuclear Regulatory Commission, Washington DC, January 1, 2010.

DHEC_01-25-1993, *Bureau of Water Pollution Control Permit to Construct, F and H-Area High-Level Radioactive Waste Tank Farms, Construction Permit #17,424-IW* , South Carolina Department of Health and Environmental Control, Columbia, SC, January 25, 1993.

DOE M 435.1-1, Chg. 1, *Radioactive Waste Management Manual,* U.S. Department of Energy, Washington DC, June 19, 2001.

DOE O 435.1, Chg. 1, *Radioactive Waste Management*, U.S. Department of Energy, Washington DC, August 28, 2001.

DOE/SRS-WD-2012-001, *Basis for Section 3116 Determination for Closure of F-Tank Farm at the Savannah River Site*, Savannah River Site, Aiken, SC., Rev. 0, March 2012.

DOE_11-10-1999, *Maintenance Guide for U.S. Department of Energy Low-Level Waste Disposal Facility Performance Assessments and Composite Analyses,* U.S. Department of Energy, Washington DC, November 10, 1999.

DOI: 10.1080/18811248.2007.9711260, (Copyright), Kobayashi, T., et al., *Solubility of Zirconium(IV) Hydrous Oxides*, Journal of Nuclear Science and Technology, 44:1, 90-94, 2007.

EPA 816-F-09-0004, *National Primary Drinking Water Regulations*, U.S. Environmental Protection Agency, Washington DC, May 2009.

EPA 816-F-10-079, *Secondary Drinking Water Regulations: Guidance for Nuisance Chemicals*, U.S. Environmental Protection Agency, Washington DC, January 7, 2011.

EPA_RSL Tbl_11-2011, *Regional Screening Level (RSL) Summary Table November 2011*, U.S Environmental Protection Agency, November 2011.

EPA-PRGs_11-13-2007, *Radionuclide Toxicity and Preliminary Remediation Goals for Superfund*, U.S. Environmental Protection Agency, Washington DC, November 13, 2007.

FR-00-9654, Federal Register/Vol. 65, No. 78, *National Primary Drinking Water Regulations; Radionuclides; Notice of Data Availability; Proposed Rule*, U.S. Environmental Protection Agency, Washington DC, April 21, 2000.

LWO-RIP-2009-00009, *Industrial Wastewater General Closure Plan for F-Area Waste Tank Systems, Industrial Wastewater Construction Permit #17,424-IW*, Savannah River Site, Aiken, SC, Rev. 3, January 24, 2011.

ML073510127, *Recommended Site-Specific Sorption Coefficients for Reviewing Non-High-Level Waste Determinations at the Savannah River Site and Idaho National Laboratory*, U.S. Nuclear Regulatory Commission, Washington DC, October 2007.

ML112371715, *Technical Evaluation Report for F-Area Tank Farm Facility, Savannah River Site, South Carolina*, U.S. Nuclear Regulatory Commission, Washington DC, October 27, 2011.

NCRP-160, (Copyright), *Ionizing Radiation Exposure of the Population of the United States* (2009), National Council on Radiation Protection and Measurements, Bethesda, MD, March 3, 2009.

NDAA_3116, *Public Law 108-375, Ronald W. Reagan National Defense Authorization Act for Fiscal Year 2005, Section 3116, Defense Site Acceleration Completion*, October 28, 2004.

PIT-MISC-0071, (Copyright), Sheppard, M.I. and Thibault, D.H., *Default Soil Solid/Liquid Partition Coefficients, K_ds, for Four Major Soil Types: A Compendium*, Health Physics, Vol. 59, No. 4, October 1990.

SCDHEC R.61-58, *State Primary Drinking Water Regulation*, South Carolina, Department Of Health and Environmental Control, Columbia, SC, August 28, 2009.

SRNL-STI-2009-00473, Kaplan, D.I., *Geochemical Data Package for Performance Assessment Calculations Related to the Savannah River Site*, Savannah River Site, Aiken, SC, Rev. 0, March 15, 2010.

SRNL-STI-2010-00493, Seaman, J.C. and Kaplan, D.I., *Chloride, Chromate, Silver, Thallium, and Uranium Sorption to SRS Soils, Sediments, and Cementitious Materials*, Savannah River Site, Aiken, SC, Rev. 0, September 29, 2010.

SRNL-STI-2011-00011, Kaplan, D.I., *Estimated Neptunium Sediment Sorption Values as a Function of pH and Measured Barium and Radium K_d Values*, Savannah River Site, Aiken, SC, Rev. 0, January 23, 2011.

SRNL-STI-2011-00672, Almond, P.M., et al., *Variability of K_d Values in Cementitious Materials and Sediments*, Savannah River Site, Aiken, SC, Rev. 0, February 2012.

SRNL-STI-2012-00404, Denham, M. and Millings, M., *Evolution of Chemical Conditions and Estimated Solubility Controls on Radionuclides in the Residual Waste Layer During Post-Closure Aging of High-Level Waste Tanks*, Savannah River Site, Aiken, SC, Rev. 0, August 2012.

SRR-CWDA-2009-00045, *F-Tank Farm Waste Tank Inventory for use in Performance Assessment Modeling*, Savannah River Site, Aiken, SC, Rev. 2, October 2012.

SRR-CWDA-2010-00124, *Tank 18/Tank 19 Special Analysis for the Performance Assessment for the F-Tank Farm at the Savannah River Site*, Savannah River Site, Aiken SC, Rev. 0, February 2012.

SRR-CWDA-2010-00128, *Performance Assessment for the H-Area Tank Farm at the Savannah River Site*, Savannah River Site, Aiken, SC, Rev. 1, November 2012.

SRR-CWDA-2012-00027, *Tank 5 Inventory Determination*, Savannah River Site, Aiken, SC, Rev. 1, August 2012.

SRR-CWDA-2012-00075, *Tank 6 Inventory Determination*, Savannah River Site, Aiken, SC, Rev. 0, August 2012.

SRR-CWDA-2012-00119, *F-Area Tank Farm Stochastic Fate and Transport Model*, Savannah River Site, Aiken SC, Rev. 0, October 2012.

SRR-CWDA-2012-00134, *Tanks 5 and 6 Special Analysis for the Performance Assessment for the F-Tank Farm at the Savannah River Site: Quality Assurance Report,* Savannah River Site, Aiken SC, Rev. 0, December 2012.

SRS-REG-2007-00002, *Performance Assessment for the F-Tank Farm at the Savannah River Site*, Savannah River Site, Aiken, SC, Rev. 1, March 31, 2010.

Title 48_Chapter 1_SC Laws, *South Carolina Pollution Control Act, Environmental Protection and Conservation*, South Carolina Legislative Council, Columbia, SC, Current through the 2008 Session.

WSRC-OS-94-42, *Federal Facility Agreement for the Savannah River Site*, Savannah River Site, Aiken, SC, August 16, 1993.

WSRC-STI-2007-00343, Farfan, E., *Air Pathway Dose Modeling for the F-Area Tank Farm*, Savannah River Site, Aiken, SC, Rev. 1, October 2012.

WSRC-STI-2007-00369, Dixon, K. and Phifer, M., *Hydraulic and Physical Properties of Tank Grouts and Base Mat Surrogate Concrete for FTF Closure*, Savannah River Site, Aiken, SC, Rev. 0, October 23, 2007

APPENDIX A

RADIOLOGICAL AND CHEMICAL CONCENTRATION TABLES TO 1,000 YEARS

Table A-1: Radiological 100-Meter Concentrations for UTRA-UZ to 1,000 Years

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs
Ac-227	N/A	1.3E-08	1,000	2.2E-08	1,000	2.2E-08	1,000	3.8E-08	872	3.0E-08	870
Al-26	N/A	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-241	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-242m	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-243	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
C-14	2,000	1.1E-11	1,000	2.8E-09	1,000	1.7E-05	1,000	4.3E-04	1,000	4.5E-05	1,000
Cf-249	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cl-36	N/A	2.5E-12	1,000	2.1E-10	1,000	9.2E-08	1,000	9.6E-07	1,000	1.1E-06	1,000
Cm-243	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-244	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-245	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-247	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-248	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Co-60	100	<1.0E-30	384	<1.0E-30	432	<1.0E-30	398	<1.0E-30	170	<1.0E-30	166
Cs-135	900	4.0E-25	1,000	6.9E-22	1,000	2.9E-14	1,000	2.1E-10	1,000	1.0E-11	1,000
Cs-137	200	2.2E-30	1,000	3.7E-27	1,000	1.8E-19	1,000	1.5E-15	1,000	2.5E-16	1,000
Eu-152	200	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	920	<1.0E-30	832	<1.0E-30	866
Eu-154	60	<1.0E-30	484	<1.0E-30	486	<1.0E-30	392	<1.0E-30	326	<1.0E-30	324
H-3	20,000	1.8E-12	558	3.2E-12	558	1.8E-11	208	2.1E-09	202	3.3E-09	202
I-129	1	3.1E-04	584	5.3E-04	584	5.1E-04	582	4.5E-04	582	2.7E-04	582
K-40	N/A	8.1E-14	1,000	1.4E-11	1,000	5.5E-08	1,000	3.1E-06	1,000	5.8E-06	1,000
Nb-93m	1,000	3.1E-20	1,000	2.9E-18	1,000	1.6E-15	1,000	2.4E-14	1,000	3.2E-14	1,000
Nb-94	N/A	4.3E-01	564	7.4E-01	562	7.1E-01	560	6.4E-01	560	3.9E-01	560

Table A-1: Radiological 100-Meter Concentrations for UTRA-UZ to 1,000 Years (Continued)

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs
Ni-59	300	4.7E-06	1,000	1.3E-06	1,000	6.2E-05	1,000	1.2E-05	1,000	1.6E-03	1,000
Ni-63	50	3.5E-07	1,000	9.1E-08	1,000	4.5E-06	1,000	7.9E-07	1,000	1.1E-04	1,000
Np-237	Total α	3.3E-02	1,000	5.7E-02	882	6.0E-02	856	3.0E-01	740	2.5E-01	738
Pa-231	Total α	1.7E-05	1,000	3.0E-05	1,000	2.9E-05	1,000	5.3E-05	844	4.2E-05	846
Pb-210	N/A	8.0E-08	1,000	1.7E-07	1,000	1.6E-06	1,000	7.0E-07	1,000	1.8E-05	1,000
Pd-107	N/A	1.0E-14	1,000	2.5E-12	1,000	1.3E-08	1,000	3.4E-07	1,000	8.5E-08	1,000
Pu-238	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-239	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-240	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-241	300	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-242	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-244	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Ra-226	Total α/Ra	6.7E-05	1,000	1.5E-04	1,000	1.2E-03	1,000	5.9E-04	1,000	1.2E-02	1,000
Ra-228	Total Ra	1.3E-28	1,000	2.2E-30	1,000	1.1E-26	1,000	1.0E-23	1,000	5.8E-24	1,000
Rn-222	N/A	6.7E-05	1,000	1.5E-04	1,000	1.2E-03	1,000	5.9E-04	1,000	1.2E-02	1,000
Se-79	N/A	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sm-151	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sn-126	N/A	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sr-90	8	6.7E-10	1,000	1.4E-09	1,000	1.3E-08	1,000	5.3E-09	1,000	1.2E-07	990
Tc-99	900	3.5E+01	746	6.0E+01	744	5.8E+01	740	2.0E+02	694	1.6E+02	694
Th-228	Total α	1.3E-28	1,000	2.2E-30	1,000	1.1E-26	1,000	1.0E-23	1,000	5.8E-24	1,000
Th-229	Total α	4.8E-10	1,000	9.0E-10	1,000	1.0E-09	1,000	4.7E-09	1,000	3.9E-09	1,000
Th-230	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	5.3E-27	1,000	1.5E-26	1,000

Table A-1: Radiological 100-Meter Concentrations for UTRA-UZ to 1,000 Years (Continued)

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Th-232	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
U-232	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	2.2E-29	1,000	1.4E-29	1,000
U-233	Total U*	1.6E-07	1,000	2.9E-07	1,000	3.1E-07	1,000	1.1E-06	1,000	9.2E-07	1,000
U-234	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	4.4E-30	1,000	7.3E-23	1,000	2.1E-22	1,000
U-235	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	6.1E-26	1,000	1.4E-25	1,000
U-236	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	1.5E-24	1,000	5.9E-25	1,000
U-238	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	7.3E-27	1,000	1.1E-26	1,000
Zr-93	N/A	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sum of beta-gamma MCL fractions		**3.9E-02**	**N/A†**	**6.7E-02**	**N/A†**	**6.4E-02**	**N/A†**	**2.3E-01**	**N/A†**	**1.8E-01**	**N/A†**
Total alpha	**15**	**3.3E-02**	**N/A†**	**5.8E-02**	**N/A†**	**6.1E-02**	**N/A†**	**3.0E-01**	**N/A†**	**2.6E-01**	**N/A†**
Total Ra	**5**	**6.7E-05**	**N/A†**	**1.5E-04**	**N/A†**	**1.2E-03**	**N/A†**	**5.9E-04**	**N/A†**	**1.2E-02**	**N/A†**

* Total uranium is evaluated in Tables A-4 through A-6.

** MCL values for beta and photon emitters are calculated in Table II-3 of FR-00-9654 based on a beta-gamma dose of 4 mrem/yr.

† The summed totals in the table conservatively used the peak radionuclide concentrations regardless of the timing of each radionuclide's peak, such that these totals are not associated with any one specific time.

Table A-2: Radiological 100-Meter Concentrations for UTRA-LZ to 1,000 Years

Radionuclide	MCL (pCi/L) **	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs
Ac-227	N/A	1.4E-08	1,000	2.7E-08	1,000	2.9E-08	1,000	4.5E-08	884	5.1E-08	878
Al-26	N/A	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-241	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-242m	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-243	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
C-14	2,000	2.0E-12	1,000	6.8E-10	1,000	3.9E-06	1,000	1.7E-04	1,000	4.6E-05	1,000
Cf-249	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cl-36	N/A	3.2E-12	1,000	2.9E-10	1,000	6.1E-08	1,000	8.4E-07	1,000	1.4E-06	1,000
Cm-243	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-244	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-245	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-247	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-248	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Co-60	100	<1.0E-30	402	<1.0E-30	448	<1.0E-30	416	<1.0E-30	184	<1.0E-30	182
Cs-135	900	7.0E-28	1,000	2.9E-24	1,000	1.5E-16	1,000	2.1E-12	1,000	1.9E-13	1,000
Cs-137	200	<1.0E-30	1,000	1.5E-29	1,000	8.6E-22	1,000	1.4E-17	1,000	4.2E-18	1,000
Eu-152	200	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	936	<1.0E-30	952
Eu-154	60	<1.0E-30	544	<1.0E-30	560	<1.0E-30	472	<1.0E-30	406	<1.0E-30	392
H-3	20,000	1.6E-12	562	3.1E-12	560	8.7E-12	210	1.0E-09	204	21.E-09	204
I-129	1	3.4E-04	586	6.4E-04	586	6.6E-04	586	6.2E-04	586	5.2E-04	584
K-40	N/A	1.8E-14	1,000	4.4E-12	1,000	1.5E-08	1,000	1.2E-06	1,000	3.9E-06	1,000
Nb-93m	1,000	3.7E-20	1,000	3.5E-18	1,000	9.8E-16	1,000	1.8E-14	1,000	3.7E-14	1,000
Nb-94	N/A	4.6E-01	564	8.6E-01	564	8.9E-01	564	8.4E-01	564	7.1E-01	564

Table A-2: Radiological 100-Meter Concentrations for UTRA-LZ to 1,000 Years (Continued)

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ni-59	300	1.7E-07	1,000	1.3E-07	1,000	1.3E-05	1,000	1.3E-06	1,000	2.6E-04	1,000
Ni-63	50	1.2E-08	1,000	8.7E-09	1,000	9.3E-07	1,000	9.6E-08	1,000	1.8E-05	1,000
Np-237	Total α	3.6E-02	1,000	7.1E-02	884	7.7E-02	872	3.3E-01	744	4.2E-01	744
Pa-231	Total α	1.9E-05	1,000	3.6E-05	1,000	3.8E-05	1,000	6.2E-05	856	7.1E-05	852
Pb-210	N/A	9.5E-09	1,000	4.4E-08	1,000	8.4E-07	1,000	1.2E-07	1,000	8.3E-06	1,000
Pd-107	N/A	2.3E-15	1,000	7.2E-13	1,000	3.1E-09	1,000	1.4E-07	1,000	5.5E-08	1,000
Pu-238	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-239	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-240	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-241	300	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-242	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-244	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Ra-226	Total α/Ra	8.9E-06	1,000	4.4E-05	1,000	7.2E-04	1,000	1.1E-04	1,000	6.5E-03	1,000
Ra-228	Total Ra	1.1E-30	1,000	<1.0E-30	1,000	3.2E-28	1,000	3.7E-25	1,000	3.2E-25	1,000
Rn-222	N/A	8.9E-06	1,000	4.4E-05	1,000	7.2E-04	1,000	1.1E-04	1,000	6.5E-03	1,000
Se-79	N/A	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sm-151	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sn-126	N/A	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sr-90	8	8.1E-11	1,000	3.8E-10	1,000	7.0E-09	1,000	9.4E-10	1,000	6.1E-08	1,000
Tc-99	900	3.8E+01	750	7.2E+01	752	7.5E+01	738	2.2E+02	704	2.7E+02	702
Th-228	Total α	1.1E-30	1,000	<1.0E-30	1,000	3.2E-28	1,000	3.7E-25	1,000	3.2E-25	1,000
Th-229	Total α	5.2E-10	1,000	1.1E-09	1,000	1.3E-09	1,000	5.0E-09	1,000	6.3E-09	1,000
Th-230	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	8.5E-31	1,000

Table A-2: Radiological 100-Meter Concentrations for UTRA-LZ to 1,000 Years (Continued)

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Th-232	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
U-232	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
U-233	Total U*	1.7E-07	1,000	3.6E-07	1,000	4.0E-07	1,000	1.3E-06	1,000	1.6E-06	1,000
U-234	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	1.2E-27	1,000	1.3E-26	1,000
U-235	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	9.3E-30	1,000
U-236	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	1.6E-29	1,000	3.8E-29	1,000
U-238	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Zr-93	N/A	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sum of beta-gamma MCL fractions		**4.2E-02**	**N/A†**	**8.0E-02**	**N/A†**	**8.4E-02**	**N/A†**	**2.4E-01**	**N/A†**	**3.0E-01**	**N/A†**
Total alpha	**15**	**3.6E-02**	**N/A†**	**7.1E-02**	**N/A†**	**7.8E-02**	**N/A†**	**3.3E-01**	**N/A†**	**4.3E-01**	**N/A†**
Total Ra	**5**	**8.9E-06**	**N/A†**	**4.4E-05**	**N/A†**	**7.2E-04**	**N/A†**	**1.1E-04**	**N/A†**	**6.5E-03**	**N/A†**

* Total uranium is evaluated in Tables A-4 through A-6.

** MCL values for beta and photon emitters are calculated in Table II-3 of FR-00-9654 based on a beta-gamma dose of 4 mrem/yr.

† The summed totals in the table conservatively used the peak radionuclide concentrations regardless of the timing of each radionuclide's peak, such that these totals are not associated with any one specific time.

Table A-3: Radiological 100-Meter Concentrations for Gordon Aquifer to 1,000 Years

Radionuclide	MCL (pCi/L) **	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ac-227	N/A	5.4E-15	1,000	3.2E-14	1,000	5.9E-14	1,000	7.1E-14	1,000	8.7E-14	1,000
Al-26	N/A	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-241	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-242m	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-243	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
C-14	2,000	6.4E-23	1,000	3.8E-20	1,000	2.8E-16	1,000	3.1E-14	1,000	3.1E-14	1,000
Cf-249	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cl-36	N/A	9.5E-16	1,000	1.2E-13	1,000	1.6E-11	1,000	3.5E-10	1,000	1.1E-09	1,000
Cm-243	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-244	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-245	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-247	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-248	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Co-60	100	<1.0E-30	452	<1.0E-30	476	<1.0E-30	448	<1.0E-30	218	<1.0E-30	218
Cs-135	900	<1.0E-30	1,000	<1.0E-30	1,000	2.2E-28	1,000	4.0E-24	1,000	2.6E-24	1,000
Cs-137	200	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	2.3E-29	1,000	3.3E-29	1,000
Eu-152	200	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Eu-154	60	<1.0E-30	634	<1.0E-30	648	<1.0E-30	562	<1.0E-30	500	<1.0E-30	474
H-3	20,000	2.7E-17	580	1.4E-16	190	4.9E-16	188	1.1E-14	226	5.0E-14	226
I-129	1	2.4E-08	734	1.3E-07	732	2.2E-07	732	2.4E-07	732	2.4E-07	736
K-40	N/A	4.2E-23	1,000	1.8E-20	1,000	7.6E-17	1,000	1.2E-14	1,000	9.2E-14	1,000
Nb-93m	1,000	9.9E-25	1,000	1.3E-22	1,000	2.2E-20	1,000	5.8E-19	1,000	2.1E-18	1,000
Nb-94	N/A	3.8E-05	614	2.1E-04	614	3.5E-04	614	3.8E-04	614	3.8E-04	614

Table A-3: Radiological 100-Meter Concentrations for Gordon Aquifer to 1,000 Years (Continued)

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ni-59	300	1.1E-17	1,000	1.2E-16	1,000	2.9E-14	1,000	4.3E-14	1,000	5.2E-12	1,000
Ni-63	50	7.5E-19	1,000	7.7E-18	1,000	2.0E-15	1,000	3.0E-15	1,000	3.5E-13	1,000
Np-237	Total α	1.8E-08	1,000	1.1E-07	1,000	2.1E-07	1,000	3.5E-07	1,000	5.2E-07	1,000
Pa-231	Total α	8.3E-12	1,000	5.1E-11	1,000	9.4E-11	1,000	1.1E-10	1,000	1.4E-10	1,000
Pb-210	N/A	1.9E-17	1,000	3.5E-16	1,000	2.1E-14	1,000	2.7E-14	1,000	7.7E-13	1,000
Pd-107	N/A	9.7E-24	1,000	5.1E-21	1,000	2.6E-17	1,000	2.4E-15	1,000	2.8E-15	1,000
Pu-238	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-239	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-240	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-241	300	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-242	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-244	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Ra-226	Total α/Ra	2.2E-14	1,000	4.0E-13	1,000	2.2E-11	1,000	2.7E-11	1,000	6.9E-10	1,000
Ra-228	Total Ra	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Rn-222	N/A	2.2E-14	1,000	4.0E-13	1,000	2.2E-11	1,000	2.7E-11	1,000	6.9E-10	1,000
Se-79	N/A	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sm-151	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sn-126	N/A	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sr-90	8	1.6E-19	1,000	2.9E-18	1,000	1.8E-16	1,000	2.3E-16	1,000	5.9E-15	1,000
Tc-99	900	2.1E-03	1,000	1.2E-02	1,000	1.9E-02	1,000	2.5E-02	1,000	3.1E-02	1,000
Th-228	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Th-229	Total α	3.5E-16	1,000	1.1E-15	1,000	1.7E-15	1,000	4.0E-15	1,000	5.5E-15	1,000
Th-230	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000

Table A-3: Radiological 100-Meter Concentrations for Gordon Aquifer to 1,000 Years (Continued)

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Th-232	Total α	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
U-232	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
U-233	Total U*	2.2E-13	1,000	7.0E-13	1,000	9.9E-13	1,000	2.5E-12	1,000	3.4E-12	1,000
U-234	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
U-235	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
U-236	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
U-238	Total U*	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Zr-93	N/A	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sum of beta-gamma MCL fractions		**2.3E-06**	**N/A†**	**1.3E-05**	**N/A†**	**2.2E-05**	**N/A†**	**2.8E-05**	**N/A†**	**3.4E-05**	**N/A†**
Total Alpha	**15**	**1.8E-08**	**N/A†**	**1.1E-07**	**N/A†**	**2.1E-07**	**N/A†**	**3.5E-07**	**N/A†**	**5.2E-07**	**N/A†**
Total Ra	**5**	**2.2E-14**	**N/A†**	**4.0E-13**	**N/A†**	**2.2E-11**	**N/A†**	**2.7E-11**	**N/A†**	**6.9E-10**	**N/A†**

* Total uranium is evaluated in Tables A-4 through A-6.

** MCL values for beta and photon emitters are calculated in Table II-3 of FR-00-9654 based on a beta-gamma dose of 4 mrem/yr.

† The summed totals in the table conservatively used the peak radionuclide concentrations regardless of the timing of each radionuclide's peak, such that these totals are not associated with any one specific time.

Table A-4: Chemical 100-Meter Concentrations for UTRA-UZ to 1,000 Years

Chemical	MCL (μg/L)	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs
Ag	N/A	1.4E-24	1,000	2.0E-22	1,000	5.7E-15	1,000	1.0E-10	1,000	4.3E-10	1,000
As	1.0E+01	1.8E-44	1,000	9.6E-42	1,000	2.3E-31	1,000	1.9E-25	1,000	6.9E-27	1,000
Ba	2.0E+03	5.7E-06	1,000	1.3E-05	1,000	1.0E-04	1,000	6.3E-05	1,000	1.3E-03	1,000
Cd	5.0E+00	3.0E-04	1,000	8.9E-04	1,000	2.1E-03	1,000	2.0E-03	1,000	9.0E-03	984
Cr	1.0E+02	1.1E-04	1,000	3.3E-04	1,000	8.0E-04	1,000	8.2E-04	1,000	3.4E-03	984
Cu	1.0E+03	2.6E-23	1,000	5.2E-21	1,000	4.9E-14	1,000	3.4E-10	1,000	1.2E-09	1,000
F	2.0E+03	8.6E-02	564	1.5E-01	562	1.4E-01	560	1.3E-01	560	7.9E-02	560
Fe	3.0E+02	9.8E-52	1,000	2.0E-49	1,000	7.7E-37	1,000	2.0E-29	1,000	4.2E-29	1,000
Hg	2.0E+00	5.3E-60	1,000	9.0E-58	1,000	2.9E-41	1,000	4.3E-31	1,000	3.5E-31	1,000
Mn	5.0E+01	9.3E-16	1,000	8.8E-17	1,000	8.1E-12	1,000	2.9E-09	1,000	7.2E-09	1,000
$NO_2 + NO_3$	1.0E+04	3.3E+00	564	5.7E+00	562	5.6E+00	560	5.0E+00	560	3.1E+00	560
Ni	N/A	2.4E-06	1,000	6.6E-07	1,000	3.1E-05	1,000	8.3E-06	1,000	1.1E-03	1,000
Pb	1.5E+01	4.4E-71	1,000	6.0E-69	1,000	1.4E-49	1,000	2.2E-37	1,000	8.8E-37	1,000
Sb	6.0E+00	1.6E-81	1,000	2.1E-79	1,000	4.0E-59	1,000	2.8E-46	1,000	9.2E-51	1,000
Se	5.0E+01	7.1E-65	1,000	1.1E-62	1,000	1.8E-45	1,000	8.4E-35	1,000	3.5E-37	1,000
U	3.0E+01	2.0E-48	1,000	4.0E-46	1,000	1.2E-33	1,000	2.2E-26	1,000	3.3E-26	1,000
Zn	5.0E+03	5.3E-36	1,000	5.3E-34	1,000	5.7E-24	1,000	3.7E-18	1,000	3.6E-17	1,000

Table A-5: Chemical 100-Meter Concentrations for UTRA-LZ to 1,000 Years

Chemical	MCL (μg/L)	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concen-tration (μg/L)	Year Peak Contri-bution Occurs	Concen-tration (μg/L)	Year Peak Contri-bution Occurs	Concen-tration (μg/L)	Year Peak Contri-bution Occurs	Concen-tration (μg/L)	Year Peak Contri-bution Occurs	Concen-tration (μg/L)	Year Peak Contri-bution Occurs
Ag	N/A	2.3E-27	1,000	8.2E-25	1,000	3.4E-17	1,000	1.3E-12	1,000	7.2E-12	1,000
As	1.0E+01	1.7E-48	1,000	2.3E-45	1,000	4.8E-35	1,000	4.2E-29	1,000	6.5E-30	1,000
Ba	2.0E+03	7.6E-07	1,000	3.8E-06	1,000	6.1E-05	1,000	1.2E-05	1,000	6.9E-04	1,000
Cd	5.0E+00	9.4E-05	1,000	5.0E-04	1,000	2.4E-03	1,000	7.5E-04	1,000	1.3E-02	1,000
Cr	1.0E+02	3.5E-05	1,000	1.9E-04	1,000	8.8E-04	1,000	2.9E-04	1,000	4.7E-03	1,000
Cu	1.0E+03	7.3E-26	1,000	3.6E-23	1,000	4.6E-16	1,000	6.7E-12	1,000	3.1E-11	1,000
F	2.0E+03	9.3E-02	564	1.7E-01	564	1.8E-01	564	1.7E-01	564	1.4E-01	564
Fe	3.0E+02	8.7E-57	1,000	3.9E-54	1,000	7.4E-42	1,000	1.5E-34	1,000	2.0E-33	1,000
Hg	2.0E+00	3.3E-67	1,000	2.8E-65	1,000	3.1E-49	1,000	2.7E-39	1,000	3.7E-38	1,000
Mn	5.0E+01	3.8E-18	1,000	5.1E-18	1,000	5.4E-13	1,000	3.1E-10	1,000	1.2E-09	1,000
$NO_2 + NO_3$	1.0E+04	3.6E+00	564	6.7E+00	564	6.9E+00	564	6.5E+00	564	5.5E+00	564
Ni	N/A	8.6E-08	1,000	6.4E-08	1,000	6.6E-06	1,000	6.8E-07	1,000	1.8E-04	1,000
Pb	1.5E+01	3.6E-80	1,000	9.7E-79	1,000	6.5E-60	1,000	5.6E-48	1,000	8.6E-46	1,000
Sb	6.0E+00	2.8E-91	1,000	5.2E-90	1,000	3.2E-70	1,000	1.2E-57	1,000	3.1E-60	1,000
Se	5.0E+01	1.5E-72	1,000	9.8E-71	1,000	5.2E-54	1,000	1.4E-43	1,000	1.8E-44	1,000
U	3.0E+01	1.9E-53	1,000	7.7E-51	1,000	1.4E-38	1,000	2.3E-31	1,000	2.2E-30	1,000
Zn	5.0E+03	6.8E-40	1,000	1.8E-37	1,000	1.9E-27	1,000	3.4E-21	1,000	4.2E-20	1,000

Table A-6: Chemical 100-Meter Concentrations for Gordon Aquifer to 1,000 Years

Chemical	MCL (μg/L)	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs
Ag	N/A	6.3E-40	1,000	3.9E-36	1,000	2.3E-28	1,000	1.8E-23	1,000	1.4E-22	1,000
As	1.0E+01	3.9E-63	1,000	1.5E-58	1,000	1.5E-48	1,000	9.7E-43	1,000	3.0E-42	1,000
Ba	2.0E+03	1.9E-15	1,000	3.5E-14	1,000	1.9E-12	1,000	2.3E-12	1,000	7.3E-11	1,000
Cd	5.0E+00	3.1E-12	1,000	4.2E-11	1,000	7.3E-10	1,000	7.8E-10	1,000	9.1E-09	1,000
Cr	1.0E+02	1.2E-12	1,000	1.6E-11	1,000	2.7E-10	1,000	2.9E-10	1,000	3.4E-09	1,000
Cu	1.0E+03	2.4E-37	1,000	1.5E-33	1,000	2.7E-26	1,000	8.1E-22	1,000	5.6E-21	1,000
F	2.0E+03	7.7E-06	614	4.2E-05	614	7.1E-05	614	7.7E-05	614	7.8E-05	614
Fe	3.0E+02	1.7E-73	1,000	2.6E-69	1,000	2.0E-57	1,000	5.1E-50	1,000	7.9E-49	1,000
Hg	2.0E+00	1.0E-87	1,000	3.3E-84	1,000	2.2E-68	1,000	1.6E-58	1,000	3.9E-57	1,000
Mn	5.0E+01	3.1E-31	1,000	1.6E-28	1,000	1.9E-23	1,000	2.3E-20	1,000	1.8E-19	1,000
$NO_2 + NO_3$	1.0E+04	3.0E-04	614	1.7E-03	614	2.7E-03	614	3.0E-03	614	3.0E-03	614
Ni	N/A	5.5E-18	1,000	5.9E-17	1,000	1.5E-14	1,000	2.2E-14	1,000	3.7E-12	1,000
Pb	1.5E+01	3.4E-104	1,000	3.8E-101	1,000	1.5E-82	1,000	2.0E-70	1,000	6.2E-68	1,000
Sb	6.0E+00	1.9E-115	1,000	1.6E-112	1,000	4.8E-93	1,000	1.5E-80	1,000	6.2E-80	1,000
Se	5.0E+01	1.4E-93	1,000	3.4E-90	1,000	1.1E-73	1,000	2.2E-63	1,000	6.7E-62	1,000
U	3.0E+01	5.8E-70	1,000	7.8E-66	1,000	7.1E-54	1,000	1.6E-46	1,000	2.0E-45	1,000
Zn	5.0E+03	2.5E-54	1,000	1.8E-50	1,000	1.1E-40	1,000	3.2E-34	1,000	4.6E-33	1,000

Table A-7: Upper Three Runs Seepline Sensitivity Run Radionuclide Concentrations to 1,000 Years

Radionuclide	Peak Seepline Concentration in 1,000 Yrs (pCi/L)	Location of Largest Contributor (Sector)	Year Largest Contribution in 1,000 Years Occurs
Am-241	<1.0E-30	UTRA-LZ	1,000
Am-243	<1.0E-30	UTRA-LZ	1,000
C-14	8.9E-12	UTRA-LZ	1,000
Cl-36	1.9E-17	UTRA-LZ	1,000
Cm-244	<1.0E-30	UTRA-LZ	1,000
Cs-135	1.9E-27	UTRA-LZ	1,000
I-129	3.5E-05	UTRA-LZ	616
Nb-93m	<1.0E-30	UTRA-LZ	1,000
Nb-94	<1.0E-30	UTRA-LZ	1,000
Np-237	1.1E-02	UTRA-LZ	844
Pa-231	2.7E-06	UTRA-LZ	964
Pu-238	<1.0E-30	UTRA-LZ	1,000
Pu-239	<1.0E-30	UTRA-LZ	1,000
Pu-240	<1.0E-30	UTRA-LZ	1,000
Pu-241	<1.0E-30	UTRA-LZ	1,000
Ra-226	1.7E-08	UTRA-LZ	1,000
Tc-99	5.9E+00	UTRA-LZ	798
Th-230	<1.0E-30	UTRA-LZ	1,000
U-234	<1.0E-30	UTRA-LZ	1,000
U-235	<1.0E-30	UTRA-LZ	1,000
U-236	<1.0E-30	UTRA-LZ	1,000
U-238	<1.0E-30	UTRA-LZ	1,000
Zr-93	<1.0E-30	UTRA-LZ	1,000

Table A-8: Fourmile Branch Seepline Sensitivity Run Radionuclide Concentrations to 1,000 Years

Radionuclide	Peak Seepline Concentration in 1,000 Yrs (pCi/L)	Location of Largest Contributor (Sector)	Year Largest Contribution in 1,000 Years Occurs
Am-241	<1.0E-30	UTRA-UZ	1,000
Am-243	<1.0E-30	UTRA-UZ	1,000
C-14	1.7E-11	UTRA-UZ	1,000
Cl-36	3.0E-17	UTRA-UZ	1,000
Cm-244	<1.0E-30	UTRA-UZ	1,000
Cs-135	3.0E-24	UTRA-UZ	1,000
I-129	5.1E-05	UTRA-UZ	610
Nb-93m	<1.0E-30	UTRA-UZ	1,000
Nb-94	<1.0E-30	UTRA-UZ	1,000
Np-237	5.7E-03	UTRA-UZ	996
Pa-231	2.8E-06	UTRA-UZ	1,000
Pu-238	<1.0E-30	UTRA-UZ	1,000
Pu-239	<1.0E-30	UTRA-UZ	1,000
Pu-240	<1.0E-30	UTRA-UZ	1,000
Pu-241	<1.0E-30	UTRA-UZ	1,000
Ra-226	5.1E-08	UTRA-UZ	1,000
Tc-99	5.4E+00	UTRA-UZ	826
Th-230	<1.0E-30	UTRA-UZ	1,000
U-234	<1.0E-30	UTRA-UZ	1,000
U-235	<1.0E-30	UTRA-UZ	1,000
U-236	<1.0E-30	UTRA-UZ	1,000
U-238	<1.0E-30	UTRA-UZ	1,000
Zr-93	<1.0E-30	UTRA-UZ	1,000

Table A-9: Radiological 1-Meter Concentrations for UTRA-UZ to 1,000 Years

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ac-227	3.4E-08	1,000	6.0E-08	1,000	1.7E-07	850	1.2E-09	1,000
Al-26	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-241	<1.0E-30	1,000	<1.0E-30	1,000	1.4E-23	1,000	<1.0E-30	1,000
Am-242m	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-243	<1.0E-30	1,000	<1.0E-30	1,000	3.6E-28	1,000	<1.0E-30	1,000
C-14	1.2E-08	1,000	1.2E-04	1,000	4.5E-03	1,000	5.8E-07	1,000
Cf-249	<1.0E-30	1,000	<1.0E-30	1,000	3.5E-30	1,000	<1.0E-30	1,000
Cl-36	1.4E-14	1,000	1.5E-07	1,000	6.6E-06	1,000	2.1E-11	1,000
Cm-243	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-244	<1.0E-30	1,000	<1.0E-30	978	<1.0E-30	928	<1.0E-30	1,000
Cm-245	<1.0E-30	1,000	<1.0E-30	1,000	7.5E-29	1,000	<1.0E-30	1,000
Cm-247	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-248	<1.0E-30	1,000	<1.0E-30	1,000	2.9E-30	1,000	<1.0E-30	1,000
Co-60	<1.0E-30	358	<1.0E-30	370	<1.0E-30	126	<1.0E-30	326
Cs-135	1.6E-24	1,000	6.5E-10	1,000	9.9E-07	1,000	2.8E-16	1,000
Cs-137	3.7E-29	1,000	5.6E-15	920	2.0E-11	786	7.6E-21	1,000
Eu-152	<1.0E-30	1,000	<1.0E-30	782	<1.0E-30	730	<1.0E-30	832
Eu-154	<1.0E-30	422	<1.0E-30	288	<1.0E-30	236	<1.0E-30	324
H-3	7.1E-12	552	1.7E-11	204	1.9E-08	196	2.8E-12	158
I-129	8.2E-04	576	1.4E-03	578	3.9E-03	534	3.8E-05	536
K-40	8.7E-15	1,000	3.0E-07	1,000	9.8E-05	1,000	1.1E-10	1,000
Nb-93m	2.8E-22	1,000	3.2E-15	1,000	2.3E-13	1,000	5.9E-19	1,000
Nb-94	1.2E+00	556	2.0E+00	556	1.9E+00	556	4.6E-02	554

Table A-9: Radiological 1-Meter Concentrations for UTRA-UZ to 1,000 Years (Continued)

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ni-59	2.2E-03	1,000	1.3E-02	1,000	2.4E+00	1,000	1.1E-02	1,000
Ni-63	1.7E-04	1,000	1.0E-03	1,000	1.8E-01	1,000	8.1E-04	1,000
Np-237	8.9E-02	1,000	1.6E-01	844	1.6E+00	722	3.2E-03	1,000
Pa-231	4.6E-05	1,000	8.0E-05	1,000	2.4E-04	826	1.8E-06	824
Pb-210	7.3E-06	1,000	4.8E-05	1,000	1.6E-03	990	1.3E-05	1,000
Pd-107	1.2E-16	1,000	9.3E-08	1,000	3.8E-06	1,000	1.6E-12	1,000
Pu-238	<1.0E-30	1,000	<1.0E-30	1,000	7.6E-26	1,000	<1.0E-30	1,000
Pu-239	<1.0E-30	1,000	<1.0E-30	1,000	1.8E-25	1,000	<1.0E-30	1,000
Pu-240	<1.0E-30	1,000	<1.0E-30	1,000	5.1E-25	1,000	<1.0E-30	1,000
Pu-241	<1.0E-30	1,000	<1.0E-30	1,000	5.7E-27	1,000	<1.0E-30	1,000
Pu-242	<1.0E-30	1,000	<1.0E-30	1,000	9.1E-27	1,000	<1.0E-30	1,000
Pu-244	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Ra-226	4.2E-03	1,000	2.7E-02	1,000	6.7E-01	960	6.0E-03	1,000
Ra-228	6.1E-25	1,000	1.5E-23	1,000	3.6E-20	1,000	5.2E-23	1,000
Rn-222	4.2E-03	1,000	2.7E-02	1,000	6.7E-01	960	6.0E-03	1,000
Se-79	<1.0E-30	1,000	1.1E-28	1,000	1.4E-17	1,000	<1.0E-30	1,000
Sm-151	<1.0E-30	1,000	<1.0E-30	1,000	1.8E-29	1,000	<1.0E-30	1,000
Sn-126	<1.0E-30	1,000	<1.0E-30	1,000	3.6E-26	1,000	<1.0E-30	1,000
Sr-90	6.4E-08	954	4.3E-07	958	4.0E-05	898	1.8E-07	912
Tc-99	9.3E+01	716	1.6E+02	730	1.1E+03	678	3.4E+00	708
Th-228	6.1E-25	1,000	1.5E-23	1,000	4.8E-20	1,000	5.2E-23	1,000
Th-229	1.5E-09	1,000	3.1E-09	1,000	2.5E-08	1,000	7.4E-11	1,000
Th-230	<1.0E-30	1,000	1.4E-24	1,000	2.2E-17	1,000	<1.0E-30	1,000

Table A-9: Radiological 1-Meter Concentrations for UTRA-UZ to 1,000 Years (Continued)

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Th-232	<1.0E-30	1,000	<1.0E-30	1,000	3.6E-25	1,000	<1.0E-30	1,000
U-232	<1.0E-30	1,000	5.2E-27	1,000	1.2E-20	1,000	<1.0E-30	1,000
U-233	4.6E-07	1,000	8.8E-07	1,000	5.6E-06	1,000	1.8E-08	1,000
U-234	<1.0E-30	1,000	1.7E-20	1,000	2.0E-13	1,000	2.9E-29	1,000
U-235	<1.0E-30	1,000	1.5E-23	1,000	1.4E-16	1,000	<1.0E-30	1,000
U-236	<1.0E-30	1,000	3.7E-22	1,000	6.3E-16	1,000	<1.0E-30	1,000
U-238	<1.0E-30	1,000	1.8E-24	1,000	1.1E-17	1,000	<1.0E-30	1,000
Zr-93	<1.0E-30	1,000	<1.0E-30	1,000	3.5E-27	1,000	<1.0E-30	1,000

Table A-10: Radiological 1-Meter Concentrations for UTRA-LZ to 1,000 Years

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ac-227	1.2E-08	1,000	3.5E-08	1,000	9.3E-08	854	7.2E-09	732
Al-26	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-241	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-242m	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-243	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
C-14	2.5E-10	1,000	2.4E-05	1,000	8.9E-04	1,000	6.2E-07	1,000
Cf-249	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cl-36	7.9E-15	1,000	6.3E-08	1,000	3.1E-06	1,000	3.9E-10	1,000
Cm-243	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-244	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-245	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-247	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-248	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Co-60	<1.0E-30	376	<1.0E-30	386	<1.0E-30	354	<1.0E-30	350
Cs-135	6.6E-27	1,000	4.5E-12	1,000	4.8E-09	1,000	1.7E-14	1,000
Cs-137	<1.0E-30	1,000	3.1E-17	1,000	9.4E-14	948	4.8E-19	1,000
Eu-152	<1.0E-30	1,000	<1.0E-30	884	<1.0E-30	816	<1.0E-30	868
Eu-154	<1.0E-30	496	<1.0E-30	364	<1.0E-30	300	<1.0E-30	352
H-3	2.1E-12	554	1.0E-11	164	3.7E-09	200	1.6E-11	158
I-129	2.8E-04	578	8.0E-04	582	1.2E-03	536	2.3E-04	536
K-40	1.6E-15	1,000	7.2E-08	1,000	3.2E-05	1,000	2.3E-09	1,000
Nb-93m	1.4E-22	1,000	1.2E-15	1,000	9.9E-14	1,000	1.1E-17	1,000
Nb-94	4.1E-01	558	1.1E+00	560	1.1E+00	560	1.4E-01	558

Table A-10: Radiological 1-Meter Concentrations for UTRA-LZ to 1,000 Years (Continued)

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ni-59	1.4E-04	1,000	2.4E-03	1,000	2.8E-01	1,000	4.6E-02	1,000
Ni-63	1.1E-05	1,000	1.8E-04	1,000	2.1E-02	1,000	3.4E-03	1,000
Np-237	3.0E-02	1,000	9.5E-02	854	8.6E-01	724	2.1E-02	686
Pa-231	1.5E-05	1,000	4.7E-05	1,000	1.3E-04	830	1.0E-05	696
Pb-210	1.2E-06	1,000	2.1E-05	1,000	4.3E-04	1,000	7.7E-05	1,000
Pd-107	2.6E-17	1,000	1.9E-08	1,000	7.1E-07	1,000	3.4E-11	1,000
Pu-238	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-239	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-240	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-241	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-242	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-244	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Ra-226	7.9E-04	1,000	1.4E-02	1,000	1.9E-01	998	3.6E-02	1,000
Ra-228	8.2E-27	1,000	2.8E-25	1,000	4.1E-22	1,000	1.9E-23	1,000
Rn-222	7.9E-04	1,000	1.4E-02	1,000	1.9E-01	998	3.6E-02	1,000
Se-79	<1.0E-30	1,000	<1.0E-30	1,000	5.7E-26	1,000	<1.0E-30	1,000
Sm-151	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sn-126	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sr-90	9.2E-09	986	1.6E-07	986	5.5E-06	926	9.4E-07	928
Tc-99	3.2E+01	724	9.2E+01	728	5.7E+02	686	2.0E+01	616
Th-228	8.2E-27	1,000	2.8E-25	1,000	4.1E-22	1,000	1.9E-23	1,000
Th-229	4.9E-10	1,000	2.0E-09	1,000	1.3E-08	1,000	4.9E-10	1,000
Th-230	<1.0E-30	1,000	7.6E-30	1,000	1.3E-22	1,000	<1.0E-30	1,000

Table A-10: Radiological 1-Meter Concentrations for UTRA-LZ to 1,000 Years (Continued)

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Th-232	<1.0E-30	1,000	<1.0E-30	1,000	2.0E-30	1,000	<1.0E-30	1,000
U-232	<1.0E-30	1,000	<1.0E-30	1,000	1.2E-25	1,000	<1.0E-30	1,000
U-233	1.5E-07	1,000	5.5E-07	1,000	3.1E-06	1,000	1.1E-07	1,000
U-234	<1.0E-30	1,000	1.1E-25	1,000	1.6E-18	1,000	8.7E-27	1,000
U-235	<1.0E-30	1,000	9.0E-29	1,000	1.1E-21	1,000	2.9E-30	1,000
U-236	<1.0E-30	1,000	2.2E-27	1,000	4.8E-21	1,000	8.5E-29	1,000
U-238	<1.0E-30	1,000	1.1E-29	1,000	8.8E-23	1,000	<1.0E-30	1,000
Zr-93	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000

Table A-11: Radiological 1-Meter Concentrations for Gordon Aquifer to 1,000 Years

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ac-227	1.2E-15	1,000	4.0E-14	1,000	5.0E-14	1,000	5.5E-15	1,000
Al-26	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-241	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-242m	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Am-243	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
C-14	8.3E-23	1,000	1.2E-15	1,000	4.0E-14	1,000	2.4E-17	1,000
Cf-249	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cl-36	1.2E-18	1,000	7.2E-12	1,000	2.6E-10	1,000	5.3E-13	1,000
Cm-243	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-244	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-245	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-247	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Cm-248	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Co-60	<1.0E-30	418	<1.0E-30	420	<1.0E-30	190	<1.0E-30	392
Cs-135	<1.0E-30	1,000	1.4E-24	1,000	1.6E-21	1,000	5.6E-25	1,000
Cs-137	<1.0E-30	1,000	8.0E-30	1,000	3.9E-26	1,000	1.3E-29	1,000
Eu-152	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	938	<1.0E-30	990
Eu-154	<1.0E-30	586	<1.0E-30	460	<1.0E-30	394	<1.0E-30	448
H-3	8.1E-18	576	3.7E-16	184	7.7E-15	222	1.1E-16	178
I-129	5.5E-09	746	1.4E-07	726	1.3E-07	722	1.7E-08	712
K-40	1.7E-24	1,000	2.8E-16	1,000	1.4E-13	1,000	2.5E-16	1,000
Nb-93m	1.6E-27	1,000	1.2E-20	1,000	7.4E-19	1,000	1.5E-21	1,000
Nb-94	8.9E-06	612	2.2E-04	610	2.1E-04	608	2.5E-05	604

Table A-11: Radiological 1-Meter Concentrations for Gordon Aquifer to 1,000 Years (Continued)

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ni-59	2.4E-14	1,000	1.6E-12	1,000	4.0E-11	1,000	2.4E-11	1,000
Ni-63	1.7E-15	1,000	1.2E-13	1,000	2.8E-12	1,000	1.7E-12	1,000
Np-237	4.0E-09	1,000	1.4E-07	1,000	2.9E-07	1,000	1.8E-08	1,000
Pa-231	1.8E-12	1,000	6.3E-11	1,000	7.8E-11	1,000	8.5E-12	1,000
Pb-210	3.3E-15	1,000	2.4E-13	1,000	2.0E-12	1,000	9.1E-13	1,000
Pd-107	6.4E-26	1,000	9.3E-17	1,000	3.6E-15	1,000	4.6E-18	1,000
Pu-238	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-239	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-240	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-241	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-242	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Pu-244	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Ra-226	3.0E-12	1,000	2.2E-10	1,000	1.6E-09	1,000	7.1E-10	1,000
Ra-228	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Rn-222	3.0E-12	1,000	2.2E-10	1,000	1.6E-09	1,000	7.1E-10	1,000
Se-79	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sm-151	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sn-126	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Sr-90	2.8E-17	1,000	2.1E-15	1,000	1.5E-14	1,000	6.6E-15	1,000
Tc-99	4.7E-04	1,000	1.2E-02	1,000	1.6E-02	1,000	1.5E-03	1,000
Th-228	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Th-229	8.9E-17	1,000	1.6E-15	1,000	5.5E-15	1,000	3.3E-16	1,000
Th-230	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000

Table A-11: Radiological 1-Meter Concentrations for Gordon Aquifer to 1,000 Years (Continued)

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Th-232	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
U-232	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
U-233	5.4E-14	1,000	9.0E-13	1,000	3.3E-12	1,000	1.8E-13	1,000
U-234	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
U-235	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
U-236	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
U-238	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000
Zr-93	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000	<1.0E-30	1,000

Table A-12: Chemical 1-Meter Concentrations for UTRA-UZ to 1,000 Years

Chemical	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs
Ag	5.2E-22	1,000	1.0E-10	1,000	3.5E-06	1,000	4.7E-16	1,000
As	3.8E-42	1,000	8.2E-24	1,000	3.0E-19	1,000	2.5E-31	1,000
Ba	3.5E-04	1,000	2.3E-03	1,000	6.7E-02	960	6.0E-04	998
Cd	5.9E-03	1,000	1.4E-02	1,000	2.5E-01	888	2.2E-03	918
Cr	2.2E-03	1,000	5.4E-03	1,000	9.2E-02	888	8.3E-04	918
Cu	4.2E-21	1,000	3.0E-10	1,000	3.8E-06	1,000	1.3E-15	1,000
F	2.5E-01	556	4.1E-01	556	5.2E-01	532	9.3E-03	554
Fe	2.1E-46	1,000	6.4E-27	1,000	8.1E-20	1,000	7.8E-36	1,000
Hg	6.0E-53	1,000	5.4E-27	1,000	2.9E-18	1,000	6.4E-38	1,000
Mn	1.2E-11	1,000	8.1E-10	1,000	1.2E-06	1,000	7.2E-09	1,000
$NO_2 + NO_3$	9.6E+00	556	1.6E+01	556	2.0E+01	532	3.6E-01	554
Ni	1.1E-03	1,000	6.7E-03	1,000	1.7E+00	1,000	7.6E-03	1,000
Pb	1.7E-61	1,000	4.6E-32	1,000	4.8E-21	1,000	2.7E-44	1,000
Sb	1.4E-77	1,000	1.5E-40	1,000	7.5E-32	1,000	1.0E-53	1,000
Se	4.2E-61	1,000	2.1E-30	1,000	4.9E-23	1,000	5.6E-41	1,000
U	5.7E-43	1,000	5.4E-24	1,000	3.2E-17	1,000	2.2E-33	1,000
Zn	9.3E-32	1,000	6.8E-17	1,000	1.5E-10	1,000	8.2E-24	1,000

Table A-13: Chemical 1-Meter Concentrations for UTRA-LZ to 1,000 Years

Chemical	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs
Ag	2.4E-24	1,000	9.7E-13	1,000	2.8E-08	1,000	2.7E-14	1,000
As	1.1E-46	1,000	1.0E-27	1,000	5.0E-23	1,000	4.0E-29	1,000
Ba	6.6E-05	1,000	1.2E-03	1,000	2.0E-02	998	3.7E-03	1,000
Cd	9.0E-04	1,000	9.0E-03	1,000	7.2E-02	918	1.3E-02	922
Cr	3.4E-04	1,000	3.4E-03	1,000	2.7E-02	918	5.0E-03	922
Cu	3.1E-23	1,000	4.4E-12	1,000	5.0E-08	1,000	6.5E-14	1,000
F	8.2E-02	558	2.3E-01	560	2.2E-01	560	3.0E-02	534
Fe	3.6E-51	1,000	2.4E-32	1,000	4.0E-25	1,000	2.5E-33	1,000
Hg	3.3E-60	1,000	1.7E-35	1,000	3.7E-26	1,000	5.1E-36	1,000
Mn	7.1E-14	1,000	6.0E-11	1,000	1.1E-07	1,000	1.4E-09	1,000
$NO_2 + NO_3$	3.2E+00	558	8.8E+00	560	8.4E+00	560	1.2E+00	534
Ni	7.3E-05	1,000	1.2E-03	1,000	2.0E-01	1,000	3.3E-02	1,000
Pb	4.9E-71	1,000	5.8E-43	1,000	5.6E-31	1,000	3.1E-43	1,000
Sb	1.5E-87	1,000	3.1E-52	1,000	5.7E-43	1,000	6.5E-53	1,000
Se	1.3E-68	1,000	1.8E-39	1,000	1.2E-30	1,000	2.8E-39	1,000
U	1.1E-47	1,000	3.2E-29	1,000	2.6E-22	1,000	6.5E-31	1,000
Zn	2.9E-35	1,000	1.9E-20	1,000	3.8E-14	1,000	1.1E-21	1,000

Table A-14: Chemical 1-Meter Concentrations for Gordon Aquifer to 1,000 Years

Chemical	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs
Ag	2.7E-36	1,000	2.1E-24	1,000	4.1E-20	1,000	4.9E-24	1,000
As	8.5E-61	1,000	3.1E-42	1,000	4.4E-37	1,000	2.0E-41	1,000
Ba	2.5E-13	1,000	1.9E-11	1,000	1.7E-10	1,000	7.5E-11	1,000
Cd	3.1E-11	1,000	2.7E-09	1,000	1.2E-08	1,000	3.6E-09	1,000
Cr	1.2E-11	1,000	1.0E-09	1,000	4.4E-09	1,000	1.4E-09	1,000
Cu	3.2E-34	1,000	8.4E-23	1,000	6.5E-19	1,000	9.6E-23	1,000
F	1.8E-06	612	4.4E-05	610	4.2E-05	608	5.4E-06	602
Fe	2.2E-67	1,000	5.4E-49	1,000	4.2E-42	1,000	1.2E-47	1,000
Hg	5.1E-80	1,000	9.9E-56	1,000	1.1E-46	1,000	8.6E-54	1,000
Mn	1.4E-26	1,000	9.3E-22	1,000	1.8E-18	1,000	2.3E-21	1,000
$NO_2 + NO_3$	6.9E-05	612	1.7E-03	610	1.6E-03	608	2.1E-04	602
Ni	1.2E-14	1,000	8.3E-13	1,000	2.9E-11	1,000	1.7E-11	1,000
Pb	2.6E-94	1,000	1.1E-66	1,000	5.6E-55	1,000	1.8E-64	1,000
Sb	7.3E-111	1,000	4.3E-76	1,000	5.6E-66	1,000	3.0E-74	1,000
Se	6.3E-89	1,000	3.1E-60	1,000	1.4E-50	1,000	1.5E-57	1,000
U	1.2E-63	1,000	1.8E-45	1,000	7.8E-39	1,000	8.2E-45	1,000
Zn	3.5E-49	1,000	1.4E-34	1,000	1.5E-28	1,000	1.3E-33	1,000

APPENDIX B

RADIOLOGICAL AND CHEMICAL CONCENTRATION TABLES TO 10,000 YEARS

Table B-1: Radiological 100-Meter Concentrations for UTRA-UZ to 10,000 Years

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs
Ac-227	N/A	3.0E-08	1,740	5.8E-08	6,144	9.4E-06	6,130	4.0E-04	6,122	5.9E-04	6,116
Al-26	N/A	<1.0E-30	10,000	2.3E-29	10,000	2.3E-18	10,000	2.4E-12	10,000	1.5E-13	10,000
Am-241	Total α	5.7E-28	10,000	2.9E-25	10,000	5.5E-15	10,000	1.7E-09	10,000	2.7E-10	10,000
Am-242m	Total α	<1.0E-30	9,424	<1.0E-30	9,874	<1.0E-30	8,476	5.8E-30	7,544	5.1E-30	7,960
Am-243	Total α	8.6E-28	10,000	4.1E-25	10,000	9.7E-15	10,000	3.6E-09	10,000	2.4E-08	10,000
C-14	2,000	2.3E-03	4,654	1.7E-01	4,500	9.4E+01	4,300	7.8E+02	4,172	1.3E+02	4,222
Cf-249	Total α	<1.0E-30	10,000	<1.0E-30	10,000	8.2E-24	10,000	2.8E-18	10,000	2.5E-19	10,000
Cl-36	N/A	6.9E-06	10,000	4.3E-05	3,692	1.8E-02	3,684	1.9E-01	3,682	2.3E-01	3,682
Cm-243	Total α	<1.0E-30	1,900	<1.0E-30	1,898	<1.0E-30	1,618	<1.0E-30	1,420	<1.0E-30	1,490
Cm-244	Total α	<1.0E-30	1,342	<1.0E-30	1,348	<1.0E-30	1,172	<1.0E-30	1,048	<1.0E-30	1,092
Cm-245	Total α	3.9E-28	10,000	1.9E-25	10,000	4.1E-15	10,000	1.4E-09	10,000	2.1E-10	10,000
Cm-247	Total α	<1.0E-30	10,000	3.5E-28	10,000	7.4E-18	10,000	2.5E-12	10,000	8.3E-14	10,000
Cm-248	Total α	3.1E-29	10,000	1.5E-26	10,000	3.3E-16	10,000	1.1E-10	10,000	3.7E-12	10,000
Co-60	100	<1.0E-30	384	<1.0E-30	432	<1.0E-30	398	<1.0E-30	168	<1.0E-30	166
Cs-135	900	8.1E-04	8,808	2.3E-03	7,524	6.7E-01	6,148	6.0E+00	5,552	2.1E+00	5,796
Cs-137	200	5.0E-29	2,076	3.6E-26	1,370	2.9E-19	1,152	1.5E-15	1,000	2.7E-16	1,048
Eu-152	200	<1.0E-30	1,034	<1.0E-30	1,038	<1.0E-30	918	<1.0E-30	832	<1.0E-30	864
Eu-154	60	<1.0E-30	482	<1.0E-30	486	<1.0E-30	390	<1.0E-30	326	<1.0E-30	324
H-3	20,000	1.8E-12	558	3.2E-12	558	1.8E-11	208	2.1E-09	202	3.3E-09	202
I-129	1	3.1E-04	584	5.3E-04	584	1.4E-02	3,792	1.3E-01	3,788	8.2E-02	3,792
K-40	N/A	3.2E-06	4,640	2.2E-04	4,552	9.4E-02	4,266	2.7E+00	4,152	4.0E+00	4,128
Nb-93m	1,000	1.8E-04	10,000	1.6E-02	9,998	8.4E+00	10,000	1.2E+02	9,998	1.6E+02	9,998
Nb-94	N/A	4.3E-01	564	7.4E-01	562	7.1E-01	560	6.4E-01	560	3.9E-01	560

Table B-1: Radiological 100-Meter Concentrations for UTRA-UZ to 10,000 Years (Continued)

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs
Ni-59	300	5.8E-01	2,060	9.9E-01	2,036	9.5E-01	2,002	1.2E+01	7,306	1.7E+01	7,288
Ni-63	50	5.2E-04	1,502	1.1E-03	1,468	1.2E-03	1,428	1.0E-03	1,478	3.3E-03	1,208
Np-237	Total α	5.7E-02	1,594	9.8E-02	1,594	9.4E-02	1,586	2.2E+00	6,054	3.3E+00	6,056
Pa-231	Total α	4.0E-05	1,712	7.6E-05	6,102	1.2E-02	6,076	5.3E-01	6,060	7.8E-01	6,060
Pb-210	N/A	1.6E-04	1,748	2.7E-04	1,730	2.7E-04	1,728	4.6E-03	10,000	6.8E-03	10,000
Pd-107	N/A	1.3E-04	5,106	1.4E-02	5,052	7.4E+00	4,872	6.4E+01	4,788	2.4E+01	4,822
Pu-238	Total α	<1.0E-30	5,310	<1.0E-30	7,998	1.3E-29	7,282	6.9E-25	6,744	3.7E-24	6,952
Pu-239	Total α	1.5E-07	10,000	4.2E-09	10,000	5.4E-06	10,000	3.6E-03	10,000	9.4E-03	10,000
Pu-240	Total α	2.5E-08	10,000	7.2E-10	10,000	9.9E-06	10,000	7.1E-03	10,000	1.0E-02	10,000
Pu-241	300	2.2E-27	10,000	1.1E-24	10,000	2.1E-14	10,000	6.6E-09	10,000	1.1E-09	10,000
Pu-242	Total α	5.7E-10	10,000	1.7E-11	10,000	3.1E-07	10,000	2.1E-04	10,000	2.8E-04	10,000
Pu-244	Total α	2.8E-13	10,000	8.2E-15	10,000	1.3E-09	10,000	8.8E-07	10,000	1.5E-07	10,000
Ra-226	Total α/Ra	6.0E-02	1,720	1.0E-01	1,702	1.0E-01	1,694	1.8E+00	10,000	2.6E+00	10,000
Ra-228	Total Ra	9.8E-14	10,000	8.7E-14	10,000	3.3E-10	10,000	6.4E-08	10,000	1.7E-07	10,000
Rn-222	N/A	6.0E-02	1,720	1.0E-01	1,702	1.0E-01	1,694	1.8E+00	10,000	2.6E+00	10,000
Se-79	N/A	1.7E-24	10,000	1.2E-22	10,000	4.9E-13	10,000	3.5E-07	10,000	2.6E-06	10,000
Sm-151	1,000	<1.0E-30	6,674	<1.0E-30	4,550	<1.0E-30	3,646	<1.0E-30	3,032	<1.0E-30	3,238
Sn-126	N/A	<1.0E-30	10,000	<1.0E-30	10,000	1.6E-20	10,000	1.7E-13	10,000	2.2E-12	10,000
Sr-90	8	9.9E-10	1,108	3.1E-09	1,084	1.4E-08	1,026	7.1E-09	1,048	1.2E-07	990
Tc-99	900	3.5E+01	746	6.0E+01	744	5.8E+01	740	2.0E+02	694	1.6E+02	694
Th-228	Total α	9.8E-14	10,000	8.7E-14	10,000	3.3E-10	10,000	6.4E-08	10,000	1.7E-07	10,000
Th-229	Total α	1.8E-07	10,000	2.7E-07	10,000	1.8E-05	10,000	2.1E-03	10,000	5.5E-03	10,000
Th-230	Total α	2.8E-09	10,000	1.4E-09	10,000	3.7E-06	10,000	2.9E-03	10,000	1.0E-02	10,000

Table B-1: Radiological 100-Meter Concentrations for UTRA-UZ to 10,000 Years (Continued)

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Th-232	Total α	2.9E-16	10,000	1.4E-16	10,000	1.1E-12	10,000	2.4E-10	10,000	6.7E-10	10,000
U-232	Total U*	<1.0E-30	4,274	<1.0E-30	2,992	2.5E-30	2,414	1.1E-25	2,020	9.7E-26	2,052
U-233	Total U*	3.6E-06	10,000	4.0E-06	10,000	1.2E-03	10,000	1.0E-01	10,000	2.5E-01	10,000
U-234	Total U*	2.0E-06	10,000	1.5E-06	10,000	2.6E-03	10,000	1.6E+00	10,000	4.6E+00	10,000
U-235	Total U*	1.9E-08	10,000	1.4E-08	10,000	4.8E-05	10,000	4.3E-03	10,000	6.1E-03	10,000
U-236	Total U*	3.7E-08	10,000	2.7E-08	10,000	1.3E-04	10,000	2.2E-02	10,000	5.6E-02	10,000
U-238	Total U*	9.1E-07	10,000	6.8E-07	10,000	1.5E-05	10,000	5.3E-04	10,000	5.6E-04	10,000
Zr-93	N/A	7.3E-25	10,000	4.7E-22	10,000	2.4E-12	10,000	2.9E-07	10,000	4.4E-08	10,000
Sum of beta-gamma MCL fractions		**4.1E-02**	**N/A†**	**7.1E-02**	**N/A†**	**1.4E-01**	**N/A†**	**9.1E-01 *** **	**N/A†**	**5.5E-01 **** **	**N/A†**
Total alpha	**15**	**1.2E-01**	**N/A†**	**2.0E-01**	**N/A†**	**2.1E-01**	**N/A†**	**4.6E+00**	**N/A†**	**6.7E+00**	**N/A†**
Total Ra	**5**	**6.0E-02**	**N/A†**	**1.0E-01**	**N/A†**	**1.0E-01**	**N/A†**	**1.8E+00**	**N/A†**	**2.6E+00**	**N/A†**

* Total uranium is evaluated in Tables B-4 through B-6.

** MCL values for beta and photon emitters are calculated in Table II-3 of FR-00-9654 based on a beta-gamma dose of 4 mrem/yr.

*** Actual Sector D Peak Concentration (pCi/L) is 4.1E-01 (documented in Appendix C) based on analysis of individual radionuclide contributions at time of occurrence rather than bounding approach of summing peak concentrations in 10,000 years, regardless of timing.

**** Actual Sector E Peak Concentration (pCi/L) is 1.8E-01 (documented in Appendix C) based on analysis of individual radionuclide contributions at time of occurrence rather than bounding approach of summing peak concentrations in 10,000 years, regardless of timing.

† The summed totals in the table conservatively used the peak radionuclide concentrations regardless of the timing of each radionuclide's peak, such that these totals are not associated with any one specific time. Appendix C shows total peaks for specific the time steps and selected radionuclides.

Table B-2: Radiological 100-Meter Concentrations for UTRA-LZ to 10,000 Years

Radionuclide	MCL (pCi/L) **	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs	Concen-tration (pCi/L)	Year Peak Contri-bution Occurs
Ac-227	N/A	4.2E-08	10,000	1.2E-07	10,000	6.4E-06	6,136	2.7E-04	6,124	7.3E-04	6,116
Al-26	N/A	<1.0E-30	10,000	<1.0E-30	10,000	2.5E-22	10,000	4.4E-16	10,000	1.0E-16	10,000
Am-241	Total α	<1.0E-30	10,000	4.7E-29	10,000	1.5E-18	10,000	9.5E-13	10,000	4.2E-13	10,000
Am-242m	Total α	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	9,268	<1.0E-30	8,440	<1.0E-30	8,628
Am-243	Total α	<1.0E-30	10,000	6.1E-29	10,000	2.4E-18	10,000	1.7E-12	10,000	3.4E-11	10,000
C-14	2,000	2.7E-03	4,750	2.2E-01	4,662	5.8E+01	4,396	6.7E+02	4,264	2.6E+02	4,310
Cf-249	Total α	<1.0E-30	10,000	<1.0E-30	10,000	2.1E-27	10,000	1.4E-21	10,000	3.7E-22	10,000
Cl-36	N/A	7.1E-06	10,000	5.9E-05	3,694	1.2E-02	3,686	1.7E-01	3,682	2.9E-01	3,682
Cm-243	Total α	<1.0E-30	2,058	<1.0E-30	2,116	<1.0E-30	1,840	<1.0E-30	1,650	<1.0E-30	1,688
Cm-244	Total α	<1.0E-30	1,458	<1.0E-30	1,492	<1.0E-30	1,314	<1.0E-30	1,192	<1.0E-30	1,218
Cm-245	Total α	<1.0E-30	10,000	3.0E-29	10,000	1.1E-18	10,000	7.2E-13	10,000	3.1E-13	10,000
Cm-247	Total α	<1.0E-30	10,000	<1.0E-30	10,000	2.0E-21	10,000	1.3E-15	10,000	1.2E-16	10,000
Cm-248	Total α	<1.0E-30	10,000	2.3E-30	10,000	8.5E-20	10,000	5.7E-14	10,000	5.5E-15	10,000
Co-60	100	<1.0E-30	402	<1.0E-30	448	<1.0E-30	416	<1.0E-30	184	<1.0E-30	182
Cs-135	900	8.8E-04	9,034	2.9E-03	7,844	4.0E-01	6,600	4.8E+00	5,976	2.6E+00	6,188
Cs-137	200	<1.0E-30	1,530	4.4E-28	1,470	2.7E-21	1,250	1.7E-17	1,106	6.2E-18	1,142
Eu-152	200	<1.0E-30	1,116	<1.0E-30	1,132	<1.0E-30	1,016	<1.0E-30	932	<1.0E-30	950
Eu-154	60	<1.0E-30	542	<1.0E-30	558	<1.0E-30	472	<1.0E-30	404	<1.0E-30	392
H-3	20,000	1.6E-12	562	3.1E-12	560	8.7E-12	210	1.0E-09	204	2.1E-09	204
I-129	1	3.4E-04	586	6.4E-04	586	9.7E-03	3,794	1.1E-01	3,790	1.1E-01	3,788
K-40	N/A	4.1E-06	4,738	3.0E-04	4,648	5.9E-02	4,362	1.8E+00	4,238	4.5E+00	4,200
Nb-93m	1,000	2.1E-04	10,000	2.0E-02	10,000	5.1E+00	10,000	9.4E+01	10,000	1.9E+02	9,998
Nb-94	N/A	4.6E-01	564	8.6E-01	564	8.9E-01	564	8.4E-01	564	7.1E-01	564

Table B-2: Radiological 100-Meter Concentrations for UTRA-LZ to 10,000 Years (Continued)

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ni-59	300	6.3E-01	2,098	1.2E+00	2,100	1.2E+00	2,090	7.8E+00	7,358	1.6E+01	7,354
Ni-63	50	2.9E-04	1,578	6.9E-04	1,530	9.9E-04	1,358	6.7E-04	1,554	3.2E-03	1,274
Np-237	Total α	6.2E-02	1,598	1.2E-01	1,600	1.2E-01	1,600	1.5E+00	6,062	3.7E+00	6,058
Pa-231	Total α	4.3E-05	1,714	1.0E-04	6,112	8.5E-03	6,084	3.6E-01	6,068	9.6E-01	6,062
Pb-210	N/A	1.7E-04	1,790	3.1E-04	1,782	3.3E-04	1,772	3.1E-03	9,998	8.2E-03	10,000
Pd-107	N/A	1.7E-04	5,176	1.8E-02	5,110	4.6E+00	4,934	5.4E+01	4,846	3.0E+01	4,874
Pu-238	Total α	<1.0E-30	5,666	<1.0E-30	8,322	<1.0E-30	7,638	1.4E-27	7,160	2.0E-26	7,270
Pu-239	Total α	1.7E-09	10,000	8.2E-11	10,000	1.8E-07	10,000	1.7E-04	10,000	6.6E-04	10,000
Pu-240	Total α	2.9E-10	10,000	1.4E-11	10,000	2.2E-07	10,000	3.6E-04	10,000	6.5E-04	10,000
Pu-241	300	<1.0E-30	10,000	1.8E-28	10,000	5.9E-18	10,000	3.7E-12	10,000	1.6E-12	10,000
Pu-242	Total α	6.7E-12	10,000	3.3E-13	10,000	6.8E-09	10,000	1.1E-05	10,000	1.8E-05	10,000
Pu-244	Total α	3.2E-15	10,000	1.6E-16	10,000	2.9E-11	10,000	4.7E-08	10,000	1.2E-08	10,000
Ra-226	Total α/Ra	6.4E-02	1,762	1.2E-01	1,752	1.3E-01	1,746	1.2E+00	10,000	3.1E+00	10,000
Ra-228	Total Ra	4.0E-15	10,000	5.3E-15	10,000	3.0E-11	10,000	9.7E-09	10,000	4.6E-08	10,000
Rn-222	N/A	6.4E-02	1,762	1.2E-01	1,752	1.3E-01	1,746	1.2E+00	10,000	3.1E+00	10,000
Se-79	N/A	2.5E-28	10,000	4.2E-26	10,000	3.1E-16	10,000	7.5E-10	10,000	8.2E-09	10,000
Sm-151	1,000	<1.0E-30	7,210	<1.0E-30	5,038	<1.0E-30	4,220	<1.0E-30	3,650	<1.0E-30	3,780
Sn-126	N/A	<1.0E-30	10,000	<1.0E-30	10,000	2.4E-25	10,000	1.0E-17	10,000	2.4E-16	10,000
Sr-90	8	3.4E-10	1,142	1.7E-09	1,116	1.0E-08	1,060	2.0E-09	1,082	6.6E-08	1,026
Tc-99	900	3.8E+01	750	7.2E+01	752	7.5E+01	738	2.2E+02	704	2.7E+02	702
Th-228	Total α	4.0E-15	10,000	5.3E-15	10,000	3.0E-11	10,000	9.7E-09	10,000	4.6E-08	10,000
Th-229	Total α	7.0E-07	10,000	1.3E-06	10,000	2.9E-06	10,000	3.6E-04	10,000	1.3E-03	10,000
Th-230	Total α	9.2E-11	10,000	6.1E-11	10,000	2.5E-07	10,000	3.2E-04	10,000	2.3E-03	10,000

Table B-2: Radiological 100-Meter Concentrations for UTRA-LZ to 10,000 Years (Continued)

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Th-232	Total α	9.1E-18	10,000	6.0E-18	10,000	7.6E-14	10,000	3.1E-11	10,000	1.5E-10	10,000
U-232	Total U*	<1.0E-30	4,546	<1.0E-30	3,254	<1.0E-30	2,690	2.7E-28	2,330	4.9E-28	2,320
U-233	Total U*	1.3E-05	10,000	2.4E-05	10,000	1.5E-04	10,000	2.1E-02	10,000	8.2E-02	10,000
U-234	Total U*	7.9E-08	10,000	7.4E-08	10,000	2.2E-04	10,000	2.2E-01	10,000	1.4E+00	10,000
U-235	Total U*	7.7E-10	10,000	7.3E-10	10,000	4.3E-06	10,000	9.1E-04	10,000	1.7E-03	10,000
U-236	Total U*	1.5E-09	10,000	1.4E-09	10,000	1.2E-05	10,000	3.7E-03	10,000	1.7E-02	10,000
U-238	Total U*	3.7E-08	10,000	3.4E-08	10,000	3.1E-06	10,000	1.2E-04	10,000	1.5E-04	10,000
Zr-93	N/A	1.0E-28	10,000	1.5E-25	10,000	1.3E-15	10,000	3.6E-10	10,000	1.2E-10	10,000
Sum of beta-gamma MCL fractions		**4.5E-02**	**N/A†**	**8.5E-02**	**N/A†**	**1.3E-01**	**N/A†**	**8.1E-01*** **	**N/A†**	**7.8E-01**** **	**N/A†**
Total alpha	**15**	**1.3E-01**	**N/A†**	**2.4E-01**	**N/A†**	**2.6E-01**	**N/A†**	**3.1E+00**	**N/A†**	**7.8E+00**	**N/A†**
Total Ra	**5**	**6.4E-02**	**N/A†**	**1.2E-01**	**N/A†**	**1.3E-01**	**N/A†**	**1.2E+00**	**N/A†**	**3.1E+00**	**N/A†**

* Total uranium is evaluated in Tables B-4 through B-6.

** MCL values for beta and photon emitters are calculated in Table II-3 of FR-00-9654 based on a beta-gamma dose of 4 mrem/yr.

*** Actual Sector D Peak Concentration (pCi/L) is 3.5E-01 (documented in Appendix C) based on analysis of individual radionuclide contributions at time of occurrence rather than bounding approach of summing peak concentrations in 10,000 years, regardless of timing.

**** Actual Sector E Peak Concentration (pCi/L) is 3.0E-01 (documented in Appendix C) based on analysis of individual radionuclide contributions at time of occurrence rather than bounding approach of summing peak concentrations in 10,000 years, regardless of timing.

† The summed totals in the table conservatively used the peak radionuclide concentrations regardless of the timing of each radionuclide's peak, such that these totals are not associated with any one specific time. Appendix C shows total peaks for specific the time steps and selected radionuclides

Table B-3: Radiological 100-Meter Concentrations for Gordon Aquifer to 10,000 Years

Radionuclide	MCL (pCi/L) **	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ac-227	N/A	3.6E-12	10,000	2.3E-11	10,000	3.8E-10	10,000	1.4E-08	10,000	8.1E-08	10,000
Al-26	N/A	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	10,000	1.2E-29	10,000	6.4E-29	10,000
Am-241	Total α	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	10,000	1.9E-27	10,000	1.6E-26	10,000
Am-242m	Total α	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	9,782	<1.0E-30	9,710
Am-243	Total α	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	10,000	3.4E-27	10,000	1.1E-24	10,000
C-14	2,000	1.6E-09	10,000	4.7E-08	9,998	6.0E-06	10,000	9.1E-05	9,998	9.8E-05	10,000
Cf-249	Total α	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	10,000
Cl-36	N/A	3.6E-09	10,000	1.6E-08	9,998	1.9E-06	3,792	4.1E-05	3,792	1.3E-04	3,794
Cm-243	Total α	<1.0E-30	2,358	<1.0E-30	2,406	<1.0E-30	2,132	<1.0E-30	1,936	<1.0E-30	1,938
Cm-244	Total α	<1.0E-30	1,642	<1.0E-30	1,674	<1.0E-30	1,498	<1.0E-30	1,374	<1.0E-30	1,352
Cm-245	Total α	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	10,000	1.4E-27	10,000	1.1E-26	10,000
Cm-247	Total α	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	10,000	2.5E-30	10,000	4.4E-30	10,000
Cm-248	Total α	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	10,000	1.1E-28	10,000	1.9E-28	10,000
Co-60	100	<1.0E-30	452	<1.0E-30	476	<1.0E-30	448	<1.0E-30	218	<1.0E-30	218
Cs-135	900	1.4E-09	10,000	9.1E-09	10,000	3.7E-07	10,000	6.8E-06	10,000	8.8E-06	10,000
Cs-137	200	<1.0E-30	1,720	<1.0E-30	1,648	<1.0E-30	1,434	1.3E-28	1,310	1.8E-28	1,340
Eu-152	200	<1.0E-30	1,240	<1.0E-30	1,262	<1.0E-30	1,138	<1.0E-30	1,058	<1.0E-30	1,042
Eu-154	60	<1.0E-30	634	<1.0E-30	648	<1.0E-30	560	<1.0E-30	498	<1.0E-30	474
H-3	20,000	2.7E-17	580	1.4E-16	190	4.9E-16	188	1.1E-14	226	5.0E-14	226
I-129	1	2.4E-08	732	1.3E-07	730	1.4E-06	4,080	2.6E-05	4,088	5.1E-05	4,098
K-40	N/A	3.0E-11	10,000	2.4E-09	9,996	2.8E-07	9,998	8.5E-06	10,000	4.2E-05	9,998
Nb-93m	1,000	7.6E-09	10,000	9.6E-07	10,000	1.6E-04	10,000	4.1E-03	10,000	1.4E-02	10,000
Nb-94	N/A	3.8E-05	614	2.1E-04	614	3.5E-04	614	3.8E-04	614	3.8E-04	614

Table B-3: Radiological 100-Meter Concentrations for Gordon Aquifer to 10,000 Years (Continued)

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ni-59	300	1.6E-05	6,440	8.8E-05	6,394	1.5E-04	7,406	5.5E-04	10,000	1.8E-03	10,000
Ni-63	50	5.1E-11	1,886	3.1E-10	1,852	7.1E-10	1,730	7.1E-10	1,768	1.7E-09	1,496
Np-237	Total α	1.7E-06	6,826	9.4E-06	6,866	1.7E-05	8,864	9.5E-05	10,000	4.3E-04	10,000
Pa-231	Total α	4.5E-09	10,000	2.9E-08	10,000	4.7E-07	10,000	1.8E-05	10,000	1.0E-04	10,000
Pb-210	N/A	3.7E-09	10,000	2.0E-08	10,000	3.8E-08	10,000	1.8E-07	10,000	8.4E-07	10,000
Pd-107	N/A	2.5E-09	9,866	3.5E-07	9,806	5.2E-05	9,718	8.8E-04	9,952	1.2E-03	9,990
Pu-238	Total α	<1.0E-30	6,390	<1.0E-30	8,890	<1.0E-30	8,298	<1.0E-30	7,940	<1.0E-30	7,806
Pu-239	Total α	1.6E-22	10,000	1.3E-22	10,000	6.6E-19	10,000	4.7E-16	10,000	3.2E-15	10,000
Pu-240	Total α	2.7E-23	10,000	2.1E-23	10,000	2.7E-19	10,000	9.9E-16	10,000	3.0E-15	10,000
Pu-241	300	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	10,000	7.0E-27	10,000	6.3E-26	10,000
Pu-242	Total α	6.3E-25	10,000	5.0E-25	10,000	7.8E-21	10,000	3.1E-17	10,000	8.3E-17	10,000
Pu-244	Total α	3.0E-28	10,000	2.4E-28	10,000	2.5E-23	10,000	9.7E-20	10,000	8.7E-20	10,000
Ra-226	Total α/Ra	1.4E-06	10,000	7.8E-06	10,000	1.4E-05	9,998	6.8E-05	10,000	3.2E-04	10,000
Ra-228	Total Ra	6.4E-25	10,000	5.9E-24	10,000	4.5E-20	10,000	4.6E-17	10,000	5.8E-16	10,000
Rn-222	N/A	1.4E-06	10,000	7.8E-06	10,000	1.4E-05	9,998	6.8E-05	10,000	3.2E-04	10,000
Se-79	N/A	<1.0E-30	10,000	<1.0E-30	10,000	1.1E-28	10,000	7.9E-22	10,000	1.2E-20	10,000
Sm-151	1,000	<1.0E-30	5,750	<1.0E-30	5,882	<1.0E-30	5,080	<1.0E-30	4,514	<1.0E-30	4,526
Sn-126	N/A	<1.0E-30	10,000	<1.0E-30	10,000	<1.0E-30	10,000	3.8E-33	10,000	<1.0E-30	10,000
Sr-90	8	1.4E-17	1,240	1.5E-16	1,218	1.5E-15	1,152	1.5E-15	1,142	1.2E-14	1,088
Tc-99	900	2.2E-03	1,078	1.2E-02	1,074	2.0E-02	1,080	2.6E-02	1,078	3.1E-02	1,076
Th-228	Total α	6.4E-25	10,000	5.9E-24	10,000	4.5E-20	10,000	4.6E-17	10,000	5.8E-16	10,000
Th-229	Total α	3.2E-10	9,998	1.8E-09	10,000	2.8E-09	10,000	5.0E-09	10,000	1.2E-08	10,000
Th-230	Total α	2.2E-20	10,000	9.8E-20	10,000	4.9E-16	10,000	1.3E-12	10,000	2.8E-11	10,000

Table B-3: Radiological 100-Meter Concentrations for Gordon Aquifer to 10,000 Years (Continued)

Radionuclide	MCL (pCi/L)**	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Th-232	Total α	2.1E-27	10,000	9.6E-27	10,000	1.4E-22	10,000	1.8E-19	10,000	1.8E-18	10,000
U-232	Total U*	<1.0E-30	5,108	<1.0E-30	3,742	<1.0E-30	3,198	<1.0E-30	2,860	<1.0E-30	2,814
U-233	Total U*	6.2E-09	10,000	3.2E-08	10,000	5.1E-08	10,000	1.3E-07	10,000	4.1E-07	10,000
U-234	Total U*	3.0E-17	10,000	1.5E-16	10,000	6.4E-13	10,000	1.4E-09	10,000	2.8E-08	10,000
U-235	Total U*	2.9E-19	10,000	1.5E-18	10,000	1.3E-14	10,000	1.0E-11	10,000	3.5E-11	10,000
U-236	Total U*	5.5E-19	10,000	2.8E-18	10,000	3.5E-14	10,000	3.6E-11	10,000	3.4E-10	10,000
U-238	Total U*	1.4E-17	10,000	7.1E-17	10,000	4.0E-14	10,000	1.5E-12	10,000	6.8E-12	10,000
Zr-93	N/A	<1.0E-30	10,000	<1.0E-30	10,000	7.5E-29	10,000	3.5E-23	10,000	1.7E-22	10,000
Sum of beta-gamma MCL fractions		**2.5E-06**	**N/A†**	**1.4E-05**	**N/A†**	**2.4E-05**	**N/A†**	**6.0E-05**	**N/A†**	**1.1E-04**	**N/A†**
Total Alpha	**15**	**3.1E-06**	**N/A†**	**1.7E-05**	**N/A†**	**3.1E-05**	**N/A†**	**1.8E-04**	**N/A†**	**8.6E-04**	**N/A†**
Total Ra	**5**	**1.4E-06**	**N/A†**	**7.8E-06**	**N/A†**	**1.4E-05**	**N/A†**	**6.8E-05**	**N/A†**	**3.2E-04**	**N/A†**

* Total uranium is evaluated in Tables B-4 through B-6.

** MCL values for beta and photon emitters are calculated in Table II-3 of FR-00-9654 based on a beta-gamma dose of 4 mrem/yr.

† The summed totals in the table conservatively used the peak radionuclide concentrations regardless of the timing of each radionuclide's peak, such that these totals are not associated with any one specific time. Appendix C shows total peaks for specific the time steps and selected radionuclides

Table B-4: Chemical 100-Meter Concentrations for UTRA-UZ to 10,000 Years

Chemical	MCL (μg/L)	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs
Ag	N/A	5.96E-04	8,898	1.13E-03	8,538	4.23E-02	6,784	6.95E-01	5,916	9.08E-01	5,776
As	1.0E+01	8.14E-06	10,000	1.63E-05	10,000	1.73E-04	10,000	1.84E-03	9,452	2.23E-03	9,232
Ba	2.0E+03	5.39E-03	1,714	9.39E-03	1,690	5.02E-02	4,974	3.95E-01	4,910	3.76E-01	6,408
Cd	5.0E+00	2.30E-02	1,414	3.94E-02	1,394	5.48E-02	7,326	2.19E+00	7,250	3.20E+00	7,258
Cr	1.0E+02	8.71E-03	1,416	1.53E-02	1,406	3.12E-02	4,740	4.04E-01	4,694	5.11E-01	4,686
Cu	1.0E+03	6.47E-04	6,342	1.14E-03	6,226	2.74E-02	6,246	4.17E-01	5,536	5.31E-01	5,418
F	2.0E+03	8.60E-02	564	1.48E-01	562	1.62E+00	3,842	1.01E+01	3,838	3.48E+00	3,840
Fe	3.0E+02	6.57E-06	10,000	4.47E-06	10,000	7.28E-03	10,000	1.45E+00	10,000	2.78E+00	10,000
Hg	2.0E+00	7.86E-23	10,000	5.72E-20	10,000	4.80E-11	10,000	1.66E-06	10,000	1.31E-06	10,000
Mn	5.0E+01	1.29E-02	5,236	2.38E-02	5,636	1.07E+00	5,276	2.73E+01	5,110	3.90E+01	5,076
$NO_2 + NO_3$	1.0E+04	3.31E+00	564	5.72E+00	562	1.60E+01	3,658	1.23E+02	3,654	9.82E+00	3,656
Ni	N/A	2.99E-01	2,066	5.11E-01	2,036	4.90E-01	2,008	4.43E-01	1,876	2.91E-01	1,668
Pb	1.5E+01	1.25E-31	10,000	2.72E-29	10,000	2.31E-17	10,000	8.87E-11	10,000	2.03E-10	10,000
Sb	6.0E+00	4.44E-35	10,000	9.41E-33	10,000	7.36E-20	10,000	1.35E-12	10,000	6.27E-15	10,000
Se	5.0E+01	3.23E-27	10,000	2.18E-24	10,000	8.92E-15	10,000	8.39E-10	10,000	5.70E-10	10,000
U	3.0E+01	2.98E-06	10,000	2.24E-06	10,000	4.69E-05	10,000	1.57E-03	10,000	1.70E-03	10,000
Zn	5.0E+03	3.02E-04	10,000	6.37E-04	10,000	2.50E-02	9,478	4.05E-01	8,002	5.26E-01	7,762

Table B-5: Chemical 100-Meter Concentrations for UTRA-LZ to 10,000 Years

Chemical	MCL (μg/L)	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concen-tration (μg/L)	Year Peak Contri-bution Occurs	Concen-tration (μg/L)	Year Peak Contri-bution Occurs	Concen-tration (μg/L)	Year Peak Contri-bution Occurs	Concen-tration (μg/L)	Year Peak Contri-bution Occurs	Concen-tration (μg/L)	Year Peak Contri-bution Occurs
Ag	N/A	6.46E-04	9,210	1.41E-03	9,016	3.06E-02	7,356	4.77E-01	6,454	7.16E-01	6,290
As	1.0E+01	3.67E-06	10,000	9.23E-06	10,000	9.06E-05	10,000	1.55E-03	10,000	2.51E-03	9,932
Ba	2.0E+03	5.84E-03	1,736	1.11E-02	1,724	3.85E-02	5,026	3.61E-01	4,950	3.64E-01	4,946
Cd	5.0E+00	2.50E-02	1,434	4.64E-02	1,440	4.81E-02	1,432	1.47E+00	7,282	3.69E+00	7,266
Cr	1.0E+02	9.46E-03	1,436	1.82E-02	1,446	2.24E-02	4,780	3.25E-01	4,728	5.20E-01	4,720
Cu	1.0E+03	6.72E-04	7,132	1.33E-03	7,272	1.99E-02	6,728	2.94E-01	5,988	4.09E-01	5,852
F	2.0E+03	9.27E-02	564	1.72E-01	564	1.31E+00	3,842	1.04E+01	3,840	6.49E+00	3,840
Fe	3.0E+02	2.62E-07	10,000	2.26E-07	10,000	3.82E-04	10,000	1.77E-01	10,000	4.74E-01	10,000
Hg	2.0E+00	2.61E-26	10,000	4.38E-23	10,000	6.64E-14	10,000	5.57E-09	10,000	8.26E-09	10,000
Mn	5.0E+01	1.41E-02	5,340	3.10E-02	5,720	7.35E-01	5,414	1.84E+01	5,228	4.07E+01	5,182
$NO_2 + NO_3$	1.0E+04	3.59E+00	564	6.68E+00	564	1.06E+01	3,660	1.12E+02	3,656	3.83E+01	3,658
Ni	N/A	3.26E-01	2,102	6.10E-01	2,118	6.32E-01	2,102	5.95E-01	2,098	4.99E-01	2,048
Pb	1.5E+01	1.56E-36	10,000	7.07E-34	10,000	6.95E-22	10,000	4.66E-15	10,000	5.28E-14	10,000
Sb	6.0E+00	1.64E-40	10,000	5.28E-38	10,000	4.97E-25	10,000	1.42E-17	10,000	4.45E-19	10,000
Se	5.0E+01	5.14E-31	10,000	7.74E-28	10,000	5.61E-18	10,000	1.28E-12	10,000	1.84E-12	10,000
U	3.0E+01	1.20E-07	10,000	1.12E-07	10,000	9.81E-06	10,000	3.46E-04	10,000	4.45E-04	10,000
Zn	5.0E+03	1.36E-04	10,000	3.58E-04	10,000	1.73E-02	10,000	2.79E-01	8,926	4.07E-01	8,620

Table B-6: Chemical 100-Meter Concentrations for Gordon Aquifer to 10,000 Years

Chemical	MCL (μg/L)	Sector A		Sector B		Sector C		Sector D		Sector E	
		Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs
Ag	N/A	1.46E-09	10,000	8.53E-09	10,000	9.93E-08	10,000	1.78E-06	10,000	4.19E-06	10,000
As	1.0E+01	2.37E-13	10,000	1.75E-12	10,000	1.57E-11	10,000	6.13E-10	10,000	1.84E-09	10,000
Ba	2.0E+03	1.76E-07	4,324	9.79E-07	4,308	5.22E-06	9,996	7.14E-05	10,000	1.15E-04	9,996
Cd	5.0E+00	8.41E-07	3,038	4.66E-06	3,018	9.30E-06	10,000	2.77E-04	10,000	1.58E-03	10,000
Cr	1.0E+02	3.17E-07	3,038	1.77E-06	3,006	4.39E-06	8,930	6.44E-05	9,766	1.77E-04	9,892
Cu	1.0E+03	1.16E-08	10,000	6.50E-08	10,000	3.29E-07	10,000	4.16E-06	10,000	9.02E-06	10,000
F	2.0E+03	7.69E-06	614	4.25E-05	614	4.96E-04	3,998	6.23E-03	3,994	7.53E-03	3,994
Fe	3.0E+02	5.47E-17	10,000	2.71E-16	10,000	3.59E-13	10,000	3.76E-10	10,000	1.66E-09	10,000
Hg	2.0E+00	9.83E-40	10,000	3.77E-35	10,000	7.01E-26	10,000	1.24E-20	10,000	4.34E-20	10,000
Mn	5.0E+01	1.23E-07	10,000	7.07E-07	10,000	2.06E-06	10,000	2.35E-05	10,000	8.93E-05	10,000
$NO_2 + NO_3$	1.0E+04	2.99E-04	614	1.65E-03	614	2.75E-03	614	1.05E-02	3,716	1.11E-02	3,716
Ni	N/A	8.47E-06	6,746	4.69E-05	6,666	7.81E-05	6,714	8.52E-05	6,760	8.82E-05	6,880
Pb	1.5E+01	2.57E-53	10,000	4.07E-49	10,000	2.37E-37	10,000	4.82E-30	10,000	7.08E-29	10,000
Sb	6.0E+00	5.73E-58	10,000	5.37E-54	10,000	9.29E-41	10,000	4.91E-33	10,000	6.08E-33	10,000
Se	5.0E+01	6.56E-45	10,000	2.69E-40	10,000	1.97E-30	10,000	7.88E-25	10,000	1.49E-23	10,000
U	3.0E+01	4.59E-17	10,000	2.32E-16	10,000	1.31E-13	10,000	4.51E-12	10,000	3.86E-11	10,000
Zn	5.0E+03	8.79E-12	10,000	6.65E-11	10,000	3.21E-09	10,000	2.06E-07	10,000	5.68E-07	10,000

Table B-7: Upper Three Runs Seepline Sensitivity Run Radionuclide Concentrations to 10,000 Years

Radionuclide	Peak Seepline Concentration in 10,000 Yrs (pCi/L)	Location of Largest Contributor (Sector)	Year Largest Contribution in 10,000 Years Occurs
Ac-227	1.6E-05	UTRA-LZ	6,220
Am-241	<1.0E-30	UTRA-LZ	10,000
Am-243	1.4E-29	UTRA-LZ	10,000
C-14	5.1E+00	UTRA-LZ	5,500
Cm-244	<1.0E-30	UTRA-LZ	1,452
Cs-135	2.1E-02	UTRA-LZ	10,000
I-129	3.4E-03	UTRA-LZ	3,820
Np-237	8.1E-02	UTRA-LZ	6,166
Pa-231	2.1E-02	UTRA-LZ	6.176
Pb-210	1.4E-04	UTRA-LZ	10,000
Pu-238	<1.0E-30	UTRA-LZ	8,060
Pu-239	1.8E-15	UTRA-LZ	10,000
Pu-240	1.1E-15	UTRA-LZ	10,000
Ra-226	5.3E-02	UTRA-LZ	10,000
Ra-228	3.4E-17	UTRA-LZ	10,000
Rn-222	5.3E-02	UTRA-LZ	10,000
Tc-99	5.9E+00	UTRA-LZ	798
Th-228	3.4E-17	UTRA-LZ	10,000
Th-229	2.8E-07	UTRA-LZ	10,000
Th-230	5.4E-13	UTRA-LZ	10,000
Th-232	2.4E-20	UTRA-LZ	10,000
U-233	5.8E-06	UTRA-LZ	10,000
U-234	7.7E-10	UTRA-LZ	10,000
U-235	1.4E-12	UTRA-LZ	10,000
U-236	6.3E-12	UTRA-LZ	10,000

Table B-8: Fourmile Branch Seepline Sensitivity Run Radionuclide Concentrations to 10,000 Years

Radionuclide	Peak Seepline Concentration in 10,000 Yrs (pCi/L)	Location of Largest Contributor (Sector)	Year Largest Contribution in 10,000 Years Occurs
Ac-227	3.3E-07	UTRA-UZ	6,266
Am-241	2.7E-30	UTRA-UZ	10,000
Am-243	6.8E-26	UTRA-UZ	10.000
C-14	6.2E-01	UTRA-UZ	5,582
Cm-244	<1.0E-30	UTRA-UZ	1,296
Cs-135	2.4E-03	UTRA-UZ	10,000
I-129	2.7E-04	UTRA-UZ	3,828
Np-237	9.5E-03	UTRA-UZ	1,694
Pa-231	4.3E-04	UTRA-UZ	6,228
Pb-210	1.8E-05	UTRA-UZ	2,200
Pu-238	<1.0E-30	UTRA-UZ	5,934
Pu-239	1.0E-12	UTRA-UZ	10,000
Pu-240	1.8E-13	UTRA-UZ	10,000
Ra-226	7.1E-03	UTRA-UZ	2,172
Ra-228	8.2E-18	UTRA-UZ	10,000
Rn-222	7.1E-03	UTRA-UZ	2,172
Tc-99	5.4E+00	UTRA-UZ	826
Th-228	8.2E-18	UTRA-UZ	10,000
Th-229	2.8E-08	UTRA-UZ	10,000
Th-230	3.0E-13	UTRA-UZ	10,000
Th-232	1.3E-20	UTRA-UZ	10,000
U-233	2.9E-07	UTRA-UZ	10,000
U-234	3.3E-10	UTRA-UZ	10,000
U-235	2.3E-12	UTRA-UZ	10,000
U-236	2.6E-12	UTRA-UZ	10,000

Table B-9: Radiological 1-Meter Concentrations for UTRA-UZ to 10,000 Years

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ac-227	8.0E-08	1,714	1.4E-05	6,122	3.4E-03	6,098	1.2E-08	6,106
Al-26	1.7E-25	10,000	8.8E-11	10,000	2.2E-06	10,000	5.3E-18	10,000
Am-241	3.5E-25	10,000	3.1E-08	10,000	6.1E-04	10,000	9.8E-15	10,000
Am-242m	<1.0E-30	8,290	8.8E-28	7,020	2.9E-21	6,626	<1.0E-30	7,990
Am-243	4.8E-23	10,000	7.1E-08	10,000	7.0E-02	10,000	8.7E-13	10,000
C-14	2.6E-03	2,178	1.7E+02	4,142	3.5E+03	4,016	1.9E-03	4,214
Cf-249	<1.0E-30	10,000	5.3E-17	9,786	1.0E-12	8,496	9.1E-24	10,000
Cl-36	2.2E-05	10,000	3.1E-02	3,680	1.3E+00	3,676	4.1E-06	3,680
Cm-243	<1.0E-30	1,828	<1.0E-30	1,302	<1.0E-30	1,188	<1.0E-30	1,416
Cm-244	<1.0E-30	1,294	<1.0E-30	978	<1.0E-30	928	<1.0E-30	1,048
Cm-245	2.0E-25	10,000	2.6E-08	10,000	5.2E-04	10,000	7.6E-15	10,000
Cm-247	1.5E-29	10,000	4.7E-11	10,000	5.1E-07	10,000	3.0E-18	10,000
Cm-248	6.5E-28	10,000	2.1E-09	10,000	2.2E-05	10,000	1.3E-16	10,000
Co-60	<1.0E-30	358	<1.0E-30	370	<1.0E-30	126	<1.0E-30	326
Cs-135	2.1E-03	6,902	1.3E+00	5,388	3.0E+01	4,826	7.8E-05	6,590
Cs-137	1.3E-24	1,918	5.6E-15	918	2.0E-11	786	7.9E-21	1,040
Eu-152	<1.0E-30	1,002	<1.0E-30	782	<1.0E-30	728	<1.0E-30	832
Eu-154	<1.0E-30	420	<1.0E-30	288	<1.0E-30	236	<1.0E-30	324
H-3	7.1E-12	552	1.7E-11	204	1.9E-08	196	2.8E-12	158
I-129	8.2E-04	576	2.4E-02	3,790	4.6E-01	3,786	3.8E-05	536
K-40	5.0E-06	10,000	1.6E-01	4,104	2.9E+01	3,924	7.0E-05	4,112
Nb-93m	1.5E-06	10,000	1.6E+01	10,000	1.1E+03	9,994	2.9E-03	10,000
Nb-94	1.2E+00	556	2.0E+00	556	1.9E+00	556	4.6E-02	554

Table B-9: Radiological 1-Meter Concentrations for UTRA-UZ to 10,000 Years (Continued)

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ni-59	1.6E+00	1,792	2.7E+00	1,884	7.7E+01	7,188	5.8E-02	1,692
Ni-63	4.3E-03	1,378	7.6E-03	1,178	2.5E-01	1,056	1.6E-03	1,102
Np-237	1.5E-01	1,564	2.6E-01	1,574	1.7E+01	6,034	5.6E-03	1,556
Pa-231	1.1E-04	1,684	1.9E-02	6,062	4.6E+00	6,038	1.6E-05	6,052
Pb-210	4.6E-04	1,634	7.7E-04	1,650	3.8E-02	10,000	1.8E-05	1,592
Pd-107	3.0E-07	4,962	1.4E+01	4,764	2.9E+02	4,686	3.9E-04	4,810
Pu-238	3.6E-28	4,738	9.9E-24	6,452	5.1E-18	6,214	3.5E-24	4,178
Pu-239	3.3E-04	10,000	3.0E-03	10,000	5.2E+00	10,000	2.1E-03	10,000
Pu-240	5.6E-05	10,000	6.9E-03	10,000	4.0E+00	10,000	2.4E-04	10,000
Pu-241	1.0E-24	10,000	1.2E-07	10,000	2.3E-03	10,000	3.8E-14	10,000
Pu-242	1.3E-06	10,000	2.1E-04	10,000	8.8E-02	10,000	8.1E-07	10,000
Pu-244	6.2E-10	10,000	7.7E-07	10,000	1.5E-04	10,000	3.8E-09	10,000
Ra-226	1.8E-01	1,606	3.0E-01	1,624	1.5E+01	10,000	6.9E-03	1,560
Ra-228	3.4E-11	10,000	9.4E-09	10,000	1.3E-05	10,000	2.9E-11	10,000
Rn-222	1.8E-01	1,606	3.0E-01	1,624	1.5E+01	10,000	6.9E-03	1,560
Se-79	2.2E-19	10,000	4.5E-07	10,000	2.9E-01	10,000	5.5E-14	10,000
Sm-151	<1.0E-30	5,942	<1.0E-30	2,688	1.0E-24	2,422	<1.0E-30	3,356
Sn-126	1.7E-29	10,000	2.5E-11	10,000	5.0E-04	10,000	9.6E-20	10,000
Sr-90	6.4E-08	954	4.3E-07	958	4.0E-05	898	1.8E-07	912
Tc-99	9.3E+01	714	1.6E+02	728	1.1E+03	678	3.4E+00	708
Th-228	3.4E-11	10,000	9.4E-09	10,000	1.3E-05	10,000	2.9E-11	10,000
Th-229	1.4E-05	10,000	6.1E-04	10,000	2.9E-01	10,000	4.0E-06	10,000
Th-230	1.2E-06	10,000	2.1E-04	10,000	6.4E-01	10,000	1.6E-06	10,000

Table B-9: Radiological 1-Meter Concentrations for UTRA-UZ to 10,000 Years (Continued)

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Th-232	1.2E-13	10,000	4.2E-11	10,000	4.9E-08	10,000	1.4E-13	10,000
U-232	<1.0E-30	3,828	1.4E-24	1,764	1.3E-19	1,470	1.7E-28	3,416
U-233	6.7E-04	10,000	1.9E-02	10,000	4.7E+00	9,138	9.0E-05	9,912
U-234	5.6E-04	10,000	8.4E-02	10,000	9.5E+01	9,122	3.6E-04	10,000
U-235	5.5E-06	10,000	9.1E-04	10,000	9.8E-02	10,000	4.4E-06	9,864
U-236	1.1E-05	10,000	2.4E-03	10,000	1.7E+00	10,000	6.1E-06	10,000
U-238	2.6E-04	10,000	1.1E-03	10,000	2.5E-02	8,484	1.9E-04	9,762
Zr-93	2.6E-23	10,000	3.2E-06	10,000	2.7E-02	10,000	1.5E-12	10,000

Table B-10: Radiological 1-Meter Concentrations for UTRA-LZ to 10,000 Years

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ac-227	2.7E-08	1,722	6.6E-06	6,122	1.6E-03	6,104	2.1E-07	6,108
Al-26	3.5E-30	10,000	1.3E-14	10,000	1.1E-09	10,000	8.8E-16	10,000
Am-241	1.7E-29	10,000	1.5E-11	10,000	9.1E-07	10,000	1.3E-12	10,000
Am-242m	<1.0E-30	9,182	<1.0E-30	7,940	1.1E-25	7,422	<1.0E-30	7,814
Am-243	2.2E-27	10,000	3.0E-11	10,000	9.0E-05	10,000	1.2E-10	10,000
C-14	8.8E-04	2,284	6.6E+01	4,224	1.0E+03	4,100	3.5E-02	4,202
Cf-249	<1.0E-30	10,000	2.4E-20	10,000	8.9E-16	10,000	1.2E-21	10,000
Cl-36	7.1E-06	10,000	1.3E-02	3,682	6.1E-01	3,678	7.8E-05	3,680
Cm-243	<1.0E-30	1,996	<1.0E-30	1,532	<1.0E-30	1,378	<1.0E-30	1,494
Cm-244	<1.0E-30	1,416	<1.0E-30	1,120	<1.0E-30	1,024	<1.0E-30	1,098
Cm-245	9.0E-30	10,000	1.2E-11	10,000	7.4E-07	10,000	9.9E-13	10,000
Cm-247	<1.0E-30	10,000	2.1E-14	10,000	2.9E-10	10,000	3.9E-16	10,000
Cm-248	<1.0E-30	10,000	9.4E-13	10,000	1.3E-08	10,000	1.7E-14	10,000
Co-60	<1.0E-30	376	<1.0E-30	386	<1.0E-30	354	<1.0E-30	350
Cs-135	7.2E-04	7,608	4.7E-01	5,780	8.1E+00	5,216	8.1E-04	5,696
Cs-137	4.6E-27	2,026	3.2E-17	1,028	9.4E-14	946	4.8E-19	1,008
Eu-152	<1.0E-30	1,088	<1.0E-30	882	<1.0E-30	814	<1.0E-30	866
Eu-154	<1.0E-30	496	<1.0E-30	364	<1.0E-30	300	<1.0E-30	352
H-3	2.1E-12	554	1.0E-11	164	3.7E-09	200	1.6E-11	158
I-129	2.8E-04	578	9.9E-03	3,790	2.1E-01	3,786	2.3E-04	536
K-40	1.5E-06	10,000	6.6E-02	4,190	1.2E+01	4,006	1.3E-03	4,094
Nb-93m	7.6E-07	10,000	6.2E+00	10,000	4.8E+02	10,000	5.6E-02	10,000
Nb-94	4.1E-01	558	1.1E+00	560	1.1E+00	560	1.4E-01	558

Table B-10: Radiological 1-Meter Concentrations for UTRA-LZ to 10,000 Years (Continued)

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ni-59	5.3E-01	1,874	1.5E+00	1,970	2.8E+01	7,250	3.1E-01	1,172
Ni-63	8.7E-04	1,446	4.4E-03	1,212	5.3E-02	1,104	9.5E-03	1,108
Np-237	5.1E-02	1,570	1.5E-01	1,582	7.4E+00	6,038	2.1E-02	686
Pa-231	3.6E-05	1,688	8.8E-03	6,066	2.1E+00	6,038	2.8E-04	6,052
Pb-210	1.5E-04	1,674	4.2E-04	1,702	1.8E-02	10,000	8.6E-05	1,034
Pd-107	1.9E-07	5,024	5.2E+00	4,818	8.4E+01	4,740	7.3E-03	4,800
Pu-238	<1.0E-30	5,142	8.8E-27	6,900	6.3E-21	6,622	4.1E-26	4,612
Pu-239	8.6E-06	10,000	1.4E-04	10,000	3.5E-01	10,000	5.4E-03	10,000
Pu-240	1.5E-06	10,000	2.5E-04	10,000	2.1E-01	10,000	6.3E-04	10,000
Pu-241	5.1E-29	10,000	5.8E-11	10,000	3.5E-06	10,000	4.8E-12	10,000
Pu-242	3.4E-08	10,000	7.8E-06	10,000	4.6E-03	10,000	2.1E-06	10,000
Pu-244	1.6E-11	10,000	2.8E-08	10,000	8.1E-06	10,000	9.8E-09	10,000
Ra-226	5.9E-02	1,650	1.6E-01	1,672	6.8E+00	10,000	3.6E-02	1,004
Ra-228	2.3E-12	10,000	1.2E-09	10,000	1.9E-06	10,000	2.2E-10	10,000
Rn-222	5.9E-02	1,650	1.6E-01	1,672	6.8E+00	10,000	3.6E-02	1,004
Se-79	1.6E-23	10,000	7.0E-10	10,000	7.4E-04	10,000	5.5E-12	10,000
Sm-151	<1.0E-30	6,570	<1.0E-30	3,324	6.3E-30	2,892	<1.0E-30	3,242
Sn-126	<1.0E-30	10,000	3.5E-16	10,000	1.1E-08	10,000	2.5E-17	10,000
Sr-90	9.2E-09	986	1.6E-07	986	5.5E-06	926	9.4E-07	928
Tc-99	3.2E+01	724	9.2E+01	728	5.7E+02	686	2.0E+01	616
Th-228	2.3E-12	10,000	1.2E-09	10,000	1.9E-06	10,000	2.2E-10	10,000
Th-229	1.0E-06	10,000	7.1E-05	10,000	6.4E-02	10,000	2.4E-05	10,000
Th-230	7.8E-08	10,000	2.2E-05	10,000	1.3E-01	10,000	1.2E-05	10,000

Table B-10: Radiological 1-Meter Concentrations for UTRA-LZ to 10,000 Years (Continued)

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Th-232	8.0E-15	10,000	4.6E-12	10,000	8.3E-09	10,000	1.0E-12	10,000
U-232	<1.0E-30	4,132	1.5E-27	2,112	7.5E-23	1,844	9.8E-29	2,068
U-233	5.3E-05	10,000	3.2E-03	10,000	1.8E+00	10,000	6.2E-04	10,000
U-234	4.2E-05	10,000	1.1E-02	10,000	3.5E+01	10,000	3.5E-03	10,000
U-235	4.2E-07	10,000	1.4E-04	10,000	2.3E-02	10,000	2.7E-05	10,000
U-236	8.2E-07	10,000	3.8E-04	10,000	4.7E-01	10,000	5.3E-05	10,000
U-238	2.0E-05	10,000	3.1E-04	10,000	6.5E-03	9,836	1.2E-03	9,950
Zr-93	8.3E-27	10,000	3.5E-09	10,000	8.0E-05	10,000	1.6E-10	10,000

Table B-11: Radiological 1-Meter Concentrations for Gordon Aquifer to 10,000 Years

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ac-227	8.1E-13	10,000	2.2E-10	10,000	2.7E-08	10,000	5.8E-11	10,000
Al-26	<1.0E-30	10,000	6.7E-29	10,000	2.9E-23	10,000	9.3E-28	10,000
Am-241	<1.0E-30	10,000	6.0E-27	10,000	1.7E-21	10,000	9.8E-26	10,000
Am-242m	<1.0E-30	10,000	<1.0E-30	9,260	<1.0E-30	8,584	<1.0E-30	9,044
Am-243	<1.0E-30	10,000	1.1E-26	10,000	1.5E-19	10,000	8.0E-24	10,000
C-14	2.9E-10	9,998	2.8E-06	9,998	2.5E-05	9,990	2.3E-08	9,990
Cf-249	<1.0E-30	10,000	<1.0E-30	10,000	1.7E-30	10,000	<1.0E-30	10,000
Cl-36	1.2E-09	10,000	8.5E-07	3,784	3.2E-05	3,778	6.4E-08	3,778
Cm-243	<1.0E-30	2,284	<1.0E-30	1,818	<1.0E-30	1,662	<1.0E-30	1,776
Cm-244	<1.0E-30	1,598	<1.0E-30	1,300	<1.0E-30	1,200	<1.0E-30	1,274
Cm-245	<1.0E-30	10,000	4.7E-27	10,000	1.4E-21	10,000	7.7E-26	10,000
Cm-247	<1.0E-30	10,000	8.5E-30	10,000	5.6E-25	10,000	3.0E-29	10,000
Cm-248	<1.0E-30	10,000	3.7E-28	10,000	2.5E-23	10,000	1.3E-27	10,000
Co-60	<1.0E-30	418	<1.0E-30	420	<1.0E-30	190	<1.0E-30	392
Cs-135	4.9E-10	10,000	2.1E-07	10,000	2.5E-06	10,000	6.0E-09	10,000
Cs-137	<1.0E-30	2,288	2.4E-29	1,242	5.8E-26	1,142	2.9E-29	1,200
Eu-152	<1.0E-30	1,204	<1.0E-30	1,004	<1.0E-30	936	<1.0E-30	986
Eu-154	<1.0E-30	586	<1.0E-30	460	<1.0E-30	392	<1.0E-30	446
H-3	8.1E-18	576	3.7E-16	184	7.7E-15	222	1.1E-16	178
I-129	5.5E-09	746	6.1E-07	4,070	1.1E-05	4,054	2.2E-08	4,052
K-40	2.6E-12	10,000	1.4E-07	10,000	1.3E-05	9,990	2.8E-08	10,000
Nb-93m	1.2E-11	9,998	8.5E-05	10,000	4.9E-03	10,000	1.0E-05	9,996
Nb-94	8.9E-06	612	2.2E-04	610	2.1E-04	608	2.5E-05	604

Table B-11: Radiological 1-Meter Concentrations for Gordon Aquifer to 10,000 Years (Continued)

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Ni-59	3.6E-06	6,830	9.2E-05	7,200	4.8E-04	10,000	1.1E-05	5,790
Ni-63	3.2E-11	1,744	8.2E-10	1,558	1.3E-09	1,414	3.5E-10	1,374
Np-237	3.9E-07	7,554	1.0E-05	8,268	1.3E-04	10,000	1.3E-06	10,000
Pa-231	1.0E-09	10,000	2.7E-07	10,000	3.4E-05	10,000	7.5E-08	10,000
Pb-210	8.8E-10	10,000	2.4E-08	10,000	2.8E-07	10,000	3.3E-09	10,000
Pd-107	1.5E-12	9,966	2.3E-05	9,346	2.6E-04	9,246	4.3E-07	9,132
Pu-238	<1.0E-30	5,942	<1.0E-30	7,612	<1.0E-30	7,250	<1.0E-30	7,490
Pu-239	4.8E-18	10,000	3.6E-16	10,000	2.7E-13	10,000	1.3E-14	10,000
Pu-240	8.2E-19	10,000	1.7E-16	10,000	1.4E-13	10,000	1.5E-15	10,000
Pu-241	<1.0E-30	10,000	2.1E-26	10,000	6.6E-21	10,000	4.0E-25	10,000
Pu-242	1.9E-20	10,000	4.9E-18	10,000	3.1E-15	10,000	6.0E-18	10,000
Pu-244	9.2E-24	10,000	1.3E-20	10,000	7.1E-18	10,000	2.8E-20	10,000
Ra-226	3.3E-07	10,000	9.1E-06	10,000	1.1E-04	10,000	1.3E-06	9,998
Ra-228	1.6E-21	10,000	1.5E-18	10,000	3.4E-15	10,000	5.9E-18	10,000
Rn-222	3.3E-07	10,000	9.1E-06	10,000	1.1E-04	10,000	1.3E-06	9,998
Se-79	<1.0E-30	10,000	1.0E-22	10,000	1.3E-16	10,000	1.3E-22	10,000
Sm-151	<1.0E-30	7,430	<1.0E-30	4,198	<1.0E-30	3,736	<1.0E-30	4,062
Sn-126	<1.0E-30	10,000	<1.0E-30	10,000	3.2E-25	10,000	<1.0E-30	10,000
Sr-90	5.8E-17	1,088	5.0E-15	1,098	1.9E-14	1,056	7.4E-15	1,034
Tc-99	5.0E-04	1,108	1.2E-02	1,056	1.6E-02	1,036	1.5E-03	1,026
Th-228	1.6E-21	10,000	1.5E-18	10,000	3.4E-15	10,000	5.9E-18	10,000
Th-229	7.1E-11	10,000	1.8E-09	10,000	5.0E-09	10,000	2.3E-10	10,000
Th-230	7.7E-17	10,000	3.3E-14	10,000	3.1E-10	10,000	4.6E-13	10,000

Table B-11: Radiological 1-Meter Concentrations for Gordon Aquifer to 10,000 Years (Continued)

Radionuclide	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs	Concentration (pCi/L)	Year Peak Contribution Occurs
Th-232	7.6E-24	10,000	7.1E-21	10,000	1.9E-17	10,000	2.8E-20	10,000
U-232	<1.0E-30	4,756	<1.0E-30	2,678	<1.0E-30	2,396	<1.0E-30	2,572
U-233	1.4E-09	10,000	3.3E-08	10,000	1.7E-07	10,000	4.2E-09	10,000
U-234	8.1E-14	10,000	3.4E-11	10,000	2.6E-07	10,000	3.7E-10	10,000
U-235	7.9E-16	10,000	4.9E-13	10,000	1.5E-10	10,000	6.3E-13	10,000
U-236	1.5E-15	10,000	1.3E-12	10,000	2.9E-09	10,000	4.2E-12	10,000
U-238	3.8E-14	10,000	2.7E-12	10,000	4.2E-11	10,000	2.3E-11	10,000
Zr-93	<1.0E-30	10,000	7.0E-23	10,000	6.5E-18	10,000	5.7E-22	10,000

Table B-12: Chemical 1-Meter Concentrations for UTRA-UZ to 10,000 Years

Chemical	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs
Ag	1.74E-03	6,918	9.69E-02	5,878	5.05E+00	4,712	6.47E-05	3,734
As	4.25E-05	10,000	3.23E-04	8,566	1.51E-02	7,008	1.62E-06	9,474
Ba	1.50E-02	1,586	9.77E-02	4,904	1.57E+00	6,308	6.05E-04	998
Cd	6.48E-02	1,266	1.08E-01	1,302	1.75E+01	7,146	2.40E-03	1,228
Cr	2.43E-02	1,264	6.07E-02	4,678	2.82E+00	4,608	9.00E-04	1,228
Cu	2.16E-03	5,296	6.24E-02	5,498	2.86E+00	4,534	8.55E-05	4,938
F	2.47E-01	556	3.33E+00	3,838	3.60E+01	3,836	9.28E-03	554
Fe	1.85E-03	10,000	6.85E-01	10,000	2.12E+02	10,000	2.86E-03	9,946
Hg	2.79E-18	10,000	9.11E-06	10,000	6.94E-02	10,000	1.23E-11	10,000
Mn	3.42E-02	4,764	1.97E+00	5,052	2.47E+02	4,808	3.71E-03	1,776
$NO_2 + NO_3$	9.63E+00	556	2.88E+01	3,654	5.29E+02	3,650	3.62E-01	554
Ni	8.07E-01	1,800	1.37E+00	1,896	4.01E+00	1,126	3.73E-02	1,166
Pb	1.54E-26	10,000	6.16E-09	10,000	5.45E-03	10,000	7.48E-17	10,000
Sb	2.54E-33	10,000	2.35E-10	10,000	2.66E-05	10,000	1.86E-19	10,000
Se	2.22E-22	10,000	7.47E-09	10,000	2.14E-04	10,000	1.99E-14	10,000
U	8.56E-04	10,000	3.66E-03	10,000	1.43E-01	8,462	1.13E-03	9,784
Zn	1.57E-03	10,000	5.73E-02	7,960	2.93E+00	5,992	5.99E-05	9,488

Table B-13: Chemical 1-Meter Concentrations for UTRA-LZ to 10,000 Years

Chemical	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs
Ag	5.81E-04	7,530	4.53E-02	6,332	1.37E+00	5,244	3.91E-04	3,786
As	1.24E-05	10,000	1.31E-04	9,620	6.30E-03	7,882	8.06E-06	5,572
Ba	5.00E-03	1,636	4.65E-02	4,928	5.47E-01	6,346	3.66E-03	1,002
Cd	2.18E-02	1,310	6.01E-02	1,360	7.66E+00	7,164	1.34E-02	922
Cr	8.17E-03	1,310	2.77E-02	4,710	1.05E+00	4,644	5.00E-03	922
Cu	7.00E-04	5,792	2.91E-02	5,882	7.42E-01	4,980	3.19E-04	3,262
F	8.23E-02	558	1.64E+00	3,838	1.47E+01	3,836	3.03E-02	534
Fe	1.46E-04	10,000	5.79E-02	10,000	3.29E+01	10,000	1.79E-02	10,000
Hg	6.14E-22	10,000	2.73E-08	10,000	4.45E-04	10,000	1.00E-09	10,000
Mn	1.16E-02	4,880	8.91E-01	5,168	9.56E+01	4,928	2.25E-02	1,790
$NO_2 + NO_3$	3.20E+00	558	1.15E+01	3,656	1.63E+02	3,652	1.18E+00	534
Ni	2.73E-01	1,876	7.76E-01	1,976	1.21E+00	1,166	2.26E-01	1,172
Pb	5.39E-31	10,000	2.95E-13	10,000	5.80E-07	10,000	1.60E-14	10,000
Sb	2.20E-38	10,000	2.18E-15	10,000	2.38E-10	10,000	5.59E-17	10,000
Se	1.65E-26	10,000	1.09E-11	10,000	8.63E-07	10,000	2.00E-12	10,000
U	6.43E-05	10,000	1.00E-03	10,000	3.75E-02	9,808	6.80E-03	9,978
Zn	4.59E-04	10,000	2.68E-02	8,722	7.65E-01	6,888	3.00E-04	5,576

Table B-14: Chemical 1-Meter Concentrations for Gordon Aquifer to 10,000 Years

Chemical	Sector 1A		Sector 1B		Sector 1C		Sector 1D	
	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs	Concentration (μg/L)	Year Peak Contribution Occurs
Ag	6.89E-10	10,000	8.96E-08	10,000	1.24E-06	10,000	4.57E-09	10,000
As	7.53E-13	10,000	3.28E-11	10,000	1.77E-09	10,000	1.22E-11	10,000
Ba	4.05E-08	4,592	2.66E-06	9,632	2.48E-05	9,662	1.26E-07	3,908
Cd	1.93E-07	3,146	5.12E-06	10,000	4.22E-04	10,000	9.70E-07	10,000
Cr	7.25E-08	3,154	2.42E-06	8,516	3.77E-05	9,336	2.27E-07	2,752
Cu	3.36E-09	10,000	2.45E-07	10,000	2.54E-06	10,000	1.52E-08	10,000
F	1.78E-06	612	2.79E-04	3,984	2.19E-03	3,978	5.40E-06	602
Fe	1.52E-13	10,000	3.42E-11	10,000	2.40E-08	10,000	1.91E-10	10,000
Hg	1.47E-35	10,000	1.06E-20	10,000	3.73E-16	10,000	5.87E-20	10,000
Mn	2.68E-08	10,000	1.29E-06	10,000	2.95E-05	10,000	1.63E-07	10,000
$NO_2 + NO_3$	6.92E-05	612	1.70E-03	610	2.55E-03	3,708	2.10E-04	602
Ni	1.95E-06	7,212	4.83E-05	6,500	4.77E-05	6,336	6.03E-06	6,042
Pb	3.19E-47	10,000	3.11E-29	10,000	8.59E-23	10,000	4.12E-28	10,000
Sb	5.75E-55	10,000	1.47E-31	10,000	1.48E-25	10,000	9.34E-31	10,000
Se	5.76E-41	10,000	1.43E-24	10,000	4.74E-19	10,000	4.54E-23	10,000
U	1.23E-13	10,000	8.68E-12	10,000	2.39E-10	10,000	1.33E-10	10,000
Zn	2.78E-11	10,000	9.81E-09	10,000	2.93E-07	10,000	6.89E-10	10,000

APPENDIX C

BETA-GAMMA PEAK CONCENTRATIONS RATIOED TO MCLS FOR INDIVIDUAL CONSTITUENTS RELATED BY TIMING AT THE 100-METER BOUNDARY

Appendix B Tables B-1 and B-2 provide a summary of beta-gamma MCL fractions summed by peak concentration in 10,000 years, regardless of timing. This approach provides conservative results as not all constituents peak at the same time within 10,000 years. Five radionuclides make up approximately 99 % of the beta-gamma inputs for the individual aquifers and sectors (C-14, I-129, Nb-93m, Ni-59, and Tc-99) within 10,000 years.

For this appendix, the peak concentrations for these five radionuclides were analyzed individually by time of occurrence against the concentration at the same time for the remaining radionuclides. For example, in UTRA-UZ Sector D, C-14 peaked in 10,000 years at 4,172 years. The remaining four radionuclides did not peak in that year, so the beta-gamma value at 4,172 years for each of these remaining radionuclides was added to the beta-gamma value for C-14 to produce an estimated beta-gamma value for the year 4,172, resulting in a beta-gamma MCL fraction of 41 % for the UTRA-UZ Sector D (Table C-1).

Tables C-1 and C-2 present the beta-gamma peak concentrations in 10,000 years by aquifer ratio to MCLs for the five radionuclides that comprise the majority of the total beta-gamma contribution utilizing concurrent time for Sectors D and E, respectively.

Table C-1: Beta-Gamma Peak Concentrations in 10,000 Years Ratio to MCLs for Selected Individual Constituents Related by Timing for Sector D

Radio-nuclide	MCL (pCi/L)	UTR-LZ			UTR-UZ		
		Conc. (pCi/L)	Beta-gamma fraction	peak (yr)	conc (pCi/L)	Beta-gamma fraction	peak (yr)
C-14	2,000	6.7E+02	3.3E-01	4,264	7.8E+02	3.9E-01	4,172
I-129	1	1.9E-02	1.9E-02	4,264	1.8E-02	1.8E-02	4,172
Nb-93m	1,000	1.9E-06	1.9E-09	4,264	1.2E-06	1.2E-09	4,172
Ni-59	300	6.7E-02	2.2E-04	4,264	1.4E-04	4.5E-07	4,172
Tc-99	900	5.1E-01	5.7E-04	4,264	5.3E-01	5.9E-04	4,172
		Total	**3.5E-01**		**Total**	**4.1E-01**	
C-14	2,000	6.4E-02	3.2E-05	3,790	8.9E-01	4.5E-04	3,788
I-129	1	1.1E-01	1.1E-01	3,790	1.3E-01	1.3E-01	3,788
Nb-93m	1,000	4.9E-08	4.9E-11	3,790	6.3E-08	6.3E-11	3,788
Ni-59	300	7.4E-02	2.5E-04	3,790	1.2E-06	3.9E-09	3,788
Tc-99	900	6.0E-01	6.7E-04	3,790	4.9E-01	5.5E-04	3,788
		Total	**1.1E-01**		**Total**	**1.3E-01**	
C-14	2,000	7.2E-01	3.6E-04	10,000	7.4E-04	3.7E-07	9,998
I-129	1	1.5E-07	1.5E-07	10,000	4.9E-08	4.9E-08	9,998
Nb-93m	1,000	9.4E+01	9.4E-02	10,000	1.2E+02	1.2E-01	9,998
Ni-59	300	1.1E+00	3.8E-03	10,000	1.6E+00	5.4E-03	9,998
Tc-99	900	1.2E+00	1.3E-03	10,000	1.3E+00	1.4E-03	9,998
		Total	**9.9E-02**		**Total**	**1.3E-01**	
C-14	2,000	1.0E+00	5.2E-04	7,358	1.4E-01	6.8E-05	7,306
I-129	1	2.3E-05	2.3E-05	7,358	9.0E-07	9.0E-07	7,306
Nb-93m	1,000	4.1E+01	4.1E-02	7,358	5.1E+01	5.1E-02	7,306
Ni-59	300	7.8E+00	2.6E-02	7,358	1.2E+01	4.0E-02	7,306
Tc-99	900	9.0E-01	1.0E-03	7,358	9.8E-01	1.1E-03	7,306
		Total	**6.8E-02**		**Total**	**9.2E-02**	
C-14	2,000	6.2E-07	3.1E-10	704	2.9E-06	1.4E-09	694
I-129	1	7.8E-05	7.8E-05	704	5.4E-08	5.4E-08	694
Nb-93m	1,000	1.6E-16	1.6E-19	704	1.7E-16	1.7E-19	694
Ni-59	300	7.3E-16	2.4E-18	704	3.0E-15	1.0E-17	694
Tc-99	900	2.2E+02	2.4E-01	704	2.0E+02	2.2E-01	694
		Total	**2.4E-01**		**Total**	**2.2E-01**	

(Peak Radionuclide Indicated in Green Shading)

Table C-2: Beta-Gamma Peak Concentrations in 10,000 Years for Individual Selected Constituents Related by Timing for Sector E

Radio-nuclide	MCL (pCi/L)	UTR-LZ			UTR-UZ		
		Conc. (pCi/L)	Beta-gamma fraction	peak (yr)	conc (pCi/L)	Beta-gamma fraction	peak (yr)
C-14	2,000	2.6E+02	1.3E-01	4,310	1.3E+02	6.5E-02	4,222
I-129	1	2.3E-02	2.3E-02	4,310	9.1E-03	9.1E-03	4,222
Nb-93m	1,000	5.4E-06	5.4E-09	4,310	2.4E-06	2.4E-09	4,222
Ni-59	300	5.8E-02	1.9E-04	4,310	5.1E-04	1.7E-06	4,222
Tc-99	900	5.4E-01	6.0E-04	4,310	4.7E-01	5.2E-04	4,222
		Total	**1.5E-01**		**Total**	**7.4E-02**	
C-14	2,000	1.4E-02	6.9E-06	3,788	4.2E-02	2.1E-05	3,792
I-129	1	1.1E-01	1.1E-01	3,788	8.3E-02	8.3E-02	3,792
Nb-93m	1,000	9.7E-08	9.7E-11	3,788	8.4E-08	8.4E-11	3,792
Ni-59	300	6.4E-02	2.1E-04	3,788	1.9E-06	6.3E-09	3,792
Tc-99	900	7.7E-01	8.5E-04	3,788	4.3E-01	4.8E-04	3,792
		Total	**1.1E-01**		**Total**	**8.3E-02**	
C-14	2,000	4.4E-01	2.2E-04	9,998	3.5E-04	1.8E-07	9,998
I-129	1	5.8E-08	5.8E-08	9,998	2.5E-09	2.5E-09	9,998
Nb-93m	1,000	1.9E+02	1.9E-01	9,998	1.6E+02	1.6E-01	9,998
Ni-59	300	2.5E+00	8.2E-03	9,998	2.2E+00	7.4E-03	9,998
Tc-99	900	1.2E+00	1.3E-03	9,998	1.1E+00	1.2E-03	9,998
		Total	**2.0E-01**		**Total**	**1.7E-01**	
C-14	2,000	6.3E-01	3.2E-04	7,354	2.8E-02	1.4E-05	7,288
I-129	1	2.9E-05	2.9E-05	7,354	6.3E-07	6.3E-07	7,288
Nb-93m	1,000	8.1E+01	8.1E-02	7,354	6.6E+01	6.6E-02	7,288
Ni-59	300	1.6E+01	5.4E-02	7,354	1.7E+01	5.7E-02	7,288
Tc-99	900	9.1E-01	1.0E-03	7,354	8.5E-01	9.5E-04	7,288
		Total	**1.4E-01**		**Total**	**1.2E-01**	
C-14	2,000	1.1E-07	5.4E-11	702	1.8E-07	9.0E-11	694
I-129	1	7.0E-05	7.0E-05	702	3.5E-08	3.5E-08	694
Nb-93m	1,000	3.0E-16	3.0E-19	702	2.2E-16	2.2E-19	694
Ni-59	300	2.1E-13	6.9E-16	702	1.4E-12	4.8E-15	694
Tc-99	900	2.7E+02	3.0E-01	702	1.6E+02	1.8E-01	694
		Total	**3.0E-01**		**Total**	**1.8E-01**	

(Peak Radionuclide Indicated in Green Shading)